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Complexity, Evolvability, and the Process of Adaptation

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complexity, evolvability, constraint, Fisher's geometric model, integration, modularity

Abstract

There is a widespread view that the process of adaptation in complex systems is made difficult due to an evolutionary cost of complexity that is reflected in lower evolvability. This line of reasoning suggests that organisms must have special properties to overcome this cost, such as integration, modularity, and robustness, and that the reduction in the rate of evolution and variational constraints could help explain why organisms might not respond to selection. Here, we discuss the issues that arise from this conviction and highlight an alternative view where complexity represents an opportunity by increasing the evolutionary potential of a population. We highlight the lack of evidence supporting the influence of complexity on evolvability. Empirical data on the patterns of contemporary selection are critical for understanding this relationship.

1. INTRODUCTION

1.1. What Is the Problem?

complexity: either the number of parts in an organism or how many hierarchical levels they encompass

evolvability: the disposition to evolve; evolvability is generated by the properties of populations or individuals in those populations

disposition: the capacity to demonstrate a behavior or state under certain conditions, e.g., glass has a disposition to shatter when struck

This paper is about the effects of organismal complexity on evolvability. We focus in particular on the very widespread conviction that there is an evolvability cost of complexity (Orr 2000) and the contrary position that complexity actually enhances evolvability (Gavrilets 2004). Our goal is to highlight the assumptions that lie behind these lines of thinking and to suggest ways in which these contrary positions can be reconciled with data.

We note at the outset that the narrow issue that we focus on is relevant for wider issues in macroevolution, such as the evolution of complexity itself (Bonner 1988, Lynch 2007b, McShea 1996), although we do not explicitly consider these here. Another such issue is how to balance our attention between the evolution of diverse organisms, exemplified by Darwin's (1859, pp. 459–60) metaphor of the tangled bank teeming with "endless forms most beautiful," or on the converse notion that "(o)rganisms in general have not done nearly as much evolving as we should reasonably expect" (Williams 1992, p. 128). This dichotomy is particularly relevant given the hypothesis that stasis can sometimes be explained by an evolutionary cost of complexity (Hansen & Houle 2004). On the other hand, we note the intriguing counter example of snake venoms. Holding et al. (2021) collected proteomes and transcriptomes from venom glands of a 20-million-year-old clade of venomous snakes. From an inferred ancestral state of 22 gene families, the number of families in contemporary taxa varies from 8 to 32, and the total number of sequences in those gene families varies tenfold. Complex traits clearly can evolve quite rapidly.

1.2. Many Complexities, Many Evolvabilities

Biologists agree that organisms are complex entities and that they vary widely in complexity. McShea (2017) distinguishes two distinct operational definitions for complexity. Horizontal complexity is the number of different entities, such as nucleotides or cell types, that are contained in an organism. Vertical complexity is the number of hierarchical levels at which an organism evolves. McShea notes that these different definitions of complexity mean that the complexity of each individual or species cannot be expressed by a single measurement. Plants generally have more complex genomes than multicellular animals but fewer cell types. Social insects have added vertical complexity over asocial organisms, even though their horizontal complexity is lower than most asocial vertebrates (Valentine et al. 1994). To measure complexity, one must first specify the definition and the level of complexity to be compared.

Evolvability is the disposition to evolve, should the evolutionary forces that could cause evolution actually occur (Hansen et al. 2023, Hansen & Pélabon 2021, Wagner & Altenberg 1996). The concept of evolvability enables us to recognize that the rate of evolution is determined by two factors—the strength and nature of the evolutionary forces that impact a population and the ability of the evolving population to produce variation for those forces to act on. In the case of adaptive evolution, the rate of evolution is determined on the one hand by the strength and nature of natural selection and on the other by the variation of the selected population. This separation between the force of selection and variation is enshrined in the fundamental equations of both population and quantitative genetics (Lande 1979, Price 1972). Under the dispositional definition, the evolutionary forces that are the cause of evolution are left unspecified. To define an appropriate evolvability, we must also specify an evolutionary force, a phenotype, and a timescale over which we would like to measure it (Houle & Pélabon 2023). Thus, like complexity, there is no single measure of evolvability that characterizes an organism or a population.

To consider the relationship between complexity and evolvability, we need to define the sort of complexity and evolvability that we are talking about. Indeed, the two canonical views of complexity mentioned above employ different aspects of complexity and evolvability.

The complexity-as-cost school of thought (e.g., Orr 2000) holds that evolvability is limited by the number of orthogonal phenotypic dimensions required to describe the phenotype (Walsh & Blows 2009). It is important to realize that a dimension is not equivalent to a measurable trait because traits are correlated with each other and because selection favors particular combinations of traits. For example, in animals, the sizes of most morphological traits are positively correlated with each other, and these allometric patterns can be very strong (Pélabon et al. 2014). Consequently, body size is generally considered to be a selectively important phenotypic dimension, even though it is measured indirectly through its association with measured traits. Similarly, shape is defined as variation in form that remains once allometry is removed (Mitteroecker & Bookstein 2007, Porto et al. 2013). Recognition of size as a critical phenotype can be justified with respect to either the striking pattern of covariation in traits with size or the likelihood that this pattern of covariation is actively selected for. Indeed, an extremely useful definition of a phenotypic dimension is that which is selected for, which is often a combination of measured traits. These arguments implicitly define complexity as the number of dimensions that are simultaneously optimized by natural selection and evolvability as the probability that novel mutations change the phenotype in a direction that increases fitness.

On the other hand, the complexity-as-opportunity school sees complexity enhancing evolvability (Gavrilets 1997, Gavrilets 2004, Wagner 2012). This argument rests on the view that organismal complexity increases evolvability by increasing the number of possible variants accessible by mutation. A greater diversity of possible mutations increases the number of neutral mutations, as well as the variety of mutations that may have higher fitness. In this view, complexity is the size of the genome or the number of traits potentially subject to selection, and evolvability includes variants subject to genetic drift, as well as selection.

orthogonal: orthogonal vectors are at right angles to each other

dimension: the minimum number of coordinates on orthogonal basis vectors necessary to specify the position of a point in a space; for example, if the values of three traits fall on a plane, the variation is in two dimensions, even when all three traits vary

adaptive landscape: the relationship between the position of an organism or a population in G or P space and their average fitness

2. SPACES AND MAPS

To discuss the evolutionary costs and opportunities generated by organismal complexity, we make use of the heuristic concepts of phenotypic (P) space, consisting of possible phenotypic states, and genotypic (G) space, the possible genomic sequences. We are all intimately familiar with the three-dimensional (3D) space in which we live. Any point in 3D space may be specified using three coordinates, x, y, and z; we can define a distance between any points and the direction that gives the shortest path between those points. Similarly, three measurable phenotypic traits specify a point in a 3D space that we can readily imagine as analogous to our 3D world. However, organisms have far more than three measurable traits. Fortunately, the idea of a space readily extends to any number of dimensions. P space, like physical space, is a continuous space, as most traits can take on a continuous range of values. G space is a discrete space because two genotypes either share the same nucleotide or indel or they do not.

Whether complexity increases or decreases evolvability depends on the relationships between genotypes, phenotypes, and fitness. The relationship between genotype and phenotype can be conceptualized as a GP map (Houle et al. 2010, Lewontin 1974). The relationship between phenotype and fitness is captured in the phenotypic adaptive landscape (Arnold et al. 2001; Fisher 1930; Simpson 1953, chapter V). The phenotypic adaptive landscape gives expected fitness as a function of an individual's position in P space, like those shown in **Figure 1**. Genotypic adaptive landscapes give the relationship between genotypes and average fitnesses directly,

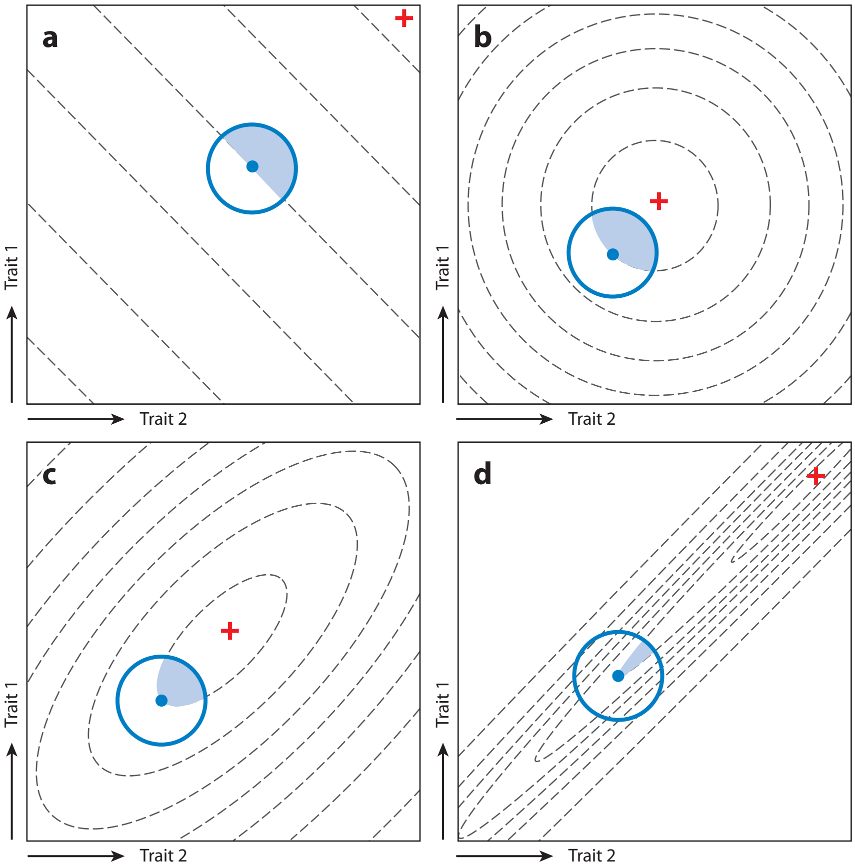

Figure 1

Proportion of beneficial mutations on four fitness landscapes. Axes represent values of two traits. In each panel, fitness contours are dashed, and the point with the highest fitness is marked with a red plus sign. The solid blue dot represents an individual genotype and the average phenotype it produces. The solid blue circle is the range of mutant phenotypes accessible from that genotype, assuming universal pleiotropy. Blue fill denotes mutant genotypes that increase fitness. (*a*) Increases in traits 1 and 2 are favored, regardless of mean phenotype. (*b*) Symmetrical fitness landscape assumed in the Fisher geometric model. (*c*) Correlated fitness landscape. (*d*) Extreme version of a correlated fitness landscape.

without reference to phenotypes. Wright's (1932) original adaptive landscapes were of this form, and this notion of a genotypic adaptive landscape is widely used in models of evolution on high-dimensional landscapes (e.g., Gavrilets 2004, Wagner 2012). The GP map and adaptive landscape are tremendously useful concepts, but they neglect the distribution of phenotypes produced by a genome and the variation in fitness, given a phenotype. Actual models or graphical representations of these maps and landscapes are at best approximations, if only because they simplify a high-dimensional reality about which our knowledge is limited. Maps and landscapes may mislead, as well as inform, when taken literally (Svensson & Calsbeek 2012).

variability: the disposition of a genotype to produce genetic and/or phenotypic variation

A population evolves when it moves in G space due to any cause, and this can occur only when there is genetic variation present in the population to enable such a move. On a longer timescale, variability, the capacity for mutation to create variation, takes the place of variation. Both variation and variability are relevant, as our reasoning applies over a range of timescales.

Variance–covariance matrices quantify the distribution of genetic variation (**G** matrix), phenotypic variation (**P** matrix), or variability (**M** matrix) in P space.

3. COST OF COMPLEXITY

3.1. Reductions in the Rate of Evolution

The thesis that there is a cost of complexity has been developed in both a population genetic (Orr 2000) and a quantitative genetic (e.g., Wagner 1988) context. In both cases, the argument depends on the form of natural selection in P space, and we first develop it intuitively using the four fitness landscapes shown in **Figure 1**. Each landscape shows fitness as a function of two measured traits whose values define the axes of each plot. As a useful null model, we assume that mutational effects show universal pleiotropy, where the distribution is symmetrical around the starting phenotype and mutations in every possible direction in P space are equally likely.

In **Figure 1*a***, the fitness landscape is a tilted plane such that the effect of each trait on fitness is independent of the other trait. Half of the mutations are favored on such a landscape. Note that this conclusion does not change when we change the direction in which the plane is tilted. In the other panels, the fitness landscape is curved, and as that curvature increases from **Figure 1*b–d***, the proportion of mutations that are advantageous is reduced. Note, however, if we allowed mutations only along the direction in which fitness changes most rapidly, the proportion of advantageous mutations would be 50%, as it is **Figure 1*a***. The vector that points uphill from the current mean is a weighted average of the measured traits and can itself be treated as a trait; we refer to this as the focal trait. Effectively, only the focal trait is under directional selection in **Figure 1*d***, while two traits are under selection in the other panels, and this explains the reduction in the probability that a mutation is advantageous. We can intuitively expand this argument by imagining a 3D bubble of mutations. If the three traits are under linear directional selection, as in **Figure 1*a***, the proportion of advantageous mutation is still one half. If the fitness landscape curves in two dimensions, that curvature reduces the proportion of advantageous mutations just as shown in **Figure 1*b,c***. If, in addition, the fitness landscape curves in all three dimensions, the additional curved dimension subtracts another slice from the proportion of mutations that are advantageous in the two-dimensional (2D) case.

Planar landscapes are unrealistic in general, as most traits must interact with others to determine fitness, thus creating curvature in the fitness landscape. Exceptions are suites of traits that interact directly with independent aspects of the external environment, rather than with other organismal traits. For example, the adaptive immune system is an evolved system of variation- and somatic selection–generating mechanisms that results in a horizontally complex repertoire of tens of millions of antibodies within the lifetime of a single organism (Rees 2020). Each antibody is evaluated for effectiveness largely independently of other antibodies, resulting in a fitness landscape that approximates a plane as the organism responds to a novel challenge.

The starting point for Orr's (2000) population genetic analysis of the cost of complexity is Fisher's geometric model (Fisher 1930, pp. 38–41), whose assumptions of a symmetrical fitness landscape with a single fitness peak and universal pleiotropy are shown in **Figure 1*b***. The simple geometry of the model allows it to be generalized to any number of selected traits. More generally, asymmetrical selection tends to form fitness ridges like those shown in **Figure 1*c***, creating correlational selection.

Far away from the optimum, the local fitness landscape around the current genotype has less curvature and can approximate the planar fitness landscape shown in **Figure 1*a***. The closer to the optimum the genotype is, the smaller the proportion of mutations that are advantageous, until at the optimum any change in the mean phenotype would be disfavored. As explained above,

variance–covariance matrices: square matrices that contain the variance of each trait along the diagonal and the covariances between traits in the remainder of the matrix

G matrix: a variance–covariance matrix that summarizes the distribution of additive genetic variation, i.e., the variation that allows a response to natural selection

P matrix: a variance–covariance matrix that summarizes the distribution of phenotypic variation within a population; this variation includes **G**

M matrix: a variance–covariance matrix that summarizes the distribution of spontaneous mutational variation and covariation

universal pleiotropy: the assumption that mutation has an equal chance of affecting all traits

correlational selection: the effect of a trait on fitness is dependent on the values of one or more other traits this probability also falls as the number of dimensions under selection, n, increases. Fisher (1930) treated the assumption that n was large as axiomatic and showed that, in the limit of large n, the rate of decrease in the probability that each mutation is favored approaches $1/\sqrt{n}$. This dimensionality effect was noted in passing; instead, Fisher emphasized the fact that mutations of large effect size have a smaller probability of being advantageous than those of small effect size. This is apparent in **Figure 1*b–d*** by the fact that the blue fill delineating advantageous mutations occupies a smaller proportion of the circumference of the circle holding mutations of smaller sizes as curvature increases. This argument was an influential justification of his infinitesimal model of adaptation (Orr 1998). Kimura (1983), however, showed that, in a finite population, the geometric model predicts that variants of intermediate effect size are the most likely to contribute to adaptation, as those with small effect size are likely to be lost due to drift.

Orr (2000) used the geometric model, as modified by Kimura, to show that the rate of increase in fitness per mutation of fixed total effect size declines faster than $1/n$ due to three effects. As n increases, the probability that each mutation is favorable, the distance that each favorable mutation moves the population toward the optimum, and the probability of fixation of each favored mutation all decline. Waxman & Welch (2005) (Welch & Waxman 2003) generalized these results for various distributions of effect sizes and for unequal strengths of selection, like the cases shown in **Figure 1*c*,*d***.

Welch & Waxman (2003) pointed out that the decline in the gain in fitness per mutation was due to Orr's assumption that the total effect size is constant, summed over all n traits. If we instead assume that there are preexisting phenotypic dimensions already subject to mutation and merely increase the number of those subject to selection, it is clearly more appropriate to hold effects constant with an increase in n. When Welch and Waxman did so, it abolished the effect of n on the rate of adaptation when the population is far from the optimum but intensified the cost of increasing n as the population approaches the optimal state. Empirical estimates of pleiotropic effects of quantitative trait loci suggest that the greater the number of traits affected by a variant, the larger the total effect on the phenotype (Porto et al. 2016, Wagner et al. 2008, Wang et al. 2010).

Most implementations of the geometric model assume that the total mutation rate is fixed, even as the number of selected traits increases, or they derive the change in fitness per mutation event (Orr 2000, Poon & Otto 2000, Welch & Waxman 2003). This makes these versions of the geometric model inappropriate to apply to the evolutionary costs of a change in organismal complexity. An actual increase in organismal complexity would often increase the phenotypic mutation rate by increasing the number of base pairs in the genome that can affect the phenotype and perhaps increase the number of interactions among existing processes, increasing the total amount of variation produced by a typical mutation (Hansen 2003). For example, there is a great deal of evidence that duplication of genes is a source of phenotypic innovation and complexity and is correlated with subsequent adaptation (Conant & Wolfe 2008, Van de Peer et al. 2009). It is therefore unclear from the geometric model how a genuine increase in the complexity of the organism would affect evolvability. Orr (2000, p. 17) raised these issues in his discussion, but his title phrase, "the cost of complexity," has remained associated with the geometric model.

Quantitative genetic models have been used to study the evolutionary impact of correlated selection in a wide variety of fitness landscapes. Wagner (1988) confirmed that on the spherical landscape assumed in the geometric model, the number of traits under selection slows the rate of adaptation due to phenotypic variance that is not in the direction of selection. This effect is intensified on correlated fitness landscapes. The extreme case is the corridor model that assumes a rising ramp of fitnesses in the direction of a focal trait, such that all mutations not in that direction are extremely deleterious (Bürger 1986, Wagner 1988). In a corridor model, the selection response can be completely halted by variation orthogonal to the focal trait.

In summary, both population genetic and quantitative genetic models show that, conditional on a level of genetic variation, the rate of evolution is often slower when more trait dimensions are subject to optimizing selection. No models that we know of demonstrate that the rate of evolution is lower in a more complex organism, just by virtue of its complexity.

eigendecomposition: rotation of the coordinate system to maximize the disparity in variation along a set of orthogonal eigenvectors (often referred to as a principal components analysis); the variance along each eigenvector is its eigenvalue

3.2. Variational Constraints

The models cited in Section 3.1 demonstrate a cost to optimizing more traits even when we assume that genetic variation is equally available in all dimensions of P space. Empirical estimates demonstrate that the amount of genetic variation varies widely among traits and suggest the possibility that some trait combinations may lack genetic variation altogether. In particular, the more complex the phenotype, the more likely it is that the available genetic variation may not allow evolution in all possible dimensions of P space (Hansen & Houle 2004, Walsh & Blows 2009), creating a variational constraint.

The form of universal pleiotropy assumed in the geometric model is widely understood to be a heuristic fiction rather than a biological reality. There is no doubt that the amount of variation and variability varies widely among traits (Houle et al. 1996). It is less clear how the directions of effects may depart from the uniform distribution assumed in the geometric model. The directions of most readily detectable effects of segregating variants and gene deletions are clearly biased toward large effects on small numbers of traits, suggesting that effects tend to be localized along trait-specific directions in P space (Wagner et al. 2008, Wang et al. 2010). To some, this suggests that pleiotropy, the number of traits affected by a variant, is, on average, low (Wagner & Zhang 2011). This result may, however, be an artifact of the low power of experiments to detect small phenotypic effects (Paaby & Rockman 2013), which is key to Boyle and coauthors' (2017) argument that pleiotropy is ubiquitous or omnigenic. Their analysis suggests that the largest genome-wide association study (GWAS), for human height, implicates 100,000 single-nucleotide polymorphisms (SNPs), affecting at least 30% of the genes in the genome. Second, other traits likely have similar patterns of effects, because the proportion of genes implicated as having effects in GWASs rises steeply with the proportion of trait heritability explained (Shi et al. 2016), which is a proxy for the power of the studies. Finally, they note that the SNPs implicated in studies of human disease genetics are not predominantly in genes whose expression is particularly important in the cells directly causing disease. Instead, they tend to map to widely expressed genes that happen to be expressed in those disease-associated cells. They argue that these results together suggest that variants must have small effects on a wide variety of traits.

Variational constraints could be transient due to finite population size. However, if variability is not available in some directions in P space, this generates a long-lived constraint. For example, growth trajectories are generally smooth functions of time (Arnold 1992), such that very rapid increases or decreases in size are not possible. Some combinations of phenotypes are involved in trade-offs, where, in a well-adapted population, the improvement of one trait is not possible without deterioration in another trait. The result is a biased spectrum of mutations, such that only deleterious variants that reduce net performance can arise (Houle 1991).

Such constraints can in principle be identified using direct measurements of genetic or mutational variation. In theory such constraints can be detected using eigendecomposition of the **G** (or **M**) matrix. With a linear GP map an absolute constraint occurs when the **G** matrix is singular, that is, has some eigenvalues that are exactly zero. The subspace defined by those eigenvectors is called a null space. In the hypothetical example of 2D variation in 3D space, the **G** matrix has three eigenvectors, two with positive eigenvalues, while the third eigenvalue is 0. Evolution is absolutely constrained in the null space (Houle 2001, Mezey & Houle 2005).

These effects can be quantified using the concepts of evolvability and conditional evolvability (Hansen et al. 2003, Hansen & Houle 2008), which measures genetic variation in the direction of a focal trait. With selection on a planar fitness landscape (**Figure 1*a***), additive genetic variation in the selected dimension alone determines the response to selection and therefore the evolvability. Conditional evolvability is the additive genetic variation in the focal trait that is uncorrelated with other phenotypic dimensions and thus allows a response to directional selection when other traits are held constant by selection, as in **Figure 1*d*** and corridor models. On correlated fitness landscapes, the realized evolvability is somewhere between the evolvability and the conditional evolvability. Conditional evolvability is always less than evolvability, except in the special case where the selection gradient lies along one of eigenvectors of the **G** matrix. Thus, conditional evolvability does not capture the costs of optimizing multiple traits that arise even in the absence of genetic correlations. Which evolvability predicts the actual response to selection better depends on the nature of the fitness landscape, illustrating the importance of specifying the conditions under which evolution may occur.

Walsh & Blows (2009) offer an empirically based argument that absolute constraints become inevitable as the number of traits under selection increases. The linchpin of their argument is Johnson & Barton's (2005) conclusion that empirical estimates of stabilizing selection on typical traits are quite strong. Given this, the total variation in fitness limits the number of independent traits that can experience such strong selection to on the order of ten traits. They conclude that selection on phenotypic variation is concentrated on a small number of phenotypic dimensions and that other dimensions are incapable of independent responses to selection. The key estimate of stabilizing selection is, however, based on a very curious interpretation of the compilations of estimates of quadratic selection, γ, compiled by Kingsolver and colleagues (Kingsolver et al. 2012, 2001). The estimates of γ are widely distributed around a modal value of 0, but Johnson and Barton's estimate of the strength of stabilizing selection is the median of the negative estimates. Negative values of γ are consistent with stabilizing selection, while positive values suggest disruptive selection. The lack of evidence for consistent stabilizing selection is surprising, given the expectation that stabilizing selection around fitness peaks should be the predominant mode of selection. While the variance of the estimates is inflated by the substantial estimation error (Morrissey & Hadfield 2012), they are unlikely to be biased, suggesting that the strength of stabilizing selection is far weaker than Johnson and Barton's estimates, allowing far more traits to be subject to optimizing selection. As a result, the quantitative aspects of Walsh & Blows' (2009) argument are very much in doubt. Direct estimates of the genetic variance in fitness show that simultaneous adaptation of multiple traits is possible in the average population (Bonnet et al. 2022), although some populations have very low fitness variation consistent with constraints.

Another avenue for demonstrating variational constraints is the identification of a null space. Unfortunately, the hypothesis of no variation is not testable, because sample sizes always limit the size of an eigenvalue that can be detected (Mezey & Houle 2005, Sztepanacz & Blows 2017). For example, a study of adult gene expression of over 12,000 transcripts in both sexes of adult *Drosophila melanogaster* yielded a **G** with just eight positive eigenvalues (Houle & Cheng 2021). While this might suggest that the null space for gene expression is very large indeed, the study included just 40 genotypes; many more dimensions would likely have statistical support in a larger study. In the case of gene expression, the known presence of distinct promoter and *cis*-regulatory elements regulating transcription makes it clear that most genes can vary independently. While it is likely that only a subset of genes actually vary in any given population, our knowledge of the molecular basis of gene expression gives us a more reliable picture of the dimensionality of variability in the transcriptome.

Biologically, the line between null and nonnull spaces is not always clear, even in theory. The amount of available variation might be so small that, relative to stochastic forces such as genetic drift or environmental variation, there is an effectively null space into which natural selection cannot drive a population (Gomulkiewicz & Houle 2009). More generally, mutationally accessible phenotypes can fall on a curved surface within the larger metric space, called a manifold. The earth's spherical surface is a 2D manifold along which humans are constrained to creep. When phenotypes fall on such a manifold, there is a null space off the manifold, but it is not detectable using linear spectral decomposition.

Practically speaking, the estimation of genetic variation in multiple traits with sufficient precision requires many times more families than *n*, an expensive and time-consuming undertaking, limiting their use to model organisms. As a result, the number of estimates of **G** matrices with more than a few traits is rather limited and is heavily skewed toward suites of traits that can be measured simultaneously using an automated approach. Studies that incorporate multiple types of traits are rare (Houle et al. 2010). Some studies of cuticular hydrocarbons (CHCs) and wing shape in various *Drosophila* species can detect variation in all or most dimensions of P space (Hine et al. 2014, Houle & Meyer 2015, Mezey & Houle 2005, Pitchers et al. 2019), although other studies of similar *Drosophila* populations sometimes detect variation in a minority of dimensions (e.g., McGuigan & Blows 2007). A study of the **M** matrix for wing shape in *D. melanogaster* revealed that at least 18 of 20 dimensions were variable (Houle & Fierst 2013). Interestingly, using the same wing data set that showed all 20 possible dimensions were variable in each sex (Houle & Meyer 2015, Mezey & Houle 2005), Sztepanacz & Houle (2019) showed that only 6 of the 20 dimensions showed sex-specific variation, suggesting the possibility of constraints on differentiation of the sexes.

Hine et al. (2014) complemented their estimates of standing variation in CHCs with selection experiments on each of the eight eigenvectors of CHC variation in *Drosophila serrata*. They detected some response in each of the eight vectors, validating the qualitative finding of variation in all trait combinations; however, one combination responded only to indirect selection on other traits. These low-variance traits may fall in a nearly null space where the effects of genetic drift are large enough to effectively prevent a selection response. The eigenvalues of the lowest ranking eigenvectors of a full-rank **G** matrix can be as much as 1,000 times lower than those of the first eigenvector (Houle & Meyer 2015).

4. EVOLVABILITY DESPITE COMPLEXITY

There is a widespread conviction that the evolvability of complex systems is a remarkable fact that calls out for an explanation (e.g., Bonner 1988, Hansen 2003, Kirschner & Gerhart 1998, Wagner & Altenberg 1996). The human solution is to decompose such complex problems into simpler pieces that can be solved in isolation from the full complexity of the system. From this perspective, the evolution of complexity suggests that evolvability itself is an evolved property of complex systems (reviewed in Hansen & Wagner 2023, Wagner et al. 2007). Others argue that the engineering analogy is misleading (Hansen 2003) and that evolvability arises as a byproduct of other evolutionary mechanisms (Hansen et al. 2011; Lynch 2005, 2007a–c). We explore these issues in regard to modularity.

4.1. The Modularity Hypotheses

One set of proposed solutions to the problem of complexity are the related properties of integration and modularity. These properties can be explained with reference to human design principles. A human designer breaks down the functions of a complex machine like an automobile into constituent processes, and then designs an element to carry out each process, resulting in, for example, the contents of the auto parts store: carburetors, spark plugs, pumps, etc. Each of these parts is an integrated unit in that each functions harmoniously and somewhat independently of other such parts—the modules. It is straightforward to independently redesign each part with relatively small modifications to, for example, produce greater output, survive greater strains, or minimize some costs. Following this logic, a functional module of an organism is composed of traits that interact together in performing some function and are relatively independent of other functional modules (Cheverud 1996, Wagner et al. 2007). For brevity, we refer to this combination of integrated parts that are relatively independent of other such parts as modularity.

variational modularity: clustering of mutational effects in P space

evolutionary modularity: clustering of evolutionary trajectories in P space

genotype–phenotype (GP) map: the correspondence between the position of an organism in G space to their position in P space, shaped by epigenesis

There is widespread consensus that organisms are modular and that this modularity is in large part responsible for the existence of evolvability (e.g., Melo et al. 2016, Wagner & Altenberg 1996, Wagner et al. 2007). For example, Raff (1996, p. 325) states that "modularity is an ineluctable feature of the biological order. It is arguably the most crucial aspect of order in living organisms and their ontogenies, and is the attribute that most strongly facilitates evolvability." For many, this also leads to the hypothesis that modularity and integration are adaptations to promote evolvability (Wagner 1996, Wagner & Altenberg 1996).

On the other hand, there is also a widespread confusion about different kinds of modularity and which sorts of modularity might matter for evolution (Zelditch & Goswami 2021). To understand this confusion, consider how modularity could alleviate the cost of optimization compared to Fisher's geometric model. The geometric model assumes universal pleiotropy, the antithesis of a modular architecture. The cost of complexity arises when the probability that a mutation has effects that are only in the direction of a focal selected trait is small. That cost would be alleviated if, instead of mutational effects having random directions, effects were concentrated in just those directions in P space that are likely to correspond to selection vectors. Variational modularity by itself is not enough to alleviate the cost of optimizing multiple traits. When selection vectors are random in direction, variational modularity has no average effect on evolvability; if the directions of selection are not along the modules, evolvability is decreased relative to the case of universal pleiotropy (Hansen 2003, Welch & Waxman 2003). Only alignment of variational modules with the selected dimensions alleviates the cost of optimizing multiple traits. If the pattern of variation and the pattern of selection over the long term are not opposed, the variation among species also comes to have a modular pattern, termed evolutionary modularity.

The concept of modularity is, however, applied to a much wider variety of situations than those that are directly conceived in terms of evolvability and evolution (Klingenberg 2008, Melo et al. 2016, Wagner et al. 2007, Zelditch & Goswami 2021), including developmental and functional modularity. Developmental modules are sets of traits whose development shares common processes and interactions, while functional modules are traits that interact to fulfill some function. It is often assumed or implied that functional or developmental modularity is relevant to variational and evolutionary modularity, without reference to a hypothesis of why this might be so.

Riedl and others (Bonner 1988; Olson & Miller 1958; Riedl 1977, 1978; Wagner 1996) argue that the modular organization of organisms is built by natural selection to enhance adaptation or evolvability, and we group these arguments under the banner of Riedl's (1978, p. 196) "imitative epigenotype." This hypothesis assumes that particular sets of traits are repeatedly challenged to adapt in a coordinated manner to changing environmental circumstances. As a consequence, the individuation and covariation of a structure within an individual, or during development, creates variational and evolutionary modularity. A component of this hypothesis is that the evolution of compensatory mechanisms that allow individuals to produce adaptive phenotypes in response to current conditions may also help to structure patterns of genetic variation as well (e.g., Draghi & Whitlock 2012). The genotype–phenotype (GP) map then evolves to shape the pattern of pleiotropic effects and plastic responses to correspond to the pattern of selection. Epistatic effects, in which the effect sizes of alleles at one locus depend on the genotype at other loci, enable such changes to the GP map (Hansen 2006).

canalization: reduction of the magnitude of environmental or genetic variation on phenotypic variation

epistasis: interactions in the effects of genetic variation at two or more sites in the genome

directional epistasis: the average tendency for alleles that change the trait mean to canalize or decanalize mutational effects

4.2. Alternative Explanations for Modularity

There are powerful arguments against a primary role for imitative epigenesis in shaping the GP map. Hansen (2011) shows that any epistatic variant that has effects on variability through its effect on the GP map also has direct effects on the traits produced by that map. Consequently, natural selection on trait values is a major force in determining how the GP map is shaped by epistatic variation. Hansen further argues that natural selection on the GP map is generally much weaker than the direct effects on the selected traits. Numerical investigations show that the results of selection on epistatic systems can indeed reshape the GP map as predicted by the imitative epigenotype hypothesis (Jones et al. 2007, 2014; Melo & Marroig 2015; Pavlicev et al. 2011). Careful theoretical studies suggest that these are special cases, and selection does not necessarily drive a system toward optimal evolvability. For example, one route to modularity is to canalize genetic effects that affect more than one module, while decanalizing effects within a module. Wagner et al. (1997) showed that there is a limit to the degree of canalization that can be achieved, as alterations that decrease variation weaken selection for further canalization. In a more general model, Hermisson et al. (2003) showed that whether canalization occurs at all depends on the properties of the epistatic interactions. If the same alleles that canalize the effects of one locus decanalize the effects of others, the net result may be no change. Hansen et al. (2006) showed that, when present, directional epistasis dominates changes in variability under directional selection and can lead to architectures that either maximize or minimize variability, regardless of the fitness consequences of variation.

There are a variety of nonadaptive hypotheses that can explain modularity and integration. Lynch (2005, 2007a–c) argues that, at the genomic level, they are likely to be the product of neutral processes of genome growth and differentiation. An analogue of this idea is that the branching structure of cell growth and differentiation within an organism naturally leads to modular patterns of variation (Altenberg 2005, West-Eberhard 2003). More closely related cells have more similar epigenomes and tend to directly interact with each other and therefore respond more similarly to stimuli than those in other parts of the somatic lineage.

Despite the appeal of intuitive arguments for the role of modularity in promoting evolvability, closer examination suggests contradictions. Traits are subject to selection throughout ontogeny, raising the possibility that different selection pressures on the same trait are typically experienced at different developmental stages (Hallgrímsson et al. 2009). The effects of variation caused early in development are retained into adulthood, resulting in a pattern of variation that cannot be solely optimized for its effects on evolvability at one developmental stage. Instead, Hallgrímsson et al. (2009) suggest the source of evolvability in complex developmental systems is in the existence of the specific developmental events that allow change in each direction in form to evolve, regardless of their quantitative impact on variation.

Biological trade-offs between different traits and biological functions are a frequent empirical finding. Trade-offs arguably occupy as large a conceptual role in evolutionary biology as modularity does, and yet they are the very antithesis of modularity. Trade-offs can have surprising impacts on a wide variety of traits. Swiderski & Zelditch (2022) highlight mechanical trade-offs between, for example, force and speed. Emlen (2001) showed that the evolution of exaggerated horns on male beetles due to sexual selection reduces the sizes of adjacent organs, which might be eyes, antennae, or wings, depending on the placement of the horns.

In many cases, the integrated phenotypic response to simple organismal changes can result in a high-fitness phenotype without further evolutionary adjustments, as in allometry. However, in cases where the integrated response is not optimal, it makes modification of that response more difficult, leading to the use of allometry as a prima facie case for nonadaptive alterations in form (e.g., Gould & Lewontin 1979). In many cases, adaptation ultimately requires both shifting the module as a whole to a new state and finer-scale adjustment of traits within the module contrary to that pattern. Whether a particular pattern of modularity achieves the right balance between these processes depends again on the form of selection.

Hansen (2003) noted that even in those cases where modularity can be advantageous, evolvability is increased by increasing the amount of variation and variability. Following up on this idea, Pavlicev & Hansen (2011) investigated patterns of variability that maximize the ability for selection to optimize two traits, when selection acts alternately to change one or the other. When the fitness landscape is not curved, universal pleiotropy is optimal. However, when one trait is subject to stabilizing selection, while the other is directionally selected, the optimal architecture maximizes the total genetic variation and minimizes genetic covariation between the two traits. Whether minimal covariance is achieved by modular or universal pleiotropy makes no difference. When maximizing genetic variance and minimizing covariance are in conflict, the optimal architecture depends on how genetic variance is maintained and can favor either modular or universal pleiotropic architectures.

4.3. Patterns of Modularity

The premier examples of trait integration are the pervasive allometric relationships between varied suites of morphological and physiological traits (Pélabon et al. 2014). Notably, allometry structures both plastic and evolutionary responses to changes in organismal size. Allometry is capable of evolving in response to selection (Bolstad et al. 2015, Voje et al. 2014) but remains remarkably consistent over long time periods in most clades, suggesting that the kind of persistent natural selection envisioned by the imitative epigenotype model is what maintains allometric relationships (Houle et al. 2019). In some cases, modular structure in more subtle aspects of form can be detected only by explicitly controlling for allometric integration (Mitteroecker & Bookstein 2007, Porto et al. 2013).

Looking beyond allometry, there are many other cases where patterns of variation and variability are consistent with the imitative epigenotype hypothesis. These need to be interpreted cautiously, given that there are both adaptive and nonadaptive explanations for modularity. In a canonical study of correlation patterns in plants, Berg (1960) showed that plants visited by specialized pollinators had lower correlations between floral and vegetative traits, but these traits remained correlated in plants pollinated by the wind or unspecialized insects. This pattern has also been reported with respect to the involucral bracts of *Dalechampia scandens*, which has a specialized pollination system based on resin-collecting bees (Hansen et al. 2007).

Patterns of variation within the vertebrate skeleton seem consistent with the imitative epigenotype hypothesis. Despite the extensive diversity of cranial morphology observed across mammals, large-scale comparative studies suggest remarkable stability in **P**, implying that **G** is also stable (Machado et al. 2018, Porto et al. 2009, Rossoni et al. 2019), but at the same time, these studies also indicate that the level of integration is highly variable (Porto et al. 2009, Rossoni et al. 2019). Similarly, highly derived tetrapods share conserved patterns of limb covariation (Hallgrímsson et al. 2002, Rolian 2009, Rolian et al. 2010, Young et al. 2010), but in some instances, species with disparate hind- and forelimb function (such as bats, gibbons, and humans) have lower interlimb covariances, compared with groups whose limbs perform more similar functions [e.g., cursorial quadrupeds (Young & Hallgrímsson 2005)]. Note that this decoupling of variational properties between structures that have evolved is also consistent with the direct effects of divergence (Hansen 2011).

Modularity is usually demonstrated by rejecting a null hypothesis using a wide variety of ad hoc descriptors of the pattern of variation (reviewed in Zelditch & Goswami 2021). Emphasis on rejection of such null models obscures the fact that modularity is rarely very strong. To give but two examples: In the saddle-backed tamarin, skull measurements within functionally and developmentally associated parts of the skull had modest genetic correlations of 0.39, while traits in unrelated segments had average correlations of 0.14 (Cheverud 1995). In *D. melanogaster* the average genetic correlations in males and females of adult traits within the same segment were 0.29 and 0.45, respectively, while the corresponding between-segment values were 0.25 and 0.34 (Cowley & Atchley 1990). This sharply limits the impact that modularity can have on evolution. We join Zelditch & Goswami (2021) in calling for more attention to the quantitative effects of patterns of variation. The important issue is the role of modularity in potentially enabling adaptation in some traits, while holding other aspects of the phenotype near their current values. The comparison of unconditional and conditional evolvability addresses just this sort of question (Hansen et al. 2003, Hansen & Houle 2008).

4.4. Does the Pattern of Selection Match the Modularity Hypothesis?

Recall that the key mechanism by which modularity is hypothesized to increase evolvability is that the pattern of modular variation corresponds to the pattern of selection. Consequently, empirical data on the relationship between the pattern of contemporary selection and modular variation are crucial to distinguishing the imitative epigenotype from alternative hypotheses. We need to characterize both the selection on integrated traits within putative modules and the selection across modules.

Surprisingly, we know of few relevant studies of contemporary selection in organisms in which modularity has been characterized. Simon et al. (2019) measured the influence of linear and correlational selection on a performance trait, bite force in lizards, as a function of the jaw dimensions. They found interesting differences in selection among species but did not relate any of these results to the variation of the constituent traits. Indirect inferences about landscapes are more common. Other studies examine the pattern of evolution in relation to modularity but lack data on the actual pattern of selection (Jamniczky et al. 2014). Given this paucity of direct evidence, there is a great deal more certainty about the importance of modularity for evolvability than is justified (Altenberg 2005). Repeated evolutionary events, such as invasions of the marine stickleback (*Gasterosteus aculeatus*) into a variety of freshwater habitats (Marques et al. 2018) and the evolution of selfing syndromes in plants (Liao et al. 2022) provide favorable, replicable opportunities to measure the relationship between contemporary selection, evolvability, and variational structure.

Anecdotally, it is easy to conceive of cases where the pattern of selection is unlikely to target single modules. Large human GWASs of body mass index suggest that variation in adiposity is supplied by genes active in the central nervous system and the immune system, as well as adipose tissues (Akiyama et al. 2017), making it easy to imagine conflicting selection pressures that might constrain the ability to optimize adiposity. Many organs simultaneously carry out multiple functions, such as the gills used for both respiration and feeding in some fishes (Roth & Wake 1989).

At the level of selected polymorphisms, population genetic data show that the human Ectodysplasin A receptor (EDAR) gene has been the target of directional selection in East Asian and Native American populations in the past 30,000 years (Kamberov et al. 2013). GWASs suggested that an amino-acid substitution in EDAR is likely to be responsible for changes in scalp-hair thickness and tooth shape. Kamberov et al. (2013) verified these phenotypic effects in a mouse model but also found that the substitution affected the number of sweat glands and the pattern of branching in the mammary gland. They then confirmed that the human variant had the predicted changes in sweat gland abundance. This one variant is thus responsible for differences in four quite different traits that are unlikely to be grouped functionally. This variant has played a role in adaptation despite having the sorts of pleiotropic effects that modularity is supposed to reduce.

4.5. Does Contemporary Variation Affect Evolution?

The biological interest of current patterns of variation, including whether they are modular or not, depends on whether they influence long-term evolution. The alternative hypothesis is that variation has no effect, because any level of variability that allows evolution in a selected direction is sufficient to enable evolution, as suggested by Hallgrímsson and colleagues' (Hallgrímsson et al. 2009) view of the GP map. Somewhat surprisingly, the data suggest that contemporary variation is correlated with the pattern of evolution.

Schluter (1996) showed that short-term responses tended to fall near the leading eigenvector of **G** but diverge farther from that vector with time, a result confirmed in other systems (e.g., Marroig & Cheverud 2010). More recently, studies of higher dimensional data suggest that current variation is correlated with long-term evolution (Bolstad et al. 2014, Voje et al. 2023). For example, Houle et al. (2017) showed a very tight association between divergence rates of wing shape and both **G** and mutational variation among Drosophilid flies that shared a common ancestor 40 million years ago.

This correspondence of variation and divergence is expected under the imitative epigenotype hypothesis. There are, broadly speaking, two alternative hypotheses. First, neutral evolution would correspond to the pattern of variation, although we would generally expect neutral evolution to be far more rapid than what is observed (Lynch 1990). This hypothesis could be rescued by assuming that the same small proportion of variation in each trait is actually neutral. Second, Schluter's (1996) explanation was that the pattern of variation remains relatively constant and the direction of selection varies randomly, resulting in faster evolution in directions with more variation. While this is plausible, the pattern of selection under which variational constraints can shape long-term evolution is quite limited and not readily explained by any of our simple models of long-term evolution (Bolstad et al. 2014, Houle et al. 2017, Voje et al. 2023).

5. COMPLEXITY AS OPPORTUNITY

While the idea of complexity as cost structures a wide swath of evolutionary thinking, contrary ideas that envision advantages to complexity are not rare.

5.1. Alternative Adaptive Landscapes

Scenic regions of the earth are filled with hills and valleys, making the metaphor of a peak and the effort and time that it requires to climb one familiar. While Fisher (1930) imagined evolution on smooth adaptive landscapes with widely separated optima, and Wright (1932) famously imagined rugged adaptive landscapes made up of many peaks of varying heights, in both types of landscapes, populations will rapidly climb an accessible peak and tend to stay there. This locates populations in the parts of the landscape where the costs of optimizing multiple traits are strongest (Welch & Waxman 2003). We consider two possible alternative views of adaptive landscapes that abolish or mitigate the cost of optimization.

5.1.1. The shape of high-dimensional landscapes. Mathematical and empirical work on high-dimensional spaces casts doubt on the peak-climbing metaphor for evolution. All existing models of evolution in high-dimensional spaces are models of evolution in G space, rather than P space, and their predictions should be evaluated with this in mind. The dimensionality, D, of G space is the number of moves that can be made away from a starting genotype, which is the length of the genome, L, multiplied by the number of alternative possible variants per site, which is 3 for nucleotides and increases further when indels are considered. Thus, G spaces are so vast that even the simplest high-dimensional landscapes are well beyond the reach of exhaustive experimental study.

neutral network: a set of genotypes with equivalent fitnesses that are connected to each other by single mutations

The key issues are how many fitness peaks there are for populations to become stranded on and how deep the fitness valleys in between peaks are. Peakiness is a function of the amount of epistasis for fitness. When there is no epistasis, landscapes are additive and the fitnesses of adjacent genotypes are highly correlated, resulting in a smooth fitness landscape. At the other extreme, when epistasis is very high, each substitution can change the fitness effects of all other variants in an arbitrary way. In a low-dimensional space, the higher the epistasis is, the more peaks the landscape has. This is the situation that Wright (1932) invented the shifting balance scenario to overcome.

In a high-dimensional landscape, however, the probability that there are isolated peaks is dramatically reduced. In an exchange of letters in 1931, Fisher wrote Wright suggesting that apparent peaks in low-dimensional space may be connected by ridges of equivalent fitness genotypes in a higher dimensional space; Wright conceded Fisher's point (Provine 1986, pp. 274–75). Conrad (1990, p. 68) called such a path around low-dimensional peaks an "extradimensional bypass."

The most accessible introductions to this general result are by Gavrilets (1997; 2004, chapter 4), who considered a series of models of evolution on random landscapes. He was concerned with the ability of genotypes to percolate widely in the G space by moving from one high-fitness genotype to another adjacent high-fitness genotype, forming links in a neutral network. In each of his simple models, he showed that the size of the neutral network increased with the dimensionality of the adaptive landscape. The most relevant version of these models randomly assigns fitnesses drawn from a uniform distribution over the interval 0, 1 to each genotype. On average, this generates a nearly neutral network of fitnesses within $1/2D$ of the starting fitness. This allows percolation across a diameter of order L substitutions or, on average, one substitution at each site in the genome. Thus, even under the random fitness assumption, increasing dimensionality enables populations to explore the breadth of G space, albeit on the slow timescale of drift. A similar argument shows that the number of adjacent genotypes that are superior in fitness to the current genotype also increases with dimensionality, although the requirement of moving strictly uphill greatly lowers the lengths of such adaptive chains.

These maximally rugged theoretical landscapes are not realistic, as there is ample evidence that many adjacent genotypes have very similar fitnesses. Several schemes for creating model landscapes that vary the correlation of fitnesses of neighboring genotypes, and thus the implied degree of epistasis, have been used to model the effects of dimensionality on landscapes (reviewed in de Visser & Krug 2014). Smoother landscapes with higher fitness correlations of adjacent genotypes broaden the width of nearly neutral networks, and enable longer adaptive walks (Agarwala & Fisher 2019).

Wagner (2012) has found that the results of these highly abstract models of random landscapes are consistent with properties of RNA molecules, proteins, and gene regulatory and metabolic networks. He used simple criteria for deciding when each of these entities possesses similar properties and showed that large neutral networks are very frequent. Wagner highlighted the fact that these neutral networks allow an initial genotype to access many novel and potentially higher fitness genotypes that are adjacent to parts of the neutral network.

5.1.2. Adaptive seascapes. The second class of models places populations on a moving adaptive landscape, rather than a static one. The original descriptions of both genotypic (Wright 1932, figure 4) and phenotypic (Simpson 1953, figure 17) adaptive landscapes included this possibility, and the idea of a population on a seascape of shifting fitness has been widely discussed since (e.g., Agarwala & Fisher 2019, Calsbeek et al. 2012, Thompson 2005, Whitlock 1997). These arguments are supported by evidence that many populations are currently experiencing strong directional selection (Hereford et al. 2004) and that natural selection frequently reverses direction (Calsbeek et al. 2012, but see Morrissey & Hadfield 2012), suggesting that fitness peaks are moving targets. Organisms with substantial geographic ranges encounter very different ecological conditions and simultaneously experience gene flow across that range, suggesting that displacement from a local selective optimum is routine and long lasting (Thompson 2005). All this evidence suggests that populations spend substantial time far from optima and thus experience parts of P space where the costs of simultaneously optimizing multiple traits are low (Welch & Waxman 2003). Seascape dynamics could help to explain the rapid, but apparently constrained, evolution of body size over millions of years (Arnold 2014, Estes & Arnold 2007, Uyeda et al. 2011).

Exploration of P space could be similar to the exploration of G space discussed in Section 5.1. The argument is that organisms with higher phenotypic complexity have more degrees of freedom to explore P space (Lauder 1981, Vermeij 1973a) and can therefore explore a wider range of phenotypes than simpler organisms. This may make it possible for evolving lineages to locate and potentially exploit a wider range of peaks in a landscape, including particularly favorable phenotypes. For example, Alfaro et al. (2004) showed that the complex four-bar lever system in the jaws of labrid fishes enabled the evolution of varied biomechanical solutions to adaptive problems.

5.2. Innovation

Evolutionary innovation spans a range of possibilities from elaboration of existing structures, such as changes in the numbers of serially homologous vertebrae (Galis 2001), through modifications to those structures, such as the differentiation of cervical, thoracic, and lumbar vertebrae in mammals (Jones et al. 2018a), to the origin of truly novel structures, such as the thoracic helmets of Membracid treehoppers (Prud'homme et al. 2011). From the perspective of evolvability, there are two important possible effects of innovation. First, innovation can change the dimensionality of the phenotype space, thus potentially altering adaptive landscapes. Second, innovation can change the variability of the phenotype through changes in the dimensionality of G space and the GP map, potentially increasing evolvability.

Turning first to changes in dimensionality, novelties by definition add traits to the phenotype space, while more modest innovations, such as changes in the number of serially homologous parts, may or may not do so, using our definition of traits as that which is under selection. For example, if the only selectable aspect of adding a lumbar vertebra is to change tail length, it just affects the variation in an existing trait, and the P space remains the same. However, if the form of each vertebra is a target of selection, P space is enlarged. Jones et al. (2018b) demonstrated that vertebrae within the lumbar region of mammals are differentiated, consistent with the possibility of adaptation of the form of each vertebra.

Under the reasoning of the geometric model, trait additions might lower evolvability as one more trait that needs to be adapted, particularly if the correlations of new with existing traits are unfavorable. Many have argued instead that increased dimensionality of phenotypes increases the rate of evolution by enabling novel solutions to adaptive problems (Alfaro et al. 2004; Galis 2001;

Vermeij 1973a,b, 2015). As support, Vermeij notes that the shell form of primitive mollusks was limited to linear forms, but their descendants evolved shells that coiled in a plane, and then shells that coiled in three dimensions. Many authors have proposed the decoupling of previously correlated traits and functions as a key factor that triggers innovation (reviewed in Zelditch & Goswami 2021). Routes to decoupling include divergence in the functions of duplicated genes (Conant & Wolfe 2008), decreased integration of previously correlated morphological traits (Lauder 1981, Roth & Wake 1989), and separation of life-history phases by metamorphosis (Moran 1994).

The added phenotypic dimensions may transform the adaptive landscape to create new accessible peaks (Welch & Waxman 2003). It might also change the locations or shapes of existing peaks to make them more (or less) accessible. Such changes would certainly change the nature of selection, but changes to evolvability are likely as well. If new traits originate to take advantage of ecological opportunities, as in the traditional model of adaptive radiations triggered by key innovations, the added evolvability might even enable the evolution of the new trait. The evidence that key innovations directly trigger adaptive radiations is weak, however (Erwin 2017).

Note that the evolvability created by innovation could either increase or decrease the rate of subsequent evolution, depending on the nature of the changes. Under the imitative genotype hypothesis, evolvability could then evolve to maximize adaptation on the new landscape. This raises the possibility that the GP map of preexisting traits may also be altered.

Innovation also can involve the reduction of trait complexity, such as the loss of serially homologous parts, or the fusion of previously individuated parts, such as the progressive reduction in the number of floral parts in some clades of plants (Armbruster et al. 2014, Stebbins 1951). Such events could also reshape the adaptive landscape, either increasing or decreasing evolvability.

6. CONCLUSIONS

The study of evolvability is dominated by an appealing set of ideas related to a cost of complexity. This idea focuses the attention of the field on those special properties of organisms that might allow the cost of complexity to be overcome or, alternatively, on the reasons why organisms might fail to evolve in response to particular challenges. We have highlighted various misconceptions and inconsistencies that arise from that world view and point to an alternative view where complexity represents an opportunity by increasing the evolutionary potential of a population. In summary, we draw the following conclusions:

1. The paradigmatic geometric model of the cost of complexity (Orr 2000) implies that, all else being equal, complex organisms adapt more slowly than simpler ones. However, the geometric model varies the number of traits that are simultaneously subjected to correlational selection not organismal complexity.
2. Integration and modularity have important effects on evolvability. They can relieve the cost of optimizing multiple traits when they match patterns of correlational selection but increase that cost when they do not. Canalization of pleiotropic effects decreases evolvability.
3. Modularity and integration can evolve to promote evolvability or for nonselective reasons, including drift, side effects of trait evolution, physical proximity, or developmental constraints.
4. The level of variational modularity is often modest. There has been no systematic attempt to quantify the potential effects of modularity on evolvability.
5. Patterns of evolution frequently correspond to variational patterns within populations, but the evolutionary pattern can be explained by either natural selection or variational constraints.

6. The assumption that integration and modularity align with the pattern of natural selection, and therefore help to explain the pattern of evolution, is largely untested. We have essentially no information on the patterns of natural selection to compare with data on variation within or among populations.
7. Models of genotypic adaptive landscapes suggest that the higher the dimension of the genome, the less likely that high-fitness genotypes represent isolated peaks. Each high-fitness genotype is likely to have some neighbors that have similar or greater fitness.
8. Evolutionary innovation can increase the dimensionality of the genotype and phenotype spaces, altering the shape of adaptive landscapes for existing traits. It can also increase the variability of the phenotype by enlarging the mutational target size.

The conviction that complexity affects evolvability positively or negatively is not supported by solid evidence. The answer is likely to be highly contingent on exactly which traits are considered, and most importantly, on the nature of natural selection. We lack good information about the nature of selection and whether organisms are generally at fitness peaks or wandering in a sea of shifting fitnesses. The harder we look at these issues, the less certain we are of how to think about evolvability in relation to complexity.

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Genomic patterns of pleiotropy and the evolution of complexity

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Pleiotropy refers to the phenomenon of a single mutation or gene affecting multiple distinct phenotypic traits and has broad implications in many areas of biology. Due to its central importance, pleiotropy has also been extensively modeled, albeit with virtually no empirical basis. Analyzing phenotypes of large numbers of yeast, nematode, and mouse mutants, we here describe the genomic patterns of pleiotropy. We show that the fraction of traits altered appreciably by the deletion of a gene is minute for most genes and the gene–trait relationship is highly modular. The standardized size of the phenotypic effect of a gene on a trait is approximately normally distributed with variable SDs for different genes, which gives rise to the surprising observation of a larger per-trait effect for genes affecting more traits. This scaling property counteracts the pleiotropy-associated reduction in adaptation rate (i.e., the "cost of complexity") in a nonlinear fashion, resulting in the highest adaptation rate for organisms of intermediate complexity rather than low complexity. Intriguingly, the observed scaling exponent falls in a narrow range that maximizes the optimal complexity. Together, the genome-wide observations of overall low pleiotropy, high modularity, and larger per-trait effects from genes of higher pleiotropy necessitate major revisions of theoretical models of pleiotropy and suggest that pleiotropy has not only allowed but also promoted the evolution of complexity.

genetics | adaptation | modularity | yeast | phenotype

Pleiotropy occurs when a single mutation or gene affects multiple distinct phenotypic traits (1). Pleiotropy has broad implications in genetics (1–3), development (4, 5), senescence (6), disease (7, 8), and many evolutionary processes such as adaptation (2, 9–13), maintenance of sex (14), preservation of redundancy (15), and stabilization of cooperation (16). For example, the antagonistic pleiotropy theory of senescence asserts that alleles beneficial to development and reproduction are deleterious after the reproductive age and cause senescence, which may explain why all species have a limited life span (6). Pleiotropy is the main theoretical reason behind the hypothesis that morphological evolution occurs more frequently through *cis*-regulatory changes than through protein sequence changes (9), as the former are thought to be less pleiotropic than the latter. Pleiotropy also has important implications in human disease, because many genetic defects each affect multiple phenotypic traits. For instance, mutations in the homeobox gene *ARX* cause ambiguous genitalia and lissencephaly (whole or parts of the surface of the brain appear smooth) (Online Mendelian Inheritance in Man no. 300215).

Due to pleiotropy's central importance in biology, several mathematical models of pleiotropy have been developed and important theoretical results have been derived from the analyses of these models (10, 11, 17, 18). For example, Fisher proposed that every mutation affects every trait and the effect size of a mutation on a trait is uniformly distributed (10). On the basis of this model and the assumption that the total effect size of a mutation is constant in different organisms, Orr derived that the rate of adaptation of a population to an environment quickly declines with the increase of the organismal complexity, which is defined by the total number of traits (12). This "cost of complexity" would likely prohibit the origins of complex organisms and hence is puzzling to evolutionary biologists (19, 20).

Although pleiotropy has been examined in detail in a few genes (21, 22), its genomic pattern is largely unknown, which seriously limits us from evaluating mathematical models of pleiotropy, verifying theoretical inferences from these models, and testing various pleiotropy-related hypotheses in many fields of biology. In this work, we compile from existing literature and databases phenotypes of large numbers of yeast, nematode, and mouse mutants. We describe the genomic patterns of pleiotropy in these organisms and show that these patterns drastically differ from any mathematical model of pleiotropy. We further demonstrate that the cost of complexity is substantially alleviated when the empirical patterns of pleiotropy are taken into consideration and that the observed value of a key parameter of pleiotropy falls in a narrow range that maximizes the optimal complexity.

Results and Discussion

Most Genes Affect only a Small Fraction of Traits. To uncover the genomic patterns of pleiotropy, we compiled three large datasets of gene pleiotropy for the baker's yeast *Saccharomyces cerevisiae*, one for the nematode worm *Caenorhabditis elegans*, and one for the house mouse *Mus musculus* (*SI Methods*). The first dataset, yeast morphological pleiotropy, is based on the measures of 279 morphological traits in haploid wild-type cells and 4,718 haploid mutant strains that each lack a different nonessential gene (23). The second dataset, yeast environmental pleiotropy, is based on the growth rates of the same collection of yeast mutants relative to the wild type in 22 different environments (24). The third dataset, yeast physiological pleiotropy, is based on 120 literature-curated physiological functions of genes recorded in the Comprehensive Yeast Genome Database (CYGD). The fourth dataset, nematode pleiotropy, is based on the phenotypes of 44 early embryogenesis traits in *C. elegans* treated with genome-wide RNA-mediated interference (25). The fifth dataset, mouse pleiotropy, is based on the phenotypes of 308 morphological and physiological traits in gene-knockout mice recorded in Mouse Genome Informatics (MGI). These five datasets provide qualitative information about the traits that are affected appreciably by each gene. In addition, the first dataset also includes the quantitative information of the effect size of each gene on each trait. Even after the removal of genes that do not affect any trait and traits that are not affected by any gene, these five datasets each include hundreds to thousands of genes and tens to hundreds of traits (Fig. 1). They are thus suitable for examining genome-wide patterns of pleiotropy.

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In all five datasets, we observed that most genes affect only a small fraction of traits and only a minority of genes affect many traits (Fig. 1). The median degree of pleiotropy varies from one to seven traits (or 1–9% of the traits considered) in these datasets. The degree of pleiotropy measured by the percentage of examined traits is expected to be more accurate for the three datasets in which the same set of traits was examined in all mutants (Fig. 1 *A*, *B*, and *D*). By bootstrapping traits, we found that the SDs of our estimated median and mean degrees of pleiotropy are generally small, indicating that these estimates are precise (Fig. 1). To examine the impact of the number of examined traits on the estimated pleiotropy, we randomly removed 50 and 90% of the traits from each dataset, respectively. We found that the mean and median degrees of pleiotropy, measured by the percentage of traits examined, remain largely unchanged (Table S1), suggesting that further additions of traits to our data would not substantially alter our results. We thus predict that the median number of traits affected by a gene is no greater than a few percent of the total number of traits in an organism. Furthermore, because gene pleiotropy is largely owing to the involvement of the same molecular function in multiple different biological processes rather than the presence of multiple molecular functions per gene (26), random mutations in a gene will likely affect the same traits as the deletion of the gene does, although the magnitude of the phenotypic effects should be much smaller. Consequently, the observable degree of pleiotropy is expected to be even lower for random mutations than for gene deletions. Our genome-wide results echo recent small-scale observations from fish and mouse quantitative trait locus (QTL) studies (27, 28) and an inference from protein sequence evolution (29) and reveal a general pattern of low pleiotropy in eukaryotes, which is in sharp contrast to some commonly used theoretically models (10, 18) that assume universal pleiotropy (i.e., every gene affects every trait) (Table S2).

Gene–Trait Relationships Are Highly Modular. The genome-wide data also allow us to test the modular pleiotropy hypothesis, which is important for a number of theories of development and evolution (30). Gene–trait relationships can be represented by a bipartite network of genes and traits, in which a link between a gene node and a trait node indicates that the gene affects the trait (Fig. 2*A*). Modular pleiotropy refers to the phenomenon that links within modules are significantly more frequent than those across modules (Fig. 2*B*). Given that cellular functions are modularly and hierarchically organized (30), modular pleiotropy likely exists, although it is not considered in commonly used models of pleiotropy (10, 12, 18) (Table S2). Using a bipartite-network–specific algorithm (31), we identified modules and estimated the modularity of each gene–trait network. Because random networks of certain structures also have nonzero modularity (32), we compared the modularity of an observed network with that of its randomly rewired networks, which have randomized links but an unchanged number of links per node (32) (Fig. 2*C*). We then calculated the scaled modularity of a network, which is the difference between the observed modularity of a network and the mean modularity of its randomly rewired networks in terms of the number of SDs (32) (*Methods*). Our results show large scaled modularity (34–238) in each of the five gene–trait networks examined (Fig. 2 *D*–*H*), providing definitive evidence for the modular pleiotropy hypothesis. Our results remain qualitatively unchanged even when a random 50% of the traits in each dataset are removed (Table S3). The modularity would be overestimated if the genetic correlations among traits are biased upward in our datasets compared with the complete datasets that include all possible traits. Although we do not know whether this bias exists, to be conservative, we merged traits whose genetic correlation coefficients are >0.7 (*Methods*). We found that highly significant modularity is still present in each of the five gene–trait networks (Table S3).

Genes Affecting More Traits Have Larger Per-Trait Effects. The yeast morphological pleiotropy data contain quantitative information about the phenotypic effect size of mutations, which is another important parameter in genetics that has never been available at the genomic scale. Using a standardized measure of effect size for all traits (*Z*-score, defined by the phenotypic difference between a mutant and the mean of the wild type for a trait in terms of the number of SDs) (*Methods*), we obtained, for each yeast gene, the frequency distribution of the effect sizes on the 279 morphological traits. As exemplified in Fig. 3*A*, this distribution is approximately normal for most genes; the actual distribution is not significantly different from a normal distribution for 85% of the genes examined (5% false discovery rate in the goodness-of-fit test). This observation is consistent with a commonly used model (18), but is in contrast to another where the distribution is assumed to be uniform (10, 12) (Table S2). In fact, the uniform distribution can be rejected for every gene at the significance level of $P = 5 \times 10^{-7}$ (goodness-of-fit test). It is notable that the SD of the effect size distribution varies greatly among genes (Fig. 3*B*), in contrast to models that assume a constant SD among genes (10, 12, 18) (Table S2). It is also notable that the typical effect size distribution has a nearly zero mean, although a minority of genes exhibit positive or negative means (Fig. S1).

If one considers only those traits that are significantly affected by a gene, the total size of the phenotypic effects of the gene can be calculated by the Euclidean distance $T_{\mathrm{E}} = \sqrt{\sum_{i=1}^{n} Z_i^2}$, where n is the gene's degree of pleiotropy defined by the number of significantly affected traits and Z_i is the gene's effect on trait i measured by the Z-score (27). We estimated from the yeast morphological pleiotropy data that the exponent b in the scaling relationship of $T_{\mathrm{E}} = an^b$ equals 0.601, with its 95% confidence interval of (0.590,

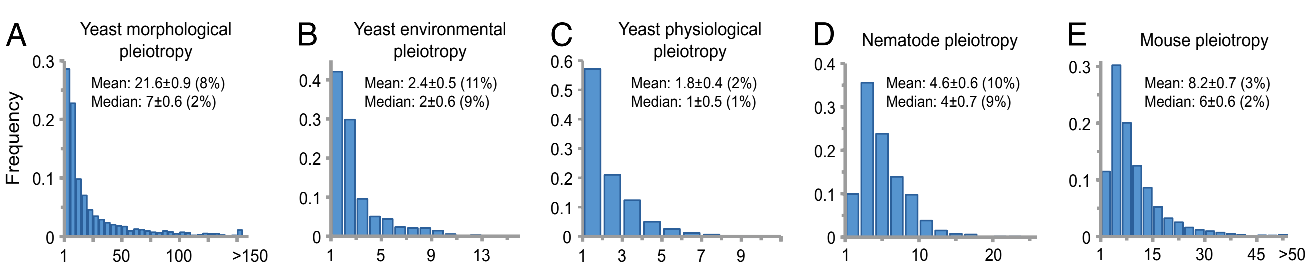

Fig. 1. Frequency distributions of degree of gene pleiotropy in (*A*) yeast morphological, (*B*) yeast environmental, (*C*) yeast physiological, (*D*) nematode, and (*E*) mouse pleiotropy data. Mean and median degrees of pleiotropy and their SDs are indicated. The numbers in parentheses are the mean and median degrees of pleiotropy divided by the total number of traits. After the removal of genes that do not affect any trait and traits that are not affected by any gene, the total numbers of genes and traits in these datasets are (*A*) 2,449 genes and 253 traits, (*B*) 774 genes and 22 traits, (*C*) 1,256 genes and 120 traits, (*D*) 661 genes and 44 traits, and (*E*) 4,915 genes and 308 traits.

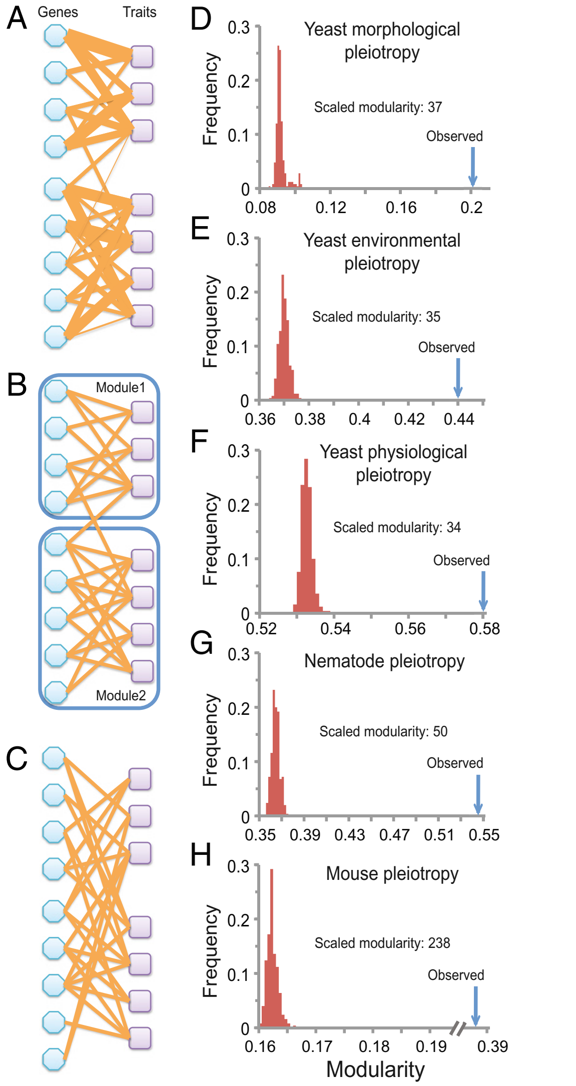

Fig. 2. High modularity of gene–trait bipartite networks. (*A*) A hypothetical gene–trait bipartite network. A link between a gene and a trait indicates that the gene affects the trait, and the thickness of the link indicates the effect size. (*B*) Two modules are identified in the hypothetical gene–trait network after the quantitative links are transformed to qualitative links (i.e., presence/absence) on the basis of whether an effect size is significantly different from 0. (*C*) A randomly rewired network that has the same degree distribution as the original hypothetical network shows no detectable modular structure. The modularity and scaled modularity of the hypothetical bipartite network are 0.41 and 3.9, respectively. *D–H* show the observed modularity (blue arrows) and distribution of modularity for 250 randomly rewired networks (red histograms) for the gene–trait networks of the (*D*) yeast morphological, (*E*) yeast environmental, (*F*) yeast physiological, (*G*) nematode, and (*H*) mouse pleiotropy datasets.

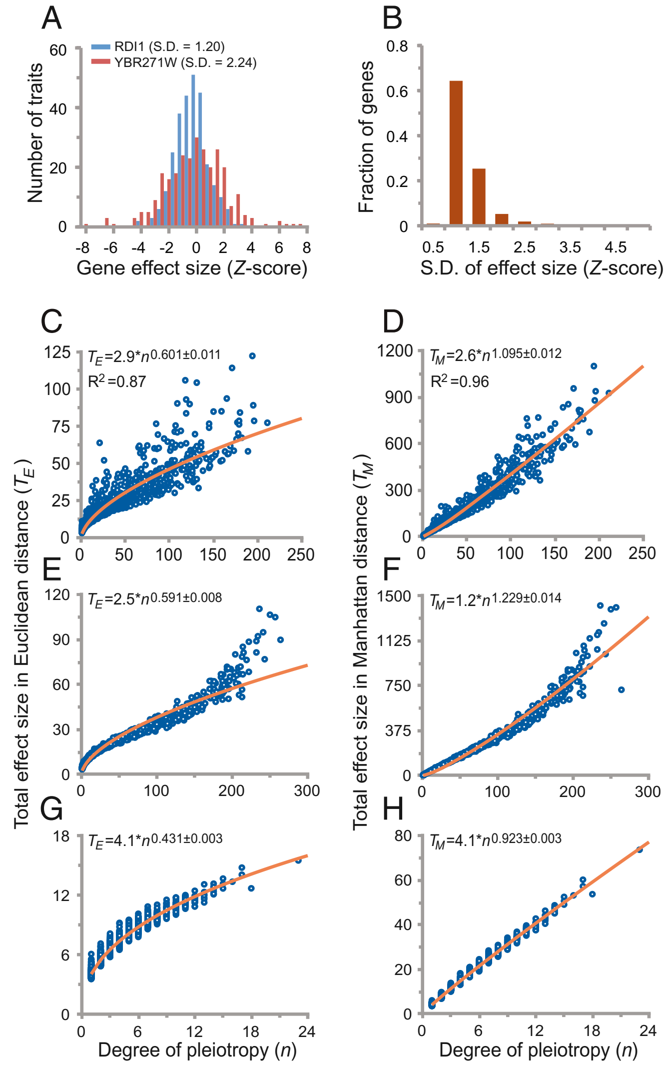

Fig. 3. Scaling relationships between the total phenotypic effect size of a gene and the degree of pleiotropy in the yeast morphological pleiotropy data. (*A*) Examples showing the normal distribution of effect size over 279 traits. Two genes are chosen to show variable SDs of the normal distributions. (*B*) Distribution of the SD of the effect size for all 4,718 genes. Observed scaling relationships between the degree of pleiotropy *n* and the total phenotypic effect of a gene are measured by (*C*) Euclidean distance or (*D*) Manhattan distance. The orange curve is the best fit to the power function whose estimated parameters are shown in the upper left. The numbers after ± show the 95% confidence interval for the estimated scaling exponent. R^2 indicates the square of the correlation coefficient. *E* and *F* are similar to *C* and *D* except that the effect sizes of each gene are randomly generated from a normal distribution with zero mean and observed SD. *G* and *H* are similar to *C* and *D* except that the effect sizes of each gene are randomly generated from a normal distribution with zero mean and a constant SD, which is the average of all SDs of all genes.

0.612) (Fig. 3*C*). This exponent is significantly greater than that assumed in any theoretical model (Table S2). For example, the invariant total effect model (12) assumes a constant total effect size ($b = 0$), whereas the Euclidian superposition model (11, 17, 18) assumes a constant effect size per affected trait ($b = 0.5$). Our results thus indicate that the per-trait effect of a gene, estimated by $T_E/\sqrt{n}$ on the basis of the definition of the Euclidian distance, is larger when the gene affects more traits. One can also measure the total effect size by the Manhattan distance (33) $T_M = \sum_{i=1}^{n}|Z_i|$. We found the exponent d in the scaling relationship of $T_M = cn^d$ to be 1.095, with its 95% confidence interval of (1.083, 1.107) (Fig. 3*D*). Again, the observed d significantly exceeds that assumed in all current models (0.5 in the invariant total effect model and 1 in the Euclidian superposition model; Table S2) and indicates larger per-trait effects estimated by T_M/n for genes affecting more traits. To examine the robustness of the above results, we randomly removed 50 and 90% of the traits from the data, respectively. Our results that b and d are slightly smaller when the number of traits used is smaller (Fig. S2 *A–D*) suggest that, when more traits are examined in the future, b and d would become slightly greater than the current estimates. Our results are also robust to merging traits with genetic correlations (Fig. S2 *E* and *F*). Because 279 morphological traits were measured in each yeast mutant, in the above analyses, a 5% false discovery rate was used as a cutoff to control for multiple testing in determining whether a trait is affected by a gene. Our results remain qualitatively unchanged when the more conservative $P = 5\%$ after Bonferroni correction is used to correct for multiple testing (Fig. S3).

We observed that the phenomenon of larger per-trait effects for genes affecting more traits disappeared when the effect sizes of all genes on all traits are randomly shuffled (Fig. S4). Thus, the phenomenon is a property of the actual data rather than an artifact of our analysis. It turns out that this phenomenon results from two genome-wide features of pleiotropy described above: (*i*) a normal distribution of effect sizes of a gene on different traits and (*ii*) variable SDs of the normal distributions among different genes. Comparing two genes both having normal distributions of effect sizes but with different SDs, we proved mathematically that the gene with the larger SD affects more traits (when a fixed effect-size cutoff is applied) and has on average a larger per-trait effect (*SI Methods*). In fact, the scaling relationships with the observed b and d values can be largely recapitulated by using randomly generated effect size data, provided that a normal distribution with the actual SD is used in generating such data for each gene (Fig. 3 *E* and *F*; *SI Methods*). By contrast, when the same SD is used in generating the random effect size data for all of the genes (*SI Methods*), we no longer observe larger per-trait effects for genes affecting more traits ($b < 0.5$ in Fig. 3*G* and $d < 1$ in Fig. 3*H*).

It is interesting to note that a recent mouse QTL study (27) also reported $b > 0.5$, but a subsequent analysis (33) showed that, owing to the likelihood of the inclusion of multiple genes per QTL, the data can establish only $b > 0$, but not $b > 0.5$. Because our yeast morphological pleiotropy data were collected from strains that each lack only one gene, they are immune from the above multiple-gene problem. Additionally, our observation that the estimated b tends to be smaller when fewer traits are used (Fig. S2) may also explain the unreliability (33) of the mouse QTL study (27), which was based on a much smaller dataset. Furthermore, because our data were generated by examining all yeast nonessential genes and a large number of traits, they are more likely to reveal the general patterns of pleiotropy. It is important to recognize that our results are based on pleiotropic effects of genes (i.e., null mutations) rather than random mutations. However, our results likely apply to random mutations because the effect sizes of random mutations in a gene are expected to be proportional to the effect sizes of the gene (*Methods*).

Cost of Complexity Is Diminished with the Actual Patterns of Pleiotropy. One of the most puzzling results from theoretical analysis of pleiotropy is the cost of complexity conundrum (12). Using Fisher's geometric model (10), Orr (12) showed that the rate of adaptation of an organism is $U = dw/dt = -(4kT_{\mathrm{E}}^2/n)Mw\ln w$, where k is the product of the effective population size and the mutation rate per generation per genome (in the functional part), T_{E} is the total effect size of a mutation defined earlier, n is the degree of pleiotropy of a mutation, w is the current mean fitness of the population relative to the optimal, and M is a function of T_{E} and n (*Methods*). Although Orr assumed that each mutation affects all traits of an organism and therefore n is also a measure of organismal complexity, we have shown that deleting a gene affects only a small fraction of traits (Fig. 1). In other words, mutational pleiotropy is much smaller than organismal complexity. Nonetheless, if the fraction of traits affected by an average mutation is similar among different organisms (Fig. 1), mutational pleiotropy is still higher in complex organisms than in simple organisms, because complex organisms have more traits than simple organisms do. Hence, n in the above formula may be interpreted as the effective organismal complexity (i.e., the number of traits affected by an average mutation), which is much lower than the actual organismal complexity (i.e., the total number of traits). Empirical evidence suggests that k may increase slightly with the level of organismal complexity, but the exact relationship between them is unclear (19). To be conservative, we here assume k to be independent of n. Note that although we are using the original formula from Orr (12) for U, which was based on a fixed mutation size T_{E} for a given n, this formula is known to be robust to variable mutation sizes (20). By comparing U among organisms of different levels of complexity, Orr implicitly assumed that the rate of adaptation is directly comparable among different organisms, which requires further theoretical and empirical support. Here, we follow Orr to compare U among different organisms.

If T_{E} is independent of n, as assumed in the invariant total effect model (12), or if T_{E} is proportional to $n^{0.5}$, as assumed in the Euclidian superposition model (11, 17, 18) (Table S2), adaptation rate U decreases with the degree of pleiotropy (or level of effective organismal complexity) n (Fig. 4*A*), creating the cost of complexity. Interestingly, the relationship between U and n changes when the scaling exponent $b > 0.5$. It can be shown mathematically that, when $b > 0.5$, an intermediate level of complexity yields the highest adaptation rate (*SI Methods*) (Fig. 4*A*). The n value that corresponds to the highest adaptation rate (n_{optimal}) depends on several parameters, including a (referred to as the mutation size, which is the expected T_{E} for mutations that affect only one trait), b, and w. Smaller a and w values lead to larger n_{optimal} (Fig. 4*B*). The null mutations in the yeast morphological pleiotropy data yield $a = 2.9$, but we expect natural random mutations to have a much smaller a, because most natural mutations affect the function of a gene only slightly and thus have on average much smaller phenotypic effects than gene deletions do (*Methods*). For example, if $a = 0.01$ for natural random mutations, $b = 0.6$ as we have shown, and $w = 0.9$, n_{optimal} becomes 9 (Fig. 4*A*).

Numerically, we found that, when a and w are given, n_{optimal} reaches its maximum at an intermediate b value (Fig. 4*C*). By examining a large parameter space ($10^{-8} \le a \le 10^{-2}$; $0.3 \le w \le 0.99$), we observed that the b value that offers the maximal n_{optimal} occurs in a narrow range between 0.56 and 0.79 (Fig. 4*D*), although b potentially can vary from negative infinity to positive infinity.

It is important to confirm two key assumptions that Orr made in deriving the formula for U (12). The first assumption is that phenotypic traits are independent from one another, as was originally described by Fisher in his geometric model (10). As questioned by Haldane and other authors, this assumption is often unmet (20, 34, 35). Nonetheless, it was later demonstrated that the geometric model can still be applied when nonindependent traits are linearly transformed to compound traits that are independent from one another (35). We therefore performed a principal component analysis of the wild-type phenotypic matrix (*SI Methods*) to make the resultant principal component traits orthogonal to one another. Applying the same linear transformation to the mutant phenotypic matrix to obtain the compound phenotypic effects for each gene (*SI Methods*), we found that our results of the Euclidian scaling coefficients $b > 0.5$ and the Manhattan scaling coefficient $d > 1.0$ remain unchanged (Fig. S5). This finding also demonstrates that the phenomenon that genes affecting more traits have larger per-trait effects holds even when the traits are independent from one another.

The second assumption of Orr is that the fitness effect of a mutation should increase with its phenotypic effect size (T_{E}). We confirmed the validity of this assumption by showing that the fitness effect of deleting a gene is significantly positively correlated with its T_{E} (Spearman's $\rho = 0.377$, $P < 10^{-10}$). This positive correlation, coupled with the positive correlation between T_{E} and the degree of pleiotropy n (Fig. 3*C*), may explain the phenomenon that deleting a more pleiotropic gene tends to cause a larger fitness reduction (36).

Concluding Remarks. In summary, our genome-wide analysis of pleiotropy in yeast, nematode, and mouse revealed a generally low level of pleiotropy for most genes in a eukaryotic genome and a highly modular structure in the gene–trait relationship.

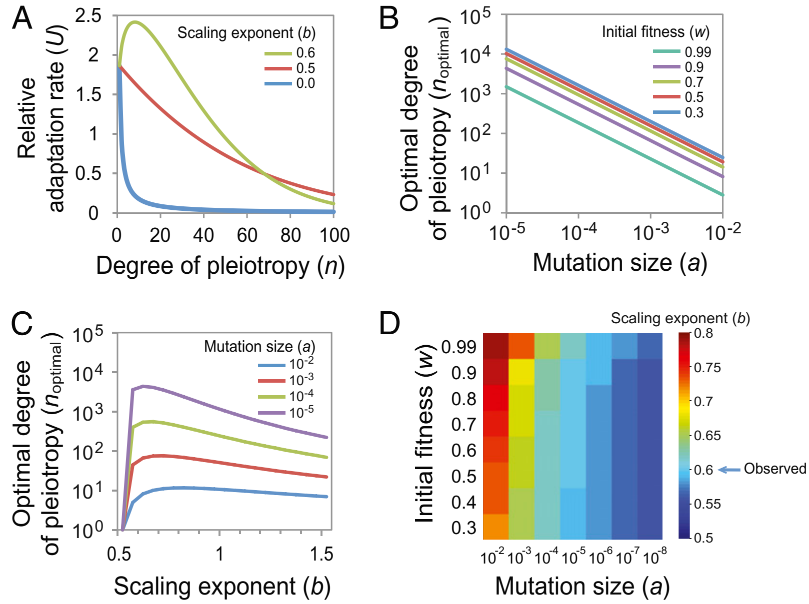

Fig. 4. The "cost of complexity" is alleviated when the scaling exponent $b > 0.5$. (*A*) The relative adaptation rate as a function of the degree of pleiotropy (n) changes with the scaling exponent b. The relative adaptation rate is calculated using Orr's formula. The initial fitness w is set at 0.9 and the mutation size a is set at 0.01. (*B*) The optimal degree of pleiotropy n_{optimal}, defined as the degree of pleiotropy that corresponds to the highest adaptation rate, changes with the mutation size a. Different curves are generated using different initial fitness (w) values but the same $b = 0.6$. (*C*) The optimal degree of pleiotropy n_{optimal} changes with different b. Different curves are generated using different a but the same $w = 0.9$. (*D*) A heat map showing the b value that provides the maximal n_{optimal}, given a and w.

Furthermore, the quantitative morphological data from yeast showed that genes affecting more traits tend to have larger per-trait phenotypic effects. Although an organism potentially contains many more traits than our data currently include, several analyses indicated that our results are robust and therefore our conclusions are expected to be largely unchanged even when most or all traits of an organism are considered. These findings necessitate a major revision of the current theoretical models that lack the above three empirical features of pleiotropy (Table S2) and require reevaluation of biological inferences derived from these models. For example, these three features substantially alleviate the cost of complexity in adaptive evolution. First, the generally low pleiotropy means that even mutations in organisms as complex as mammals do not normally affect many traits simultaneously. Second, high modularity reduces the probability that a random mutation is deleterious, because the mutation is likely to affect a set of related traits in the same direction rather than a set of unrelated traits in random directions (20, 37). These two properties substantially lower the effective complexity of an organism. Third, the greater per-trait effect size for more pleiotropic mutations (i.e., $b > 0.5$) causes a greater probability of fixation and a larger amount of fitness gain when a beneficial mutation occurs in a more complex organism than in a less complex organism. These effects, counteracting lower frequencies of beneficial mutations in more complex organisms (10), result in intermediate levels of effective complexity having the highest rate of adaptation. Together, they explain why complex organisms could have evolved despite the cost of complexity. Because organisms of intermediate levels of effective complexity have greater adaptation rates than organisms of low levels of effective complexity due to the scaling property of pleiotropy, pleiotropy may have promoted the evolution of complexity. Whether the intriguing finding that the empirically observed scaling exponent b falls in a narrow range that offers the maximal optimal complexity is the result of natural selection for evolvability or a by-product of other evolutionary processes (38) requires further exploration.

Methods

Modularity of Gene–Trait Bipartite Networks. Gene–trait relationships can be represented by a bipartite network where the genes form one type of nodes and traits form the second type of nodes. A link between a gene node and a trait node indicates that the gene affects the trait. To separate modules, we used BRIM (31), which is modified from the widely used Newman definition of modularity (39) for bipartite networks. For a given module partition, this algorithm calculates the difference between the density of within-module links and its random expectation. It then attempts to find the module partition that yields the highest difference, which is called the modularity of the network. The raw modularity score obtained from the algorithm has a theoretical range between 0 and 1, with 0 meaning no modularity and 1 meaning very high modularity. However, because even a random network may have a nonzero modularity depending on the network size and degree distribution (32, 40), we measure the level of network modularity by scaled modularity (32), which is the difference between the observed modularity and the mean modularity of its randomly rewired networks divided by the SD in modularity among the randomly rewired networks. The randomly rewired networks were generated by randomly linking nodes while conserving the number of links of each node and the total number of nodes of the original network. We also calculated scaled modularity after merging traits whose genetic correlation coefficient is >0.7. We chose this cutoff because, after the merge, no trait can genetically explain more than one-half of the variance of another trait ($0.7^2 = 0.49$).

Scaling Relationships Between the Degree of Pleiotropy and the Total Effect Size. Using the yeast morphological pleiotropy data, we calculated the number (n) of traits that are significantly affected by each gene. We then measured a gene's total phenotypic effect on these n traits, using either the Euclidian distance (T_E) or the Manhattan distance (T_M). We expect the scaling relationships of $T_E = an^b$ and $T_M = cn^d$. We estimated a, b, c, and d using the curve-fitting toolbox in MATLAB, which employs a nonlinear least-squares method to fit the observations with the power function. The confidence intervals of the estimated parameters were also calculated by MATLAB, which uses the decomposition of the Jacobian, the degree of freedom, and the root mean squared error.

Because gene pleiotropy is largely owing to the involvement of the same molecular function in multiple different biological processes rather than the presence of multiple molecular functions per gene (26), random mutations in a gene will likely affect the same traits as the deletion of the gene does, although the magnitude of the phenotypic effects should be much smaller. For simplicity, let us assume that the effect on a trait from a random mutation in a gene is on average h times the effect on the same trait from a null mutation in the gene, where the effect is again measured by Z-score and $0 < h << 1$. Let $T_{E'}$ be the total effect size of the random mutation in Euclidean distance. It can be shown that $T_{E'} = hT_E = (ah)n^b$. Thus, the scaling relationship between the total effect size of a random mutation and pleiotropy is the same as that between the total effect size of a null mutation and pleiotropy, except that the mutation size parameter for random mutations is h times that for null mutations.

Calculating the Rate of Adaptation. Assuming Fisher's geometric model, Orr (12) derived the formula for the rate of fitness increase during an adaptive walk to the optimal to be $U = dw/dt = -(4kT_E^2/n)Mw \ln w$, where $M = (1/\sqrt{2\pi})\int_x^{+\infty}(y-x)^2 e^{-y^2/2}dy$, $x = T_E\sqrt{n}/(2\sqrt{-2 \ln w})$, and the other variables are defined in the main text. In Orr's calculation (12), T_E was assumed to be independent of n. In our model, T_E scales with the degree of pleiotropy by $T_E = an^b$, where a is the mutation size parameter that corresponds to the mutation size when the degree of pleiotropy is 1 and b is the scaling exponent. We implemented numerical calculations of the above formulas in MATLAB.

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An Evolutionary Model for the Origin of Modularity in a Complex Gene Network

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ABSTRACT Scale-free cellular networks are organized into a complex topology by massive interactions (links) between nodes, which can be typically characterized by a power-law degree. In contrast, almost all cellular networks show the feature of modularity. The popular BA model (Barabasi and Albert) demonstrated the origin of scale-free property by the attachment preference, but not for the origin of modularity. We propose a BBA model (Biological BA) by introducing the random link-loss mechanism under the original BA model, showing that scale-free and modularity can emerge as a derived property of the BBA model. Data analysis has shown that, roughly, the rate of gene network evolution can be described as a 2-2-1 pattern (adding two new genes and two new links with the cost of one link-loss). *J. Exp. Zool. (Mol. Dev. Evol.) 312B:75–82, 2009.* © 2008 Wiley-Liss, Inc.

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One of the central issues in systems biology is to understand the origin of gene network complexity (Hartwell et al., '99; Barabasi and Oltvai, 2004; Wagner et al., 2007). Many biological systems, from metabolic pathway to protein–protein interactions, can be represented as (gene) networks (Barabasi and Oltvai, 2004). In a typical network, many nodes are organized into a complex topology by massive interactions (links) between nodes. For example, in the yeast protein–protein interaction network (Jeong et al., 2001), a node represents a protein, and a link between two proteins indicates their interaction. The high complexity of a gene network can be characterized by a power-law degree, i.e., the number of links for a node (called degree) is right-skewedly distributed; many nodes with low degrees, while a small number with high degrees (Albert and Barabasi, 2002). Substantial genome-wide analyses (Barabasi and Albert, '99; Jeong et al., 2000, 2001; Hahn et al., 2004; Han et al., 2004) have shown that cellular networks can be featured by (*i*) the power-law of degree distribution; (*ii*) the small-world property with occasional long-range links; and (*iii*) network centrality with hubs (nodes with many links). They are called scale-free because these network features are independent of the size of the network.

Another important feature of a cellular network is the modularity, which refers to a network of interactions if it can be subdivided into relatively autonomous, internally highly connected components (Hartwell et al., '99). Though there is a general agreement that organisms have a modular organization as a fundamental rule, current study of modularity seeks to capture the various levels and kinds of functional heterogeneity found in organisms, which has emerged as a hot research subject in developmental and evolutionary biology, as well as systems biology (Gu and Su, 2007; Su et al., 2007; Wagner et al., 2007). The main open problem is about the origin and evolution of modularity, e.g., it has been highly controversial whether modules arise through the action of natural selection or because of nonadaptive processes, such as biased mutational mechanisms (Lynch, 2007; Wagner et al., 2007).

The pioneer work of Barabasi and Albert ('99), refereed as the BA model below, provides an

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elegant yet simple evolutionary model for the origin of scale-free network (power-law). It was based on two mechanisms: (*i*) Network growth is a continuous process during the course of evolutionand (*ii*) hubs that already have many interactions have high chances to increase the connectivity, called the attachment preference. Though the BA model successfully predicts the scale-free property, it was unable to predict the property of modularity (Barabasi and Oltvai, 2004). In contrast, Wagner and Mezey (2004) suggested that the evolution of modularity could be driven by the selection for robustness, such as differential elimination of pleiotropic effects. This idea can be generalized to a ''differential erosion model'' for the evolution of modularity (von Mering et al., 2002; Wagner, 2003; Wagner et al., 2007). We have recognized that from the gene network point-of-view, elimination of pleiotropic effects can be viewed as the process of link-losses. Because the origin of new links has the attachment preference to hubs that leads to the scale-free property, we speculate that a random link-loss (no hub preference) may provide a mechanism for the origin of modularity. In this article, we propose an evolutionary model of gene network that may lead to the emergence of scale-free property and modularity simultaneously.

RESULTS

Scale-Free Property and Power-Law

In a scale-free network characterized by highly connected genes (nodes), or hubs, the number of genes with a given degree (number of links) follows a power-law. That is, the probability that a chosen gene node has exactly k links follows

$$P(k) \sim k^{-\gamma} \tag{1}$$

where γ is the degree component, which is usually in the range of $2<\gamma<3$ (Albert and Barabasi, 2002). The power-law indicates that there is a high heterogeneity of node degrees and no typical value in the network such as the mean degree that could be used to characterize the level (or scale) of node connectivity. Thus, a network with the absence of a typical degree is usually called ''scale-free'', in which a few hubs dominate network features. For instance, a common feature of scale-free networks is the small-world property that any two nodes can be connected with a much shorter chain of links (through hubs) than that expected by the random network. Analysis of yeast protein–protein interaction network indicated that about 60% of the hub proteins (defined as >15 interactions) are essential, whereas only 10% of few connected proteins with less than five links (Jeong et al., 2001).

Modularity and Clustering Coefficient

Meanwhile, many studies (e.g., Ravasz et al., 2002; Giot et al., 2003; Almaas et al., 2004; Barabasi and Oltvai, 2004; Vazquez et al., 2004; Wagner et al., 2007) have unveiled that almost all cellular networks from protein–protein interactions, metabolic to regulatory networks show the feature of modularity. Moreover, several authors (Ravasz et al., 2002; Barabasi and Oltvai, 2004; Vazquez et al., 2004) used the node-specific clustering coefficient, the cohesiveness of the neighborhood of a node (A) that has k links, to examine the modularity in a scale-free network. From this view, gene A's neighbor is the set (k) of genes that are directly linked to gene A. Mathematically, the clustering coefficient $C(k)$ is defined by

$$C(k) = 2T(k)/[k(k-1)] \tag{2}$$

where $T(k)$ is the number of direct links between any two gene As neighbors, and $k(k-1)/2$ is the total possible number for links among neighbors. In fact, $C(k)$ measures how close the local neighborhood of a node is to being part of a clique (module), a region of the graph where every node is connected to every other node. In practice, we used the averaged clustering coefficients of nodes that have the same degree k to characterize the network modularity.

It has been claimed (Barabasi and Oltvai 2004) that in a typical scale-free network, the mean clustering coefficient $C(k)$ is independent of the node degree k, i.e., $C(k) \sim C_0$, a feature of no modularity that is similar to a random network. This indicates that, on average, the size of a local neighborhood, determined by the node degree (k), does not influence the cohesiveness. Using the power-law as a measure of scale-free property and the clustering coefficient as modularity, Ravasz et al. (2002) found that the prediction of constant clustering coefficient was rejected by several cellular networks. This finding implies that a pure scale-free network generated by the BA model (Barabasi and Albert, '99) may not be sufficient to explain the complexity of cellular networks. Further, Ravasz et al. (2002) observed a log–log inverse relationship between $C(k)$ and the node degree (k), indicating that nodes with small links (k) and a high $C(k)$ belong to highly interconnected small modules, while highly connected hubs with a low $C(k)$ link different modules. Then, they proposed the concept of hierarchical networks that integrate a scale-free topology with an inherent modular structure, hence it not only has a power-law degree distribution $P(k)$, but also a power-law scaling of the clustering coefficient with node degree k, that is

$$C(k) \sim k^{-\theta} \quad (3)$$

In short, in a hierarchical network, sparsely connected nodes are part of highly clustered subnets, with communication between the different highly clustered neighborhoods around a few hubs. However, it seems a puzzle how evolution can lead to various biological networks that are characterized by the ''contrasted'' scale-free and modular features.

How a Scale-Free Network With Modularity to be Evolved?

Cellular networks have long been thought to be modular, composed of functionally separable subnetworks corresponding to specific biological functions (Hartwell et al., '99; Wagner et al., 2007). In addition, there is a high degree of overlap and crosstalk between modules (Han et al., 2004). Hence, the hierarchical network, or the scale-free network with modularity, is biologically sound. A fundamental issue then is how a hierarchical network can be evolved, which remains unsolved.

Barabasi and Albert ('99) developed an elegant and concise framework (the BA model for short) to describe the origin of a scale-free network. It incorporates two mechanisms: the growth (i.e., an increase in the numbers of nodes and links over the time) and the preferential attachment (i.e., a new gene favorably linking to the existed ''hub'' gene). Barabasi and Albert ('99) have shown that the generated network has a power-law $P(k) \sim k^{-3}$, while the clustering coefficient is a constant (Ravasz et al., 2002). A number of modified BA models that included various network constraints and system-specific preferential attachments have been proposed, see, for a review, Albert and Barabasi (2002) to explain gene networks with degree component $\gamma = 2 \sim 3$ but cannot predict the modularity.

Alternatively, Ravasz et al. (2002) proposed a growing network model by iterative subnetwork duplications and integrations to its original seed core. This growth algorithm leads to a power-law of node degree with $\gamma \approx 2$, and a scaling power function of clustering coefficient $C(k) \sim k^{-1}$ when the size of the seed graph of duplication is four. However, Ravasz et al.'s (2002) model has no intrinsic relationship with the BA model, and the biological meaning of the proposed mechanism was unclear. The purpose of our study is to develop a unified evolutionary model that can explain the following network features: (*i*) the scale-free property characterized by a power-law of Equation (1), agreeing with the range of observed degree exponent $\gamma = 1.5 - 2.5$ and (*ii*) the decreasing of clustering coefficient $C(k)$ with node degree k as Equation (3), agreeing with the range of observed clustering-degree exponent $\theta = 0.5 - 1.5$. As will be shown below, our result is surprisingly simple: by adding the random link-loss (no preference) mechanism to the classical BA model, we show that the network can evolve to be scale-free with the feature of modularity.

Biological BA Model (BBA): Evolution by Preference Attachment and Random-Loss of Links

We propose a revised BA model, coined by the term biological BA (BBA). Under the new BBA model, network evolution is driven by three parameters: λ and μ are the gain rate and loss rate of a link, respectively, and g is the origin rate of a new gene (node). Following Barabasi and Albert ('99), we have implemented a linear preferential linking, i.e., the probability (π) of any gene A to have a new link to another gene B is proportional to the connectivity of gene B, or, $\pi(k) \sim k$, where k is the number of links of gene B. As shown in the Appendix, we have shown that the generated network is scale-free. The node-degree distribution $P(k)$ follows a power-law, $P(k) \sim k^{-\gamma}$, with the degree component

$$\gamma = 3 - 2\mu/\lambda \quad (4)$$

Therefore, the classical BA model is a special case of $\mu = 0$, i.e., no link-loss, resulting in $\gamma = 3$. Note that the assumption of network growth requires the constraint that the loss rate of links must be less than the gain rate, i.e., $\mu < \lambda$, consistent to the fact of $\gamma > 1$ for most biological networks. Interestingly, the loss–gain ratio of connectivity evolution (μ/λ) determines the shape of any scale-free network with $1 < \gamma \leq 3$. For cellular networks with $\gamma \approx 2$, Equation (4) indicates that $\mu \approx 0.5\lambda$, i.e., the rate of link-gain is approximately two-fold larger than the rate of link-loss.

Random Link-Loss Mechanism Generating Modularity of Gene Network

We envisage a scenario of complex network evolution to generate the modularity through the random process of link-losses. The key is that link-loss has no preference between high-degree nodes and low-degree nodes. Consequently, while the number of links (degree) in hubs is less affected by the random link-loss, the degree in some nonhub nodes can be reduced to a very few number of links, creating a hierarchical network randomly.

In the following, we derive the power-scaling of clustering coefficient $C(k)$, with the assumption that gene duplication is the main mechanism for the origin of new genes. First, we consider $T(k)$, the number of direct links between k-neighbors of gene A. Suppose that one of gene As neighbors, say, the ith neighbor, was duplicated; the duplicated copy was numbered as the $k+1$th neighbor of gene A. Because the clustering coefficient $C(k)$ is the probability that two neighbors are linked, the expected number of links between the ith neighbor and other neighbors is therefore given by $(k-1)C(k)$. After the gene duplication, the increased number of direct links between neighbors is thus expected to be $(k-1)C(k)$. In contrast, random link-loss may reduce the number of direct links between neighbors. During the time period (Δt) for gene duplication and preservation, the number of link-losses is expected to be $(k-1)C(k)\times \mu\Delta t$. Typically, Δt can be characterized by $1/g$; here g is the rate of gene duplication, i.e., $\Delta t \approx 1/g$. Therefore, the expected loss number of direct links is given by $(k-1)C(k)\times \mu/g$. Together, the net change in the number of direct links, denoted by $\Delta T(k) = T(k+1) - T(k)$, is given by

$$\begin{aligned}\Delta T(k) &= (k-1)C(k) - (k-1)[\mu/g]C(k)\\ &= 2(1-\mu/g)\frac{T(k)}{k}\end{aligned} \tag{5}$$

Using the continuous approximation, we claim that $T(k)$ satisfies the following differential equation

$$\frac{\mathrm{d}T(k)}{\mathrm{d}k} = 2(1-\mu/g)\frac{T(k)}{k} \tag{6}$$

Solving with the initial condition $T(2) = \frac{1}{2}$ results in

$$T(k) = 0.5 \times k^{2(1-\mu/g)} \tag{7}$$

and we obtain a power-law relation between $C(k)$ and k as follows:

$$C(k) = \frac{2T(k)}{k(k-1)} \sim k^{-2\mu/g} \tag{8}$$

Therefore, we have derived that the clustering coefficient has a power-law, i.e., $C(k) \sim k^{-\theta}$, where θ is given by

$$\theta = 2\mu/g \tag{9}$$

In the case of the classical BA model with no link-loss ($\mu = 0$), $\theta = 0$ that indicates no inherent modularity of the scale-free gene network. In other words, when highly clustered neighborhoods around several hubs that can be generated by preferential linking, random-losses of links may provide an evolutionary mechanism to create sparsely connected genes that connect between the different highly clustered subnets of hubs.

Data Analysis

We have analyzed protein–protein interaction networks from ten organisms (Rain et al., 2001; Giaever et al., 2002; Rives and Galitski, 2003; Ito et al., 2001; Li et al., 2004; Uetz et al., 2000; Giot et al., 2003; Han et al., 2004; Stelzl et al., 2005) (see Table 1). In each data set, we counted the distribution of node degrees, $P(k)$, and calculated the average clustering coefficients, $C(k)$, for each node degree $k \geq 2$. Then we estimated both degree components (γ, θ) by the $\ln P(k) \sim \ln k$ and $\ln C(k) \sim \ln k$ regressions, respectively. Other methods including nonlinear approaches gave similar results (not shown). For instance, Figure 1 shows the $P(k) \sim k$ and $C(k) \sim k$ relationships in the human protein–protein interaction network. In addition, we included the estimates of two other cellular networks (metabolic and regulatory) in two organisms (*E.coli* and yeast) (Vazquez et al., 2004).

It is impressive that all these gene or cellular networks show the scale-free property and modularity. The range of degree component (γ) is from 1.40 to 2.53, while the range of clustering-component (θ) is from 0.63 to 1.50. Roughly speaking, the power-law degree of $P(k)$ is around 2 ($\gamma = 1.86 \pm 0.07$), while that of $C(k)$ is around 1 ($\theta = 0.96 \pm 0.08$). We found no evidence for any difference due to the taxon level, network type, or network size. Furthermore, according to Equations (4) and (9), we estimated the relative gain rate of links (λ/g) and the relative loss rate of links (μ/g).

TABLE 1. Estimated evolutionary parameters of gene networks under the BBA (Biological BA) model

Network types	Organisms	γ	θ	λ/g	μ/g	λ/μ
Protein interaction	Yeast	1.80	1.02	0.85	0.51	1.67
	Human	1.73	1.07	0.84	0.54	1.57
	Worm	1.59	0.63	0.45	0.32	1.42
	House mouse	2.53	0.91	1.94	0.46	4.26
	Helicobactor	1.64	0.45	0.33	0.23	1.47
	E.coli	1.84	1.38	1.19	0.69	1.72
	Norway rat	2.03	1.19	1.23	0.60	2.06
	Arabidopsis	1.63	1.50	1.09	0.75	1.46
	Rice	1.77	0.63	0.51	0.32	1.63
	Cow	1.40	1.12	0.70	0.56	1.25
Metabolic	Yeast	2.00	0.70	0.70	0.35	2.00
	E.coli	2.00	0.80	0.80	0.40	2.00
Regulatory	Yeast	2.00	1.00	1.00	0.50	2.00
	E.coli	2.10	1.00	1.11	0.50	2.22
Mean $\pm$ s.e.		1.86 ± 0.07	0.96 ± 0.08	0.91 ± 0.11	0.48 ± 0.04	1.91 ± 0.20

γ: power-law degree component; θ: clustering coefficient component; λ/g: the rate of new link relative to the rate of new gene; μ/g: the rate of link-loss relative to the rate of new gene; and λ/μ: the rate ratio of new links and link losses.

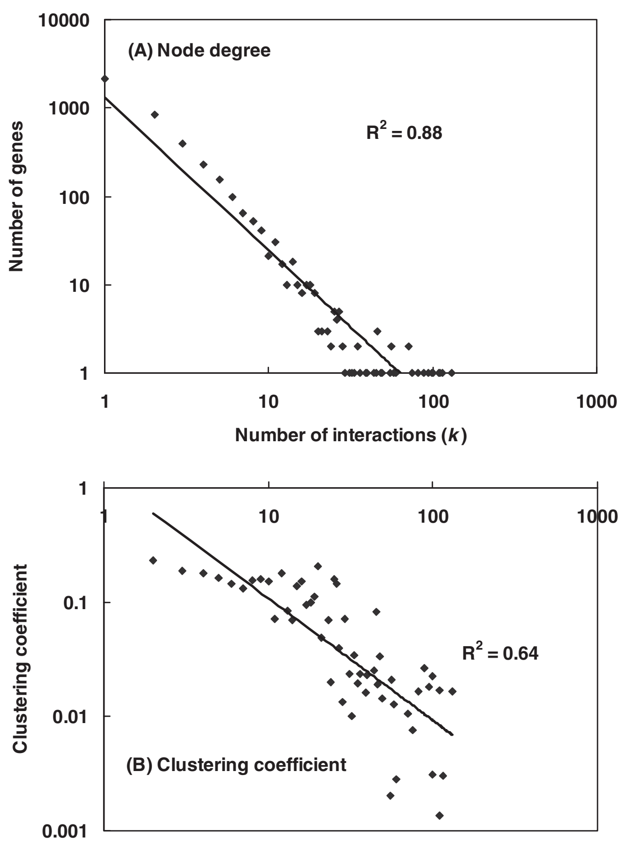

Fig. 1. The $P(k) - C(k)$ analysis for human protein–protein interactions. (**A**) Number of genes plotting against the number of interactions (links), showing a power-law of $P(k)$. (**B**) (Mean) clustering coefficient plotting against the number of interactions (links), showing a power-law of $C(k)$.

Taking average over 14 network data sets, we obtained $\lambda/g = 0.91 \pm 0.11$ and $\mu/g = 0.48 \pm 0.04$, respectively. We interprete these data as follows: When the network increases with two nodes (genes), on average, two new links are created and one existed link is lost.

DISCUSSION

We have developed the BBA model for gene network evolution, providing a unifying explanation for the origin of modularity and scale-free property. That is, the scale-free property is the result of preferential new-link origins, while the modularity is the result of random-loss of existed links. Given the complexity of gene networks, our BBA model reveals the connection between network features and basic evolutionary mechanisms that can be potentially tested by high-throughput genomics data.

Gene Network Growth

The BBA model assumes that the growth of gene network (through gene duplications) is a constant. Therefore, the number of genes in a genome increases linearly with the evolutionary time. This assumption is only the first-order approximation for the trend of long-term gene network evolution. For an evolutionary lineage during a particular geological time period, however, the size of gene network can be either static, reduced, or expanded (Gu and Zhang, 2004). In this sense, like the original BA model, our BBA model only represents a theoretical theme of gene network evolution from simple life forms to advanced organisms.

Attachment Preference and Random-Loss of Links

The central premise of the BA model is the attachment preference for new links. Under the BBA model, we tentatively demonstrate its biological interpretations. If the new link is through the process of developmental rewiring, the attachment preference means that rewiring is to enhance the existed major developmental pathways. Though new genes were mainly from gene duplications, follow-up functional divergence, either subfunctionalization (complementary loss of links between two duplicate copies), or neo-functionalization (creating some new links) cannot directly result in the scale-free property (Pastor-Satorras et al., 2003; Wu and Gu, 2005). We suggest that neo-functionalization after gene duplications may tend to have a new link to an existed molecular pathway characterized by some hubs. In contrast, subfunctionalization resulting in the loss of existed links may be at random without any preference. In fact, random link-losses without double-loss in duplicate genes can effectively mimic the DDC process (duplication, degeneration, and complementation).

The Random-Loss Links and the Erosion Model of Modularity

We have realized the intrinsic relationship between our BBA model (the random-loss mechanism of existed links) and the erosion model of differential elimination of pleiotropic effects (Wagner and Mezey, 2004). Under BBA, emergence of modularity is the result of random link-loss (reducing the gene pleiotropy), while hubs are formed by the attachment preference. What is the difference between these two models is that the erosion model implies a transition from a relatively uniform interaction to modularity by differential link-losses, while the BBA model suggested a gradual process for the origin of modularity in evolution. Further research will be focused on how to test these theoretical models. For instance, we recently developed a statistical method for estimating gene pleiotropy from protein sequence analysis (Gu, 2007), which may provide some insights for the relationship between modularity, pleiotropy, and network complexity.

The 2-2-1 Hypothesis

The pattern for gene network evolution we found can be concisely called the 2-2-1 hypothesis, which is short for two new genes, two new links, and one lost link, may articulate the evolutionary pace of gene networks. One may argue that this hypothesis may lead to many isolated nodes in the network. In practice, however, it will not happen if new genes have been mostly generated through gene duplications. Moreover, we found that two-degree components that characterize the power-law (γ) and modularity (θ) are influenced by the same underlying evolutionary mechanisms. Hence, when the relative gain rate of links $\lambda/g \approx 1$, one can easily shown from Equations (4) and (8), the following evolutionary constraint of the scale-free property and modularity:

$$\gamma + \theta \approx 3 \tag{10}$$

Hence, our model is helpful to have a coherent understanding of the emergence of gene network and functional organization from the insight of evolutionary biology.

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APPENDIX
DERIVATION OF EQUATION (4) UNDER THE BBA MODEL

Our study starts from the original BA model (Barabasi and Albert, '99). That is, the evolutionary mechanism for a gene network can be described as follows: (1) *Gene Network Growth*: We assume that at the dawn of the origin of life, there was a small number (n_0) of nodes (RNA/DNA/proteins) to form the most preliminary genetic network. These initial connections, in spite of small numbers (I_0), should be very fundamental for the meaning of life. Since then, as new genes are continuously added into the genome, new connections have been created with a rate of λ between new and existed genes. (2) *Gene Connectivity Preference:* The rule of ''rich-gets-richer'' means that a new node may have a higher probability to be linked to a node that already has a large number of connections.

The original BA model results in a fixed power-law degree, i.e., $\gamma = 3$. We have noticed that the BA model assumes that a connection in the gene network cannot be lost once it is generated. In fact, loss of connections between nodes in a gene network (or more generally, loss-of-function) has played an important role during the course of evolution. Therefore, we make the following additional assumption. (3) *Random-Loss of Connections*:

Let x_i be the number of interactions of node i at time t. Under the assumptions (1)–(3), the original BA model can be extended as the following differential equation

$$\dot{x}_i = \lambda(1-\omega)\pi(x_i,t) + \frac{\omega\lambda f}{n(t)} - \frac{\mu}{n(t)} \tag{A1}$$

where $\dot{x} = dx_i/dt$, λ and μ are the growth rate and loss rate of interactions, respectively, the probability $\pi(x_i,t)$ corresponds to the preference of node i to be linked at time t, and $n(t)$ is the total number of genes (nodes) in the gene network at time t. The constant ω $(0 \le \omega \le 1)$ represents the chance for the random selection of node i to be interaction-generator; here we set $\omega = \frac{1}{2}$.

Barabasi and Albert ('99) suggested a linear function for $\pi(x_i,t)$ to measure the preferential attachment, i.e.,

$$\pi(x_i) = \frac{x_i}{\sum_{j=1}^{n(t)} x_j} = \frac{x_i}{2I(t)} \tag{A2}$$

where $I(t)$ is the total number of interactions in the gene network at time t. Because an interaction between two genes (nodes) will be counted twice, we have $2I(t) = \sum_{j=1}^{n(t)} x_j$. From Equation (10), one can easily verify that $2\dot{I} = \sum_{i=1}^{n(t)} \dot{x}_i$, leading to $\dot{I} = (\lambda - \mu)/2$. Moreover, the simplest evolutionary model for the network growth is given by $\dot{n}(t) = g$, where g is the origin rate of a new node. Together, the growth of gene network and the growth of network interactions are both linear with respect to the evolutionary time, that is

$$n(t) = gt + n_0$$

$$I(t) = (\lambda - \mu)t/2 + I_0 \tag{A3}$$

where n_0 and I_0 are the initial numbers of genes and interactions, respectively. Because the initial numbers n_0 and I_0 are usually small, they can be neglected for the long-term evolution (large t). Thus, combining Equations (A1)–(A2), Equation (10) can be expressed as follows:

$$\dot{x}_i = \frac{x_i}{bt} + \frac{h}{t} \tag{A4}$$

where $b = 2(1-\mu/\lambda)$ and $h = (\lambda/2 - \mu)/g$.

The solution of Equation (A3) relies on the corresponding initial condition: when gene (node) i was generated at the evolutionary time τ_i, this newly born gene i had x_0 interactions, i.e., $x_i(\tau_i) = x_{0,i}$, leading to the result $x_i(t) = (x_{0,i} + a)(t/\tau_i)^{1/b} - a$, where $a = bh$. It indicates that the expected number of interactions of a gene increases with the evolutionary time t. In other words, given the current time-point by $t = T$, the expected interaction difference between genes is determined by their difference in age. Because the gene age is measured by the initial time-point τ_i, the frequency of gene connectivity, $P(k)$, can be derived from the age distribution of τ_i. To this end, we rewrite the solution of Equation (A3) as follows:

$$\tau_i = \left[\frac{x_{0,i} + a}{x_i + a}\right]^b T \tag{A5}$$

In fact, the linear growth of gene network, as illustrated by Equation (3), means that genes are added constantly to the network during the course of evolution from the time period $[0, T]$, implying a uniform distribution of gene age (τ_i), that is, for any given r $(0<r<1)$, we have $P(\tau_i < rT) = r$. Consider the probability of a node i whose interactions are smaller than a given number k, $P(x_i < k)$. From Equation (A4), one can verify that $x_i < k$ means $\tau_i > [(x_{0,i} + a)/(k+a)]^b T$. Under the assumption of uniform distribution for τ_i in the time period $[0, T]$, we obtain

$$\begin{aligned} P(x_i < k) &= P\left(\tau_i > \left[\frac{x_{0,i}+a}{k+a}\right]^b T\right) \\ &= 1 - P\left(\tau_i < \left[\frac{x_{0,i}+a}{k+a}\right]^b T\right) \\ &= 1 - \left[\frac{x_{0,i}+a}{k+a}\right]^b \end{aligned} \tag{A6}$$

It follows that the probability $P(k) = P(x_i < k+1) - P(x_i < k)$ can be approximately obtained by $P(k) \sim \partial P(x_i < k)/\partial k$, resulting in a general form of the power-law, $P(k) \sim (k+a)^{-\gamma}$, with the power-law degree $\gamma = b + 1$

$$\gamma = 3 - 2\mu/\lambda \tag{A7}$$

where $a = bh$ is called the shift factor; when $a = 0$, it is reduced to the canonical form.

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RESEARCH ARTICLE

Network Modules of the Cross-Species Genotype-Phenotype Map Reflect the Clinical Severity of Human Diseases

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Abstract

Recent advances in genome sequencing techniques have improved our understanding of the genotype-phenotype relationship between genetic variants and human diseases. However, genetic variations uncovered from patient populations do not provide enough information to understand the mechanisms underlying the progression and clinical severity of human diseases. Moreover, building a high-resolution genotype-phenotype map is difficult due to the diverse genetic backgrounds of the human population. We built a cross-species genotype-phenotype map to explain the clinical severity of human genetic diseases. We developed a data-integrative framework to investigate network modules composed of human diseases mapped with gene essentiality measured from a model organism. Essential and nonessential genes connect diseases of different types which form clusters in the human disease network. In a large patient population study, we found that disease classes enriched with essential genes tended to show a higher mortality rate than disease classes enriched with nonessential genes. Moreover, high disease mortality rates are explained by the multiple comorbid relationships and the high pleiotropy of disease genes found in the essential gene-enriched diseases. Our results reveal that the genotype-phenotype map of a model organism can facilitate the identification of human disease-gene associations and predict human disease progression.

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Introduction

Disease mortality provides important information for the assessment of the clinical severity of a disease within a patient population. Doctors use the mortality information about a specific disease to decide whether to hospitalize patients or improve prevention and early intervention [1]. Diseases with high mortality rates need to be carefully controlled for the overall public health of a country. Policymakers have utilized mortality information to allocate health resources and identify individuals who need urgent health care [2].

The identification of genetic variations associated with clinically severe diseases would allow the prediction of life expectancy. However, we do not have a clear understanding of the genotype-phenotype relationship to predict genetic variations associated with disease mortality. Recent advances in genome sequencing techniques have enabled the identification of genetic variants within an individual. However, the discovery of genetic variations associated with human disease is difficult due to the diverse genetic background of the human population [3,4]. Disease-associated genes detected on the basis of population statistics can explain only a small proportion of disease heritability. Moreover, the detailed molecular mechanisms underlying diseases cannot be studied directly in human subjects due to ethical reasons [3,5,6]. Therefore, the genotype-phenotype map of a model organism could play a complementary role to human population studies, because the genetic background in model organisms can be controlled through the breeding of isogenic lines.

Although model organisms are used frequently for human disease studies, the phenotypic relevance of model organism genes that are orthologous to human disease genes remains unclear [7]. In particular, mutations in an orthologous pair of genes do not always exhibit similar phenotypes in different species [8,9]. For instance, the knockout of *Hprt* exhibits no observable phenotypes in mice, whereas mutations in the human ortholog of *Hprt*, *HPRT1*, cause Lesch-Nyhan syndrome, whose symptoms include the overproduction of uric acid, nervous system impairment, and self-injurious behavior [10].

Modular network analysis of the cross-species genotype-phenotype relationship is helpful to understand the mechanisms underlying human disease progression. It has been suggested that disease phenotypes can be caused by alterations in several genes within a module rather than a single gene mutation [11]. In this concept, a disease module represents a group of disease-associated genes involved in similar phenotypes as well as having molecular connections, such as co-expression, protein interactions, metabolic pathways, co-localizations, and evolutionary constraints [12–14]. Therefore, equivalent phenotypes between different species, called phenologs, might arise from the conservation of functionally related modules that are composed of highly interconnected groups of genes in gene networks [15,16].

Here, we have taken a data-integrative approach to understand the molecular reasons underlying the clinical severity of human diseases. To understand the genotype-phenotype relationship of human diseases, we mapped gene essentiality information from the model organism to human disease genes via their orthology relationship. We discovered that essential and nonessential genes connect diseases in different types, and form clusters in the human disease network. Using a large patient population study, we found that essential gene-enriched disease classes exhibited a higher mortality rate and clinical severity than disease classes enriched with nonessential genes. Moreover, high mortality rate of essential gene-enriched diseases is associated with the high comorbidity of multiple diseases. We found that clinically severe pathological symptoms may be associated with the pleiotropy and high network degree of essential genes in the protein interaction network. Our results suggest that a module-based approach based on the genotype-phenotype map may facilitate the understanding of the progression of human diseases from genetic variation.

Results

Relationship between gene essentiality and human diseases

To investigate whether gene essentiality in a model organism is related to human disease phenotypes, we mapped essential mouse genes to human ortholog disease genes (Fig 1; see *Methods*). We selected 2,526 essential genes from mutant gene phenotypes in the Mouse Genome Informatics database (MGI) [17]. We used the mouse as a model organism to define

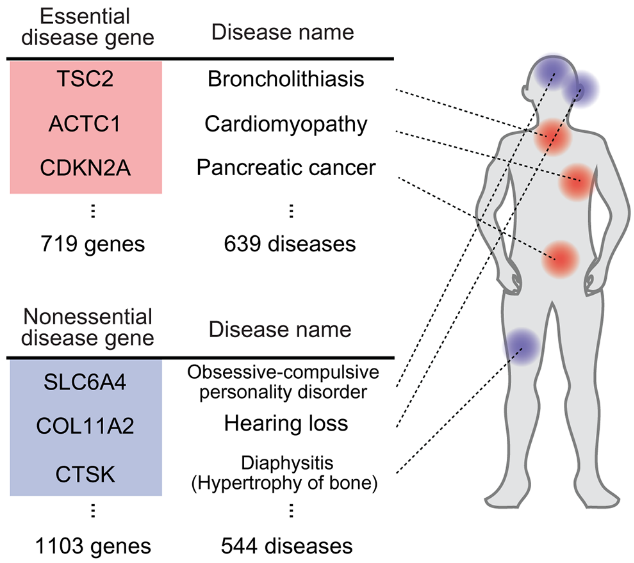

Fig 1. Gene essentiality and human disease genes. Mapping mouse essential and nonessential genes to human disease genes through gene orthologs.

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gene essentiality because mutant mouse gene phenotypes are well-annotated from genome-wide analyses of gene deletion models such as knockouts [18]. A gene is considered essential if a mutation of the gene causes a lethal phenotype, which is defined as a developmental failure or a lifespan of less than 50 days [8]. We mapped the essential/nonessential mouse genes to human disease genes in the Online Mendelian Inheritance in Man database (OMIM). Using an ortholog mapping between human and mouse genes, a total of 1,822 disease genes are orthologous with mouse genes. The disease genes are mapped to 713 essential and 1,103 nonessential genes (Fig 1). This procedure enabled us to classify human disease-associated genes that are orthologous to mouse essential and nonessential genes. We categorized these genes as ***essential*** and ***nonessential disease genes***, respectively. These essential and nonessential disease genes are listed in S1 Table.

Modular organization of the genotype-phenotype map of human disease based on gene essentiality

We found that essential/nonessential disease genes organize human diseases into a modular structure (Fig 2A). To investigate how the genotype-phenotype relationship organizes disease phenotypes based on gene essentiality, we constructed a human disease network (HDN) linked by essential or nonessential disease genes. HDN nodes represent diseases and links connect them if two diseases have any shared genetic origin [12]. HDN links were classified into essential or nonessential links when the shared genes of disease pair are exclusively essential or nonessential disease genes. HDN links were classified 'others' when the shared genes of disease pair include both essential and nonessential disease genes. We supplied the list of essential/nonessential links and shared genes in the HDN (S2 Table).

We found that clusters of diseases are enriched with either essential or nonessential links (Fig 2B). For example, in a disease cluster composed of cancer diseases (panel I), most diseases in the cluster have essential links between them. Indeed, among the genes associated with the disease cluster, 18 of the 23 are essential. Also, the associated genes include *BRCA1*, *BRCA2*, *RAD51* and *TP53*, which are strongly associated with breast cancer [19]. In a disease cluster

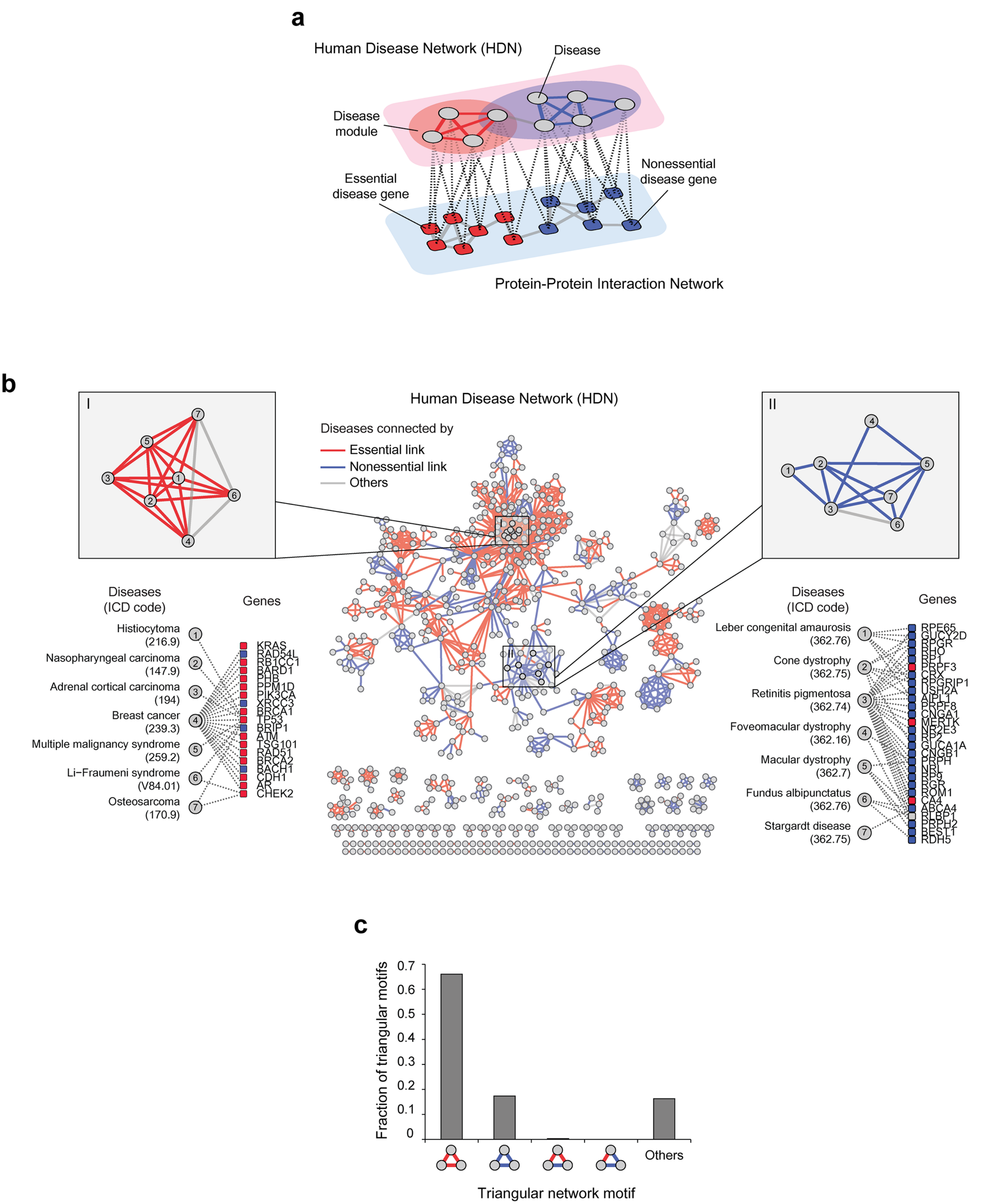
a
Human Disease Network (HDN)
Disease
Disease module
Essential disease gene
Nonessential disease gene
Protein-Protein Interaction Network
b
Human Disease Network (HDN)
Diseases connected by
Essential link
Nonessential link
Others
I
II
Diseases (ICD code)
Genes
Histiocytoma (216.9)
Nasopharyngeal carcinoma (147.9)
Adrenal cortical carcinoma (194)
Breast cancer (239.3)
Multiple malignancy syndrome (259.2)
Li−Fraumeni syndrome (V84.01)
Osteosarcoma (170.9)
KRAS
RAD54L
RB1CC1
BARD1
PHB
PPM1D
PIK3CA
XRCC3
BRCA1
TP53
BRIP1
ATM
TSG101
RAD51
BRCA2
BACH1
CDH1
AR
CHEK2
Diseases (ICD code)
Genes
Leber congenital amaurosis (362.76)
Cone dystrophy (362.75)
Retinitis pigmentosa (362.74)
Foveomacular dystrophy (362.16)
Macular dystrophy (362.7)
Fundus albipunctatus (362.76)
Stargardt disease (362.75)
RPE65
GUCY2D
RPGR
RHO
RP1
PRPF3
CRX
RPGRIP1
USH2A
AIPL1
PRPF8
CNGA1
MERTK
NR2E3
RP2
GUCA1A
CNGB1
PRPH
NRL
RP9
RGR
ROM1
CA4
ABCA4
RLBP1
PRPH2
BEST1
RDH5
c
Fraction of triangular motifs
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Others
Triangular network motif

Fig 2. Mapping essential and nonessential disease genes to the Human Disease Network (HDN). (a) The modular architecture of human diseases and their gene essentiality in the HDN. (b) The HDN with the gene essentiality of shared genes is highlighted. Essential/nonessential/other links are colored in red, blue and gray, respectively. Panels I and II show examples of essential and nonessential disease clusters, respectively. (c) The fraction of triangular network motifs connected by essential or nonessential genes. Others are triangular network motifs, where both essential and nonessential disease genes connect minimum two diseases in the network motifs.

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composed of ophthalmological diseases (panel II), most diseases in the cluster have nonessential links. In this cluster, 24 of the 28 associated-genes are nonessential and include *RP1*, *RPO*, and *RPGR*, which are well-known genes to be associated with sensory perception for light stimuli [20].

Next, we quantified the modularity of the HDN linked by essential/nonessential disease genes. Specifically, we measured the essential and nonessential links in the network motifs. We found that most network triangles were comprised exclusively of either essential or nonessential links (approximately 80% of the total) (Fig 2C). The triangular network motif is a basic component of network clusters and the smallest network motif that comprises a complete subgraph [21]. To validate if the pattern of network motifs is affected by gene essentiality, we compared the fraction of network motifs with random control. Gene essentiality information is randomly shuffled across disease genes. We confirmed that the observed fractions of network motifs connected by essential links are indeed higher than expected (S1 Fig).

Gene essentiality and the clinical severity of disease classes

Gene essentiality configures the genotype-phenotype map of human diseases in a modular manner, as shown in Fig 2. Thus, we transferred gene essentiality onto the disease classes, because disease classes are clustered in human disease network and grouped in terms of phenotypic similarity based on the affected physiological system [12]. We discovered that disease classes are biased toward the enrichment of essential or nonessential genes. Specifically, genes in different disease classes are differentially enriched in either essential or nonessential disease genes (Fig 3A). Genes in cancer, cardiovascular, endocrine, developmental, respiratory, and gastrointestinal disease classes as well as diseases that involve multiple systems are enriched with essential disease genes ($P < 0.05$ in the hypergeometric distribution). In contrast, genes in ear-nose-throat, connective tissue, ophthalmological, psychiatric, and immunological disease classes are enriched with nonessential disease genes ($P < 0.05$ in the hypergeometric distribution).

We investigated disease mortality in patients who carried diseases from disease classes enriched with essential or nonessential disease genes (see *Methods*). We found that diseases enriched with essential genes were associated with a higher crude death rate than diseases enriched with nonessential genes (Fig 3B). The crude death rate for a specific disease was based on an analysis of the number of deaths per 100,000 individuals [22]. Among the six disease classes that were identified to be enriched with essential genes, five disease classes (cardiovascular, cancer, respiratory, gastrointestinal, and endocrine) were associated with a high crude death rate. Developmental diseases, which were also identified to be enriched with essential genes, were associated with low crude death rates. However, only live births were included in the calculation of death rates, which introduces a selection bias because many cases that could have led to death may have been aborted at the fetal stage [22]. The total crude death rates due to developmental diseases may therefore be much higher than current studies suggest.

We also found that diseases enriched with essential genes have higher case fatality rates than diseases enriched with nonessential genes. Case fatality rate is the proportion of patients that die from a particular disease within a specified period of time (see *Methods*). For example,

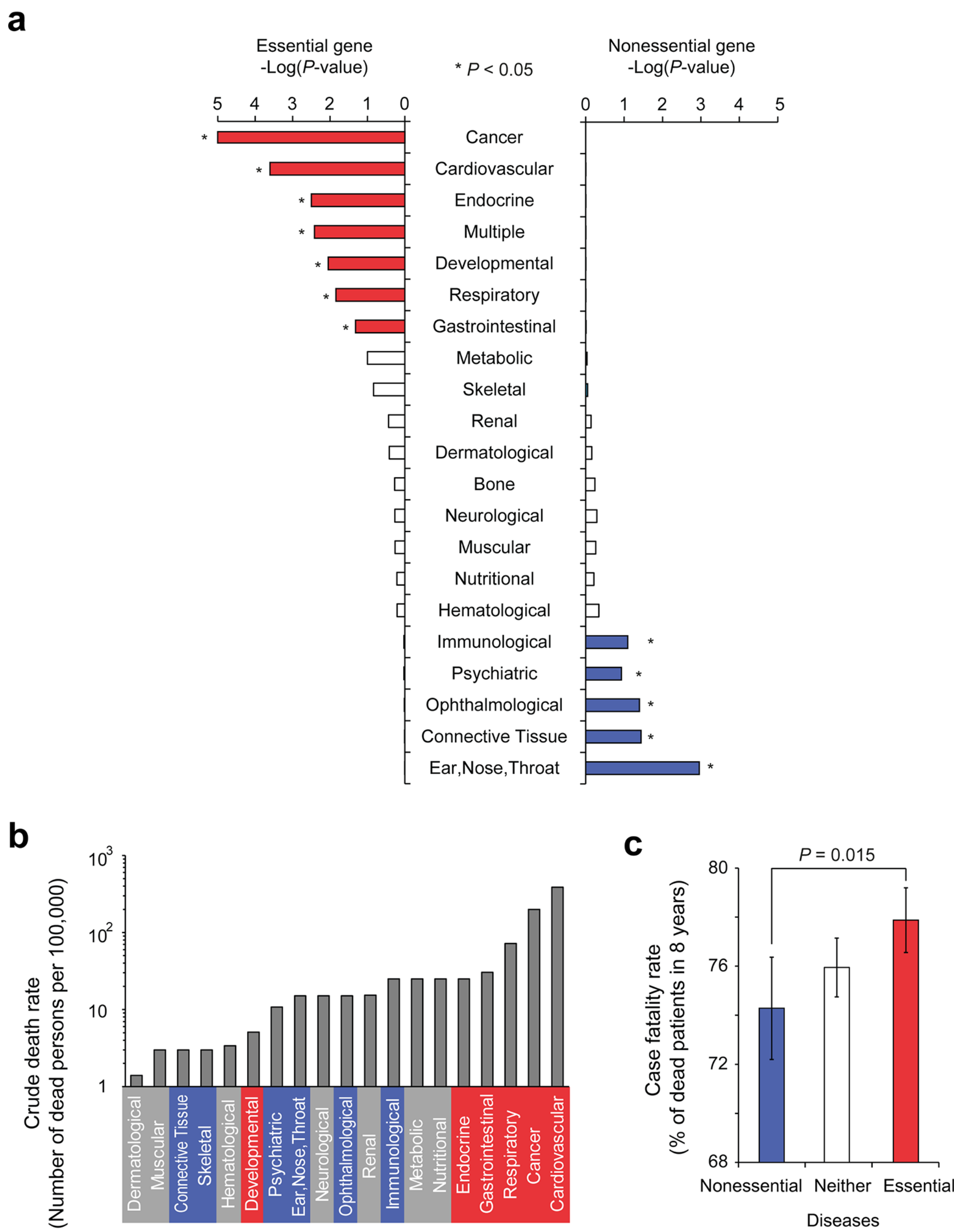

Fig 3. The mortality of human disease classes according to gene essentiality. (a) Human disease classes sorted by gene essentiality. The enrichment of essential and nonessential genes in 21 human disease classes is shown. * indicates $P < 0.05$ (Hypergeometric P-values). (b) The crude death rate of different disease classes. The crude death rate is calculated as the number of deaths reported each calendar year per 100,000 individuals. Disease classes that are enriched with essential and nonessential genes are colored in red and blue, respectively. (c) The case fatality rate of diseases differentially enriched with essential and nonessential genes. The case fatality rates of diseases enriched with neither essential nor nonessential genes are also shown as "Neither".

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patients with essential gene-enriched diseases were more likely to be deceased within 8 years of the initial diagnosis (Fig 3C). According to Medicare records, on average 76% of patients were deceased within 8 years for all disease types. However, essential gene-enriched diseases were associated with a higher clinical severity than nonessential gene-enriched diseases ($P = 0.015$, Mann-Whitney *U* Test). The list of essential and nonessential gene-enriched diseases and their clinical severity is provided in S3 Table. These results suggest that the gene essentiality map of a model organism can be used to predict the clinical severity of human diseases.

We found that the diseases enriched with essential genes have higher case fatality rates (Fig 3C). Among them we discovered cancers and cardiovascular diseases, which are often manifested in elderly patient group. Because we used medical claims associated with elderly patients ($\geq$ age 65), one might ask whether the fatality rate data could have a bias toward disease classes frequently found in elderly patients. Therefore, we tested the potential bias in the dataset by counting the number of the patients from the disease classes enriched in essential/nonessential genes. We confirmed that the number of patients do not have bias towards diseases associated with essential or nonessential genes (S2 Fig; $P = 0.23$, Mann-Whitney *U* Test). Furthermore, the conclusion is reconfirmed by the crude death rate which comes from a different data source (Fig 3A). Thus, we believe that the association between disease fatality and gene essentiality to be true, although the case fatality data should be interpreted with care.

Essential gene-enriched diseases are associated with high comorbidity and pleiotropy

Why do essential genes tend to associate with mortal diseases? The high disease mortality in essential gene-enriched diseases also may be due to comorbid diseases. Disease comorbidity is the co-occurrence of other diseases with a primary disease. It has been shown that the patients affected by disease having many comorbid diseases tended to die sooner [23].

We therefore investigated the relationship between comorbidity and gene essentiality. We found that diseases enriched with essential genes have a higher number of comorbid disease pairs compared to diseases enriched with nonessential genes (Fig 4A; $P = 8.12\times10^{-5}$, Mann-Whitney *U* test). To quantify the number of comorbid disease pairs, we counted the number of diseases that co-occurred with a particular disease compared to random expectation in a patient population (see *Methods*). This result suggests that diseases caused by mutations in essential genes are more likely to progress to clinically severe conditions with multiple comorbid diseases.

Next, we analyzed disease pleiotropy in human genetic diseases as another potential cause of multiple pathological symptoms. Genes with disease pleiotropy are associated with two or more genetic disorders and may cause multiple pathological defects due to the involvement of more diverse cellular functions [24]. We quantified the genes that exhibited disease pleiotropy and found that essential genes tend to exhibit disease pleiotropy more frequently than nonessential genes (Fig 4B; $P = 2.58\times10^{-6}$, Fisher's exact test). High comorbidity and disease pleiotropy suggest that essential genes connect a larger number of disease phenotypes via molecular connections than nonessential genes.

If a gene has more connections in the cellular network (*i.e.*, the gene is a hub gene), then its perturbation tends to result in the disconnection of multiple cellular function, which leads to disease pleiotropy. Jeong et al. [25] previously reported that essential genes in another model organism, *Saccharomyces cerevisiae* (yeast), have more PPI partners than nonessential genes. We expanded this analysis to examine the PPIs of human disease genes, and found that essential disease genes have more PPI partners than nonessential disease genes (Fig 4C; $P = 3.18\times 10^{-14}$, Mann-Whitney *U* Test). The higher pleiotropy and connectivity of essential disease genes supports the observation that perturbations of these genes cause more lethal effects in the cellular network and explains the clinical severity of essential gene-enriched diseases.

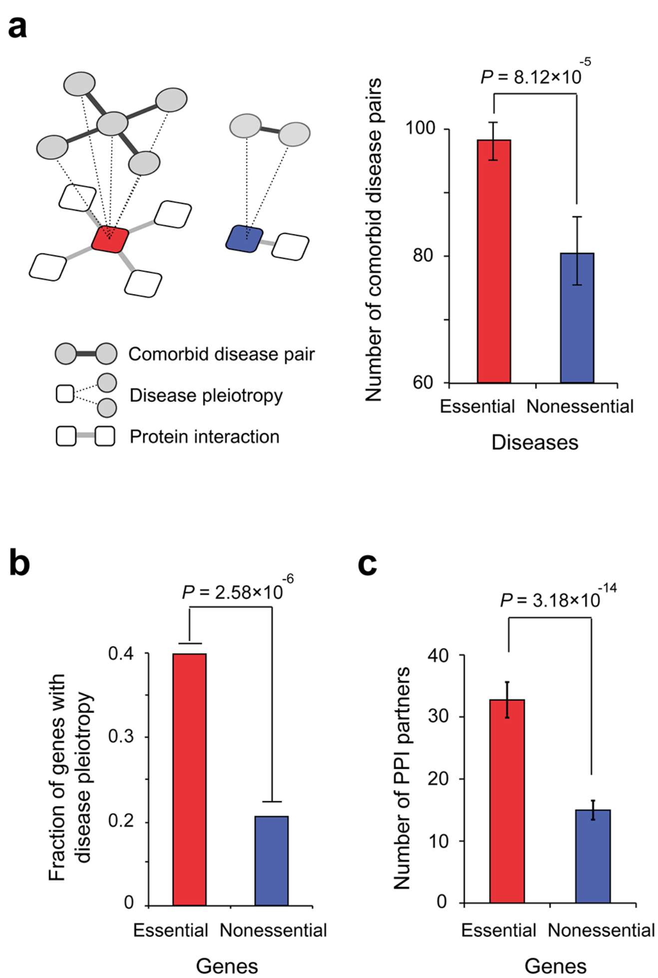

Fig 4. The comorbidity and pleiotropy of essential and nonessential gene-enriched diseases. (a) The comorbid disease pairs of essential and nonessential gene-enriched diseases. (b) The fraction of pleiotropic diseases connected by essential and nonessential genes. (c) The number of PPI partners of essential and nonessential disease genes.

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Discussion

We found that gene essentiality derived from the mouse model organism is correlated with the clinical severity of human diseases despite the evolutionary distance between mice and humans. Our results suggest that a module-based approach using genotype-phenotype mapping of a model organism may provide a better understanding of the genetic variations that lead to human diseases. Specifically, human disease classes tend to be more clinically severe if their associated genes are enriched in essential genes (Figs 2 and 3).

A modular architecture was found in the genotype-phenotype map of human diseases along with gene essentiality (Fig 2). Although we utilized gene-to-gene orthology to map the mouse essential genes to human disease genes, the observed modular architecture of human diseases might be rooted in a conservation of modules in the genotype-phenotype map in evolution. The modules in the genotype-phenotype map are known to have emerged during the course of evolution [26,27] because the modular architecture may reduce the potential deleterious effects of mutations, which otherwise may spread and threaten the survival of the organism [27,28].

We found that nonessential gene-enriched diseases were less clinically severe than essential gene-enriched diseases (Fig 3). This result indicates that nonessential gene-enriched diseases may affect quality of life rather than mortality. This observation implies that it may be necessary to systematically screen phenotypes in the model organism in adulthood to better understand disease phenotypes. For example, the serotonin neurotransmitter transporter *SLC6A4* is a mouse nonessential gene, and mutations to the human ortholog of this gene cause obsessive-compulsive disorder through a defect in the regulation of serotonin levels [29]. Only an extensive screening of behavior enabled the detection of a phenotype in which the *SLC6A4* knockout mouse exhibited obsessive behavior, including continuous and progressive grooming. A mutation in another mouse nonessential gene, *FAM107B*, also was recently reported to display a deafness phenotype. Extensive screening of the signal to the brain in response to auditory stimulation was necessary to detect such a phenotype [30]. Currently, several mouse phenotypes are being systematically screened, and diverse phenotypes in the adult stage of model organisms have been identified [31–33]. Therefore, we anticipate that our approach can be expanded toward understanding diseases that affect quality of life.

Methods

Essential and nonessential gene data set

Essential and nonessential genes were compiled from the mutant phenotypes listed in the Mouse Genome Informatics database (www.informatics.jax.org, downloaded in 2012) [17]. These mutant phenotypes were documented from knockout, trapping, or random mutagenesis studies. Genes were classified as essential genes if the mutant phenotype exhibited a severe effect, namely "embryonic lethality" (MP: 0002080), "prenatal lethality" (MP: 0002081), "postnatal survival lethality" (MP: 0002082), "abnormal reproductive system morphology" (MP: 0002160), or "abnormal reproductive system physiology" (MP: 0001919). The remaining genes in the mouse genome were classified as nonessential genes. A total of 2,526 essential and 15,014 nonessential genes were identified from the mouse.

Comparative genomic analysis between humans and mice

The orthologous relationship between human and mouse genes was predicted by the sequence homology. The human and mouse genomes were curated from the National Center for Biotechnology Information NCBI36 and NCBIM36 databases. The mouse orthologs, including sequences, of the human genes were extracted using EnsemblCompara GeneTrees [34], which is downloaded in 2012 via Biomart (http://www.biomart.org). Consequently, one-to-one orthologs of 14,223 genes were detected between humans and mice.

Human disease genes and phenotypes

Human gene and disease phenotype associations were curated from the OMIM database (http://www.ncbi.nlm.nih.gov/omim/, 2009 version) [35]. The OMIM database provided gene-disease associations between the 2,929 disease types in the Morbid Map and 1,777 disease-associated genes. According to Goh et al [12], disease subtypes were combined into a single disease based on a string match of disease annotations because some disease types have minor differences in their names. A total of 1,228 unique diseases were extracted from 2,161 disease annotations [12]. Human diseases were classified by Goh et al [12] into 21 different disease classes based on the physiological systems associated with each disease. The disease classification is based on the phenotype level because this type of classification has been shown to discriminate phenotypic similarities and differences based on disease symptoms [36].

Mortality rate from human patient population studies

The case fatality rate of a disease was measured as the percentage of patients who were deceased within 8 years from the initial diagnosis of the disease. According to Hidalgo et al [23], the disease progression data were compiled from 13,039,018 patients in MedPAR, a database that includes elderly Americans aged 65 or older who were enrolled in the Medicare program from 1990 to 1993. The crude death rate was measured as the number of patients deceased from that disease per 100,000 individuals in the U.S. population. The crude death rate data were extracted from the compressed mortality data from 1979 to 1998 with ICD-9 codes (http://wonder.cdc.gov/cmf-icd9.html) reported by the U.S. Centers for Disease Control and Prevention (CDC) [22].

Identification of comorbid disease pairs

The number of comorbid disease pairs was quantified by relative risk (*RR*). If a disease pair had an $RR \geq 2$, then the pair was counted as a comorbid disease pair. $\mathrm{RR} \geq 2$ was known as a baseline of comorbidity value whose disease pairs are linked in the HDN [37]. The *RR* quantifies the co-occurrence of two diseases compared to random expectation in the human patient population. The *RR* of diseases *i* and *j* is given by:

$$RR = \frac{C_{ij}}{C^{*}_{ij}} \tag{1}$$

where C_{ij} is the number of patients who had both disease *i* and disease *j*, and C_{ij}^{*} is equal to $I_i I_j / N$, which represents random expectation. I_i is the incidence of disease *i*. *N* is the total number of patients (13,039,018) in the Medicare record.

Construction of the PPI network

We compiled human protein interactions from a total of 22 existing protein interaction databases: the Bio-molecular Interaction Network Database (BIND) [38], the Human Protein Reference Database (HPRD) [39], the Molecular Interaction database (MINT) [40], the Database of Interacting Proteins (DIP) [41], IntAct [42], BioGRID [43], Reactome [44], the Protein-Protein Interaction Database (PPID), BioVerse [45], CCS-HI1 [46], the Comprehensive Resource of Mammalian protein complexes (CORUM) [47], IntNetDB [48], the Mammalian Protein-Protein Interaction Database (MIPS) [49], the Online Predicted Human Interaction Database (OPHID) [50], Ottowa [51], PC/Ataxia [52], Sager [53], Transcriptome Complex [54], Unilever, a protein-protein interaction database for PDZ domains (PDZBase), and a protein interaction data set from the literature [55]. We removed low-confidence interactions that were not supported by direct experimental evidence. The final network included 101777 interactions between 11,043 human proteins. We supplied the integrated PPI network (S4 Table).

Supporting Information

S1 Fig. The distribution of triangular network motifs connected by (a) all essential link, (b) nonessential link, (c, d) both essential and nonessential link and (e) others with random control.
(TIF)

S2 Fig. Comparison of the number of patients of (a) diseases associated with essential or nonessential genes (b) diseases class enriched with essential, nonessential or neither gene.
(TIF)

S1 Table. The list of essential and nonessential disease genes.
(XLS)

S2 Table. The list of essential/nonessential links and shared genes in HDN.
(XLS)

S3 Table. The case fatality rate of human diseases enriched with essential and nonessential genes.
(XLS)

S4 Table. The list of protein interactions from integration of 22 protein interaction databases.
(TXT)

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Author Contributions

Conceived and designed the experiments: SKH SK. Performed the experiments: SKH JH. Analyzed the data: SKH SK. Wrote the paper: SKH IK SK.

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ANNUAL REVIEWS

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Patterns and Evolutionary Consequences of Pleiotropy

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Abstract

Pleiotropy refers to the phenomenon of one gene or one mutation affecting multiple phenotypic traits. While the concept of pleiotropy is as old as Mendelian genetics, functional genomics has finally allowed the first glimpses of the extent of pleiotropy for a large fraction of genes in a genome. After describing conceptual and operational difficulties in quantifying pleiotropy and the pros and cons of various methods for measuring pleiotropy, I review empirical data on pleiotropy, which generally show an L-shaped distribution of the degree of pleiotropy (i.e., the number of traits affected), with most genes having low pleiotropy. I then review the current understanding of the molecular basis of pleiotropy. In the rest of the review, I discuss evolutionary consequences of pleiotropy, focusing on advances in topics including the cost of complexity, regulatory versus coding evolution, environmental pleiotropy and adaptation, evolution of ageing and other seemingly harmful traits, and evolutionary resolution of pleiotropy.

1. INTRODUCTION

Pleiotropy refers to the phenomenon of one gene or one mutation affecting multiple traits. For example, mutations causing cystic fibrosis, primarily a lung disease, also cause male infertility (Sokol 2001). The phenomenon of pleiotropy had been recognized long before its formal definition in 1910 by the German geneticist Ludwig Plate (Plate 1910). For instance, some medical syndromes were known to have a suite of symptoms and a simple familial component (Pyeritz 1989, Stearns 2010). Another example is in the foundational paper of genetics in which Gregor Mendel reported the cosegregation of three traits in garden peas—brown seed coat, violet flowers, and axial spots versus white seed coat, white flowers, and no spots—and considered them to be controlled by a single locus (Mendel 1866). For two reasons, pleiotropy is commonly thought to be widespread. First, any organism has many more phenotypic traits than genes, so at least some genes are pleiotropic. Second, given the high prevalence of interactions among various parts of an organism and interactions among genes/mutations (Tyler et al. 2009, Zhang 2017), it is extremely unlikely that a mutation would affect only one trait.

Pleiotropy has significant implications for many areas of biology and is frequently invoked in theories and explanations of a broad spectrum of phenomena, including genetic disease (Carter & Nguyen 2011, Byars & Voskarides 2020), cancer (Rodier et al. 2007, Bien & Peters 2019), senescence (Williams 1957, Promislow 2004, Flatt & Partridge 2018), between-trait trade-offs (Guillaume & Otto 2012, Yang et al. 2014, Burmeister et al. 2020, Mauro & Ghalambor 2020), sexual conflict (Rice 1992, Bonduriansky & Chenoweth 2009, Innocenti & Morrow 2010), generalism versus specialism (Jerison et al. 2020, Bakerlee et al. 2021), cooperation (Foster et al. 2004, Bentley et al. 2022), evolutionary constraints on genes or traits (Barton 1990, Waxman & Peck 1998, Carroll 2005, He & Zhang 2006, Jiang & Zhang 2020), adaptation (Fisher 1930, Orr 2000, Otto 2004), neofunctionalization (Hughes 1994, Francino 2005, Bergthorsson et al. 2007, Hittinger & Carroll 2007), and speciation (Berlocher & Feder 2002, Kirkpatrick & Ravigne 2002, Rundle & Nosil 2002). For example, a mutation may improve one trait but worsen another, causing compromises among adaptations of different traits. It has been proposed that the above situation creates a cost for complexity—complex organisms are less adaptable than simple organisms because mutations are inherently more pleiotropic in the former than the latter (Fisher 1930, Orr 2000). Pleiotropy is the main theoretical reason behind the hypothesis that morphological evolution occurs more frequently via *cis*-regulatory sequence changes than protein sequence changes, because the former are thought to be less pleiotropic than the latter (Carroll 2005). It has been suggested that a number of disease mutations are maintained in the population because of their benefits to other aspects of life that are often hidden (Carter & Nguyen 2011).

For a long time, theoreticians have assumed universal pleiotropy, a notion that every gene/mutation affects every trait, despite the availability of only limited data on pleiotropy. Thanks to the development of high-throughput biology, the last two decades have witnessed a rapid accumulation of empirical data that allow the assessment of general patterns and various evolutionary consequences of pleiotropy. Here, I first review classifications of pleiotropy, conceptual and operational difficulties in measuring pleiotropy, methods for quantifying pleiotropy, empirical patterns of pleiotropy, the molecular basis of pleiotropy, and causal relationships among multiple effects of a mutation. I then review our understanding of some important evolutionary consequences of pleiotropy and discuss outstanding questions.

2. CLASSIFICATION OF PLEIOTROPY

Strictly speaking, pleiotropy is a characteristic of mutations. However, when the mutation of interest is a null mutation of a gene, which abolishes all functions of the gene, the mutational pleiotropy is also known as gene pleiotropy, a property of genes.

German biologist Hans Grüneberg was among the first to study the mechanistic basis of pleiotropy when researching a genetic skeletal abnormality in rats in the 1930s. He divided pleiotropy into genuine and spurious; genuine pleiotropy results from multiple products of a single gene locus, whereas spurious pleiotropy is caused by a single product that is used in multiple ways or initiates a series of events with multiple phenotypic effects (Grüneberg 1938). While Grüneberg's genuine and spurious categories are both considered true biological pleiotropy today, attempts to distinguish types of pleiotropy by different underlying causes have not stopped.

He & Zhang (2006) classified gene pleiotropy based on whether it is conferred by multiple molecular functions of the gene product (type I pleiotropy) or by multiple phenotypic consequences of a single molecular function (type II pleiotropy). A typical example of type I pleiotropy involves the mammalian serum albumin, which binds to fatty acids and toxic metabolites but is also involved in the oxidation of nitric oxide (Rafikova et al. 2002), a clear indication of multiple molecular functions. By contrast, the yeast gene HIS7 exhibits type II pleiotropy, because its product glutamine amidotransferase uses a single catalytic activity in both histidine biosynthesis and purine nucleotide monophosphate biosynthesis. Type I and type II pleiotropy loosely correspond to genuine and spurious pleiotropy, respectively, although He and Zhang emphasized the number of molecular functions of the gene product, while Grüneberg emphasized the number of gene products.

Hadorn (1961) distinguished mosaic from relational pleiotropy. Mosaic pleiotropy arises when a single locus separately affects multiple phenotypic traits, while relational pleiotropy arises when a single locus starts off a cascade of events influencing multiple traits, similar to Grüneberg's second form of spurious pleiotropy. Mosaic and relational pleiotropy were subsequently renamed horizontal and vertical pleiotropy, respectively (Tyler et al. 2009), two terms that are now commonly used in human genetics. For example, mutations truncating the human αB-crystallin (*CryAB*) gene are associated with both cataracts (Liu et al. 2006) and dilated cardiomyopathy (Inagaki et al. 2006) because the gene is expressed in both the eye and heart and performs apparently unrelated functions in the two organs; these mutations are said to exhibit horizontal pleiotropy. By contrast, the sickle cell mutation in the β-globin gene causes hemoglobin to polymerize when deoxygenated, deforming red blood cells into the sickle shape, which hinders red blood cells from flowing in capillaries and causes ischemia in organs and peripheral tissues (Tyler et al. 2009). This series of phenotypic effects follow directly from one to the next, so the sickle cell mutation is said to exhibit vertical pleiotropy.

A mutation exhibits synergistic or concordant pleiotropy if its phenotypic effects on two traits are in the same direction (typically in terms of fitness or its proxies) and antagonistic pleiotropy if they are in opposite directions. For example, a mutation that accelerates the sexual maturation of an animal but reduces its fecundity is antagonistically pleiotropic.

The same phenotypic trait measured in different environments is often regarded as different traits, especially in microbial studies (Dudley et al. 2005). Hence, a mutation with an effect on a trait in multiple environments can be considered pleiotropic, and such pleiotropy is referred to as environmental pleiotropy. Environmental pleiotropy is related to but does not equal gene by environment interaction (G×E), which refers to the phenomenon of a mutation exerting different effects on a trait in different environments (Wei & Zhang 2017, Li & Zhang 2018). Antagonistic environmental pleiotropy means that the mutational effects are in opposite directions in two environments, so antagonistic environmental pleiotropy is by definition G×E. By contrast, with concordant environmental pleiotropy, the mutational effects in two environments may or may not differ so may or may not exhibit G×E. Similarly, G×E may or may not imply environmental pleiotropy because the mutational effect could be nonzero in both environments or nonzero in only one environment.

When the phenotypic variations of two traits among genotypes are genetically mapped to the same genomic region, it is usually unknown (due to limited mapping resolutions) whether these variations are both caused by the same genetic polymorphism. In such cases, the genomic region is considered to exhibit statistical pleiotropy, which may or may not represent biological pleiotropy, depending on whether the multiple phenotypic effects are caused by the same polymorphism or not (Wagner & Zhang 2011, Hackinger & Zeggini 2017, Watanabe et al. 2019). In this review, unless otherwise mentioned, pleiotropy refers to biological pleiotropy.

In addition, because mutational effects may be environment dependent, mutational pleiotropy (excluding environmental pleiotropy) can also be environment dependent. For example, whether a mutation has concordant or antagonistic pleiotropic effects on the maximum growth rate and carrying capacity of yeast's logistic population growth is influenced by the growth medium (Wei & Zhang 2019).

3. QUANTIFYING PLEIOTROPY

3.1. Conceptual Difficulties

A phenotypic trait is an observable characteristic of an organism such as the height, weight, eye color, blood pressure, ABO blood type, oxygen saturation, mRNA concentration of a particular gene in a particular tissue, educational attainment, or personality of a human. An organism has almost an infinite number of traits, although many traits are correlated with one another, such as the lengths of the left and right arms of a human. In the most extreme case, two traits may be perfectly correlated (e.g., the height and twice the height of a tree). To quantify the pleiotropy of a mutation, we must measure the phenotypic effects of the mutation on multiple traits. What traits should be measured and counted when quantifying pleiotropy? Obviously, the height and twice the height of a tree should not be regarded as two traits in quantifying pleiotropy, but how about other traits that are highly correlated with one another without the correlation being as obvious as in the above example (e.g., the tree height and a complex mathematical function of the tree height)? Because of the above difficulty, pleiotropy is often said to be the phenomenon of a mutation affecting multiple seemingly unrelated traits, but what exactly are seemingly unrelated traits? Are the lengths of the left and right arms of a human seemingly related? What about the lengths of the left arm and left leg (or right leg)—are they seemingly related or not? Different people would have different answers. Because the correlation between traits can be anywhere from −1 to 1, it is difficult to objectively define whether two traits are seemingly unrelated. In practice, when quantifying pleiotropy, researchers generally include all traits measured, excluding those with a correlation of 1 or −1 (Wagner & Zhang 2011).

3.2. Operational Difficulties

Two operational difficulties in quantifying pleiotropy exist. First, there is no way to phenotype all traits of an organism, so it is impossible to quantify the pleiotropy of any mutation comprehensively. As a result, all estimates of pleiotropy are based on a limited set of traits. Furthermore, the traits examined in a mutant are often related to the trait initially found to be impacted by the mutation. For example, a heart-disease-causing mutation tends to be examined for potential effects on the heart or cardiovascular system. Pleiotropy estimated from such data is typically not comparable across mutations. Alternatively, a fixed set of traits may be investigated in many mutants, usually because these traits are relatively easy to phenotype. As a result, the estimated pleiotropy is commonly considered comparable for the involved mutations. However, a mutation may have detectable effects only on a group of related traits, for example, pertaining to a specific organ, so inclusion/exclusion of this group of traits in phenotyping drastically alters the estimated pleiotropy of the mutation. Because the traits phenotyped are usually such a small fraction of all traits, caution is needed in comparing pleiotropy among mutations.

Second, the phenotypic effect of a mutation on a trait may be too small to detect. Measuring the mutational effect on a trait means comparing the phenotypic trait between two genotypes that differ only by the mutation of interest. All measurements have errors; if the magnitude of a mutational effect is smaller than that of the measurement error, the mutational effect cannot be detected, causing an underestimation of pleiotropy. How sensitive need phenotyping methods be? Ideally, if we are concerned with the evolutionary fate of a mutation, a sensitive method should be able to detect a phenotypic effect on a trait (E, measured by the fractional change in trait value) that has a fitness effect (s) whose absolute value exceeds the magnitude of genetic drift, because the fate of the mutation would be influenced by natural selection arising from this phenotypic effect. The magnitude of genetic drift is one divided by the effective population size (N_e), which is $\sim 10^5$ for vertebrates, $\sim 10^6$ for invertebrates and land plants, $\sim 10^7$ for unicellular eukaryotes, and 10^8 or greater for free-living prokaryotes (Lynch et al. 2011). Let us assume that, when $|E|$ is small, $|s| = k|E|$, where k is a trait-specific parameter that may be referred to as trait importance (Ho et al. 2017); k is 100 times the fitness effect caused by a 1% change in trait value. Therefore, a sensitive phenotyping method should be able to detect a phenotypic effect size of $1/(kN_e)$. The greater the product of trait importance and effective population size, the higher the demand for phenotyping sensitivity. Phenotyping sensitivity often cannot reach $1/(kN_e)$, causing an underestimation of pleiotropy. For example, when $k = 0.1$ and $N_e = 10^6$, $1/(kN_e) = 0.001\%$, but almost no phenotyping can detect an effect equal to 0.001% of the wild-type trait value. However, when k is very small (e.g., 10^{-4}), a detected phenotypic effect (e.g., $E = 10^{-2}$) may not have a fitness effect exceeding $1/N_e$, especially in species with relatively small N_e such as large mammals. Although pleiotropy is defined according to the number of phenotypic traits affected regardless of whether each phenotypic effect has an associated fitness effect, a phenotypic effect without a fitness effect or with a fitness effect whose size is smaller than $1/N_e$ is of minimal evolutionary importance. Hence, a statistically significant phenotypic effect may or may not be evolutionarily significant.

3.3. Forward Genetic Approaches

Pleiotropy is quantified by either forward or reverse genetic approaches. Forward genetic approaches determine the genetic loci—genes or mutations—responsible for the phenotypic variation of a trait of interest. The most common forward genetic approaches are quantitative trait locus (QTL) mapping and genome-wide association (GWA) mapping. QTL mapping analyzes the genotypes and phenotypes of relatives in a known pedigree or from a designed cross, while GWA mapping analyzes the genotypes and phenotypes of nonrelatives. In both cases, however, a locus identified to be responsible for a phenotypic variation may be causally responsible for the phenotypic variation or linked with a locus causally responsible for the phenotypic variation. When a locus is found by forward genetics to be responsible for the variations of multiple phenotypic traits, the locus is said to exhibit statistical pleiotropy. Depending on the resolution of the QTL or GWA mapping, which is determined by multiple factors such as the density of genetic markers used, recombination rate in the relevant genomic region of the species, and sample size, the distance between a causal locus and the locus identified from forward genetics may be quite large (i.e., low mapping resolution). Increasing the mapping resolution in forward genetics lowers the statistical pleiotropy of a locus, because the locus would represent fewer linked causal loci. Therefore, when comparing the level of statistical pleiotropy across loci, it is important that all loci compared are subject to the same set of forward genetic analysis with similar mapping resolutions.

3.4. Reverse Genetic Approaches

Reverse genetic approaches determine the phenotypic consequences of known mutations and involve generating mutations and measuring their effects on a set of traits. A mutation is said to be pleiotropic when it is found to affect multiple traits. Because the phenotypic effects determined in reverse genetics are causal, the estimated pleiotropy is biological. Therefore, reverse genetics is preferred over forward genetics for estimating (biological) pleiotropy. One caution is that, because the sensitivity of phenotyping varies among traits, if different mutants are phenotyped for different traits, the estimated pleiotropy may not be directly comparable among mutations. The type of mutations generated in reverse genetics is another variable to consider when comparing pleiotropy. Certain types of mutations are expected to be more pleiotropic than other types. For example, null mutations of a gene presumably abolish all functions of a gene whereas missense mutations may impact the gene function only partially. By convention, gene pleiotropy is measured by phenotyping null mutants.

4. PATTERNS OF PLEIOTROPY

4.1. Fisher's Geometric Model and Universal Pleiotropy

In an influential model of phenotypic adaptation known as the geometric model, Fisher (1930) imagined a high-dimensional space in which each dimension represents a phenotypic trait and each position in this space represents a genotype and corresponding phenotype. The relative fitness of a phenotype is determined by its distance in the space from the optimal phenotype; the smaller the distance, the higher the fitness. A mutation moves an organism from one position to another in this space. Just as a random line in a two-dimensional space is extremely unlikely to be parallel to a predefined axis, any random mutation in the high-dimensional space is expected to affect all phenotypic traits, albeit by different amounts. Therefore, the geometric model suggests universal pleiotropy—every mutation affects every trait. However, it is important to recognize that the geometric model is a toy model not expected to reflect biological realities. Furthermore, it appears that Fisher assumed that different genotypes must have different phenotypes, although at least in principle, different genotypes could have the same phenotype. Under the latter scenario, a mutation may not affect any phenotypic trait.

Nonetheless, the notion that every gene or every mutation affects every trait is often assumed in theoretical population genetics. Philosophically, this notation seems to make sense, because all traits of an organism are related to some extent as they together make up the organism; consequently, a mutation would affect all traits when it affects one trait. In the most extreme case, one could argue that a mutation from one nucleotide to another at a genomic position influences the cellular concentrations of the two nucleotides, which could have ripple effects on all traits of the organism even though the effect size is presumably exceedingly small for almost all traits. The notion of universal pleiotropy makes the statement that a particular gene or mutation affects a trait meaningless and both forward and reverse genetics futile, because every gene or mutation is already assumed to affect every trait. More fundamentally, universal pleiotropy is unfalsifiable and therefore not a scientific hypothesis, because one can always argue that the statistical power is insufficient whenever the mutational effect on a trait is not found to be statistically significant (Wagner & Zhang 2012).

4.2. Patterns Revealed by Forward Genetics

Statistical pleiotropy has been estimated by a number of forward genetic analyses in the last 15 years. For instance, Kenney-Hunt et al. (2008) and Wagner et al. (2008) mapped 120 QTLs

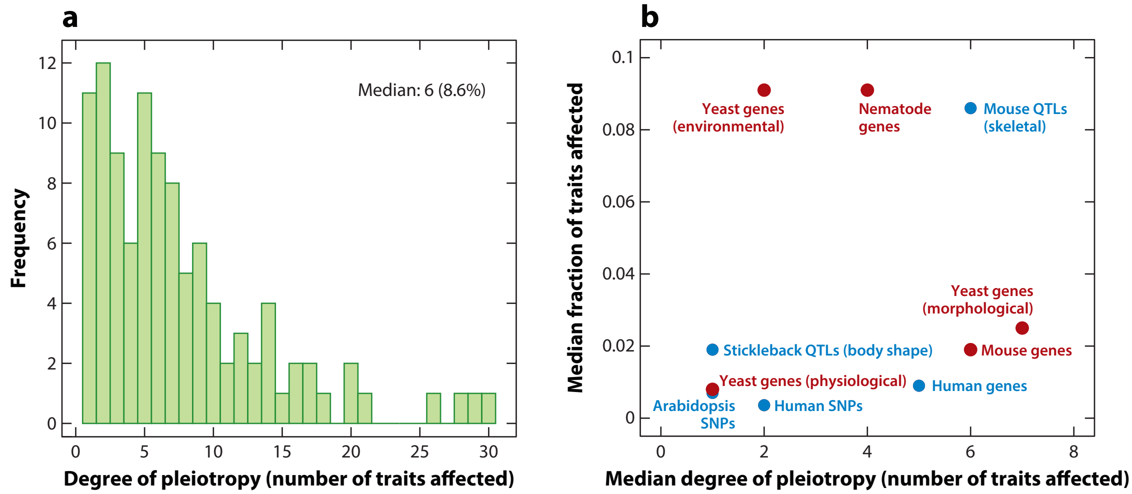

Figure 1

Frequency distribution of the degree of pleiotropy (i.e., the number of traits affected). (*a*) Distribution of pleiotropy of 120 mouse QTLs underlying 70 skeletal traits. The median degree is 6 traits, corresponding to 8.6% of all traits examined. (*b*) Median degrees of pleiotropy and median fractions of traits affected from forward and reverse genetic surveys of pleiotropy in human, mouse, stickleback, nematode, Arabidopsis, and yeast. Blue dots represent forward genetic surveys; red dots represent reverse genetic surveys. Abbreviations: QTL, quantitative trait locus; SNP, single nucleotide polymorphism. Panel *a* is based on data from Kenney-Hunt et al. (2008, table 3). Panel *b* is based on results from Albert et al. (2008), Wagner et al. (2008), Wang et al. (2010), Frachon et al. (2017), and Watanabe et al. (2019).

underlying the variations in 70 traits of the bony skeleton between inbred mice selected for either small or large body sizes. They reported that the distribution of pleiotropy is L shaped, with most QTLs affecting a few traits and a few QTLs affecting many traits (**Figure 1*a***). The median pleiotropy is six traits, or 8.6% of the traits examined. Similarly, Albert et al. (2008) mapped 26 QTLs underlying the variations in 54 position traits of the body shape between two closely related three-spined stickleback species: a Pacific marine species and a benthic lake species. Again, the distribution of pleiotropy is L shaped, with the median pleiotropy being only 1 trait, or 1.9% of the traits examined (**Figure 1*b***). By today's standard, the mapping resolution and sample size were both moderate in these two studies: The mouse study genotyped 1,040 individuals for 471 genetic markers, while the fish study genotyped 372 individuals for 248 genetic markers. A low mapping resolution due to the use of relatively few genetic markers can lead to an overestimation of (statistical) pleiotropy while a small sample size makes small effects undetectable and so leads to an underestimation of pleiotropy.

Using GWA mapping, Frachon et al. (2017) mapped single nucleotide polymorphisms (SNPs) underlying 144 eco-phenotypes in the model plant *Arabidopsis thaliana*, again reporting an L-shaped distribution of pleiotropy with a median of 1 trait (**Figure 1*b***).

GWA mapping has been extensively performed in humans for many diseases and other phenotypic variations (Welter et al. 2014, Chesmore et al. 2018). A meta-analysis of 558 traits from well-powered GWA studies (with sample sizes exceeding 50,000 individuals) offered a global view of statistical pleiotropy in humans (Watanabe et al. 2019). At the gene level, the distribution of pleiotropy is L shaped (when including only genes with at least one significant association), with a median pleiotropy of 5 traits or 0.90% of the traits examined (**Figure 1*b***). Two genes, *TNXB* (encoding tenascin-X) and *ATF6B* (encoding activating transcription factor 6 beta), are the most pleiotropic genes, each affecting 123 traits. At the SNP level, the distribution of pleiotropy is also L shaped (when including only SNPs with at least one significant association), with a median pleiotropy of 2 traits or 0.36% of the traits examined (**Figure 1*b***). The most pleiotropic SNP, located in an intron of *ATXN2* (encoding ataxin 2), affects 79 traits; despite this observation, exonic SNPs are generally more pleiotropic than intronic SNPs (Watanabe et al. 2019).

4.3. Patterns Revealed by Reverse Genetics

Wang et al. (2010) performed an analysis of gene pleiotropy using reverse genetic data from yeast, nematodes, and mice. They analyzed three data sets in the yeast *Saccharomyces cerevisiae*: (*a*) measures of 279 cellular morphological traits of single-gene-deletion strains (Ohya et al. 2005), (*b*) growth rates of single-gene-deletion strains in 22 different environments (Dudley et al. 2005), and (*c*) 120 literature-curated physiological functions of genes recorded in the Comprehensive Yeast Genome Database. In all three cases, the distribution of pleiotropy is L shaped, with the median pleiotropy equal to 7 (2% of the traits examined), 2 (9%), and 1 (1%) for the morphological, environmental, and physiological pleiotropy data sets, respectively (**Figure 1*b***). The nematode data set was based on the phenotypes of 44 early embryogenesis traits in *Caenorhabditis elegans* treated with genome-wide RNA-mediated interference (Sonnichsen et al. 2005). The mouse data set was based on the phenotypes of 308 morphological and physiological traits in gene-knockout mice recorded in Mouse Genome Informatics. In both species, the distribution of pleiotropy is approximately L shaped, with the median equal to 4 (9% of traits examined) and 6 traits (2%), respectively (**Figure 1*b***). Note that all of the above pleiotropy estimates could be underestimates as a result of a limited statistical power for detecting mutational effects (Hill & Zhang 2012, Paaby & Rockman 2013a).

The relationships between a set of genes and a set of traits can be described by a bipartite network of genes and traits, in which a link between a gene node and a trait node indicates that the gene affects the trait. A question of interest about pleiotropy is its potential modular structure (Wagner et al. 2007), where links within modules are significantly more frequent than those across modules. Indeed, the modularity is dozens of standard deviations (SDs) greater for each of the observed pleiotropy networks than for the mean of the corresponding rewired networks where gene–trait links are randomly drawn under the condition that the number of links for each gene and the number of links for each trait are unchanged (Wang et al. 2010).

4.4. Size Distribution of Mutational Effects on Multiple Traits

Although mutational pleiotropy is commonly measured by the number or fraction of traits affected by the mutation, the effect sizes also matter. Wang et al. (2010) measured the mutational effect size by Z score, which is the absolute effect size divided by the SD of the trait value among wild-type individuals. They found that the size distribution of the effects of a yeast gene deletion on 279 morphological traits is approximately normal, with the mean close to 0, for most gene deletions examined. However, the SD of the normal distribution varies substantially across different gene deletions; the greater the SD, the greater the number of traits significantly impacted by the deletion (Wang et al. 2010). This phenomenon suggests the possibility of quantifying pleiotropy by the SD, which is presumably less influenced by phenotyping sensitivity than is the degree of pleiotropy.

5. MOLECULAR BASIS OF PLEIOTROPY

First, a mutation can be pleiotropic because it affects multiple genes that overlap in their regulatory or coding sequences. Overlapping genes, commonly defined as genes that overlap in their mRNA sequences, have been extensively documented in viral, prokaryotic, and eukaryotic genomes (Wright et al. 2022). More common than overlapping genes is the scenario where multiple genes share regulatory nucleotides that are untranscribed, such as enhancers controlling multiple linked genes, bidirectional promoters, and promoters of operons. For example, the opsin locus control region (LCR) regulates the expression of red and green opsin genes on the human X chromosome; a dysfunctional LCR can cause the loss of both red and green opsins, leading to blue cone monochromacy (Carroll et al. 2010).

Second, a mutation can be pleiotropic because it affects multiple RNA or protein products of a gene. It is common for a gene to have multiple products, for example, due to alternative splicing, RNA editing, alternative transcriptional initiations, and alternative translational initiations, although not all products necessarily have distinct functions (Zhang & Xu 2022). For instance, human *LEF1* encodes lymphoid enhancer binding factor 1, which regulates the transcription of Wingless/Integrated (Wnt)/β-catenin genes. Two different protein isoforms are produced as a result of alternative transcriptional initiations: The longer isoform recruits β-catenin to Wnt target genes, whereas the shorter isoform cannot interact with β-catenin and instead suppresses the Wnt regulation of target genes (Arce et al. 2006). Thus, a mutation occurring in the region shared by the two isoforms potentially affects both isoforms and their respective functions.

Third, a mutation can be pleiotropic because it affects one protein product that has multiple molecular functions such as the mammalian serum albumin mentioned in Section 2 (He & Zhang 2006).

Fourth, a mutation can be pleiotropic simply because a single molecular function of a protein is directly or indirectly involved in multiple biological processes. For example, a protein may have only one molecular function as a transcription factor, but this transcription factor may directly regulate many target genes. Hence, a mutation reducing the DNA-binding affinity of the transcription factor could affect the expression of multiple target genes. In addition, the resulting expression changes of the target genes could impact the expression of their target genes, causing indirect effects of the mutation concerned. For example, a natural polymorphism in an intron of the gene encoding the transcription factor FLM influences *Arabidopsis thaliana* growth and color by affecting *FLM* splicing (Hanemian et al. 2020).

He & Zhang (2006) asked whether pleiotropy is more likely to be caused by the third or fourth mechanisms listed above. They compared yeast genes with different levels of environmental pleiotropy but found them to have similar mean numbers of molecular functions. By contrast, yeast genes with higher degrees of environmental pleiotropy tend to participate in more biological processes. Further, their protein products are found in more cellular components and have more protein-interaction partners. The authors concluded that yeast gene pleiotropy is more often attributable to the same molecular function involved in multiple biological processes (type II pleiotropy) than to multiple molecular functions (type I pleiotropy).

6. CAUSAL RELATIONSHIPS AMONG MULTIPLE EFFECTS OF A MUTATION

As mentioned, pleiotropy can be classified based on whether a single mutation separately affects multiple phenotypic traits (horizontal pleiotropy) or a single mutation starts off a cascade of events that sequentially influence multiple traits (vertical pleiotropy). This classification requires delineating the relationships among the phenotypic effects of a mutation on multiple traits. For example, if a mutation increases the chance of hypertension and reduces life span, it is possible that the reduction in the mutant's life span results entirely, partly, or not at all from the mutational impact on the probability of hypertension. If we no longer find a difference in life span between wild-type and mutant individuals after controlling for hypertension (i.e., among individuals with similar hypertension status), we can conclude that the mutational effect on life span is entirely due to its effect on hypertension (i.e., vertical pleiotropy). Alternatively, if the life span difference between wild-type and mutant individuals remains unchanged after controlling for hypertension, we can conclude that the mutational effect on life span is not due to its effect on hypertension (i.e., horizontal pleiotropy). It is also possible that the life span difference between wild-type and mutant individuals is reduced (but not to zero) after controlling for hypertension, suggesting that the pleiotropy has both horizontal and vertical components.

Note that distinguishing between horizontal and vertical pleiotropy requires finding the relationships among the mutational effects on multiple traits, not the relationships among these traits per se, as analyzed recently (Geiler-Samerotte et al. 2020). In the context of the above example, the relationship between hypertension and life span is not the same as the relationship between the effects of a mutation on hypertension and life span, which can be different for different mutations. While it remains generally difficult to distinguish between horizontal and vertical pleiotropy, progress is expected given the recent advancement in causal inference by statistical genetics (Pingault et al. 2018, Verbanck et al. 2018).

7. EVOLUTIONARY CONSEQUENCES OF PLEIOTROPY

7.1. Maintenance of Seemingly Deleterious Traits

Mutations causing Huntington's disease, a neurodegenerative disorder in which symptoms typically manifest after the reproductive age, also increase fecundity (Carter & Nguyen 2011). As mentioned, homozygous mutations causing cystic fibrosis also cause male infertility, but heterozygotes show higher fecundity than the wild type (Carter & Nguyen 2011). Such conflicting effects on different traits weaken the negative selection against disease mutations, cause the maintenance of these mutations in the population, and increase the disease incidence. It has been suggested that mutations causing Huntington's disease, cystic fibrosis, sickle-cell anemia, glucose-6-phosphate dehydrogenase deficiency, cancer, and many other diseases are kept in the population because of their benefits to other aspects of life such as development, fecundity, and host defense (Carter & Nguyen 2011). Note that a disease mutation could also hitchhike on linked beneficial mutations; that is, the above phenomenon extends to statistical pleiotropy. For example, mutations associated with myopia were found to be selected for in the current British population due to (presumably statistical) pleiotropic effects on reproduction (Long & Zhang 2021).

Ageing refers to a gradual deterioration of bodily functions that manifests as an increase in the death rate with age after sexual maturity. Ageing is widespread among multicellular organisms and demands an explanation, because natural selection should favor mutations conferring an extended reproductive life span (Reichard 2017). The antagonistic pleiotropy hypothesis posits that mutations contributing to ageing could be positively selected because of their pleiotropic beneficial effects early in life, such as promoting development or reproduction (Williams 1957). Indeed, a recent analysis of human genotype–phenotype data revealed a negative genetic correlation between reproduction and life span (i.e., the same or linked genetic variants have opposite effects on reproduction and life span), supporting the antagonistic pleiotropy hypothesis of ageing (Long & Zhang 2023).

7.2. Cost of Complexity in Adaptation

Orr found that the rate of adaptation of a population (U), measured by its fitness increase per generation, decreases precipitously with mutational pleiotropy (n) (Orr 2000). Because Orr assumed universal pleiotropy under Fisher's geometric model, n is the total number of traits of the organism, which can be regarded as a measure of organismal complexity. Hence, Orr concluded that the rate of adaptation decreases with organismal complexity, a result dubbed the cost of complexity.

However, Orr made several simplifying assumptions in his formulation. First, as mentioned, empirical data suggest that a mutation typically affects only a small proportion of traits (**Figure 1**). Nevertheless, if we assume that the proportion of traits affected by a mutation is comparable between species of different levels of complexity, Orr's formula still predicts a cost of complexity, albeit a lessened one.

Second, the scaling relationship between the total size (T) of all phenotypic effects of a mutation and n can be written as $T = an^b$, where a and b are two parameters. Orr assumed that T is independent of n (i.e., $b = 0$), while some authors believe that the per trait effect size is independent of n (i.e., $b = 0.5$) (Waxman & Peck 1998). The rate at which U declines with n is much lower under $b = 0.5$ than under $b = 0$ (Wang et al. 2010). In their QTL mapping of skeletal traits of mice, however, Wagner et al. (2008) observed a greater per trait effect size with larger n (i.e., $b > 0.5$). A similar trend was evident from human GWA mapping results (Chesmore et al. 2018). This pattern was also observed for yeast morphological traits, with b estimated at 0.601 ± 0.011 (95% confidence interval) (Wang et al. 2010). Similarly, $b = 0.724 \pm 0.0035$ for a set of eco-phenotypes of *Arabidopsis thaliana* (Frachon et al. 2017). When $b > 0.5$, U no longer monotonically declines with n but is an inverted U-shaped function of n, with U reaching its maximum at an intermediate n (Wang et al. 2010). Thus, with an empirically estimated b value, complexity is much less costly than Orr's original finding and may even promote adaptation. Let n_{optimal} be the level of pleiotropy corresponding to the highest U. It is intriguing that n_{optimal} reaches the highest value when b is in a narrow range from 0.56 to 0.79, which happens to encompass the empirically observed b values (Wang et al. 2010, Frachon et al. 2017). In the *Arabidopsis* study, Frachon et al. further observed the strongest signal of positive selection on genetic variants with intermediate levels of pleiotropy, consistent with the theoretical prediction that an intermediate n yields the highest U.

7.3. Gene Pleiotropy and Evolutionary Rate

It has been hypothesized that the functional constraint of a gene rises with its level of pleiotropy (Hodgkin 1998). Indeed, compared with yeast genes with relatively low morphological or environmental pleiotropy, those with relatively high pleiotropy cause greater fitness reductions when deleted (Cooper et al. 2007) and are more likely to have detectable fly or nematode homologs (He & Zhang 2006). Furthermore, the nonsynonymous nucleotide substitution rate (d_N) of a gene decreases with gene pleiotropy (He & Zhang 2006), even after controlling for the most important d_N determinant—gene expression level (Zhang & Yang 2015).

However, when gene pleiotropy is measured by the number of protein–protein interaction (PPI) partners (N_{PPI}) that the gene has, the result is mixed. A negative correlation was initially reported between the evolutionary rate of a protein and its N_{PPI} (Fraser et al. 2002), but the correlation was later shown to be caused at least in part by a confounder—gene expression level (Jordan et al. 2003). The controversy was subsequently resolved by the discovery of two types of proteins: The first type uses the same set of residues for interacting with different partners such that the increase in N_{PPI} does not alter the fraction of functionally constrained residues, while the second type uses different residues for interacting with different partners such that the increase in N_{PPI} raises the fraction of functionally constrained residues. Consequently, pleiotropy measured by N_{PPI} constrains sequence evolution for the second but not the first type of proteins (Kim et al. 2006).

7.4. Pleiotropy and Mechanisms of Phenotypic Evolution

Based on a summary of case studies and some theoretical reasoning, Carroll proposed different genetic mechanisms for the evolution of morphological and physiological traits (Carroll 2005). Specifically, he posited that morphological evolution more often proceeds by *cis*-regulatory changes than protein functional changes, while the opposite may be true for physiological evolution. The central consideration in his hypothesis is pleiotropy. A protein functional change affects all tissues where the protein is expressed, so it could be too pleiotropic to tolerate in morphological evolution. By contrast, a regulatory change may affect only one tissue, so it could be more tolerable. While the above makes sense, Carroll did not explain why physiological evolution would be different. Regardless, Liao et al. (2010) compared mouse genes that, when knocked out, affect only morphological traits (dubbed morphogenes) and those that affect only physiological traits (dubbed physiogenes). Interestingly, they found many differences between the two groups of genes. For example, morphogenes are enriched with transcriptional regulators, while physiogenes are enriched with channels, transporters, receptors, and enzymes. Compared with physiogenes, morphogenes are significantly more pleiotropic, less tissue specific in expression, and evolve more slowly in their protein sequences but faster in tissue-expression profiles. These results support Carroll's thesis and suggest that physiological evolution can probably tolerate protein functional changes because physiogenes tend to be tissue specific in their expressions.

A change in the expression of a gene can be caused by a *cis*-effect mutation, which is on the same DNA molecule as the gene (e.g., in the promoter or enhancers of the gene), or a *trans*-effect mutation, which can be anywhere in the genome because it acts through diffusible molecules such as transcription factors. It has repeatedly been shown that the expression divergence between orthologous genes is more often attributable to *cis*-effect mutations than *trans*-effect mutations and that this trend becomes more and more prominent when the divergence between orthologs gets larger (Hill et al. 2021). These patterns are explained by the fact that *cis*-effect mutations are less pleiotropic (i.e., affect fewer downstream genes) and therefore less deleterious than *trans*-effect mutations (Vande Zande et al. 2022).

7.5. Antagonistic Pleiotropy and Evolution in Changing Environments

The environment of virtually every natural population changes over time. For a neutral mutation destined for fixation, the time from its initial appearance to its fixation takes on average $4N_e$ generations for diploids. Even for beneficial mutations destined for fixation, the expected time to fixation is $2 \ln (2N_e-1)/s$ (Otto & Whitlock 2013), which equals ~2,000 generations under $N_e = 10^4$ and ~3,400 generations under $N_e = 10^7$ when $s = 0.01$. Because it is highly likely that the environment varies during the above time, it is important to consider the possibility of environmental pleiotropy of mutations when discussing their behaviors in a population. Of particular interest is antagonistic environmental pleiotropy, under which a mutation would be selected for under some environments but selected against under other environments. Because beneficial mutations are far less common than deleterious mutations, it is likely that any beneficial mutation observed in an environment is deleterious in most other environments. Consequently, these mutations are rarely fixed, although they contribute to the adaptation of the population to the environment in which they are beneficial. Adaptive evolution would then be underestimated by the rate of nonsynonymous substitution relative to that of synonymous substitution (d_N/d_S).

To verify the above prediction, Chen & Zhang (2020) performed yeast experimental evolution in a changing environment that rotated among five media, with a duration of 224 generations per medium, totaling 1,120 generations. The five media were chosen based on the fact that the growth rates of a set of strains tend to be negatively correlated across the five media. For comparison, they performed experimental evolution in each of the five media separately for 1,120 generations. Because the selection intensity imposed by a medium diminishes as the population adapts to it, the average selection intensity in the constant environments over 1,120 generations should be lower than that in the changing environment where a new selection was imposed every 224 generations. Yet, by comparing the genome sequences of the progenitor and evolved strains, the authors found d_N/d_S in the changing environment to be significantly lower than the average d_N/d_S in the five constant environments. Examining the dynamics of mutations in the changing environment, the authors observed the expected patterns of antagonistic pleiotropy—allele frequencies rose quickly in one medium but dropped precipitously in the next medium.

Given that genome-wide d_N/d_S typically exceeds 1 in microbial experimental evolution performed in constant environments (Tenaillon et al. 2012, Lang et al. 2013, Kryazhimskiy et al. 2014, Tenaillon et al. 2016) and that the environment varies more frequently in nature than in these constant-environment experimental evolution scenarios, we predict that positive selection is frequent in nature. Hence, it is plausible that the observation of $d_N/d_S \ll 1$ between (even very closely related) strains sampled from natural environments (Rocha et al. 2006, Wolf et al. 2009, Maclean et al. 2017) does not reflect a lack of positive selection in nature but rather is due to antagonistic pleiotropy that conceals signals of positive selection (Chen & Zhang 2020).

7.6. Evolution of Pleiotropy

Several mechanisms are known to resolve adaptation-impeding pleiotropy. First, if the expression of a gene is beneficial in some environments or tissues but detrimental in other environments or tissues, gene regulation that promotes gene expression in the former but suppresses gene expression in the latter would be selectively favored. Such regulation reduces gene pleiotropy, because the gene no longer has an effect in the latter environments or tissues. In a yeast study, Qian et al. (2012) reported that deleting the *PRD17* gene from a lab strain is deleterious in ETH medium but beneficial in YPG medium, indicating that *PRD17* in the lab strain exhibits environmental antagonistic pleiotropy. PDR17 is a phosphatidylinositol transfer protein participating in phospholipid synthesis and transport and is involved in resistance to multiple drugs. If such pleiotropy is (at least partially) evolutionarily resolved in natural strains that experience environments similar to ETH or YPG, the expression of *PRD17* is expected to be lower in YPG than in ETH in these natural strains. This was indeed the case. Furthermore, *PRD17* expression in YPG and the ratio of *PRD17* expression in YPG to that in ETH are both lower in a strain adapted to YPG than in a strain adapted to ETH. Over 80% of genes tested showed signs of evolutionary resolution of antagonistic pleiotropy by gene regulation (Qian et al. 2012).

Second, Qian et al. (2012) discovered that pleiotropy can also be resolved by protein subcellular relocalization. Specifically, MIG1 is a transcription factor functioning exclusively inside the nucleus in glucose repression. Its functional allele was found to be fitter than the null allele in YPD medium, but the opposite is true in OAK medium. In the strain adapted to an environment mimicked by OAK medium, MIG1 is localized to the nucleus under YPD. However, under OAK, where MIG1 would be deleterious, MIG1 is localized to the cytoplasm and hence imposes no harm.

Third, a multi-functional protein may not be able to optimize all of its functions due to tradeoffs, and gene duplication followed by functional specialization is a well-recognized mechanism to resolve such adaptive conflicts (Hughes 1994). For example, mammalian pancreatic ribonuclease (RNASE1) is an enzyme that typically digests both single-stranded RNA (ssRNA) and double-stranded RNA (dsRNA); while the former activity is primarily used for food digestion, the latter is probably involved in host defense against RNA viruses. *RNASE1* was duplicated in an ancestor of the leaf-eating monkey *Pygathrix nemaeus*; upon duplication, one daughter gene has retained its activity for degrading dsRNA while the other daughter gene (*RNASE1B*) has evolved an elevated activity for digesting ssRNA in the acidic small intestine of leaf monkeys but has almost completely lost its activity toward dsRNA (Zhang et al. 2002). There have been nine amino acid substitutions in the mature peptide of RNASE1B since the gene duplication, and each of them reduced its activity toward dsRNA (Zhang et al. 2002). Therefore, without gene duplication and the relaxation of the functional constraint from the activity toward dsRNA, none of the nine substitutions, some of which are necessary for the improved activity toward ssRNA (Zhang 2006), would have been allowed in RNASE1B (Zhang et al. 2002). In another example, Näsvall et al. (2012) showed in an experimental evolution setting that a gene in *Salmonella enterica* with low levels of two distinct activities was duplicated and each ancestral activity increased in one of the daughter genes in subsequent evolution. The authors found that, although it was not impossible to evolve a single generalist gene with both activities increased, the high rate of gene duplication made it more likely to evolve specialist genes in their study.

Fourth, in principle, mechanisms that generate multiple distinct products from one gene could also help resolve adaptation-impeding pleiotropy. For instance, two functions of a gene may require slightly different protein sequences, which may be accomplished by RNA editing, alternative splicing, alternative translational initiation, and so on (Zhang & Xu 2022).

Because the optimal phenotype can differ between males and females, a mutation may exhibit antagonistic pleiotropy between the two sexes. Evidence for such pleiotropy is abundant (Bonduriansky & Chenoweth 2009). For example, male fitness and female fitness tend to show negative correlations across different laboratory-constructed genotypes of *Drosophila melanogaster* (Innocenti & Morrow 2010). Several mechanisms have been found to alleviate or resolve such sexual antagonism. First, a gene that is beneficial to one sex but deleterious to the other can be differentially expressed in the two sexes (Ellegren & Parsch 2007, Mank 2009, VanKuren & Long 2018). Second, genomic imprinting, where only one parental allele is expressed in the offspring, is believed to reflect a compromise in the conflict of interest between the two parents (Moore & Haig 1991, Wilkins & Haig 2003). For example, for genes that promote embryonic growth, such as the insulin-like growth factor 2 gene in mice, the paternal copy tends to be expressed, while the maternal copy is silenced. Third, sex-specific dominance reversal—the allele beneficial to males is dominant in males but recessive in females and vice versa—can alleviate sexual antagonism (Grieshop & Arnqvist 2018). Fourth, relocating genes beneficial to males but deleterious to females to the (male-specific) Y chromosome should alleviate sexual antagonism. It is, however, debated whether the X chromosome enriches sexually antagonistic genes when compared with autosomes (Rice 1984, Mank 2009, Ruzicka & Connallon 2020, Ruzicka & Connallon 2022).

It is important to distinguish mutations that alter gene pleiotropy from those that alter the pleiotropy of future mutations. For example, a mutation that suppresses *PRD17* expression in YPG medium reduces the pleiotropy of *PRD17* and is advantageous in YPG. There are also mutations that can be called pleiotropy modifiers, which modify the pleiotropy of future mutations (Pavlicev et al. 2008). Natural selection on pleiotropy modifiers is a second-order selection, so it is expected to be very weak because the modifiers do not alter the fitness of the individuals carrying the modifiers but that of the offspring of these individuals.

8. OUTSTANDING QUESTIONS

First, as discussed, reliably quantifying pleiotropy is difficult, both conceptually and operationally. Consequently, it is debated whether the empirically measured pleiotropy reflects the truth or is an artifact of limited phenotyping sensitivity (Wagner & Zhang 2011, Hill & Zhang 2012, Wagner

& Zhang 2012, Paaby & Rockman 2013a, Paaby & Rockman 2013b, Zhang & Wagner 2013). Theoretical work is needed to determine whether pleiotropy can be reliably measured without substantially increasing phenotyping sensitivity and the number of traits examined. For example, can the SD of the distribution of the effect sizes of a mutation on a moderate number of traits (Wang et al. 2010) serve as a reliable measure of pleiotropy?

Second, as mentioned, adaptation-impeding pleiotropy may be resolved by many mechanisms such as gene duplication and gene regulation. Furthermore, at least in theory, pleiotropy modifiers that minimize the negative impact of the pleiotropy of future mutations are selectively favored. Thus, does pleiotropy constrain evolution only temporarily? Under what conditions is pleiotropy unresolvable, and to what extent does pleiotropy permanently constrain evolution? Theory shows that, when the scaling exponent $b > 0.5$, the adaptation rate U is maximized at an intermediate degree of pleiotropy. However, the pleiotropy that yields the highest U depends on several parameters, so it is unclear whether the observed pleiotropy is larger or smaller than the pleiotropy that yields the highest U. Answering this question will help determine whether there is a cost of complexity and on what type of organisms the cost is imposed.

Third, laboratory microbial evolution in a changing environment showed that antagonistic environmental pleiotropy of mutations causes underestimation of positive selection and conceals adaptive signals. How widespread is this phenomenon in nature? Answering this question will be important for estimating the relative prevalence of neutral and adaptive evolution.

Fourth, how do we reliably separate horizontal from vertical pleiotropy? If a mutation affects trait B because of its effect on trait A (vertical pleiotropy), one could minimize the mutational effect on B by prohibiting the change of A (by any means influencing A). This would not work if the mutational pleiotropy were horizontal. Hence, distinguishing vertical from horizontal pleiotropy can be important in medicine and other applications.

In summary, in less than two decades, pleiotropy research has evolved from a largely theoretical realm with little systematic, empirical data to a rapidly expanding domain with large-scale data, multiple experimental tools and model systems, and ample opportunities for empirical tests of relevant hypotheses. With these developments and the fundamental roles of pleiotropy in so many biological phenomena, pleiotropy research is poised to bring many new insights in the future.

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ANNUAL REVIEWS

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Patterns and Evolutionary Consequences of Pleiotropy

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Abstract

Pleiotropy refers to the phenomenon of one gene or one mutation affecting multiple phenotypic traits. While the concept of pleiotropy is as old as Mendelian genetics, functional genomics has finally allowed the first glimpses of the extent of pleiotropy for a large fraction of genes in a genome. After describing conceptual and operational difficulties in quantifying pleiotropy and the pros and cons of various methods for measuring pleiotropy, I review empirical data on pleiotropy, which generally show an L-shaped distribution of the degree of pleiotropy (i.e., the number of traits affected), with most genes having low pleiotropy. I then review the current understanding of the molecular basis of pleiotropy. In the rest of the review, I discuss evolutionary consequences of pleiotropy, focusing on advances in topics including the cost of complexity, regulatory versus coding evolution, environmental pleiotropy and adaptation, evolution of ageing and other seemingly harmful traits, and evolutionary resolution of pleiotropy.

1. INTRODUCTION

Pleiotropy refers to the phenomenon of one gene or one mutation affecting multiple traits. For example, mutations causing cystic fibrosis, primarily a lung disease, also cause male infertility (Sokol 2001). The phenomenon of pleiotropy had been recognized long before its formal definition in 1910 by the German geneticist Ludwig Plate (Plate 1910). For instance, some medical syndromes were known to have a suite of symptoms and a simple familial component (Pyeritz 1989, Stearns 2010). Another example is in the foundational paper of genetics in which Gregor Mendel reported the cosegregation of three traits in garden peas—brown seed coat, violet flowers, and axial spots versus white seed coat, white flowers, and no spots—and considered them to be controlled by a single locus (Mendel 1866). For two reasons, pleiotropy is commonly thought to be widespread. First, any organism has many more phenotypic traits than genes, so at least some genes are pleiotropic. Second, given the high prevalence of interactions among various parts of an organism and interactions among genes/mutations (Tyler et al. 2009, Zhang 2017), it is extremely unlikely that a mutation would affect only one trait.

Pleiotropy has significant implications for many areas of biology and is frequently invoked in theories and explanations of a broad spectrum of phenomena, including genetic disease (Carter & Nguyen 2011, Byars & Voskarides 2020), cancer (Rodier et al. 2007, Bien & Peters 2019), senescence (Williams 1957, Promislow 2004, Flatt & Partridge 2018), between-trait trade-offs (Guillaume & Otto 2012, Yang et al. 2014, Burmeister et al. 2020, Mauro & Ghalambor 2020), sexual conflict (Rice 1992, Bonduriansky & Chenoweth 2009, Innocenti & Morrow 2010), generalism versus specialism (Jerison et al. 2020, Bakerlee et al. 2021), cooperation (Foster et al. 2004, Bentley et al. 2022), evolutionary constraints on genes or traits (Barton 1990, Waxman & Peck 1998, Carroll 2005, He & Zhang 2006, Jiang & Zhang 2020), adaptation (Fisher 1930, Orr 2000, Otto 2004), neofunctionalization (Hughes 1994, Francino 2005, Bergthorsson et al. 2007, Hittinger & Carroll 2007), and speciation (Berlocher & Feder 2002, Kirkpatrick & Ravigne 2002, Rundle & Nosil 2002). For example, a mutation may improve one trait but worsen another, causing compromises among adaptations of different traits. It has been proposed that the above situation creates a cost for complexity—complex organisms are less adaptable than simple organisms because mutations are inherently more pleiotropic in the former than the latter (Fisher 1930, Orr 2000). Pleiotropy is the main theoretical reason behind the hypothesis that morphological evolution occurs more frequently via *cis*-regulatory sequence changes than protein sequence changes, because the former are thought to be less pleiotropic than the latter (Carroll 2005). It has been suggested that a number of disease mutations are maintained in the population because of their benefits to other aspects of life that are often hidden (Carter & Nguyen 2011).

For a long time, theoreticians have assumed universal pleiotropy, a notion that every gene/mutation affects every trait, despite the availability of only limited data on pleiotropy. Thanks to the development of high-throughput biology, the last two decades have witnessed a rapid accumulation of empirical data that allow the assessment of general patterns and various evolutionary consequences of pleiotropy. Here, I first review classifications of pleiotropy, conceptual and operational difficulties in measuring pleiotropy, methods for quantifying pleiotropy, empirical patterns of pleiotropy, the molecular basis of pleiotropy, and causal relationships among multiple effects of a mutation. I then review our understanding of some important evolutionary consequences of pleiotropy and discuss outstanding questions.

2. CLASSIFICATION OF PLEIOTROPY

Strictly speaking, pleiotropy is a characteristic of mutations. However, when the mutation of interest is a null mutation of a gene, which abolishes all functions of the gene, the mutational pleiotropy is also known as gene pleiotropy, a property of genes.

German biologist Hans Grüneberg was among the first to study the mechanistic basis of pleiotropy when researching a genetic skeletal abnormality in rats in the 1930s. He divided pleiotropy into genuine and spurious; genuine pleiotropy results from multiple products of a single gene locus, whereas spurious pleiotropy is caused by a single product that is used in multiple ways or initiates a series of events with multiple phenotypic effects (Grüneberg 1938). While Grüneberg's genuine and spurious categories are both considered true biological pleiotropy today, attempts to distinguish types of pleiotropy by different underlying causes have not stopped.

He & Zhang (2006) classified gene pleiotropy based on whether it is conferred by multiple molecular functions of the gene product (type I pleiotropy) or by multiple phenotypic consequences of a single molecular function (type II pleiotropy). A typical example of type I pleiotropy involves the mammalian serum albumin, which binds to fatty acids and toxic metabolites but is also involved in the oxidation of nitric oxide (Rafikova et al. 2002), a clear indication of multiple molecular functions. By contrast, the yeast gene HIS7 exhibits type II pleiotropy, because its product glutamine amidotransferase uses a single catalytic activity in both histidine biosynthesis and purine nucleotide monophosphate biosynthesis. Type I and type II pleiotropy loosely correspond to genuine and spurious pleiotropy, respectively, although He and Zhang emphasized the number of molecular functions of the gene product, while Grüneberg emphasized the number of gene products.

Hadorn (1961) distinguished mosaic from relational pleiotropy. Mosaic pleiotropy arises when a single locus separately affects multiple phenotypic traits, while relational pleiotropy arises when a single locus starts off a cascade of events influencing multiple traits, similar to Grüneberg's second form of spurious pleiotropy. Mosaic and relational pleiotropy were subsequently renamed horizontal and vertical pleiotropy, respectively (Tyler et al. 2009), two terms that are now commonly used in human genetics. For example, mutations truncating the human αB-crystallin (*CryAB*) gene are associated with both cataracts (Liu et al. 2006) and dilated cardiomyopathy (Inagaki et al. 2006) because the gene is expressed in both the eye and heart and performs apparently unrelated functions in the two organs; these mutations are said to exhibit horizontal pleiotropy. By contrast, the sickle cell mutation in the β-globin gene causes hemoglobin to polymerize when deoxygenated, deforming red blood cells into the sickle shape, which hinders red blood cells from flowing in capillaries and causes ischemia in organs and peripheral tissues (Tyler et al. 2009). This series of phenotypic effects follow directly from one to the next, so the sickle cell mutation is said to exhibit vertical pleiotropy.

A mutation exhibits synergistic or concordant pleiotropy if its phenotypic effects on two traits are in the same direction (typically in terms of fitness or its proxies) and antagonistic pleiotropy if they are in opposite directions. For example, a mutation that accelerates the sexual maturation of an animal but reduces its fecundity is antagonistically pleiotropic.

The same phenotypic trait measured in different environments is often regarded as different traits, especially in microbial studies (Dudley et al. 2005). Hence, a mutation with an effect on a trait in multiple environments can be considered pleiotropic, and such pleiotropy is referred to as environmental pleiotropy. Environmental pleiotropy is related to but does not equal gene by environment interaction (G×E), which refers to the phenomenon of a mutation exerting different effects on a trait in different environments (Wei & Zhang 2017, Li & Zhang 2018). Antagonistic environmental pleiotropy means that the mutational effects are in opposite directions in two environments, so antagonistic environmental pleiotropy is by definition G×E. By contrast, with concordant environmental pleiotropy, the mutational effects in two environments may or may not differ so may or may not exhibit G×E. Similarly, G×E may or may not imply environmental pleiotropy because the mutational effect could be nonzero in both environments or nonzero in only one environment.

When the phenotypic variations of two traits among genotypes are genetically mapped to the same genomic region, it is usually unknown (due to limited mapping resolutions) whether these variations are both caused by the same genetic polymorphism. In such cases, the genomic region is considered to exhibit statistical pleiotropy, which may or may not represent biological pleiotropy, depending on whether the multiple phenotypic effects are caused by the same polymorphism or not (Wagner & Zhang 2011, Hackinger & Zeggini 2017, Watanabe et al. 2019). In this review, unless otherwise mentioned, pleiotropy refers to biological pleiotropy.

In addition, because mutational effects may be environment dependent, mutational pleiotropy (excluding environmental pleiotropy) can also be environment dependent. For example, whether a mutation has concordant or antagonistic pleiotropic effects on the maximum growth rate and carrying capacity of yeast's logistic population growth is influenced by the growth medium (Wei & Zhang 2019).

3. QUANTIFYING PLEIOTROPY

3.1. Conceptual Difficulties

A phenotypic trait is an observable characteristic of an organism such as the height, weight, eye color, blood pressure, ABO blood type, oxygen saturation, mRNA concentration of a particular gene in a particular tissue, educational attainment, or personality of a human. An organism has almost an infinite number of traits, although many traits are correlated with one another, such as the lengths of the left and right arms of a human. In the most extreme case, two traits may be perfectly correlated (e.g., the height and twice the height of a tree). To quantify the pleiotropy of a mutation, we must measure the phenotypic effects of the mutation on multiple traits. What traits should be measured and counted when quantifying pleiotropy? Obviously, the height and twice the height of a tree should not be regarded as two traits in quantifying pleiotropy, but how about other traits that are highly correlated with one another without the correlation being as obvious as in the above example (e.g., the tree height and a complex mathematical function of the tree height)? Because of the above difficulty, pleiotropy is often said to be the phenomenon of a mutation affecting multiple seemingly unrelated traits, but what exactly are seemingly unrelated traits? Are the lengths of the left and right arms of a human seemingly related? What about the lengths of the left arm and left leg (or right leg)—are they seemingly related or not? Different people would have different answers. Because the correlation between traits can be anywhere from −1 to 1, it is difficult to objectively define whether two traits are seemingly unrelated. In practice, when quantifying pleiotropy, researchers generally include all traits measured, excluding those with a correlation of 1 or −1 (Wagner & Zhang 2011).

3.2. Operational Difficulties

Two operational difficulties in quantifying pleiotropy exist. First, there is no way to phenotype all traits of an organism, so it is impossible to quantify the pleiotropy of any mutation comprehensively. As a result, all estimates of pleiotropy are based on a limited set of traits. Furthermore, the traits examined in a mutant are often related to the trait initially found to be impacted by the mutation. For example, a heart-disease-causing mutation tends to be examined for potential effects on the heart or cardiovascular system. Pleiotropy estimated from such data is typically not comparable across mutations. Alternatively, a fixed set of traits may be investigated in many mutants, usually because these traits are relatively easy to phenotype. As a result, the estimated pleiotropy is commonly considered comparable for the involved mutations. However, a mutation may have detectable effects only on a group of related traits, for example, pertaining to a specific organ, so inclusion/exclusion of this group of traits in phenotyping drastically alters the estimated pleiotropy of the mutation. Because the traits phenotyped are usually such a small fraction of all traits, caution is needed in comparing pleiotropy among mutations.

Second, the phenotypic effect of a mutation on a trait may be too small to detect. Measuring the mutational effect on a trait means comparing the phenotypic trait between two genotypes that differ only by the mutation of interest. All measurements have errors; if the magnitude of a mutational effect is smaller than that of the measurement error, the mutational effect cannot be detected, causing an underestimation of pleiotropy. How sensitive need phenotyping methods be? Ideally, if we are concerned with the evolutionary fate of a mutation, a sensitive method should be able to detect a phenotypic effect on a trait (E, measured by the fractional change in trait value) that has a fitness effect (s) whose absolute value exceeds the magnitude of genetic drift, because the fate of the mutation would be influenced by natural selection arising from this phenotypic effect. The magnitude of genetic drift is one divided by the effective population size (N_e), which is $\sim 10^5$ for vertebrates, $\sim 10^6$ for invertebrates and land plants, $\sim 10^7$ for unicellular eukaryotes, and 10^8 or greater for free-living prokaryotes (Lynch et al. 2011). Let us assume that, when $|E|$ is small, $|s| = k|E|$, where k is a trait-specific parameter that may be referred to as trait importance (Ho et al. 2017); k is 100 times the fitness effect caused by a 1% change in trait value. Therefore, a sensitive phenotyping method should be able to detect a phenotypic effect size of $1/(kN_e)$. The greater the product of trait importance and effective population size, the higher the demand for phenotyping sensitivity. Phenotyping sensitivity often cannot reach $1/(kN_e)$, causing an underestimation of pleiotropy. For example, when $k = 0.1$ and $N_e = 10^6$, $1/(kN_e) = 0.001\%$, but almost no phenotyping can detect an effect equal to 0.001% of the wild-type trait value. However, when k is very small (e.g., 10^{-4}), a detected phenotypic effect (e.g., $E = 10^{-2}$) may not have a fitness effect exceeding $1/N_e$, especially in species with relatively small N_e such as large mammals. Although pleiotropy is defined according to the number of phenotypic traits affected regardless of whether each phenotypic effect has an associated fitness effect, a phenotypic effect without a fitness effect or with a fitness effect whose size is smaller than $1/N_e$ is of minimal evolutionary importance. Hence, a statistically significant phenotypic effect may or may not be evolutionarily significant.

3.3. Forward Genetic Approaches

Pleiotropy is quantified by either forward or reverse genetic approaches. Forward genetic approaches determine the genetic loci—genes or mutations—responsible for the phenotypic variation of a trait of interest. The most common forward genetic approaches are quantitative trait locus (QTL) mapping and genome-wide association (GWA) mapping. QTL mapping analyzes the genotypes and phenotypes of relatives in a known pedigree or from a designed cross, while GWA mapping analyzes the genotypes and phenotypes of nonrelatives. In both cases, however, a locus identified to be responsible for a phenotypic variation may be causally responsible for the phenotypic variation or linked with a locus causally responsible for the phenotypic variation. When a locus is found by forward genetics to be responsible for the variations of multiple phenotypic traits, the locus is said to exhibit statistical pleiotropy. Depending on the resolution of the QTL or GWA mapping, which is determined by multiple factors such as the density of genetic markers used, recombination rate in the relevant genomic region of the species, and sample size, the distance between a causal locus and the locus identified from forward genetics may be quite large (i.e., low mapping resolution). Increasing the mapping resolution in forward genetics lowers the statistical pleiotropy of a locus, because the locus would represent fewer linked causal loci. Therefore, when comparing the level of statistical pleiotropy across loci, it is important that all loci compared are subject to the same set of forward genetic analysis with similar mapping resolutions.

3.4. Reverse Genetic Approaches

Reverse genetic approaches determine the phenotypic consequences of known mutations and involve generating mutations and measuring their effects on a set of traits. A mutation is said to be pleiotropic when it is found to affect multiple traits. Because the phenotypic effects determined in reverse genetics are causal, the estimated pleiotropy is biological. Therefore, reverse genetics is preferred over forward genetics for estimating (biological) pleiotropy. One caution is that, because the sensitivity of phenotyping varies among traits, if different mutants are phenotyped for different traits, the estimated pleiotropy may not be directly comparable among mutations. The type of mutations generated in reverse genetics is another variable to consider when comparing pleiotropy. Certain types of mutations are expected to be more pleiotropic than other types. For example, null mutations of a gene presumably abolish all functions of a gene whereas missense mutations may impact the gene function only partially. By convention, gene pleiotropy is measured by phenotyping null mutants.

4. PATTERNS OF PLEIOTROPY

4.1. Fisher's Geometric Model and Universal Pleiotropy

In an influential model of phenotypic adaptation known as the geometric model, Fisher (1930) imagined a high-dimensional space in which each dimension represents a phenotypic trait and each position in this space represents a genotype and corresponding phenotype. The relative fitness of a phenotype is determined by its distance in the space from the optimal phenotype; the smaller the distance, the higher the fitness. A mutation moves an organism from one position to another in this space. Just as a random line in a two-dimensional space is extremely unlikely to be parallel to a predefined axis, any random mutation in the high-dimensional space is expected to affect all phenotypic traits, albeit by different amounts. Therefore, the geometric model suggests universal pleiotropy—every mutation affects every trait. However, it is important to recognize that the geometric model is a toy model not expected to reflect biological realities. Furthermore, it appears that Fisher assumed that different genotypes must have different phenotypes, although at least in principle, different genotypes could have the same phenotype. Under the latter scenario, a mutation may not affect any phenotypic trait.

Nonetheless, the notion that every gene or every mutation affects every trait is often assumed in theoretical population genetics. Philosophically, this notation seems to make sense, because all traits of an organism are related to some extent as they together make up the organism; consequently, a mutation would affect all traits when it affects one trait. In the most extreme case, one could argue that a mutation from one nucleotide to another at a genomic position influences the cellular concentrations of the two nucleotides, which could have ripple effects on all traits of the organism even though the effect size is presumably exceedingly small for almost all traits. The notion of universal pleiotropy makes the statement that a particular gene or mutation affects a trait meaningless and both forward and reverse genetics futile, because every gene or mutation is already assumed to affect every trait. More fundamentally, universal pleiotropy is unfalsifiable and therefore not a scientific hypothesis, because one can always argue that the statistical power is insufficient whenever the mutational effect on a trait is not found to be statistically significant (Wagner & Zhang 2012).

4.2. Patterns Revealed by Forward Genetics

Statistical pleiotropy has been estimated by a number of forward genetic analyses in the last 15 years. For instance, Kenney-Hunt et al. (2008) and Wagner et al. (2008) mapped 120 QTLs

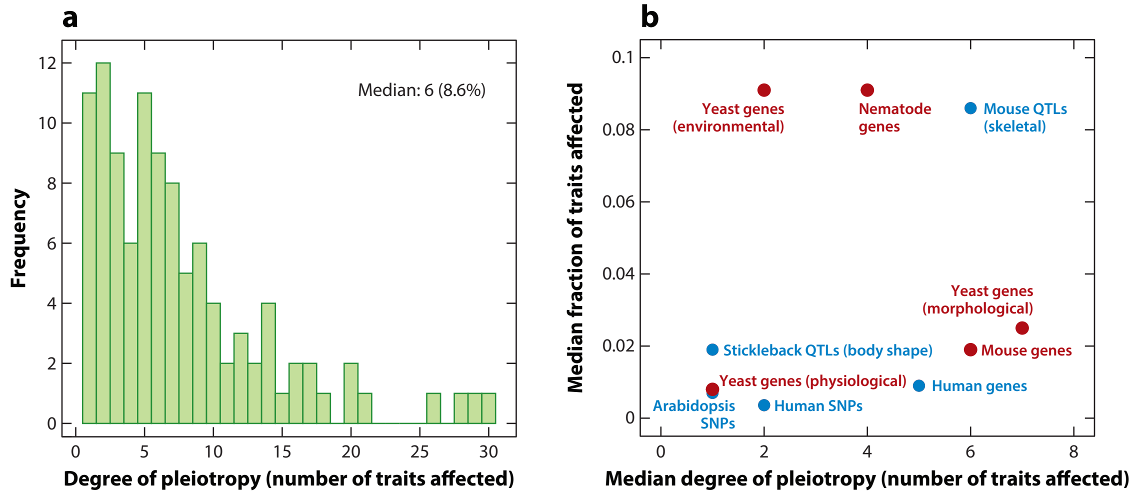

Figure 1

Frequency distribution of the degree of pleiotropy (i.e., the number of traits affected). (*a*) Distribution of pleiotropy of 120 mouse QTLs underlying 70 skeletal traits. The median degree is 6 traits, corresponding to 8.6% of all traits examined. (*b*) Median degrees of pleiotropy and median fractions of traits affected from forward and reverse genetic surveys of pleiotropy in human, mouse, stickleback, nematode, Arabidopsis, and yeast. Blue dots represent forward genetic surveys; red dots represent reverse genetic surveys. Abbreviations: QTL, quantitative trait locus; SNP, single nucleotide polymorphism. Panel *a* is based on data from Kenney-Hunt et al. (2008, table 3). Panel *b* is based on results from Albert et al. (2008), Wagner et al. (2008), Wang et al. (2010), Frachon et al. (2017), and Watanabe et al. (2019).

underlying the variations in 70 traits of the bony skeleton between inbred mice selected for either small or large body sizes. They reported that the distribution of pleiotropy is L shaped, with most QTLs affecting a few traits and a few QTLs affecting many traits (**Figure 1*a***). The median pleiotropy is six traits, or 8.6% of the traits examined. Similarly, Albert et al. (2008) mapped 26 QTLs underlying the variations in 54 position traits of the body shape between two closely related three-spined stickleback species: a Pacific marine species and a benthic lake species. Again, the distribution of pleiotropy is L shaped, with the median pleiotropy being only 1 trait, or 1.9% of the traits examined (**Figure 1*b***). By today's standard, the mapping resolution and sample size were both moderate in these two studies: The mouse study genotyped 1,040 individuals for 471 genetic markers, while the fish study genotyped 372 individuals for 248 genetic markers. A low mapping resolution due to the use of relatively few genetic markers can lead to an overestimation of (statistical) pleiotropy while a small sample size makes small effects undetectable and so leads to an underestimation of pleiotropy.

Using GWA mapping, Frachon et al. (2017) mapped single nucleotide polymorphisms (SNPs) underlying 144 eco-phenotypes in the model plant *Arabidopsis thaliana*, again reporting an L-shaped distribution of pleiotropy with a median of 1 trait (**Figure 1*b***).

GWA mapping has been extensively performed in humans for many diseases and other phenotypic variations (Welter et al. 2014, Chesmore et al. 2018). A meta-analysis of 558 traits from well-powered GWA studies (with sample sizes exceeding 50,000 individuals) offered a global view of statistical pleiotropy in humans (Watanabe et al. 2019). At the gene level, the distribution of pleiotropy is L shaped (when including only genes with at least one significant association), with a median pleiotropy of 5 traits or 0.90% of the traits examined (**Figure 1*b***). Two genes, *TNXB* (encoding tenascin-X) and *ATF6B* (encoding activating transcription factor 6 beta), are the most pleiotropic genes, each affecting 123 traits. At the SNP level, the distribution of pleiotropy is also L shaped (when including only SNPs with at least one significant association), with a median pleiotropy of 2 traits or 0.36% of the traits examined (**Figure 1*b***). The most pleiotropic SNP, located in an intron of *ATXN2* (encoding ataxin 2), affects 79 traits; despite this observation, exonic SNPs are generally more pleiotropic than intronic SNPs (Watanabe et al. 2019).

4.3. Patterns Revealed by Reverse Genetics

Wang et al. (2010) performed an analysis of gene pleiotropy using reverse genetic data from yeast, nematodes, and mice. They analyzed three data sets in the yeast *Saccharomyces cerevisiae*: (*a*) measures of 279 cellular morphological traits of single-gene-deletion strains (Ohya et al. 2005), (*b*) growth rates of single-gene-deletion strains in 22 different environments (Dudley et al. 2005), and (*c*) 120 literature-curated physiological functions of genes recorded in the Comprehensive Yeast Genome Database. In all three cases, the distribution of pleiotropy is L shaped, with the median pleiotropy equal to 7 (2% of the traits examined), 2 (9%), and 1 (1%) for the morphological, environmental, and physiological pleiotropy data sets, respectively (**Figure 1*b***). The nematode data set was based on the phenotypes of 44 early embryogenesis traits in *Caenorhabditis elegans* treated with genome-wide RNA-mediated interference (Sonnichsen et al. 2005). The mouse data set was based on the phenotypes of 308 morphological and physiological traits in gene-knockout mice recorded in Mouse Genome Informatics. In both species, the distribution of pleiotropy is approximately L shaped, with the median equal to 4 (9% of traits examined) and 6 traits (2%), respectively (**Figure 1*b***). Note that all of the above pleiotropy estimates could be underestimates as a result of a limited statistical power for detecting mutational effects (Hill & Zhang 2012, Paaby & Rockman 2013a).

The relationships between a set of genes and a set of traits can be described by a bipartite network of genes and traits, in which a link between a gene node and a trait node indicates that the gene affects the trait. A question of interest about pleiotropy is its potential modular structure (Wagner et al. 2007), where links within modules are significantly more frequent than those across modules. Indeed, the modularity is dozens of standard deviations (SDs) greater for each of the observed pleiotropy networks than for the mean of the corresponding rewired networks where gene–trait links are randomly drawn under the condition that the number of links for each gene and the number of links for each trait are unchanged (Wang et al. 2010).

4.4. Size Distribution of Mutational Effects on Multiple Traits

Although mutational pleiotropy is commonly measured by the number or fraction of traits affected by the mutation, the effect sizes also matter. Wang et al. (2010) measured the mutational effect size by Z score, which is the absolute effect size divided by the SD of the trait value among wild-type individuals. They found that the size distribution of the effects of a yeast gene deletion on 279 morphological traits is approximately normal, with the mean close to 0, for most gene deletions examined. However, the SD of the normal distribution varies substantially across different gene deletions; the greater the SD, the greater the number of traits significantly impacted by the deletion (Wang et al. 2010). This phenomenon suggests the possibility of quantifying pleiotropy by the SD, which is presumably less influenced by phenotyping sensitivity than is the degree of pleiotropy.

5. MOLECULAR BASIS OF PLEIOTROPY

First, a mutation can be pleiotropic because it affects multiple genes that overlap in their regulatory or coding sequences. Overlapping genes, commonly defined as genes that overlap in their mRNA sequences, have been extensively documented in viral, prokaryotic, and eukaryotic genomes (Wright et al. 2022). More common than overlapping genes is the scenario where multiple genes share regulatory nucleotides that are untranscribed, such as enhancers controlling multiple linked genes, bidirectional promoters, and promoters of operons. For example, the opsin locus control region (LCR) regulates the expression of red and green opsin genes on the human X chromosome; a dysfunctional LCR can cause the loss of both red and green opsins, leading to blue cone monochromacy (Carroll et al. 2010).

Second, a mutation can be pleiotropic because it affects multiple RNA or protein products of a gene. It is common for a gene to have multiple products, for example, due to alternative splicing, RNA editing, alternative transcriptional initiations, and alternative translational initiations, although not all products necessarily have distinct functions (Zhang & Xu 2022). For instance, human *LEF1* encodes lymphoid enhancer binding factor 1, which regulates the transcription of Wingless/Integrated (Wnt)/β-catenin genes. Two different protein isoforms are produced as a result of alternative transcriptional initiations: The longer isoform recruits β-catenin to Wnt target genes, whereas the shorter isoform cannot interact with β-catenin and instead suppresses the Wnt regulation of target genes (Arce et al. 2006). Thus, a mutation occurring in the region shared by the two isoforms potentially affects both isoforms and their respective functions.

Third, a mutation can be pleiotropic because it affects one protein product that has multiple molecular functions such as the mammalian serum albumin mentioned in Section 2 (He & Zhang 2006).

Fourth, a mutation can be pleiotropic simply because a single molecular function of a protein is directly or indirectly involved in multiple biological processes. For example, a protein may have only one molecular function as a transcription factor, but this transcription factor may directly regulate many target genes. Hence, a mutation reducing the DNA-binding affinity of the transcription factor could affect the expression of multiple target genes. In addition, the resulting expression changes of the target genes could impact the expression of their target genes, causing indirect effects of the mutation concerned. For example, a natural polymorphism in an intron of the gene encoding the transcription factor FLM influences *Arabidopsis thaliana* growth and color by affecting *FLM* splicing (Hanemian et al. 2020).

He & Zhang (2006) asked whether pleiotropy is more likely to be caused by the third or fourth mechanisms listed above. They compared yeast genes with different levels of environmental pleiotropy but found them to have similar mean numbers of molecular functions. By contrast, yeast genes with higher degrees of environmental pleiotropy tend to participate in more biological processes. Further, their protein products are found in more cellular components and have more protein-interaction partners. The authors concluded that yeast gene pleiotropy is more often attributable to the same molecular function involved in multiple biological processes (type II pleiotropy) than to multiple molecular functions (type I pleiotropy).

6. CAUSAL RELATIONSHIPS AMONG MULTIPLE EFFECTS OF A MUTATION

As mentioned, pleiotropy can be classified based on whether a single mutation separately affects multiple phenotypic traits (horizontal pleiotropy) or a single mutation starts off a cascade of events that sequentially influence multiple traits (vertical pleiotropy). This classification requires delineating the relationships among the phenotypic effects of a mutation on multiple traits. For example, if a mutation increases the chance of hypertension and reduces life span, it is possible that the reduction in the mutant's life span results entirely, partly, or not at all from the mutational impact on the probability of hypertension. If we no longer find a difference in life span between wild-type and mutant individuals after controlling for hypertension (i.e., among individuals with similar hypertension status), we can conclude that the mutational effect on life span is entirely due to its effect on hypertension (i.e., vertical pleiotropy). Alternatively, if the life span difference between wild-type and mutant individuals remains unchanged after controlling for hypertension, we can conclude that the mutational effect on life span is not due to its effect on hypertension (i.e., horizontal pleiotropy). It is also possible that the life span difference between wild-type and mutant individuals is reduced (but not to zero) after controlling for hypertension, suggesting that the pleiotropy has both horizontal and vertical components.

Note that distinguishing between horizontal and vertical pleiotropy requires finding the relationships among the mutational effects on multiple traits, not the relationships among these traits per se, as analyzed recently (Geiler-Samerotte et al. 2020). In the context of the above example, the relationship between hypertension and life span is not the same as the relationship between the effects of a mutation on hypertension and life span, which can be different for different mutations. While it remains generally difficult to distinguish between horizontal and vertical pleiotropy, progress is expected given the recent advancement in causal inference by statistical genetics (Pingault et al. 2018, Verbanck et al. 2018).

7. EVOLUTIONARY CONSEQUENCES OF PLEIOTROPY

7.1. Maintenance of Seemingly Deleterious Traits

Mutations causing Huntington's disease, a neurodegenerative disorder in which symptoms typically manifest after the reproductive age, also increase fecundity (Carter & Nguyen 2011). As mentioned, homozygous mutations causing cystic fibrosis also cause male infertility, but heterozygotes show higher fecundity than the wild type (Carter & Nguyen 2011). Such conflicting effects on different traits weaken the negative selection against disease mutations, cause the maintenance of these mutations in the population, and increase the disease incidence. It has been suggested that mutations causing Huntington's disease, cystic fibrosis, sickle-cell anemia, glucose-6-phosphate dehydrogenase deficiency, cancer, and many other diseases are kept in the population because of their benefits to other aspects of life such as development, fecundity, and host defense (Carter & Nguyen 2011). Note that a disease mutation could also hitchhike on linked beneficial mutations; that is, the above phenomenon extends to statistical pleiotropy. For example, mutations associated with myopia were found to be selected for in the current British population due to (presumably statistical) pleiotropic effects on reproduction (Long & Zhang 2021).

Ageing refers to a gradual deterioration of bodily functions that manifests as an increase in the death rate with age after sexual maturity. Ageing is widespread among multicellular organisms and demands an explanation, because natural selection should favor mutations conferring an extended reproductive life span (Reichard 2017). The antagonistic pleiotropy hypothesis posits that mutations contributing to ageing could be positively selected because of their pleiotropic beneficial effects early in life, such as promoting development or reproduction (Williams 1957). Indeed, a recent analysis of human genotype–phenotype data revealed a negative genetic correlation between reproduction and life span (i.e., the same or linked genetic variants have opposite effects on reproduction and life span), supporting the antagonistic pleiotropy hypothesis of ageing (Long & Zhang 2023).

7.2. Cost of Complexity in Adaptation

Orr found that the rate of adaptation of a population (U), measured by its fitness increase per generation, decreases precipitously with mutational pleiotropy (n) (Orr 2000). Because Orr assumed universal pleiotropy under Fisher's geometric model, n is the total number of traits of the organism, which can be regarded as a measure of organismal complexity. Hence, Orr concluded that the rate of adaptation decreases with organismal complexity, a result dubbed the cost of complexity.

However, Orr made several simplifying assumptions in his formulation. First, as mentioned, empirical data suggest that a mutation typically affects only a small proportion of traits (**Figure 1**). Nevertheless, if we assume that the proportion of traits affected by a mutation is comparable between species of different levels of complexity, Orr's formula still predicts a cost of complexity, albeit a lessened one.

Second, the scaling relationship between the total size (T) of all phenotypic effects of a mutation and n can be written as $T = an^b$, where a and b are two parameters. Orr assumed that T is independent of n (i.e., $b = 0$), while some authors believe that the per trait effect size is independent of n (i.e., $b = 0.5$) (Waxman & Peck 1998). The rate at which U declines with n is much lower under $b = 0.5$ than under $b = 0$ (Wang et al. 2010). In their QTL mapping of skeletal traits of mice, however, Wagner et al. (2008) observed a greater per trait effect size with larger n (i.e., $b > 0.5$). A similar trend was evident from human GWA mapping results (Chesmore et al. 2018). This pattern was also observed for yeast morphological traits, with b estimated at 0.601 ± 0.011 (95% confidence interval) (Wang et al. 2010). Similarly, $b = 0.724 \pm 0.0035$ for a set of eco-phenotypes of *Arabidopsis thaliana* (Frachon et al. 2017). When $b > 0.5$, U no longer monotonically declines with n but is an inverted U-shaped function of n, with U reaching its maximum at an intermediate n (Wang et al. 2010). Thus, with an empirically estimated b value, complexity is much less costly than Orr's original finding and may even promote adaptation. Let n_{optimal} be the level of pleiotropy corresponding to the highest U. It is intriguing that n_{optimal} reaches the highest value when b is in a narrow range from 0.56 to 0.79, which happens to encompass the empirically observed b values (Wang et al. 2010, Frachon et al. 2017). In the *Arabidopsis* study, Frachon et al. further observed the strongest signal of positive selection on genetic variants with intermediate levels of pleiotropy, consistent with the theoretical prediction that an intermediate n yields the highest U.

7.3. Gene Pleiotropy and Evolutionary Rate

It has been hypothesized that the functional constraint of a gene rises with its level of pleiotropy (Hodgkin 1998). Indeed, compared with yeast genes with relatively low morphological or environmental pleiotropy, those with relatively high pleiotropy cause greater fitness reductions when deleted (Cooper et al. 2007) and are more likely to have detectable fly or nematode homologs (He & Zhang 2006). Furthermore, the nonsynonymous nucleotide substitution rate (d_N) of a gene decreases with gene pleiotropy (He & Zhang 2006), even after controlling for the most important d_N determinant—gene expression level (Zhang & Yang 2015).

However, when gene pleiotropy is measured by the number of protein–protein interaction (PPI) partners (N_{PPI}) that the gene has, the result is mixed. A negative correlation was initially reported between the evolutionary rate of a protein and its N_{PPI} (Fraser et al. 2002), but the correlation was later shown to be caused at least in part by a confounder—gene expression level (Jordan et al. 2003). The controversy was subsequently resolved by the discovery of two types of proteins: The first type uses the same set of residues for interacting with different partners such that the increase in N_{PPI} does not alter the fraction of functionally constrained residues, while the second type uses different residues for interacting with different partners such that the increase in N_{PPI} raises the fraction of functionally constrained residues. Consequently, pleiotropy measured by N_{PPI} constrains sequence evolution for the second but not the first type of proteins (Kim et al. 2006).

7.4. Pleiotropy and Mechanisms of Phenotypic Evolution

Based on a summary of case studies and some theoretical reasoning, Carroll proposed different genetic mechanisms for the evolution of morphological and physiological traits (Carroll 2005). Specifically, he posited that morphological evolution more often proceeds by *cis*-regulatory changes than protein functional changes, while the opposite may be true for physiological evolution. The central consideration in his hypothesis is pleiotropy. A protein functional change affects all tissues where the protein is expressed, so it could be too pleiotropic to tolerate in morphological evolution. By contrast, a regulatory change may affect only one tissue, so it could be more tolerable. While the above makes sense, Carroll did not explain why physiological evolution would be different. Regardless, Liao et al. (2010) compared mouse genes that, when knocked out, affect only morphological traits (dubbed morphogenes) and those that affect only physiological traits (dubbed physiogenes). Interestingly, they found many differences between the two groups of genes. For example, morphogenes are enriched with transcriptional regulators, while physiogenes are enriched with channels, transporters, receptors, and enzymes. Compared with physiogenes, morphogenes are significantly more pleiotropic, less tissue specific in expression, and evolve more slowly in their protein sequences but faster in tissue-expression profiles. These results support Carroll's thesis and suggest that physiological evolution can probably tolerate protein functional changes because physiogenes tend to be tissue specific in their expressions.

A change in the expression of a gene can be caused by a *cis*-effect mutation, which is on the same DNA molecule as the gene (e.g., in the promoter or enhancers of the gene), or a *trans*-effect mutation, which can be anywhere in the genome because it acts through diffusible molecules such as transcription factors. It has repeatedly been shown that the expression divergence between orthologous genes is more often attributable to *cis*-effect mutations than *trans*-effect mutations and that this trend becomes more and more prominent when the divergence between orthologs gets larger (Hill et al. 2021). These patterns are explained by the fact that *cis*-effect mutations are less pleiotropic (i.e., affect fewer downstream genes) and therefore less deleterious than *trans*-effect mutations (Vande Zande et al. 2022).

7.5. Antagonistic Pleiotropy and Evolution in Changing Environments

The environment of virtually every natural population changes over time. For a neutral mutation destined for fixation, the time from its initial appearance to its fixation takes on average $4N_e$ generations for diploids. Even for beneficial mutations destined for fixation, the expected time to fixation is $2 \ln (2N_e - 1)/s$ (Otto & Whitlock 2013), which equals ~2,000 generations under $N_e = 10^4$ and ~3,400 generations under $N_e = 10^7$ when $s = 0.01$. Because it is highly likely that the environment varies during the above time, it is important to consider the possibility of environmental pleiotropy of mutations when discussing their behaviors in a population. Of particular interest is antagonistic environmental pleiotropy, under which a mutation would be selected for under some environments but selected against under other environments. Because beneficial mutations are far less common than deleterious mutations, it is likely that any beneficial mutation observed in an environment is deleterious in most other environments. Consequently, these mutations are rarely fixed, although they contribute to the adaptation of the population to the environment in which they are beneficial. Adaptive evolution would then be underestimated by the rate of nonsynonymous substitution relative to that of synonymous substitution (d_N/d_S).

To verify the above prediction, Chen & Zhang (2020) performed yeast experimental evolution in a changing environment that rotated among five media, with a duration of 224 generations per medium, totaling 1,120 generations. The five media were chosen based on the fact that the growth rates of a set of strains tend to be negatively correlated across the five media. For comparison, they performed experimental evolution in each of the five media separately for 1,120 generations. Because the selection intensity imposed by a medium diminishes as the population adapts to it, the average selection intensity in the constant environments over 1,120 generations should be lower than that in the changing environment where a new selection was imposed every 224 generations. Yet, by comparing the genome sequences of the progenitor and evolved strains, the authors found d_N/d_S in the changing environment to be significantly lower than the average d_N/d_S in the five constant environments. Examining the dynamics of mutations in the changing environment, the authors observed the expected patterns of antagonistic pleiotropy—allele frequencies rose quickly in one medium but dropped precipitously in the next medium.

Given that genome-wide d_N/d_S typically exceeds 1 in microbial experimental evolution performed in constant environments (Tenaillon et al. 2012, Lang et al. 2013, Kryazhimskiy et al. 2014, Tenaillon et al. 2016) and that the environment varies more frequently in nature than in these constant-environment experimental evolution scenarios, we predict that positive selection is frequent in nature. Hence, it is plausible that the observation of $d_N/d_S \ll 1$ between (even very closely related) strains sampled from natural environments (Rocha et al. 2006, Wolf et al. 2009, Maclean et al. 2017) does not reflect a lack of positive selection in nature but rather is due to antagonistic pleiotropy that conceals signals of positive selection (Chen & Zhang 2020).

7.6. Evolution of Pleiotropy

Several mechanisms are known to resolve adaptation-impeding pleiotropy. First, if the expression of a gene is beneficial in some environments or tissues but detrimental in other environments or tissues, gene regulation that promotes gene expression in the former but suppresses gene expression in the latter would be selectively favored. Such regulation reduces gene pleiotropy, because the gene no longer has an effect in the latter environments or tissues. In a yeast study, Qian et al. (2012) reported that deleting the *PRD17* gene from a lab strain is deleterious in ETH medium but beneficial in YPG medium, indicating that *PRD17* in the lab strain exhibits environmental antagonistic pleiotropy. PDR17 is a phosphatidylinositol transfer protein participating in phospholipid synthesis and transport and is involved in resistance to multiple drugs. If such pleiotropy is (at least partially) evolutionarily resolved in natural strains that experience environments similar to ETH or YPG, the expression of *PRD17* is expected to be lower in YPG than in ETH in these natural strains. This was indeed the case. Furthermore, *PRD17* expression in YPG and the ratio of *PRD17* expression in YPG to that in ETH are both lower in a strain adapted to YPG than in a strain adapted to ETH. Over 80% of genes tested showed signs of evolutionary resolution of antagonistic pleiotropy by gene regulation (Qian et al. 2012).

Second, Qian et al. (2012) discovered that pleiotropy can also be resolved by protein subcellular relocalization. Specifically, MIG1 is a transcription factor functioning exclusively inside the nucleus in glucose repression. Its functional allele was found to be fitter than the null allele in YPD medium, but the opposite is true in OAK medium. In the strain adapted to an environment mimicked by OAK medium, MIG1 is localized to the nucleus under YPD. However, under OAK, where MIG1 would be deleterious, MIG1 is localized to the cytoplasm and hence imposes no harm.

Third, a multi-functional protein may not be able to optimize all of its functions due to trade-offs, and gene duplication followed by functional specialization is a well-recognized mechanism to resolve such adaptive conflicts (Hughes 1994). For example, mammalian pancreatic ribonuclease (RNASE1) is an enzyme that typically digests both single-stranded RNA (ssRNA) and double-stranded RNA (dsRNA); while the former activity is primarily used for food digestion, the latter is probably involved in host defense against RNA viruses. *RNASE1* was duplicated in an ancestor of the leaf-eating monkey *Pygathrix nemaeus*; upon duplication, one daughter gene has retained its activity for degrading dsRNA while the other daughter gene (*RNASE1B*) has evolved an elevated activity for digesting ssRNA in the acidic small intestine of leaf monkeys but has almost completely lost its activity toward dsRNA (Zhang et al. 2002). There have been nine amino acid substitutions in the mature peptide of RNASE1B since the gene duplication, and each of them reduced its activity toward dsRNA (Zhang et al. 2002). Therefore, without gene duplication and the relaxation of the functional constraint from the activity toward dsRNA, none of the nine substitutions, some of which are necessary for the improved activity toward ssRNA (Zhang 2006), would have been allowed in RNASE1B (Zhang et al. 2002). In another example, Näsvall et al. (2012) showed in an experimental evolution setting that a gene in *Salmonella enterica* with low levels of two distinct activities was duplicated and each ancestral activity increased in one of the daughter genes in subsequent evolution. The authors found that, although it was not impossible to evolve a single generalist gene with both activities increased, the high rate of gene duplication made it more likely to evolve specialist genes in their study.

Fourth, in principle, mechanisms that generate multiple distinct products from one gene could also help resolve adaptation-impeding pleiotropy. For instance, two functions of a gene may require slightly different protein sequences, which may be accomplished by RNA editing, alternative splicing, alternative translational initiation, and so on (Zhang & Xu 2022).

Because the optimal phenotype can differ between males and females, a mutation may exhibit antagonistic pleiotropy between the two sexes. Evidence for such pleiotropy is abundant (Bonduriansky & Chenoweth 2009). For example, male fitness and female fitness tend to show negative correlations across different laboratory-constructed genotypes of *Drosophila melanogaster* (Innocenti & Morrow 2010). Several mechanisms have been found to alleviate or resolve such sexual antagonism. First, a gene that is beneficial to one sex but deleterious to the other can be differentially expressed in the two sexes (Ellegren & Parsch 2007, Mank 2009, VanKuren & Long 2018). Second, genomic imprinting, where only one parental allele is expressed in the offspring, is believed to reflect a compromise in the conflict of interest between the two parents (Moore & Haig 1991, Wilkins & Haig 2003). For example, for genes that promote embryonic growth, such as the insulin-like growth factor 2 gene in mice, the paternal copy tends to be expressed, while the maternal copy is silenced. Third, sex-specific dominance reversal—the allele beneficial to males is dominant in males but recessive in females and vice versa—can alleviate sexual antagonism (Grieshop & Arnqvist 2018). Fourth, relocating genes beneficial to males but deleterious to females to the (male-specific) Y chromosome should alleviate sexual antagonism. It is, however, debated whether the X chromosome enriches sexually antagonistic genes when compared with autosomes (Rice 1984, Mank 2009, Ruzicka & Connallon 2020, Ruzicka & Connallon 2022).

It is important to distinguish mutations that alter gene pleiotropy from those that alter the pleiotropy of future mutations. For example, a mutation that suppresses *PRD17* expression in YPG medium reduces the pleiotropy of *PRD17* and is advantageous in YPG. There are also mutations that can be called pleiotropy modifiers, which modify the pleiotropy of future mutations (Pavlicev et al. 2008). Natural selection on pleiotropy modifiers is a second-order selection, so it is expected to be very weak because the modifiers do not alter the fitness of the individuals carrying the modifiers but that of the offspring of these individuals.

8. OUTSTANDING QUESTIONS

First, as discussed, reliably quantifying pleiotropy is difficult, both conceptually and operationally. Consequently, it is debated whether the empirically measured pleiotropy reflects the truth or is an artifact of limited phenotyping sensitivity (Wagner & Zhang 2011, Hill & Zhang 2012, Wagner

& Zhang 2012, Paaby & Rockman 2013a, Paaby & Rockman 2013b, Zhang & Wagner 2013). Theoretical work is needed to determine whether pleiotropy can be reliably measured without substantially increasing phenotyping sensitivity and the number of traits examined. For example, can the SD of the distribution of the effect sizes of a mutation on a moderate number of traits (Wang et al. 2010) serve as a reliable measure of pleiotropy?

Second, as mentioned, adaptation-impeding pleiotropy may be resolved by many mechanisms such as gene duplication and gene regulation. Furthermore, at least in theory, pleiotropy modifiers that minimize the negative impact of the pleiotropy of future mutations are selectively favored. Thus, does pleiotropy constrain evolution only temporarily? Under what conditions is pleiotropy unresolvable, and to what extent does pleiotropy permanently constrain evolution? Theory shows that, when the scaling exponent $b > 0.5$, the adaptation rate U is maximized at an intermediate degree of pleiotropy. However, the pleiotropy that yields the highest U depends on several parameters, so it is unclear whether the observed pleiotropy is larger or smaller than the pleiotropy that yields the highest U. Answering this question will help determine whether there is a cost of complexity and on what type of organisms the cost is imposed.

Third, laboratory microbial evolution in a changing environment showed that antagonistic environmental pleiotropy of mutations causes underestimation of positive selection and conceals adaptive signals. How widespread is this phenomenon in nature? Answering this question will be important for estimating the relative prevalence of neutral and adaptive evolution.

Fourth, how do we reliably separate horizontal from vertical pleiotropy? If a mutation affects trait B because of its effect on trait A (vertical pleiotropy), one could minimize the mutational effect on B by prohibiting the change of A (by any means influencing A). This would not work if the mutational pleiotropy were horizontal. Hence, distinguishing vertical from horizontal pleiotropy can be important in medicine and other applications.

In summary, in less than two decades, pleiotropy research has evolved from a largely theoretical realm with little systematic, empirical data to a rapidly expanding domain with large-scale data, multiple experimental tools and model systems, and ample opportunities for empirical tests of relevant hypotheses. With these developments and the fundamental roles of pleiotropy in so many biological phenomena, pleiotropy research is poised to bring many new insights in the future.

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The Importance of Regulatory Network Structure for Complex Trait Heritability and Evolution

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Abstract

Complex traits are determined by many loci—mostly regulatory elements—that, through combinatorial interactions, can affect multiple traits. Such high levels of epistasis and pleiotropy have been proposed in the omnigenic model and may explain why such a large part of complex trait heritability is usually missed by genome-wide association studies, while raising questions about the possibility for such traits to evolve in response to environmental constraints. To explore the molecular bases of complex traits and understand how they can adapt, we systematically analyzed the distribution of SNP heritability for 11 traits across 29 tissue-specific expression quantitative trait locus networks. We find that heritability is clustered in a small number of tissue-specific, functionally relevant SNP–gene modules and that the greatest heritability occurs in local "hubs" that are both the cornerstone of the network's modules and tissue-specific regulatory elements. The network structure could thus both amplify the genotype–phenotype connection and buffer the deleterious effect of the genetic variations on other traits. We confirm that this structure has allowed complex traits to evolve in response to environmental constraints, with the local "hubs" being the preferential targets of past and ongoing directional selection. Together, these results provide a conceptual framework for understanding complex trait architecture and evolution.

Keywords: GTEx, eQTL, bipartite networks, GWAS, heritability, complex traits, polygenic selection

Introduction

In many organisms, including yeasts, insects, worms, plants, and mammals, adaptation often leverages polygenic traits to respond to new environmental challenges (Daub et al. 2013; He et al. 2016; Zan and Carlborg 2019). Such "complex" traits have an interconnected genetic architecture involving from a small number to potentially thousands of partly independent loci (Salomé et al. 2011; Pais et al. 2013; Peiffer et al. 2014; Shi et al. 2016). Up to 90% of the SNPs associated with such traits are localized outside of coding regions, according to genome-wide association studies (GWAS) (Ward and Kellis 2012; Tak and Farnham 2015). They are thus likely to affect traits by modifying gene expression—often by altering the sequence of *cis*-regulatory elements. This is consistent with the observation that GWAS-significant regions colocalize with SNPs that are linked to the expression of nearby genes (expression quantitative trait loci, *cis*-eQTLs) (Hormozdiari et al. 2016; Zhu et al. 2016). Partitioning heritability for these traits across various annotations while correcting for linkage disequilibrium has confirmed that *cis*-eQTLs as a group explain more of complex trait heritability than would be expected by chance (Torres et al. 2014; Gamazon et al. 2018; Hormozdiari et al. 2018) but still fail to explain a majority of trait heritability (Yao et al. 2020). Others have shown that including *trans*-eQTLs may capture additional heritability and explain important pathological mechanisms (Luningham et al. 2020), but this is rarely done because *trans*-eQTL studies are generally underpowered.

Advances in systems biology and functional genomics have highlighted the fact that at the molecular level, complex traits are the results of many regulatory interactions between actors of different biological processes within a complex cellular gene regulatory network (Hallgrimsson et al. 2014; Sonawane et al. 2019). This high level of pleiotropy coupled with significant epistatic interactions may explain the missing heritability observed in quantitative genetics studies and has led to the development of the omnigenic model (Boyle et al. 2017; Liu et al. 2019). In this model, trait association signals are spread across most of the genome in a way that includes many genes lacking an obvious connection to a particular trait. The model posits that most heritability can be explained by effects on genes outside of "core pathways," often acting in *trans*, but acting in ways that alter the functioning of those pathways and thus account for most of a given trait's heritability. The amount of pleiotropy implied by this model may limit, at first glance, the possibility for a given complex trait to adapt in response to an environmental change without interfering with other unrelated traits. However, the evidence for polygenic adaption processes in SNPs associated with complex traits in several species highlights the need to dig deeper into the molecular bases of complex traits to understand their architecture and how they adapt (Barreiro and Quintana-Murci 2010;

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Jakobson et al. 2019; Fagny and Austerlitz 2021; Mathieson 2021).

Given the number of loci and the number of potential interactions involved, gene regulatory networks provide an efficient tool for understanding the biological processes behind trait heritability (Kim et al. 2019). Combining GWAS results with gene coexpression networks have proven useful to identify the causal gene within a candidate region in many species (Chan et al. 2011; Lee et al. 2011; Schaefer et al. 2014; Taşan et al. 2015; Calabrese et al. 2017; Schaefer et al. 2018). By integrating SNP–gene or SNP–trait association studies results with coexpression networks, these studies have highlighted the fact that *cis*- and *trans*-regulatory relationships involved in complex trait determination are clustering within coexpression subnetworks enriched for functionally relevant molecular pathways. Due to power issues, integration methods generally do not account directly *trans*-regulatory effects (Siewert-Rocks et al. 2022) and focus on *cis*-regulatory associations that only explain a small fraction of trait heritability (Yao et al. 2020), relying on the coexpression network to provide indirect information on *trans*-regulatory relationships. However, *trans*-regulatory factors that have been shown to carry an important share of the heritability (Westra et al. 2013; Torres et al. 2014; Brynedal et al. 2017), as reflected in the omnigenic model (Boyle et al. 2017; Liu et al. 2019). We previously proposed a way to integrate both *cis*- and *trans*-eQTL results using a bipartite eQTL network representation (Platig et al. 2016; Fagny et al. 2017) that relies on summary statistics from eQTL studies and is relatively insensitive to a high false discovery rate, allowing one to partially compensate for the lack of power of *trans*-eQTL studies; an update to this method allowed us to weigh SNP–gene regulatory relationships by eQTL effect sizes (Gaynor et al. 2022). These analyses have shown that highly structured eQTL networks can reliably identify, in a tissue-specific manner, the biological functions disrupted by traits-associated SNPs (Fagny et al. 2017, 2019), and further defined network topological features that are useful in explaining in part the link between genotype and phenotype in complex traits.

In this study, we build on eQTL networks to investigate how complex trait heritability is spread within and among the tissue-specific eQTL networks to better understand both the genetic architecture of complex traits and how they may evolve under directional selection. We used GWAS summary statistics from eleven complex traits and diseases and built 29 tissue-specific *cis*- and *trans*-eQTL networks using the Genotype-Tissue Expression (GTEx) dataset. We then partitioned heritability across various features, including network node topological summary statistics, to identify key determinants of trait heritability. We investigated whether heritability for each trait is clustered in particular subparts (regulatory modules, also sometimes referred to as communities) of the eQTL networks, and identified network communities regulating biological functions that explain most of the heritability of each trait. Finally, in order to assess how complex traits may evolve, we looked for signatures of past polygenic adaptive processes in key regulatory loci considered in relationship with their role in the eQTL network. We found that heritability is not scattered uniformly across the genome but rather "clustered" in eQTL modules that represent trait-relevant functions and that loci playing a key role in regulating tissue-specific biological processes were not only more likely to explain trait heritability but also to have been preferential targets of past polygenic adaptation events.

Results

SNPs Associated at the Genome-Wide Level with Traits or Diseases are Clustered in a Few Modules

We wanted to understand how heritability is distributed across the eQTL network. We had previously reported that SNPs significantly associated with chronic obstructive pulmonary disease, cancers, and other traits in GWAS are concentrated in a small number of modules (Platig et al. 2016) and wanted to verify that this result can be extended beyond genome-wide significantly associated SNPs to SNPs explaining a larger part of genetic heritability. We thus used RNA-seq and genotyping data from GTEx to perform eQTL analyses and built 29 tissue-specific weighted bipartite eQTL networks. To compensate for the fact that *trans*-eQTL detection is notoriously underpowered compared with *cis*-eQTL detection, we used a lose false discovery rate (FDR) threshold of 0.2 (see Materials and Methods and supplementary text, Supplementary Material online). We identified modules containing highly connected SNPs and genes within each network based on CONDOR's bipartite modularity maximization (Platig et al. 2016) (see Materials and Methods, supplementary text, Supplementary Material online, supplementary fig. S1, Supplementary Material online, and supplementary table S1, Supplementary Material online). We also confirmed that our eQTL networks are summarizing information related to the genetic bases of gene expression coregulation rather than merely gene co-expression profiles (see Materials and Methods, supplementary text, Supplementary Material online, and supplementary table S3, Supplementary Material online) and that they are enriched for long-range *cis* 3D chromatin interactions (see Materials and Methods, supplementary text, Supplementary Material online, and supplementary table S4, Supplementary Material online).

We considered a total of eleven traits and diseases chosen for their medical or evolutionary relevance in populations of European descent. Breast cancer (BRC), ovarian cancer (OVC), and prostate cancer (PRC), Alzheimer's disease (ALZ), multiple sclerosis, Schizophrenia (SCZ), high-density lipoprotein levels (HDL), and type 2 diabetes (TIID; diabetes) are all increasing in populations of European descent and are known to exhibit partial genetic heritability. Smoking cessation is a measure of smoking dependency that is partly explained by genetic factors and is linked to a host of pulmonary and other diseases. SCZ is a highly heritable but poorly understood polygenic disease. Finally, height (HGT) is the canonical example of a polygenic trait, and some consider it a natural selection target in populations of European descent. These traits and diseases also represent a wide range of global genetic heritability, as reported in the literature, from 25% for TIID, to up to 80% to 90% for SCZ and Alzheimer's. These traits and diseases also represent a wide range of global genetic heritability, as reported in the literature, from 25% for TIID, to up to 80% to 90% for SCZ and Alzheimer's. We conditioned on European descent because of the number of GWAS data available and the demographics of GTEx. A complete description of these traits and diseases and their GWAS summary statistics is reported in the supplementary table S2, Supplementary Material online. For each trait or disease, we obtained summary statistics (see Materials and Methods).

We used the summary statistic Z^2 as a proxy for the per-SNP heritability, and SNPs with a GWAS Z^2 in the top 5% were named "high heritability SNPs." We found that the high heritability SNPs cluster in a small number of network modules

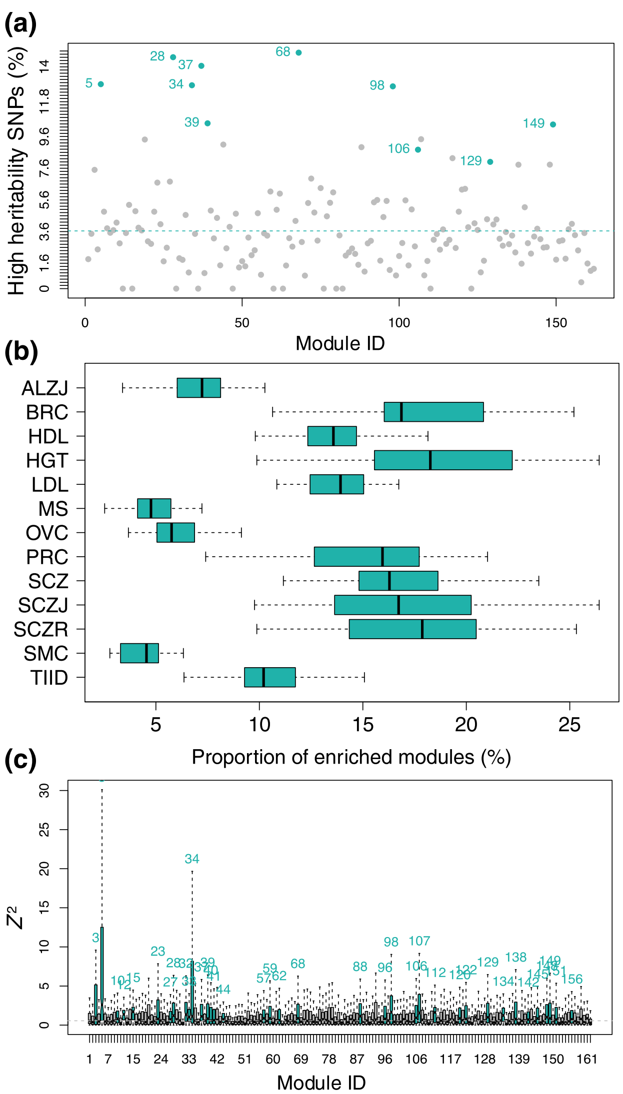

Fig. 1. High heritability SNPs are clustered in a few trait-specific modules. (a) Proportion of high heritability SNPs for BRC in each module of the skin—not sun-exposed (lower leg) eQTL network. Modules significantly enriched for high heritability SNPs are highlighted in color, with module number printed on the left (Benjamini–Hochberg corrected $P \leq 0.01$ using a χ^2-test). The dotted line represents the expected proportion of high heritability SNPs by chance in each module. (b) Distribution of the proportion of modules enriched for high heritability SNPs for each trait or disease among all tissue-specific eQTL networks. (c) Distribution of Z^2 values in each module of brain—nucleus accumbens (basal ganglia) for Alzheimer's disease (Kruskal–Wallis $P < 2 \times 10^{-16}$). Modules with a distribution skewed towards high values are represented in color, with module number printed on top (modules with a Benjamini–Hochberg-corrected $P \leq 0.01$ using one-sided Mann–Whitney U tests for each module vs. the rest of the network were considered as significantly enriched in high Z^2). The dotted blue line represents the expected Z^2 median across the whole network.

(supplementary table S5, Supplementary Material online). As an example, the breast cancer-associated SNPs appear in only 29 (12.6%) of the 230 network modules in the SKN (skin—not sun-exposed—suprapubic) eQTL network (Fig. 1a) representing a substantial concentration of heritability. We found similar results for other diseases and other tissues, with some combinations exhibiting even greater clustering of heritability in a limited number of network modules (Fig. 1b). We validated that these results were not dependent on the method we use to infer the clusters in our networks, because we obtained the same results using a different starting point for the same louvain-clustering based algorithm, and two other clustering algorithm altogether (see Materials and Methods, supplementary text, Supplementary Material online, and supplementary fig. S2, Supplementary Material online).

Because the top 5% SNPs only explain a small proportion of the heritability of the associated traits (particularly for highly polygenic traits), we examined the distribution of per-SNP heritability across the different eQTL network modules. We found that heritability was not distributed evenly across modules (Kruskal–Wallis test $P = 0$ for all pairs of tissue-specific networks and traits tested). As an example, consider Z^2 calculated for ALZ in brain—nucleus accumbens (basal ganglia) networks, which is plotted in Fig. 1c; all the results are in supplementary table S5, Supplementary Material online). Comparing the distribution of Z^2 for each module with the rest of the network, we found that less than a third of the modules contain high heritability SNPs (55 [5 to 85] modules representing about 23.8% [5.6 to 33.1] of the total number of modules depending on tissue-specific network/trait pair, with Benjamini–Hochberg-corrected Mann–Whitney U tests $P \leq 0.01$; see supplementary table S6, Supplementary Material online).

Heritability Is Enriched in Trait-Specific, Functionally Relevant Modules

We investigated whether the heritability for different, uncorrelated traits was clustered in the same or different modules in the tissue-specific eQTL networks. Depending on the tissue-specific network, about 38% [26 to 52] of the modules were not enriched for high heritability SNPs associated with any traits, 28% [22 to 33] were enriched for high heritability SNPs from only one trait and can be considered trait-specific, and 2% [1 to 6] were enriched for high heritability SNPs from at least six of the eleven traits and can be considered as shared across many traits (Fig. 2a and supplementary fig. S3, Supplementary Material online).

We performed a pairwise comparison of traits in each of the tissue-specific networks between modules enriched for high heritability SNPs. Using 10,000 resamplings, we found that top-heritability SNPs for two independent traits did not cluster in the same modules more than expected by chance in most cases. Indeed, among the 1,595 pairwise comparisons possible—$\binom{11}{2}$ comparisons ×29 tissues—only 54 show significant enrichment in overlap compared to what would expected by chance (Benjamini–Hocheberg-corrected $P \leq 0.01$). In more than half of these cases (33), the excess of overlap was observed between HDL and low-density lipoprotein (LDL), HGT, and/or and TIID in various tissue-specific networks (see supplementary table S7, Supplementary Material online). This co-occurence of HDL and LDL is also found in all four ADV network clustering (LCS1, LCS2, FG, and LEC), with $FDR \leq 0.05$ (supplementary table S8, Supplementary Material online).

We also investigated the biological functions represented by genes with the modules enriched for high heritability SNPs. We performed a GO term enrichment analysis for each module using Bioconductor R topGO package (see Materials and Methods); the results are presented in supplementary table S9, Supplementary Material online. This allowed us to identify modules that were tissue-specific (see Materials and Methods). The list of tissue-specific modules can change marginally depending on the Jaccard Index cutoff chosen. Based on the distribution of Jaccard indexes for pairwise comparison of tissue-specific networks, we chose two different thresholds: 0.3 and 0.4 (see supplementary fig. S4, Supplementary Material online for all

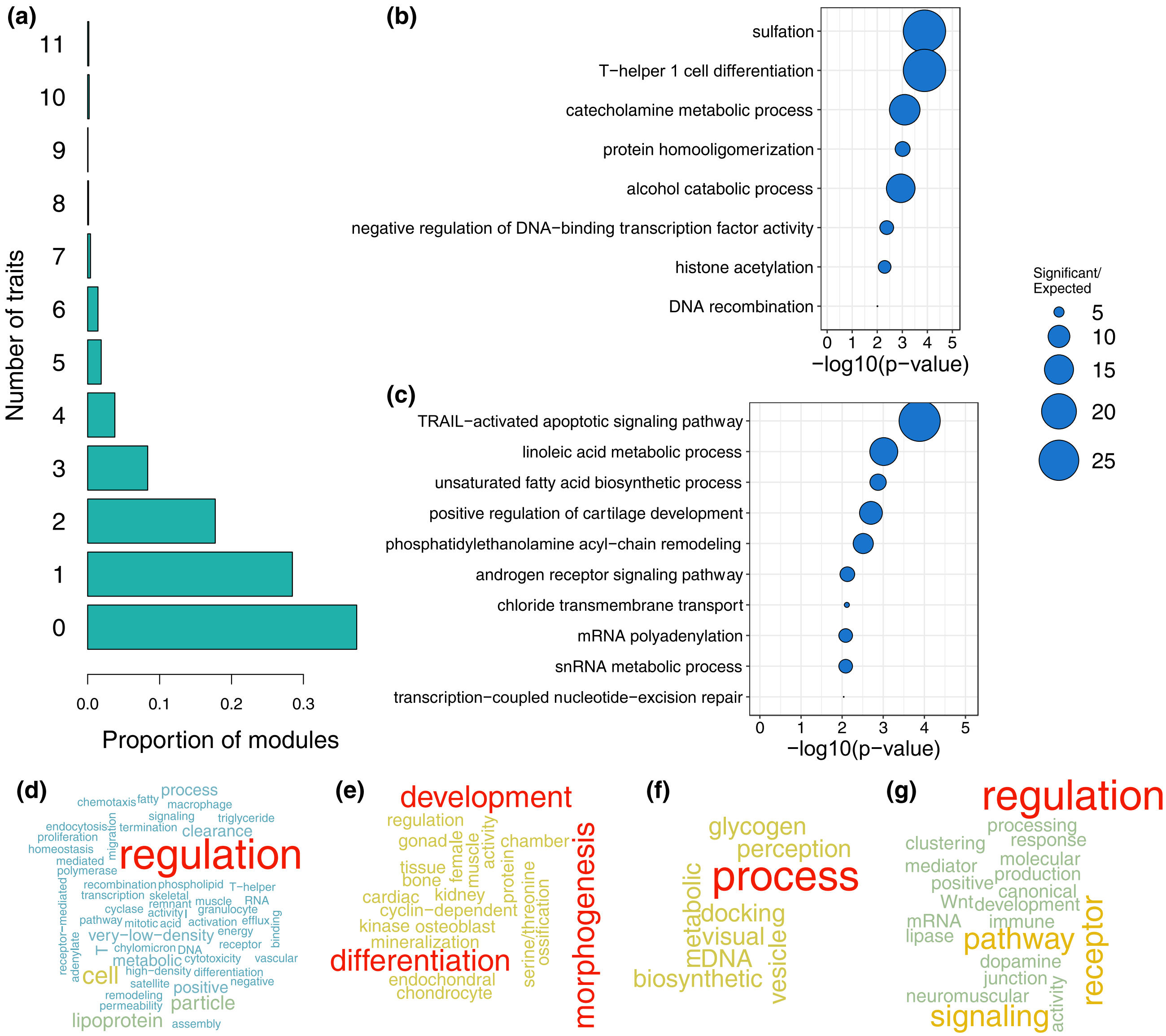

Fig. 2. Heritability is clustered in trait-specific, biologically significant modules. a) Average proportion of modules enriched in high heritability SNPs for one to eleven traits or diseases across all tissue-specific modules. To remove artifacts due to the high genetic correlation between the three schizophrenia studies, only one schizophrenia study was taken into account in the analysis. The 0 category represents the modules that are not enriched in high heritability SNPs for any trait or disease. For a split by tissue-specific network, see supplementary fig. S3, Supplementary Material online. b to g) Gene Ontology enrichment analyses results on gene content for a few tissue-specific modules enriched for high heritability SNPs for one or two traits and diseases. Bubble plots for top ten terms or terms with *P*-value ¡0.01 for b) brain (nucleus accumbens—basal ganglia) module 5, enriched for high heritability SNPs for Alzheimer's disease and multiple sclerosis and c) colon—transverse module 149, enriched for high heritability SNPs for prostate cancer. Word clouds representing word frequencies in gene ontology terms enriched in d) adipose—visceral omentum, module 142, enriched for high heritability SNPs for HDL; e) muscle—skeletal, module 90, enriched for high heritability SNPs for height; f) whole blood, module 17, enriched for high heritability SNPs for type 2 diabetes; g) brain (nucleus accumbens—basal ganglia) module 100, enriched for high heritability SNPs for schizophrenia.

comparisons and supplementary fig. S5, Supplementary Material online for tissue-by-tissue comparisons with the adipose visceral (omentum) network. If choosing a more stringent 0.3 threshold mechanically decreases the number of tissue-specific modules from 2,748 to 1,615, it does not change our main conclusions. We then focused on trait- and tissue-specific modules, defined as those enriched for high heritability SNPs from one to three traits and enriched for tissue-specific GO Terms (supplementary table S10, Supplementary Material online)—to investigate the specificity of each trait and/or disease heritability and explicit the underlying molecular mechanisms. We found that these modules were enriched in genes involved in biologically and trait-relevant functions.

For example, heritability for ALZ and multiple sclerosis were both clustered in module 5 of the brain—nucleus accumbens (basal ganglia) network; module 5 is enriched in genes involved in catecholamine metabolic process (Fig. 2b), a class of molecules whose concentrations are altered in symptomatic Alzheimer's and multiple sclerosis diseases, both of which are amyloid plaque diseases (Cercignani et al. 2021; Henjum et al. 2022). Heritability for SCZ was most strongly clustered in module 100 of the same tissue, a molecule that is enriched for dopamine receptor signaling pathway genes (Fig. 2g), and this pathway is known to be functionally disrupted in the brain striatum in SCZ patients (McCutcheon et al. 2019).

Unsurprisingly, heritability for cancers, including BRC, OVC, and PRC, was enriched in epithelial tissues (skin—not sun-exposed exposed, skin—sun-exposed, colon—transverse) in network modules consisting of genes enriched for immune response (PRC and module 76 of skin, not sun-exposed), response to cellular hypoxia (BRC, prostate cancer and module 157 of skin—sun-exposed), DNA break repair, cell cycle, apoptosis, and epithelium differentiation and growth (breast cancer and module 149 of skin—sun-exposed, and module 191 of skin (not sun-exposed), OVC and module 199 and skin, prostate cancer and modules 78, 97 of skin and module 149 of skin—sun-exposed). Particularly interesting, PRC heritability is clustered in module 149 of the colon-transverse network, enriched for TRAIL-activated apoptotic signaling pathway genes, a long-known signaling pathway involved in cancer progression (Johnstone et al. 2008) (Fig. 2c). Breast and OVC heritability are also clustered in modules involved in estrogen metabolism and signaling (BRC and module 180 of skin—sun-exposed, OVC and module 182 of skin—not sun-exposed).

Metabolic traits such as high blood levels of HDL or LDL and TIID tend to cluster in modules enriched for lipid and carbohydrate metabolism. More generally, heritability for HDL and LDL tend to cocluster in several tissue-specific networks, including in adipose–visceral omentum, where almost half of the modules enriched for one trait being also enriched for the other one, the gastro-intestinal networks (CLS-, EGJ, EMC), the heart-related networks (HRA, HRV), the liver network, and the esophagus muscularis and fibroblasts networks. Heritability for high HDL and LDL levels are for example enriched in module 142 of adipose–visceral omentum, enriched for genes involved in very-low-density lipoprotein and HDL particle assembly, remodeling, and clearance (Fig. 2d). They also cocluster in modules enriched for genes involved in lipid metabolism (modules 92 and 197), and cholesterol efflux (module 206). High HDL and LDL heritability can also cluster in different modules enriched for genes belonging to similar pathways: inflammation (modules 4, 133, and 189 for HDL—modules 5, 71, and 128 for LDL) and phosphorylation of STAT protein (module 179 for HDL, 49 for LDL). High HDL heritability is also found in modules related to cholesterol homeostasis (module 189) and ketone metabolism (module 133), while high LDL heritability is found in modules related to lipoprotein lipase activity (module 144), response to stresses (modules 156 and 239) and redox metabolism (modules 57 and 109). These results are robust to the clustering algorithm, in particular the enrichment of both high HDL and high LDL heritability in ADV modules with genes belonging to the lipoprotein metabolic and clearance pathway (module 71 in LCS2; 44 in FG; and modules 1 and 52, with a less significant $P \leq 0.05$ for enrichment of heritabilility, in LEC, supplementary tables S11 and S12, Supplementary Material online).

Finally, TIID heritability clusters in module 17 of whole blood, enriched in genes involved in glycogen metabolism (Fig. 2f). Finally, heritability for HGT, a trait primarily related to development and, in particular, bone and muscle growth, is clustered in module 90 of muscle–skeletal, enriched for genes involved in muscle tissue morphogenesis, endochondral ossification, bone mineralization, and chondrocyte and osteoblast differentiation (Fig. 2e).

Local and Global Hubs Carry a Large Part of Trait Heritability

We then explored how heritability is distributed across the SNPs nodes in the eQTL network and whether there are particular network topological features that correlate with a greater-than-expected ability to explain the heritability of a particular trait. We thus computed two different summary statistics characterizing the topological properties of SNPs within each network: outdegree and core score (Platig et al. 2016). The outdegree measures the centrality of an SNP within the entire network. SNPs within the top 25% of outdegree distribution were considered as global hubs. The core score measures the contribution of SNPs to the modularity of the network module in which it arises. SNPs within the top 25% of core score distribution were considered local hubs (core SNPs).

We investigated whether trait heritability was distributed evenly across all SNPs or instead concentrated in SNPs with local or global centrality. Using a likelihood-ratio test accounting for linkage disequilibrium and module size (see Materials and Methods), we found for almost every trait we considered that the SNPs explaining the greatest portion of the heritability (top 5% of Z^2, or high heritability SNPs) are more likely to have both high outdegree and core scores (supplementary fig. S6, Supplementary Material online for outdegrees and supplementary fig. S7c, Supplementary Material online for core scores). The enrichment in high heritability among local "hubs" appears stronger and more systematic than for global "hubs"; in artery coronary and liver, the global hubs do not show any significant enrichment at all. Using the adipose visceral (omentum) and the whole blood networks as a model, we found that enrichment of high heritability SNPs in local hubs increases with the threshold chosen to define high outdegree and core score [from the top 90% to the top 5%, supplementary fig. S8a, Supplementary Material online for whole blood and supplementary fig. S8c, Supplementary Material online for adipose visceral (omentum)]. This result is also robust to clustering method (see Materials and Methods and supplementary figs. S8*c* to S8*f*, Supplementary Material online and detailed results for threshold =0.75 in supplementary fig. S9a, Supplementary Material online).

Given the demonstrated importance of eQTL network topology, we explored whether the increased heritability we found was driven simply by being in the network (a proxy for being an eQTL), being a global hub, or being a local hub. These annotations are overlapping and reflect potentially confounding factors such as underlying chromatin annotations (e.g. global hubs are enriched for non-genic enhancers, while local hubs are enriched for genic enhancers and promoters (Fagny et al. 2017)). For this reason, we partitioned heritability across various functional annotations while accounting for linked markers using stratified LD score regression (Bulik-Sullivan et al. 2015; Finucane et al. 2015) (see Materials and Methods) using the 97-level baseline annotation model, to which we added our three annotations: belonging to eQTL network, being in the top quartile of outdegrees (global hubs), or of core scores (local hubs). The total proportion of heritability explained by SNPs as estimated using the LDSC software is reported in supplementary table S2, Supplementary Material online. We found that these proportions are not always perfectly correlated with the estimated global genetic heritability reported in published twin and pedigree studies (see supplementary table S2, Supplementary Material online), but our estimate of the total heritability of traits explained by SNPs as computed by the LD-score regression is coherent with previous reports (see Lindström et al. 2017 for breast, ovarian, and prostate cancer examples).

The LD-score regression confirmed the results we obtained above using likelihood-ratio tests: SNPs with high core scores or high outdegrees are enriched for trait heritability, and this enrichment increases with the threshold chosen to define high scores (supplementary fig. S8b, Supplementary Material online). The detailed results that show enrichment in h^2 among SNPs within each annotation for each of the 11 traits and diseases are reported in supplementary table S13, Supplementary Material online. These results are once again independent of the chosen clustering algorithm (supplementary fig. S9b, Supplementary Material online).

We performed a meta-analysis across the 11 uncorrelated traits and diseases for each tissue-specific network. An example of enrichment in h^2 among each annotation for the Whole-Blood network is shown in Fig. 3a. As noted earlier, SNPs that are within the eQTL networks tend to be significantly enriched in h^2 independent of tissue (Fig. 3b). However, the enrichment in h^2 is even greater for both high outdegree SNPs and high core score SNPs in all networks except artery—coronary and liver. In many ways, this trend is exactly what one would expect: SNPs that fall within the eQTL networks have increased heritability because they are potentially capable of affecting the expression of multiple genes, a trend that increases as the SNPs become increasingly connected in the eQTL networks. These results are robust to clustering method (see Materials and Methods and supplementary fig. S9b, Supplementary Material online).

Local Hubs Evolve Under Polygenic Adaptation

The clustering of complex traits and diseases heritability in a few modules and in hubs in eQTL networks, together with previous results suggesting that global hubs are likely to evolve under negative selection (Wollenberg Valero 2024) suggest that directional selection on complex traits may preferentially be mediated by local hubs. The importance of the local hubs makes logical sense. Indeed, selection acts at the level of the traits. As each specific trait is likely to be under the control of one or several network modules consisting of genes involved in the same biological function, the local hubs play an important role in their determination. Targeted selection acting on local hubs would moreover mitigate the potential perturbative effects on the network as a whole or other functional modules. To test for this hypothesis, we searched for enrichment among global or local hubs of high scores for different statistics measuring three types of selection signatures. Genomic evolutionary rate profiling (GERP) score detects regions that lack substitutions, which can indicate negative selection (Cooper et al. 2005). Integrated haplotype score (iHS) detects longer-than-expected haplotypes around one of the alleles at one locus, a signature of recent, strong directional selection event or selective sweeps (Voight et al. 2006). F_{ST} measures population differentiation that may reflect older and sometimes milder directional selection events and has proven to be powerful in detecting the shifts in beneficial allele frequencies observed in polygenic selection (Wright 1965; Barghi et al. 2020).

We found that both global and local hubs, while enriched for complex traits and disease heritability, were more likely to be located in constrained genomic regions with high GERP scores in all the tissue-specific networks (see Fig. 4a and supplementary fig. S10a, Supplementary Material online). Global hubs were even more likely than local hubs to belong to constrained regions (supplementary fig. S11, Supplementary Material online). Both types of hubs are globally evolving under negative selection, in particular global hubs.

Neither local nor global hubs presented a clear pattern of enrichment for high |iHS| in all tissue-specific networks (see Fig. 4b and supplementary fig. S10b, Supplementary Material online). However, local hubs were significantly enriched for high |iHS| in more tissue-specific networks than global hubs (18 of the 29 networks for local hubs vs. 13 of the 29 networks for global hubs, see Fig. 4b and supplementary fig. S10b, Supplementary Material online). Local hubs were also enriched for high F_{ST} between European and both African and East Asian populations in almost all networks. Conversely, global hubs were depleted for high F_{ST} in almost all networks. Local hubs thus showed more signatures corresponding to polygenic selection than the other SNPs in the network, while global hubs seemed to be evolving under strict negative selection. These results were robust to clustering method (see Materials and Methods and supplementary figs. S12 and S13, Supplementary Material online). Using different thresholds to define high outdegree and core score (from the top 90% to the top 5%), we show that the results obtained reinforce our conclusion on the selection dynamic in the global and local hubs (see Materials and Methods, supplementary text, Supplementary Material online, and supplementary fig. S14, Supplementary Material online).

Discussion

Of the SNPs found through GWAS to be associated with complex traits, most (up to 90%) fall outside of gene coding regions suggesting that they likely play a regulatory role. Bipartite eQTL networks have allowed us to explore how these SNP loci work together to regulate the many genes involved in the biological processes underlying the traits, in a tissue- or cell-type-specific manner (Platig et al. 2016; Fagny et al. 2017, 2019). Using eQTL networks, we found that not only do SNPs act in both *cis* and *trans* to influence the expression of complex networks of genes, but that these networks have a robust, modular structure consisting of SNP–gene modules with properties that help explain the polygenic determinism that underlie most common traits and provide clues on how they can evolve despite the high degree of pleiotropy. Specifically, the modules in eQTL networks are enriched for functionally related groups of genes and nucleated around "core SNPs" that are both local (module) hubs and the SNPs most likely to exhibit strong GWAS associations with various phenotypes (including diseases) due to their association with module-driven, trait-related processes (Platig et al. 2016; Fagny et al. 2017).

The omnigenic model was important in that it helped bridge the gap between the missing heritability found in many diseases and the growing number of traits for which hundreds, if not thousands, of small effect-size genetic variants contribute to a given trait. Our eQTL network-based model is complimentary to the omnigenic model but provides a more nuanced view of the importance of SNP–gene "regulatory" associations in defining phenotypes, identifying disease associations and functions, and explaining both missing heritability and selection. Although it has been reported that global hubs in eQTL networks are enriched for tissue-relevant trait heritability (Gaynor et al. 2022), as are gene modules appearing in various types of networks (Kim et al. 2019), there has not been a

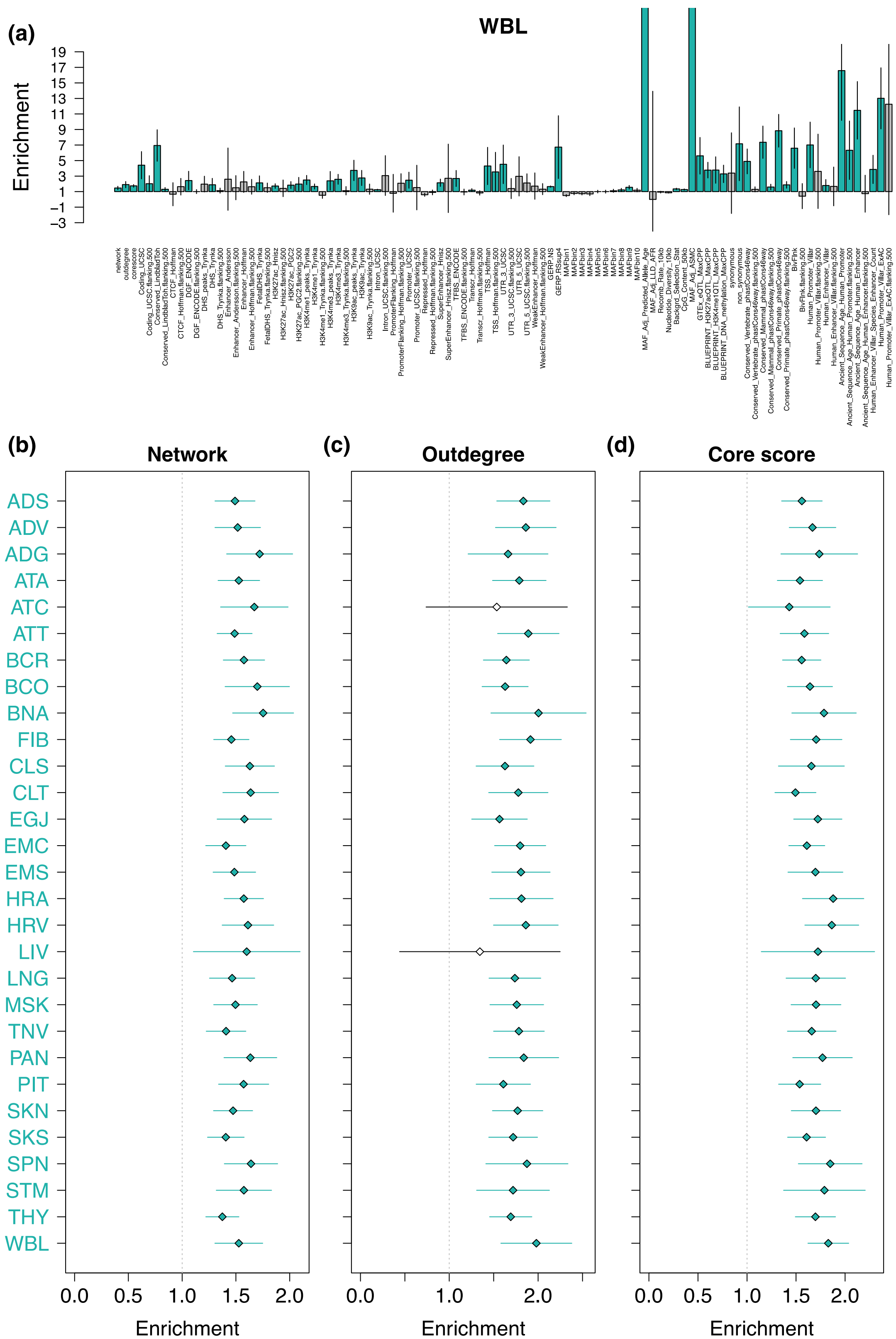

Fig. 3. Global and local hubs explain a larger share of heritability than expected by chance. a) Meta-analysis of h^2 enrichment among various annotations across 10 uncorrelated traits and diseases for the Whole Blood eQTL network. Meta-analysis of h^2 enrichment among SNP subcategories across 10 uncorrelated traits and diseases for each tissue-specific eQTL network: b) among SNPs within the eQTL network; c) among global hubs (high degrees); and d) among local hubs (high core scores.)

systematic exploration of these two complementary and important conceptual advances in understanding genetic effects in complex traits.

We addressed that gap in understanding by performing an analysis of eleven polygenic traits that represent a wide range of genetic heritability to determine the distribution of trait

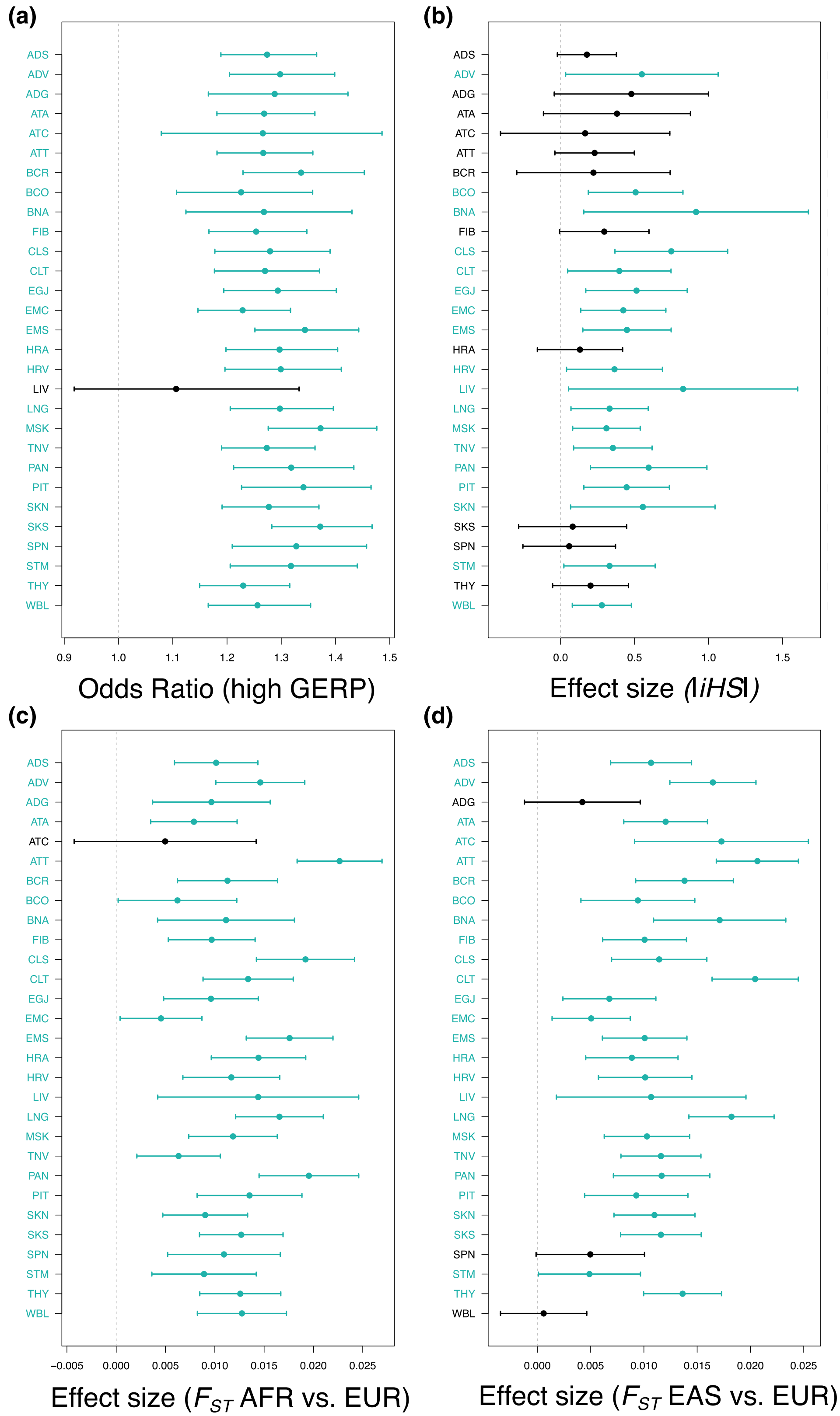

Fig. 4. While overall under negative selection, local hubs are enriched for positive selection signals in almost all tissue-specific eQTL networks. a) Odds ratio of being conserved when the SNP is a local hub vs. a leaf. Bars indicate confidence intervals. Confidence intervals were computed using a logistic regression. b) iHS that detect recent selective sweeps signals. c) F_{ST}(AFR, EUR) measuring population differentiation between African (AFR) and European (EUR) samples from 1000 genomes. d) F_{ST}(EAS, EUR) measuring population differentiation between East-Asian (EAS) and European (EUR) samples from 1000 genomes. b to d) Effect size of being a local hub vs a leaf on the value of several statistics sensitive to positive selection. Bars indicate $2 \times$ SE. Standard error (SE) were computed using a linear regression. a to d) For each tissue-specific network, significant enrichment (an odds ratio significantly different from 1 or effect size significantly different from 0) are highlighted in blue. Significance was called when $P \leq 0.05$.

heritability across 29 tissue-specific eQTL networks. We found that although heritability is widely distributed across loci, as suggested by the omnigenic model, the distribution of heritability is far from homogeneous. Instead, there is an uneven distribution with the greatest heritability concentrated in network modules containing genes that represent trait-specific and biologically relevant functions. This makes sense as variations of a given polygenic trait arise through alterations of the expression of specific biological processes relevant to this trait. Further, we found that trait heritability was more likely to be explained by SNPs occupying key positions in the eQTL networks, especially among the "core SNPs" that are local hubs in their functional modules. The only exception are the artery coronary and liver networks, for which this enrichment is not observed. They also behave differently from the other tissue-specific networks in other aspects (tissue-specificity of modules, enrichment of local hubs in negative and polygenic selection signals). This is likely due the relatively small sample size for RNA-seq data in these tissues, leading to smaller eQTL networks, despite their high modularity. These same SNPs, which we had previously shown to be enriched in tissue-specific activated regulatory elements (Fagny et al. 2017), are thus likely to determine a significant proportion of the heritability of complex traits.

The clustering of heritability in modules is also not evenly distributed across traits and differs among tissues. We found that heritability in each phenotype tended to be clustered in a small number of biologically relevant modules within the highly modular eQTL networks that are relevant for understanding the phenotype in question. As previously noted, these modules tend to be trait-specific, even when traits are not genetically correlated. This module-dependent concentration of heritability, coupled with the enrichment of heritability in core SNPs of those same modules, could help explain why even highly connected regulatory networks (Boyle et al. 2017; Liu et al. 2019) are robust to the genetic perturbations of deleterious genetic variants. Indeed, the highly modular structure of eQTL networks provides a means by which the disruptive effect of regulatory mutations can be buffered in a tissue-specific manner against altering the broader functionality of the wider regulatory networks active in living cells.

We also found that global hubs are evolving under strong constraints, as observed in previous studies (Wollenberg Valero 2024). However, our results revealed that another category of nodes, the local hubs that regulate many genes involved in the same biological processes and articulate modules, have been preferential targets of past and ongoing polygenic selection. Because of their strategic position in the eQTL regulatory network, these loci greatly influence the regulation of biological processes in a tissue-specific manner, and, because their effect is limited beyond their "home" module, they are ideal candidates for adaptation. Finally, our results confirm the hypothesis, first presented in an opinion paper by Fagny and Austerlitz (Fagny and Austerlitz 2021), that local hubs are preferential targets for polygenic selection, and provide a means to understand how complex traits may evolve despite the strong pleiotropy that exists in biological networks.

All our results were obtained on bipartite eQTL networks inferred using statistical correlations between individual *cis* and *trans* SNPs to each gene independently. However, other relationships are hidden in our network, including spurious gene co-expression patterns, SNPs-SNPs interactions caused by linkage disequilibrium, and other SNP–SNP interactions such as epistasis. For the first part, we have demonstrated using comparison with co-expression networks that our approach manage to root out a lot of the spurious co-expression patterns observed in co-expression networks, focusing instead on the portion of gene expression correlated with genetic factors. We also controlled for linkage disequilibrium in our analyses, using two different strategies: (i) treating SNPs within a linkage block as one marker in our downstream analyses and (ii) using the LD-score software, a power tool to account for both SNP linkage and annotations overlaps. Finally, regularized regression models have been recently developed that take these interactions into account. However, the genome-wide application of such methods to a large genome like the human genome one is still limited by computation time, in particular when the eQTL detection is repeated across 29 tissue-specific datasets. Indeed, regularized regression models are generally applied on filtered datasets to limit the number of tests to perform and reduce the computational burde. They are either applied exclusively to *cis*-eQTLs (Marchetti-Bowick et al. 2019; Yan et al. 2020), or applied to heavily filtered and pruned data, such as functionally annotated regions with prior pruning of SNPs for linkage disequilibrium (Banerjee et al. 2021). This may lead to miss some important regulatory SNPs. We consequently arbitrated our pipeline choices in favor of increasing the number of candidates and to the detriment of the potential interactions, and choose a voluntarily loose *FDR* threshold to balance for the multiple testing issues of *trans*-eQTL detection.

Conclusions

Overall, our results demonstrate a synergy between the eQTL network and omnigenic models in explaining how genetic variants work together to influence traits while addressing their respective shortcomings. The value of such a synthesis can be seen in the results derived from the collection of complex traits we chose to analyze, including cancers, metabolic diseases, and auto-immune neurodegenerative disorders. In each of these traits we can see genetic risk factors identified through GWAS perturbing tissue-specific functional modules (while affecting others), while those modules are simultaneously affected by many other genetic variants of smaller overall effect size. These complex relationships define a conceptual framework for understanding disease risk, helps to define phenotypes for health and disease, and provide a means by which one could potentially prioritize therapeutic targets and design treatment protocols that account for the network architecture. Our findings also allow a better understanding of how complex traits can change as populations adapt to local changes through selection acting on traits mediated at the genetic and molecular level by key regulatory elements. This conceptual framework also provides an explanation, at the molecular level, of how polygenic selection of regulatory mutations beneficial in one environment could lead to the phenomenon of maladaptation that many suspect is the origin of many complex diseases.

Finally, the transferability of these results obtained on human data to other organisms is an open question. One main issue is the impact of the genome structure on the results. In plants genomes, there are many more structural variants than in the human genome (Marroni et al. 2014; Saxena et al. 2014). These variants are more likely to have a greater impact on the value of the phenotype and are also often associated to adaptation to the environment (Gui et al. 2022; Kang et al. 2023; Li et al. 2023). While they could be integrated in our eQTL network, it is not clear how their presence would affect the conclusions. Keeping in mind these caveats, studying the transferability of these results to other organisms could be of tremendous value to breeders by pointing out which loci are the more likely to improve the tolerance of crops or cattle to environmental stresses without jeopardizing their agricultural values.

Materials and Methods

GTEx Dataset

We used genotyping and gene expression level data from the NHGRI GTEx project version 8.0 (GTEx Consortium 2015). For appropriate statistical power in downstream analyses and network stability, we filtered out tissues for which the number of individuals with both genotyping and RNA-seq data available was less than 200; sex-specific tissues were not included. This left 29 tissues for analyses, as can be found in supplementary table S1, Supplementary Material online. Genotyping data were downloaded from the database of genotypes and phenotypes (dbGaP): phs000424.v8.p2. Genotyping data were preprocessed on the Bridges system at the Pittsburgh Supercomputing Center (PSC) and the Cannon cluster supported by the Faculty of Arts and Sciences Division of Science, Research Computing Group at Harvard University (see Gaynor et al. 2022).

The sequencing data were processed in plink 1.90 to retain only SNPs, and we removed variants with genotype missingness greater than 10% or minor allele frequency less than 0.1 (Purcell et al. 2007). SNP imputation was then performed using Eagle2 (Loh et al. 2016).

Fully processed, filtered, and normalized RNA-seq data were obtained from the GTEx Portal (www.gtexportal.org). The GENCODE 26 model was used to collapse transcripts and quantify expression using RNA-SeQC (https://www.gencodegenes.org/human/release_26.html#).

Bipartite eQTL Network Inference

eQTLs were obtained from Gaynor et al. (2022). Rapidly, with the R MatrixEQTL package (Shabalin 2012), the association between SNP genotypes and gene expression was modeled using linear regression (Eq. 1) that included potential confounding factors as covariates (Kendziorski et al. 2006): the two first principal components for population structure, sex, age and RIN that measures RNA quality. If $\mathbf{G}$ is an $r \times m$ matrix of gene expression and $\mathbf{S}$ is an $r \times n$ matrix of SNP genotypes, each with r rows representing observations and columns representing n SNPs and m genes, respectively, $\mathbf{X}$ is a covariate matrix, the eQTL of a particular SNP i on a locus's gene expression j is then :

$$\mathbf{G}_{\mathrm{j}} = \mathbf{X}^{\mathrm{T}}\boldsymbol{\alpha} + t_{ij}\mathbf{S}_{\mathrm{i}}. \tag{1}$$

Associations were evaluated for SNPs in both *cis*—SNPs within 1MB of a gene's transcription start site—and *trans*. The eQTL associations between all pairs of SNPs and genes were then represented as a sparse, weighted bipartite network (Fig. 5). Each SNP and gene was considered a node in the network. Using a fixed cutoff $q = 0.2$ on the FDR of the eQTL regression (Platig et al. 2016; Fagny et al. 2017), the edge weight $a_{i,j}$ between SNP i and gene j was defined by the function $I_{i,j}\{\mathrm{FDR} <= q\}|t_{ij}|$, where t_{ij} is the value of the t statistic computed by the MatrixEQTL package. Thus, when the estimated FDR of the eQTL regression was below the threshold of 0.2, then $a_{i,j} = |t_{ij}|$, indicating that there was an edge connecting the nodes, and $a_{i,j} = 0$ otherwise.

Overlap with Hi-C Data

3D chromatin loop data stemming from Hi-C experiments for 9 different tissues corresponding to nine of the 29 explored tissues were downloaded from the 3D genome website (see Data Availability). Loops had a resolution of 10 kb, with intrachromosomal loops length spanning between 70 and 2,880 kb. To match our *cis*-eQTL data, we retained only loops spanning a maximum of 1 Mb (on average 99.5% [97.1 to 99.9] of all loops, corresponding to 13,775 [7,812 to 18,749] loops), and for which one side of the loop's 10 kb block was containing a gene TSS. This filtered dataset corresponded to our "null" dataset.

For this analysis, we merged eQTLs by blocks of linkage disequilibrium (with an FDR cutoff of 0.2 for SNP–gene association significance, we retained an average of 749,672 [403,759 to 1,104,908] significant block–gene associations. We retrieved overlaps of Hi-C loops and eQTL data, keeping only cases where the 10-kb block containing the eQTL's LD block was on one side of one loop and the 10-kb block containing the TSS of the associated gene was on the other side of the same loop (on average 646 [304 to 1,068]). We then used a χ^2 test to compare proportions of Hi-C loops in LD blocks that carried significant eQTL and in LD blocks that did not.

Network Summary Statistics

The module structure of each tissue-specific network was determined using the bipartite modularity maximization approach (Platig et al. 2016) implemented in the netZooR Bioconductor package. This is a two-step approach : (i) project the eQTL network on the gene space and infer a first cluster structure using one of the three available algorithms : louvain clustering (LCS), leading eigenvector community detection (LEC), and the fastgreedy community detection (FG); and (ii) improve the clustering by adding SNPs back in the network and searching for improved modularity by reattributing SNPs to modules to obtain the best modularity possible and then genes. The second step is iterated as many times as necessary to reach convergence. We tested all three algorithms on step (i) to ensure that our results are robust to clustering method and implemented a new version of the second step of the algorithm (CONDOR:condorSplitMatrixModularity) that allows for a balance between computation time and memory usage (Fig. 5) and has been published in the R netzooR package fork of https://github.com/maudf/.

$$k_i = \sum_{j=1}^{g} I_{i,j}\{\mathrm{FDR} < q\}|t_{ij}| \tag{2}$$

The SNP core score was defined as the SNP's contribution to the modularity of its module, it measures the centrality of the SNP in the module (Platig et al. 2016). If $m = s \times g$ is the total number of possible edges in a network made of s SNPs and g genes, $\tilde{a}_{ij}$ is the observed edge value between SNP i and gene j. Here, d_j is the gene i indegree defined as $d_j = \sum_{i=1}^{s} I_{i,j}\{\mathrm{FDR} < q\}|t_{ij}|$, then, for SNP i in module h, its core score, Q_{ih}, is defined by Eq. 3:

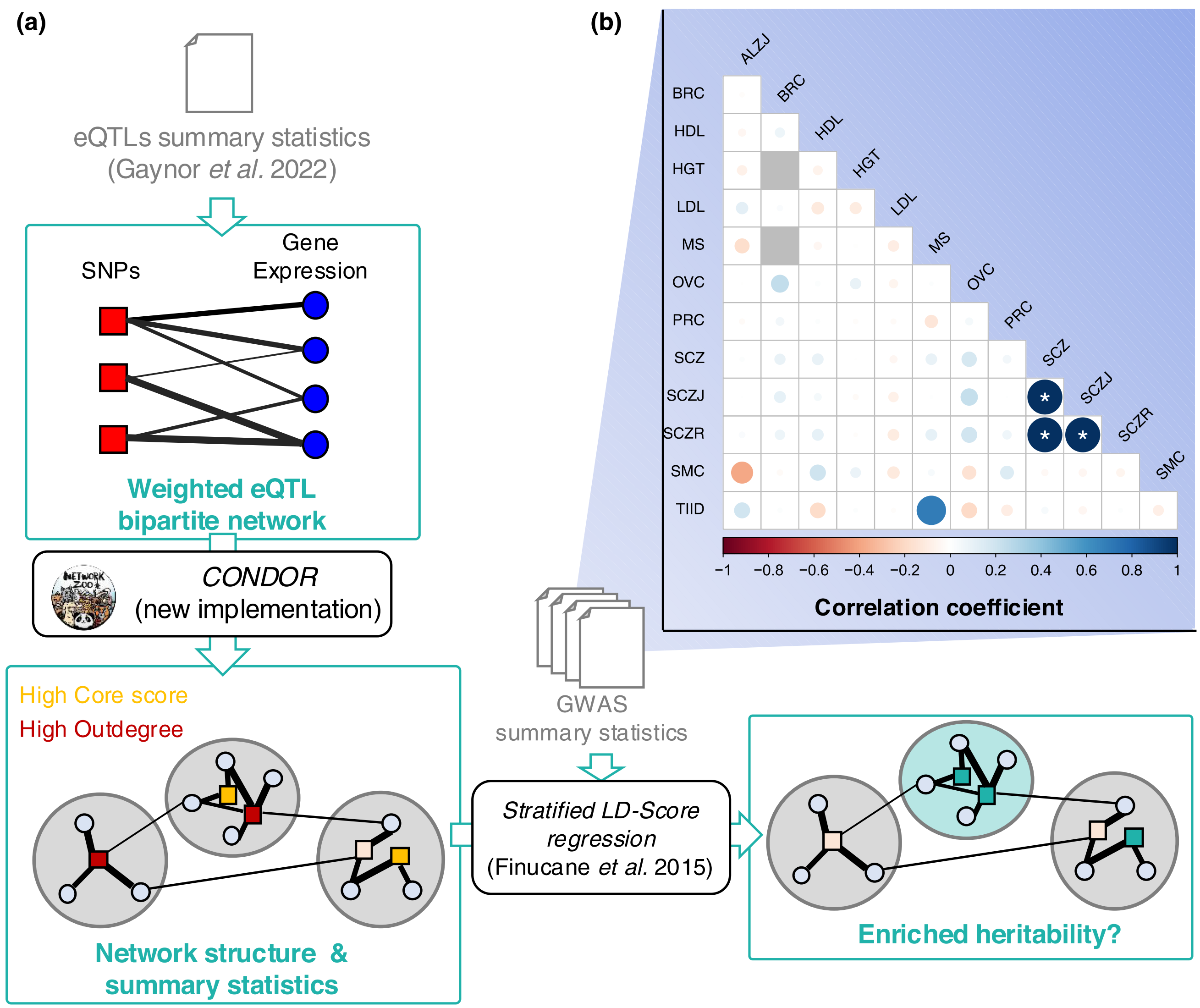

Fig. 5. Pipeline and traits correlation. a) Pipeline of data analyses. Input data are in grey. GTEx eQTL summary statistics were obtained from a previous study (Gaynor et al. 2022) and are described in Materials and Methods and supplementary table S1, Supplementary Material online. GWAS summary statistics were downloaded from https://console.cloud.google.com/storage/browser/broad-alkesgroup-public-requester-pays and are described in supplementary table S2, Supplementary Material online. eQTL network summary statistics are presented in supplementary fig. S1, Supplementary Material online. b) Pairwise genetic correlations between traits based on GWAS summary statistics.

$$Q_{ih} = \frac{1}{m}\sum_{j}\left(\tilde{a}_{ij} - \frac{k_i \times d_j}{m}\right)\delta(C_i, h)\delta(C_j, h) \qquad (3)$$

GWAS Data

GWAS data were obtained from the Alkes group (see supplementary table S2, Supplementary Material online). We considered a total of eleven traits and diseases presenting varying levels of estimated genetic heritability. For each trait or disease, we obtained summary statistics including SNP chromosome, position, alleles 1 and 2, χ^2, and Z-score. Here, the relationship between an individual's i genotype at SNP s and its phenotype Y, accounting for covariates M, is usually tested using the following regression: $Y_{i,s} = \beta_{1,s}X_{i,s} + \beta_2 M_i + \epsilon_{i,s}$, where $\epsilon_{i,s}$ is the error term. Then, $Z = \beta_{1,s}/\mathrm{se}(\beta_{1,s})$ and $\chi^2 = qnorm((P/2)^2)$ where $P = 2 \times pnorm(-|Z|)$.

We used Z^2 as a proxy for normalized heritability explained by each SNP. High heritability SNPs were defined as those with a Z^2 in the 95th percentile of the distribution.

GWAS Outdegree and Core Score Enrichment among High Heritability SNPs

We compared the distribution of SNP outdegrees and core scores between the high heritability SNPs and the rest of the SNPs in the network using a likelihood-ratio test (LRT), correcting for linkage disequilibrium. To control for LD between SNPs, we generated lists of SNPs falling into the same LD block, using the plink1.9—blocks option, a 5-Mb maximum block size, and an r^2 of 0.8. In each module, for each LD block, we extracted the median of either outdegrees (k_i) or core scores (Q_{ih}) for high and nonhigh heritability SNPs separately and used these values as input in the linear regressions.

The LRT we used assesses whether a linear model that includes GWAS status (Eq. (5)) fits the observed data better than a linear model that does not include this variable (Eq. (4)). As the distribution of SNP Q_{ih} is not uniform across modules, we added module identity as a covariate in the linear regression when computing LTR for core scores. In Eqs. (4) and (5), Score$_i$ is the score of SNP i

(either outdegree or core score), $I(GWAS = 1)$ is an indicator function equal to 1 if the SNP is a high heritability SNP and equal to 0 otherwise, and $I(C_k = 1)$ is an indicator function equal to 1 if the SNP belongs to module k and equal to 0 otherwise:

$$Score_i \sim \left[\sum_{k=1}^{n-1} I(C_k = 1)\right] + \epsilon \tag{4}$$

$$Score_i \sim I(GWAS = 1) + \left[\sum_{k=1}^{n-1} I(C_k = 1)\right] + \epsilon \tag{5}$$

Genetic Correlation Between Traits and Heritability Enrichment Analyses among Local and Global Hubs

We computed partitioned heritability with the 97 annotation baseline-LD model from Gazal et al. (2018) and Hujoel et al. (2019) for each of the eleven traits and diseases selected above (Fig. 5). This allowed us to estimate the enrichment and standardized effect size of the baseline annotations and three additional parameters on the heritability: belonging to the network (being an eQTL), having a high outdegree, and having a high core score (Gazal et al. 2018; Hujoel et al. 2019). We considered a binary annotation to ensure sufficiently stable estimates. For outdegrees and core scores, annotations were set equal to 1 if k_i and Q_{ih} were in the top quartile of the distribution and 0 otherwise.

Given that approximately 85% of the GTEx study population consists of individuals of European descent, we used LD scores computed from the 1000 Genomes Project data from individuals with European ancestry using the GRCh38 genome version and regression weights that exclude the HLA region; P-values were computed using a block-jacknife.

Meta-analyses were then performed across uncorrelated traits using the meta.summaries function from the R rmeta package v.3.0, with random-effect weights. To identify uncorrelated traits and diseases, pairwise genetic correlations between the eleven traits and diseases were first computed using stratified LD-score regression (S-LDSC). Traits and diseases with a pairwise genetic correlation of less than 0.3 were considered uncorrelated.

High Heritability SNP Enrichment Among Modules

High heritability SNP enrichment among modules was performed using a χ^2-test on data corrected for linkage disequilibrium. Using the same LD blocks as previously described, we considered that an LD block was a high heritability block if at least one SNP in this LD block had a Z^2 in the top 95th percentile. For each module, we counted the number of high and nonhigh heritability LD blocks in and outside of the modules and used these data to perform a χ^2-test.

Gene Ontology Enrichment Analyses and Identification of Tissue-Specific Modules

We performed Gene Ontology enrichment analyses using the Bioconductor R topGO package v.2.44 (Alexa and Rahnenfuhrer 2016), using the elim method; this method is more efficient than Fisher's exact test (Alexa et al. 2006). The GO categories are tested sequentially, following the GO tree structure from bottom to top: if one GO category is found to be significant, the genes involved are removed from the parent nodes before they are tested. The tests are thus not independent, and no multiple testing correction can be applied. Following the guidelines in the topGO users' manual, we filtered uncorrected P-values using a stringent threshold of 0.01. We also filtered out GO categories that did not include at least three genes in the gene set of interest. The gene ontology database used in this analysis was the one from the R bioconductor org.Hs.eg.db package v.3.13.0. For each test, the background gene set contains all the genes of the network and the gene set of interest contains all the genes from the module of interest.

We identified common and tissue-specific modules in the eQTL networks based on pairwise comparisons of GO term assignments. For each module of a first network, the GO ID enriched in the module was compared to the GO ID enriched in each of the modules in a second network using Jaccard Index. The best matching module was determined based on the highest Jaccard Index, and if the best Jaccard Index was ≥0.3, the modules were considered as similar and otherwise different. Then, for each module in each tissue-specific network, we counted the number of similar modules in the other 28 networks. If this number of similar functional modules was ≤3, we considered it a tissue-specific module. To evaluate the impact of the jaccard index threshold, we also investigated a threshold of 0.4.

Selection Scores Enrichment Analyses

Conserved SNPs were determined by intersecting the annotations of the SNPs present in our eQTL networks with conserved regions determined using GERP scores downloaded from Ensembl 111 release (https://ftp.ensembl.org/pub/release-111/bed/ensembl-compara/91_mammals.gerp_constrained_element/gerp_constrained_elements.homo_sapiens.bb Martin et al. 2023). Enrichment in conserved SNPs among local and/or global hubs were computed using the following logistic regression model, where $I(GERP = 1)$ is an indicator function equal to 1 if the SNP falls in a conserved region and equal to 0 otherwise, $I(Cat = 1)$ is an indicator function equal to 1 if the SNP belongs to a given category (local or global hubs) and equal to 0 otherwise, and $I(C_k = 1)$ is an indicator function equal to 1 if the SNP belongs to module k and equal to 0 otherwise:

$$\text{Logit}(I(GERP_i = 1)) \sim I(Cat_i = 1) + \left[\sum_{k=1}^{n-1} I(C_{ik} = 1)\right] + \epsilon$$

iHS, which measures the differences in haplotype lengths between haplotypes carrying the ancestral and the derived SNPs, detect signatures of recent selective sweeps (Voight et al. 2005). We downloaded iHS standardized scores (Johnson and Voight 2018) computed for 26 populations for 1000 genomes project Phase 3 dataset from https://zenodo.org/records/7842512. We extracted scores computed on the CEU population (Utah residents (CEPH) with Northern and Western European ancestry) samples as being the closest population from the European descent samples that represent 85% of the GTEx dataset v8 release.

F_{ST} between African, Eurasian, and European samples from the 1000 genomes data were computed using plink2.0 (Purcell et al. 2007) on genotypic data for samples from the EUR, AFR, and EAS populations from the 1000 genomes Phase 3 dataset. Genotyping data were downloaded from ftp://ftp.1000genomes.ebi.ac.uk/vol1/ftp/data_collections/1000G_2504_high_coverage/working/20201028_3202_raw_GT_with_annot/.

Effect sizes of being a local or global hub on iHS or F_{ST} were computed using the following linear regression model, where $Score_i$ is the value of the statistics for SNP i, $I(\mathrm{Cat} = 1)$ is an indicator function equal to 1 if the SNP belongs to a given category (local or global hubs) and equal to 0 otherwise, and $I(C_k = 1)$ is an indicator function equal to 1 if the SNP i belongs to module k and equal to 0 otherwise:

$$Score_i \sim I(Cat_i = 1) + \left[\sum_{k=1}^{n-1} I(C_{ik} = 1)\right] + \epsilon_i$$

The community ID terms were only added when local hubs were one of the two categories compared, as the q-score determining local hubs depends on community ID.

Supplementary Material

Supplementary material is available at *Molecular Biology and Evolution* online.

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Author Contributions

K.L.S., J.P., J.Q., and M.F. contributed to the conception of the work. K.L.S., J.P., J.Q., and M.F. contributed to the study design and method development. K.L.S., J.P., and M.F. contributed analysis, verified the data, and drafted the manuscript. All authors reviewed the manuscript and approved the submitted work.

Ethics Approval

This work was conducted under dbGaP-approved protocol # 9112.

Conflict of Interest

The authors declare no competing interests.

Data Availability

All the code used to analyze the data is available at https://github.com/maudf/heritability. The new CONDOR: condorSplitMatrixModularity is available at https://github.com/maudf/netZooR. The data used for the analyses described in this manuscript were obtained from the GTEx Portal on 17 December 2019 and dbGaP accession number phs000424.v8 on 12/17 19 for RNA-seq and Genotyping data, from https://console.cloud.google.com/storage/browser/broad-alkesgroup-public-requester-pays for GWAS data, and from https://3dgenome.fsm.northwestern.edu/downloads/loops-hg38.zip on 01/02/2025 for the Hi-C data.

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The Importance of Regulatory Network Structure for Complex Trait Heritability and Evolution

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Abstract

Complex traits are determined by many loci—mostly regulatory elements—that, through combinatorial interactions, can affect multiple traits. Such high levels of epistasis and pleiotropy have been proposed in the omnigenic model and may explain why such a large part of complex trait heritability is usually missed by genome-wide association studies, while raising questions about the possibility for such traits to evolve in response to environmental constraints. To explore the molecular bases of complex traits and understand how they can adapt, we systematically analyzed the distribution of SNP heritability for 11 traits across 29 tissue-specific expression quantitative trait locus networks. We find that heritability is clustered in a small number of tissue-specific, functionally relevant SNP–gene modules and that the greatest heritability occurs in local "hubs" that are both the cornerstone of the network's modules and tissue-specific regulatory elements. The network structure could thus both amplify the genotype–phenotype connection and buffer the deleterious effect of the genetic variations on other traits. We confirm that this structure has allowed complex traits to evolve in response to environmental constraints, with the local "hubs" being the preferential targets of past and ongoing directional selection. Together, these results provide a conceptual framework for understanding complex trait architecture and evolution.

Keywords: GTEx, eQTL, bipartite networks, GWAS, heritability, complex traits, polygenic selection

Introduction

In many organisms, including yeasts, insects, worms, plants, and mammals, adaptation often leverages polygenic traits to respond to new environmental challenges (Daub et al. 2013; He et al. 2016; Zan and Carlborg 2019). Such "complex" traits have an interconnected genetic architecture involving from a small number to potentially thousands of partly independent loci (Salomé et al. 2011; Pais et al. 2013; Peiffer et al. 2014; Shi et al. 2016). Up to 90% of the SNPs associated with such traits are localized outside of coding regions, according to genome-wide association studies (GWAS) (Ward and Kellis 2012; Tak and Farnham 2015). They are thus likely to affect traits by modifying gene expression—often by altering the sequence of *cis*-regulatory elements. This is consistent with the observation that GWAS-significant regions colocalize with SNPs that are linked to the expression of nearby genes (expression quantitative trait loci, *cis*-eQTLs) (Hormozdiari et al. 2016; Zhu et al. 2016). Partitioning heritability for these traits across various annotations while correcting for linkage disequilibrium has confirmed that *cis*-eQTLs as a group explain more of complex trait heritability than would be expected by chance (Torres et al. 2014; Gamazon et al. 2018; Hormozdiari et al. 2018) but still fail to explain a majority of trait heritability (Yao et al. 2020). Others have shown that including *trans*-eQTLs may capture additional heritability and explain important pathological mechanisms (Luningham et al. 2020), but this is rarely done because *trans*-eQTL studies are generally underpowered.

Advances in systems biology and functional genomics have highlighted the fact that at the molecular level, complex traits are the results of many regulatory interactions between actors of different biological processes within a complex cellular gene regulatory network (Hallgrimsson et al. 2014; Sonawane et al. 2019). This high level of pleiotropy coupled with significant epistatic interactions may explain the missing heritability observed in quantitative genetics studies and has led to the development of the omnigenic model (Boyle et al. 2017; Liu et al. 2019). In this model, trait association signals are spread across most of the genome in a way that includes many genes lacking an obvious connection to a particular trait. The model posits that most heritability can be explained by effects on genes outside of "core pathways," often acting in *trans*, but acting in ways that alter the functioning of those pathways and thus account for most of a given trait's heritability. The amount of pleiotropy implied by this model may limit, at first glance, the possibility for a given complex trait to adapt in response to an environmental change without interfering with other unrelated traits. However, the evidence for polygenic adaption processes in SNPs associated with complex traits in several species highlights the need to dig deeper into the molecular bases of complex traits to understand their architecture and how they adapt (Barreiro and Quintana-Murci 2010;

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Jakobson et al. 2019; Fagny and Austerlitz 2021; Mathieson 2021).

Given the number of loci and the number of potential interactions involved, gene regulatory networks provide an efficient tool for understanding the biological processes behind trait heritability (Kim et al. 2019). Combining GWAS results with gene coexpression networks have proven useful to identify the causal gene within a candidate region in many species (Chan et al. 2011; Lee et al. 2011; Schaefer et al. 2014; Taşan et al. 2015; Calabrese et al. 2017; Schaefer et al. 2018). By integrating SNP–gene or SNP–trait association studies results with coexpression networks, these studies have highlighted the fact that *cis*- and *trans*-regulatory relationships involved in complex trait determination are clustering within coexpression subnetworks enriched for functionally relevant molecular pathways. Due to power issues, integration methods generally do not account directly *trans*-regulatory effects (Siewert-Rocks et al. 2022) and focus on *cis*-regulatory associations that only explain a small fraction of trait heritability (Yao et al. 2020), relying on the coexpression network to provide indirect information on *trans*-regulatory relationships. However, *trans*-regulatory factors that have been shown to carry an important share of the heritability (Westra et al. 2013; Torres et al. 2014; Brynedal et al. 2017), as reflected in the omnigenic model (Boyle et al. 2017; Liu et al. 2019). We previously proposed a way to integrate both *cis*- and *trans*-eQTL results using a bipartite eQTL network representation (Platig et al. 2016; Fagny et al. 2017) that relies on summary statistics from eQTL studies and is relatively insensitive to a high false discovery rate, allowing one to partially compensate for the lack of power of *trans*-eQTL studies; an update to this method allowed us to weigh SNP–gene regulatory relationships by eQTL effect sizes (Gaynor et al. 2022). These analyses have shown that highly structured eQTL networks can reliably identify, in a tissue-specific manner, the biological functions disrupted by traits-associated SNPs (Fagny et al. 2017, 2019), and further defined network topological features that are useful in explaining in part the link between genotype and phenotype in complex traits.

In this study, we build on eQTL networks to investigate how complex trait heritability is spread within and among the tissue-specific eQTL networks to better understand both the genetic architecture of complex traits and how they may evolve under directional selection. We used GWAS summary statistics from eleven complex traits and diseases and built 29 tissue-specific *cis*- and *trans*-eQTL networks using the Genotype-Tissue Expression (GTEx) dataset. We then partitioned heritability across various features, including network node topological summary statistics, to identify key determinants of trait heritability. We investigated whether heritability for each trait is clustered in particular subparts (regulatory modules, also sometimes referred to as communities) of the eQTL networks, and identified network communities regulating biological functions that explain most of the heritability of each trait. Finally, in order to assess how complex traits may evolve, we looked for signatures of past polygenic adaptive processes in key regulatory loci considered in relationship with their role in the eQTL network. We found that heritability is not scattered uniformly across the genome but rather "clustered" in eQTL modules that represent trait-relevant functions and that loci playing a key role in regulating tissue-specific biological processes were not only more likely to explain trait heritability but also to have been preferential targets of past polygenic adaptation events.

Results

SNPs Associated at the Genome-Wide Level with Traits or Diseases are Clustered in a Few Modules

We wanted to understand how heritability is distributed across the eQTL network. We had previously reported that SNPs significantly associated with chronic obstructive pulmonary disease, cancers, and other traits in GWAS are concentrated in a small number of modules (Platig et al. 2016) and wanted to verify that this result can be extended beyond genome-wide significantly associated SNPs to SNPs explaining a larger part of genetic heritability. We thus used RNA-seq and genotyping data from GTEx to perform eQTL analyses and built 29 tissue-specific weighted bipartite eQTL networks. To compensate for the fact that *trans*-eQTL detection is notoriously underpowered compared with *cis*-eQTL detection, we used a lose false discovery rate (FDR) threshold of 0.2 (see Materials and Methods and supplementary text, Supplementary Material online). We identified modules containing highly connected SNPs and genes within each network based on CONDOR's bipartite modularity maximization (Platig et al. 2016) (see Materials and Methods, supplementary text, Supplementary Material online, supplementary fig. S1, Supplementary Material online, and supplementary table S1, Supplementary Material online). We also confirmed that our eQTL networks are summarizing information related to the genetic bases of gene expression coregulation rather than merely gene co-expression profiles (see Materials and Methods, supplementary text, Supplementary Material online, and supplementary table S3, Supplementary Material online) and that they are enriched for long-range *cis* 3D chromatin interactions (see Materials and Methods, supplementary text, Supplementary Material online, and supplementary table S4, Supplementary Material online).

We considered a total of eleven traits and diseases chosen for their medical or evolutionary relevance in populations of European descent. Breast cancer (BRC), ovarian cancer (OVC), and prostate cancer (PRC), Alzheimer's disease (ALZ), multiple sclerosis, Schizophrenia (SCZ), high-density lipoprotein levels (HDL), and type 2 diabetes (TIID; diabetes) are all increasing in populations of European descent and are known to exhibit partial genetic heritability. Smoking cessation is a measure of smoking dependency that is partly explained by genetic factors and is linked to a host of pulmonary and other diseases. SCZ is a highly heritable but poorly understood polygenic disease. Finally, height (HGT) is the canonical example of a polygenic trait, and some consider it a natural selection target in populations of European descent. These traits and diseases also represent a wide range of global genetic heritability, as reported in the literature, from 25% for TIID, to up to 80% to 90% for SCZ and Alzheimer's. These traits and diseases also represent a wide range of global genetic heritability, as reported in the literature, from 25% for TIID, to up to 80% to 90% for SCZ and Alzheimer's. We conditioned on European descent because of the number of GWAS data available and the demographics of GTEx. A complete description of these traits and diseases and their GWAS summary statistics is reported in the supplementary table S2, Supplementary Material online. For each trait or disease, we obtained summary statistics (see Materials and Methods).

We used the summary statistic Z^2 as a proxy for the per-SNP heritability, and SNPs with a GWAS Z^2 in the top 5% were named "high heritability SNPs." We found that the high heritability SNPs cluster in a small number of network modules

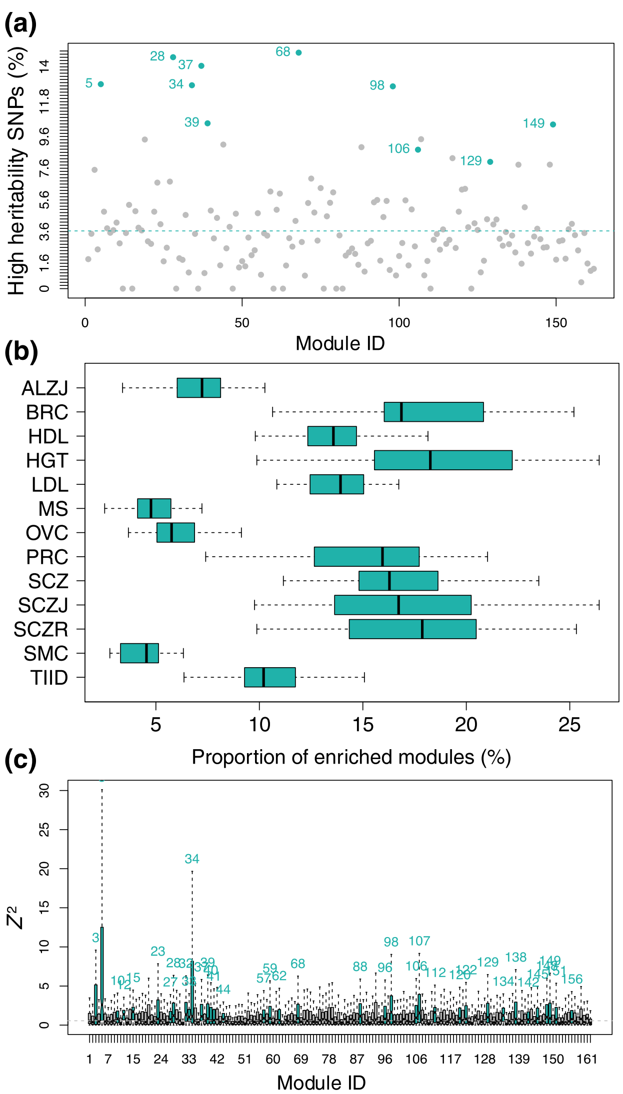

Fig. 1. High heritability SNPs are clustered in a few trait-specific modules. (a) Proportion of high heritability SNPs for BRC in each module of the skin—not sun-exposed (lower leg) eQTL network. Modules significantly enriched for high heritability SNPs are highlighted in color, with module number printed on the left (Benjamini–Hochberg corrected $P \leq 0.01$ using a χ^2-test). The dotted line represents the expected proportion of high heritability SNPs by chance in each module. (b) Distribution of the proportion of modules enriched for high heritability SNPs for each trait or disease among all tissue-specific eQTL networks. (c) Distribution of Z^2 values in each module of brain—nucleus accumbens (basal ganglia) for Alzheimer's disease (Kruskal–Wallis $P < 2 \times 10^{-16}$). Modules with a distribution skewed towards high values are represented in color, with module number printed on top (modules with a Benjamini–Hochberg-corrected $P \leq 0.01$ using one-sided Mann–Whitney U tests for each module vs. the rest of the network were considered as significantly enriched in high Z^2). The dotted blue line represents the expected Z^2 median across the whole network.

(supplementary table S5, Supplementary Material online). As an example, the breast cancer-associated SNPs appear in only 29 (12.6%) of the 230 network modules in the SKN (skin—not sun-exposed—suprapubic) eQTL network (Fig. 1a) representing a substantial concentration of heritability. We found similar results for other diseases and other tissues, with some combinations exhibiting even greater clustering of heritability in a limited number of network modules (Fig. 1b). We validated that these results were not dependent on the method we use to infer the clusters in our networks, because we obtained the same results using a different starting point for the same louvain-clustering based algorithm, and two other clustering algorithm altogether (see Materials and Methods, supplementary text, Supplementary Material online, and supplementary fig. S2, Supplementary Material online).

Because the top 5% SNPs only explain a small proportion of the heritability of the associated traits (particularly for highly polygenic traits), we examined the distribution of per-SNP heritability across the different eQTL network modules. We found that heritability was not distributed evenly across modules (Kruskal–Wallis test $P = 0$ for all pairs of tissue-specific networks and traits tested). As an example, consider Z^2 calculated for ALZ in brain—nucleus accumbens (basal ganglia) networks, which is plotted in Fig. 1c; all the results are in supplementary table S5, Supplementary Material online). Comparing the distribution of Z^2 for each module with the rest of the network, we found that less than a third of the modules contain high heritability SNPs (55 [5 to 85] modules representing about 23.8% [5.6 to 33.1] of the total number of modules depending on tissue-specific network/trait pair, with Benjamini–Hochberg-corrected Mann–Whitney U tests $P \leq 0.01$; see supplementary table S6, Supplementary Material online).

Heritability Is Enriched in Trait-Specific, Functionally Relevant Modules

We investigated whether the heritability for different, uncorrelated traits was clustered in the same or different modules in the tissue-specific eQTL networks. Depending on the tissue-specific network, about 38% [26 to 52] of the modules were not enriched for high heritability SNPs associated with any traits, 28% [22 to 33] were enriched for high heritability SNPs from only one trait and can be considered trait-specific, and 2% [1 to 6] were enriched for high heritability SNPs from at least six of the eleven traits and can be considered as shared across many traits (Fig. 2a and supplementary fig. S3, Supplementary Material online).

We performed a pairwise comparison of traits in each of the tissue-specific networks between modules enriched for high heritability SNPs. Using 10,000 resamplings, we found that top-heritability SNPs for two independent traits did not cluster in the same modules more than expected by chance in most cases. Indeed, among the 1,595 pairwise comparisons possible—$\binom{11}{2}$ comparisons ×29 tissues—only 54 show significant enrichment in overlap compared to what would expected by chance (Benjamini–Hocheberg-corrected $P \leq 0.01$). In more than half of these cases (33), the excess of overlap was observed between HDL and low-density lipoprotein (LDL), HGT, and/or and TIID in various tissue-specific networks (see supplementary table S7, Supplementary Material online). This co-occurence of HDL and LDL is also found in all four ADV network clustering (LCS1, LCS2, FG, and LEC), with $FDR \leq 0.05$ (supplementary table S8, Supplementary Material online).

We also investigated the biological functions represented by genes with the modules enriched for high heritability SNPs. We performed a GO term enrichment analysis for each module using Bioconductor R topGO package (see Materials and Methods); the results are presented in supplementary table S9, Supplementary Material online. This allowed us to identify modules that were tissue-specific (see Materials and Methods). The list of tissue-specific modules can change marginally depending on the Jaccard Index cutoff chosen. Based on the distribution of Jaccard indexes for pairwise comparison of tissue-specific networks, we chose two different thresholds: 0.3 and 0.4 (see supplementary fig. S4, Supplementary Material online for all

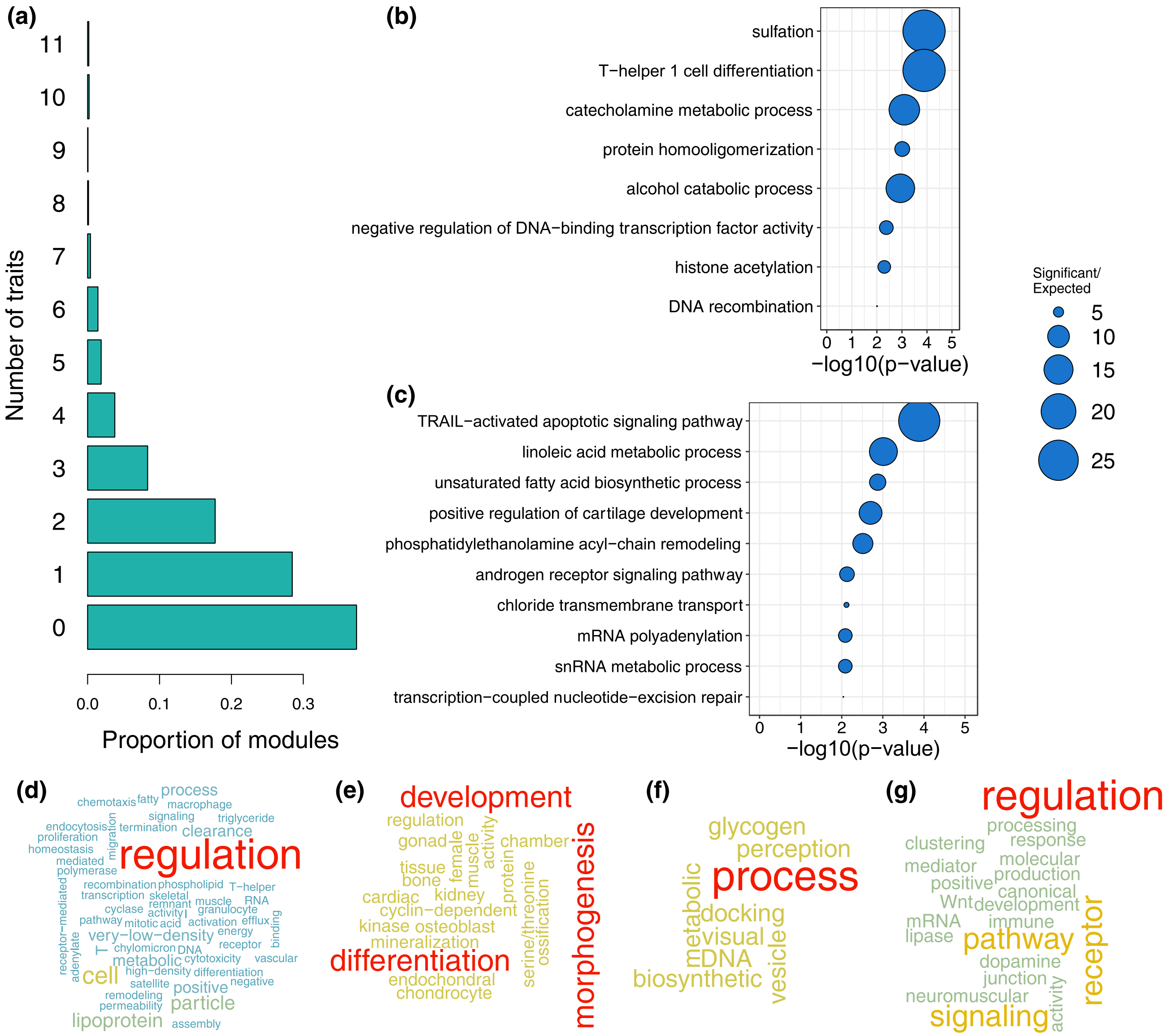

Fig. 2. Heritability is clustered in trait-specific, biologically significant modules. a) Average proportion of modules enriched in high heritability SNPs for one to eleven traits or diseases across all tissue-specific modules. To remove artifacts due to the high genetic correlation between the three schizophrenia studies, only one schizophrenia study was taken into account in the analysis. The 0 category represents the modules that are not enriched in high heritability SNPs for any trait or disease. For a split by tissue-specific network, see supplementary fig. S3, Supplementary Material online. b to g) Gene Ontology enrichment analyses results on gene content for a few tissue-specific modules enriched for high heritability SNPs for one or two traits and diseases. Bubble plots for top ten terms or terms with *P*-value ¡0.01 for b) brain (nucleus accumbens—basal ganglia) module 5, enriched for high heritability SNPs for Alzheimer's disease and multiple sclerosis and c) colon—transverse module 149, enriched for high heritability SNPs for prostate cancer. Word clouds representing word frequencies in gene ontology terms enriched in d) adipose—visceral omentum, module 142, enriched for high heritability SNPs for HDL; e) muscle—skeletal, module 90, enriched for high heritability SNPs for height; f) whole blood, module 17, enriched for high heritability SNPs for type 2 diabetes; g) brain (nucleus accumbens—basal ganglia) module 100, enriched for high heritability SNPs for schizophrenia.

comparisons and supplementary fig. S5, Supplementary Material online for tissue-by-tissue comparisons with the adipose visceral (omentum) network. If choosing a more stringent 0.3 threshold mechanically decreases the number of tissue-specific modules from 2,748 to 1,615, it does not change our main conclusions. We then focused on trait- and tissue-specific modules, defined as those enriched for high heritability SNPs from one to three traits and enriched for tissue-specific GO Terms (supplementary table S10, Supplementary Material online)—to investigate the specificity of each trait and/or disease heritability and explicit the underlying molecular mechanisms. We found that these modules were enriched in genes involved in biologically and trait-relevant functions.

For example, heritability for ALZ and multiple sclerosis were both clustered in module 5 of the brain—nucleus accumbens (basal ganglia) network; module 5 is enriched in genes involved in catecholamine metabolic process (Fig. 2b), a class of molecules whose concentrations are altered in symptomatic Alzheimer's and multiple sclerosis diseases, both of which are amyloid plaque diseases (Cercignani et al. 2021; Henjum et al. 2022). Heritability for SCZ was most strongly clustered in module 100 of the same tissue, a molecule that is enriched for dopamine receptor signaling pathway genes (Fig. 2g), and this pathway is known to be functionally disrupted in the brain striatum in SCZ patients (McCutcheon et al. 2019).

Unsurprisingly, heritability for cancers, including BRC, OVC, and PRC, was enriched in epithelial tissues (skin—not sun-exposed exposed, skin—sun-exposed, colon—transverse) in network modules consisting of genes enriched for immune response (PRC and module 76 of skin, not sun-exposed), response to cellular hypoxia (BRC, prostate cancer and module 157 of skin—sun-exposed), DNA break repair, cell cycle, apoptosis, and epithelium differentiation and growth (breast cancer and module 149 of skin—sun-exposed, and module 191 of skin (not sun-exposed), OVC and module 199 and skin, prostate cancer and modules 78, 97 of skin and module 149 of skin—sun-exposed). Particularly interesting, PRC heritability is clustered in module 149 of the colon-transverse network, enriched for TRAIL-activated apoptotic signaling pathway genes, a long-known signaling pathway involved in cancer progression (Johnstone et al. 2008) (Fig. 2c). Breast and OVC heritability are also clustered in modules involved in estrogen metabolism and signaling (BRC and module 180 of skin—sun-exposed, OVC and module 182 of skin—not sun-exposed).

Metabolic traits such as high blood levels of HDL or LDL and TIID tend to cluster in modules enriched for lipid and carbohydrate metabolism. More generally, heritability for HDL and LDL tend to cocluster in several tissue-specific networks, including in adipose–visceral omentum, where almost half of the modules enriched for one trait being also enriched for the other one, the gastro-intestinal networks (CLS-, EGJ, EMC), the heart-related networks (HRA, HRV), the liver network, and the esophagus muscularis and fibroblasts networks. Heritability for high HDL and LDL levels are for example enriched in module 142 of adipose–visceral omentum, enriched for genes involved in very-low-density lipoprotein and HDL particle assembly, remodeling, and clearance (Fig. 2d). They also cocluster in modules enriched for genes involved in lipid metabolism (modules 92 and 197), and cholesterol efflux (module 206). High HDL and LDL heritability can also cluster in different modules enriched for genes belonging to similar pathways: inflammation (modules 4, 133, and 189 for HDL—modules 5, 71, and 128 for LDL) and phosphorylation of STAT protein (module 179 for HDL, 49 for LDL). High HDL heritability is also found in modules related to cholesterol homeostasis (module 189) and ketone metabolism (module 133), while high LDL heritability is found in modules related to lipoprotein lipase activity (module 144), response to stresses (modules 156 and 239) and redox metabolism (modules 57 and 109). These results are robust to the clustering algorithm, in particular the enrichment of both high HDL and high LDL heritability in ADV modules with genes belonging to the lipoprotein metabolic and clearance pathway (module 71 in LCS2; 44 in FG; and modules 1 and 52, with a less significant $P \leq 0.05$ for enrichment of heritabilility, in LEC, supplementary tables S11 and S12, Supplementary Material online).

Finally, TIID heritability clusters in module 17 of whole blood, enriched in genes involved in glycogen metabolism (Fig. 2f). Finally, heritability for HGT, a trait primarily related to development and, in particular, bone and muscle growth, is clustered in module 90 of muscle–skeletal, enriched for genes involved in muscle tissue morphogenesis, endochondral ossification, bone mineralization, and chondrocyte and osteoblast differentiation (Fig. 2e).

Local and Global Hubs Carry a Large Part of Trait Heritability

We then explored how heritability is distributed across the SNPs nodes in the eQTL network and whether there are particular network topological features that correlate with a greater-than-expected ability to explain the heritability of a particular trait. We thus computed two different summary statistics characterizing the topological properties of SNPs within each network: outdegree and core score (Platig et al. 2016). The outdegree measures the centrality of an SNP within the entire network. SNPs within the top 25% of outdegree distribution were considered as global hubs. The core score measures the contribution of SNPs to the modularity of the network module in which it arises. SNPs within the top 25% of core score distribution were considered local hubs (core SNPs).

We investigated whether trait heritability was distributed evenly across all SNPs or instead concentrated in SNPs with local or global centrality. Using a likelihood-ratio test accounting for linkage disequilibrium and module size (see Materials and Methods), we found for almost every trait we considered that the SNPs explaining the greatest portion of the heritability (top 5% of Z^2, or high heritability SNPs) are more likely to have both high outdegree and core scores (supplementary fig. S6, Supplementary Material online for outdegrees and supplementary fig. S7c, Supplementary Material online for core scores). The enrichment in high heritability among local "hubs" appears stronger and more systematic than for global "hubs"; in artery coronary and liver, the global hubs do not show any significant enrichment at all. Using the adipose visceral (omentum) and the whole blood networks as a model, we found that enrichment of high heritability SNPs in local hubs increases with the threshold chosen to define high outdegree and core score [from the top 90% to the top 5%, supplementary fig. S8a, Supplementary Material online for whole blood and supplementary fig. S8c, Supplementary Material online for adipose visceral (omentum)]. This result is also robust to clustering method (see Materials and Methods and supplementary figs. S8*c* to S8*f*, Supplementary Material online and detailed results for threshold =0.75 in supplementary fig. S9a, Supplementary Material online).

Given the demonstrated importance of eQTL network topology, we explored whether the increased heritability we found was driven simply by being in the network (a proxy for being an eQTL), being a global hub, or being a local hub. These annotations are overlapping and reflect potentially confounding factors such as underlying chromatin annotations (e.g. global hubs are enriched for non-genic enhancers, while local hubs are enriched for genic enhancers and promoters (Fagny et al. 2017)). For this reason, we partitioned heritability across various functional annotations while accounting for linked markers using stratified LD score regression (Bulik-Sullivan et al. 2015; Finucane et al. 2015) (see Materials and Methods) using the 97-level baseline annotation model, to which we added our three annotations: belonging to eQTL network, being in the top quartile of outdegrees (global hubs), or of core scores (local hubs). The total proportion of heritability explained by SNPs as estimated using the LDSC software is reported in supplementary table S2, Supplementary Material online. We found that these proportions are not always perfectly correlated with the estimated global genetic heritability reported in published twin and pedigree studies (see supplementary table S2, Supplementary Material online), but our estimate of the total heritability of traits explained by SNPs as computed by the LD-score regression is coherent with previous reports (see Lindström et al. 2017 for breast, ovarian, and prostate cancer examples).

The LD-score regression confirmed the results we obtained above using likelihood-ratio tests: SNPs with high core scores or high outdegrees are enriched for trait heritability, and this enrichment increases with the threshold chosen to define high scores (supplementary fig. S8b, Supplementary Material online). The detailed results that show enrichment in h^2 among SNPs within each annotation for each of the 11 traits and diseases are reported in supplementary table S13, Supplementary Material online. These results are once again independent of the chosen clustering algorithm (supplementary fig. S9b, Supplementary Material online).

We performed a meta-analysis across the 11 uncorrelated traits and diseases for each tissue-specific network. An example of enrichment in h^2 among each annotation for the Whole-Blood network is shown in Fig. 3a. As noted earlier, SNPs that are within the eQTL networks tend to be significantly enriched in h^2 independent of tissue (Fig. 3b). However, the enrichment in h^2 is even greater for both high outdegree SNPs and high core score SNPs in all networks except artery—coronary and liver. In many ways, this trend is exactly what one would expect: SNPs that fall within the eQTL networks have increased heritability because they are potentially capable of affecting the expression of multiple genes, a trend that increases as the SNPs become increasingly connected in the eQTL networks. These results are robust to clustering method (see Materials and Methods and supplementary fig. S9b, Supplementary Material online).

Local Hubs Evolve Under Polygenic Adaptation

The clustering of complex traits and diseases heritability in a few modules and in hubs in eQTL networks, together with previous results suggesting that global hubs are likely to evolve under negative selection (Wollenberg Valero 2024) suggest that directional selection on complex traits may preferentially be mediated by local hubs. The importance of the local hubs makes logical sense. Indeed, selection acts at the level of the traits. As each specific trait is likely to be under the control of one or several network modules consisting of genes involved in the same biological function, the local hubs play an important role in their determination. Targeted selection acting on local hubs would moreover mitigate the potential perturbative effects on the network as a whole or other functional modules. To test for this hypothesis, we searched for enrichment among global or local hubs of high scores for different statistics measuring three types of selection signatures. Genomic evolutionary rate profiling (GERP) score detects regions that lack substitutions, which can indicate negative selection (Cooper et al. 2005). Integrated haplotype score (iHS) detects longer-than-expected haplotypes around one of the alleles at one locus, a signature of recent, strong directional selection event or selective sweeps (Voight et al. 2006). F_{ST} measures population differentiation that may reflect older and sometimes milder directional selection events and has proven to be powerful in detecting the shifts in beneficial allele frequencies observed in polygenic selection (Wright 1965; Barghi et al. 2020).

We found that both global and local hubs, while enriched for complex traits and disease heritability, were more likely to be located in constrained genomic regions with high GERP scores in all the tissue-specific networks (see Fig. 4a and supplementary fig. S10a, Supplementary Material online). Global hubs were even more likely than local hubs to belong to constrained regions (supplementary fig. S11, Supplementary Material online). Both types of hubs are globally evolving under negative selection, in particular global hubs.

Neither local nor global hubs presented a clear pattern of enrichment for high |iHS| in all tissue-specific networks (see Fig. 4b and supplementary fig. S10b, Supplementary Material online). However, local hubs were significantly enriched for high |iHS| in more tissue-specific networks than global hubs (18 of the 29 networks for local hubs vs. 13 of the 29 networks for global hubs, see Fig. 4b and supplementary fig. S10b, Supplementary Material online). Local hubs were also enriched for high F_{ST} between European and both African and East Asian populations in almost all networks. Conversely, global hubs were depleted for high F_{ST} in almost all networks. Local hubs thus showed more signatures corresponding to polygenic selection than the other SNPs in the network, while global hubs seemed to be evolving under strict negative selection. These results were robust to clustering method (see Materials and Methods and supplementary figs. S12 and S13, Supplementary Material online). Using different thresholds to define high outdegree and core score (from the top 90% to the top 5%), we show that the results obtained reinforce our conclusion on the selection dynamic in the global and local hubs (see Materials and Methods, supplementary text, Supplementary Material online, and supplementary fig. S14, Supplementary Material online).

Discussion

Of the SNPs found through GWAS to be associated with complex traits, most (up to 90%) fall outside of gene coding regions suggesting that they likely play a regulatory role. Bipartite eQTL networks have allowed us to explore how these SNP loci work together to regulate the many genes involved in the biological processes underlying the traits, in a tissue- or cell-type-specific manner (Platig et al. 2016; Fagny et al. 2017, 2019). Using eQTL networks, we found that not only do SNPs act in both *cis* and *trans* to influence the expression of complex networks of genes, but that these networks have a robust, modular structure consisting of SNP–gene modules with properties that help explain the polygenic determinism that underlie most common traits and provide clues on how they can evolve despite the high degree of pleiotropy. Specifically, the modules in eQTL networks are enriched for functionally related groups of genes and nucleated around "core SNPs" that are both local (module) hubs and the SNPs most likely to exhibit strong GWAS associations with various phenotypes (including diseases) due to their association with module-driven, trait-related processes (Platig et al. 2016; Fagny et al. 2017).

The omnigenic model was important in that it helped bridge the gap between the missing heritability found in many diseases and the growing number of traits for which hundreds, if not thousands, of small effect-size genetic variants contribute to a given trait. Our eQTL network-based model is complimentary to the omnigenic model but provides a more nuanced view of the importance of SNP–gene "regulatory" associations in defining phenotypes, identifying disease associations and functions, and explaining both missing heritability and selection. Although it has been reported that global hubs in eQTL networks are enriched for tissue-relevant trait heritability (Gaynor et al. 2022), as are gene modules appearing in various types of networks (Kim et al. 2019), there has not been a

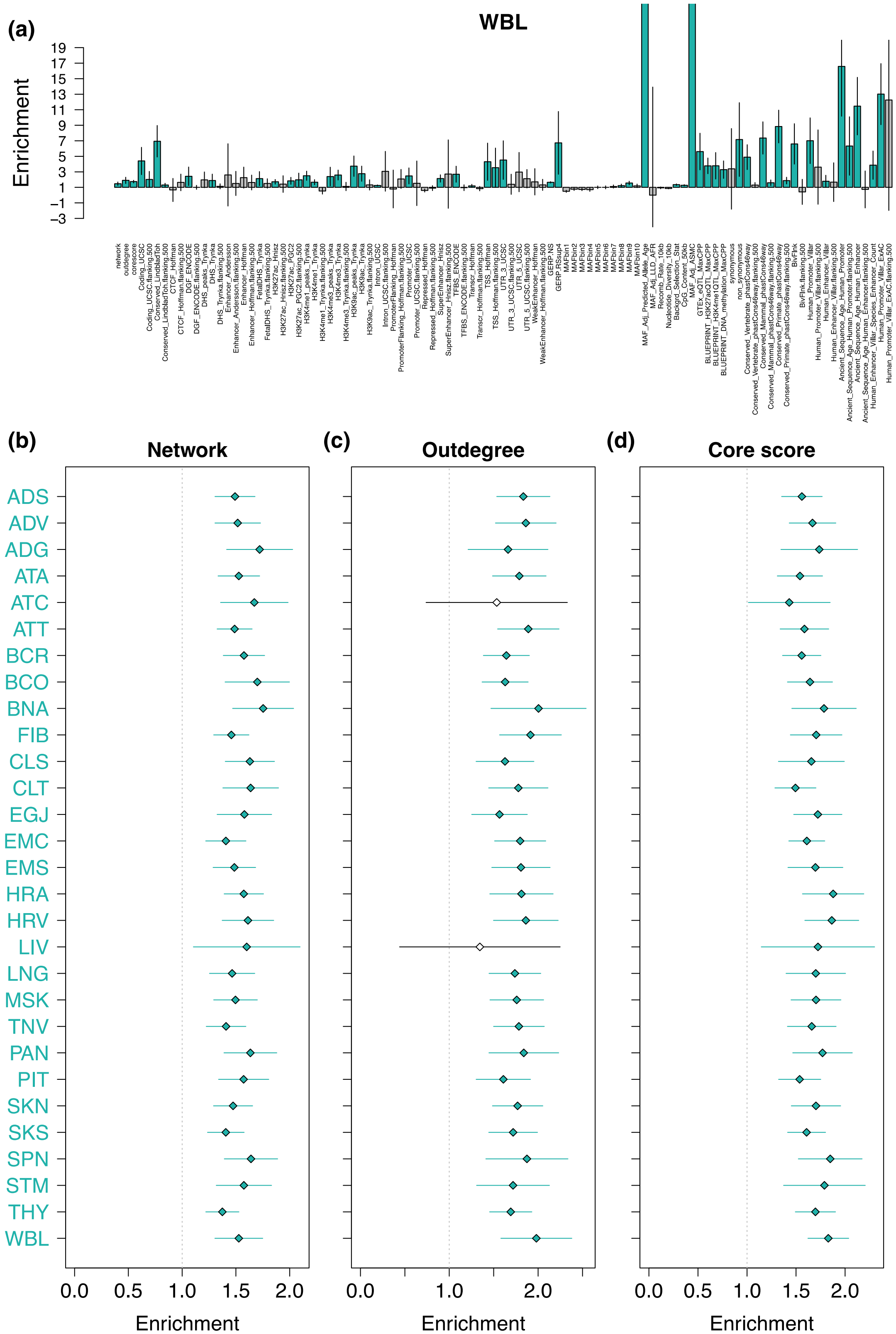

Fig. 3. Global and local hubs explain a larger share of heritability than expected by chance. a) Meta-analysis of h^2 enrichment among various annotations across 10 uncorrelated traits and diseases for the Whole Blood eQTL network. Meta-analysis of h^2 enrichment among SNP subcategories across 10 uncorrelated traits and diseases for each tissue-specific eQTL network: b) among SNPs within the eQTL network; c) among global hubs (high degrees); and d) among local hubs (high core scores.)

systematic exploration of these two complementary and important conceptual advances in understanding genetic effects in complex traits.

We addressed that gap in understanding by performing an analysis of eleven polygenic traits that represent a wide range of genetic heritability to determine the distribution of trait

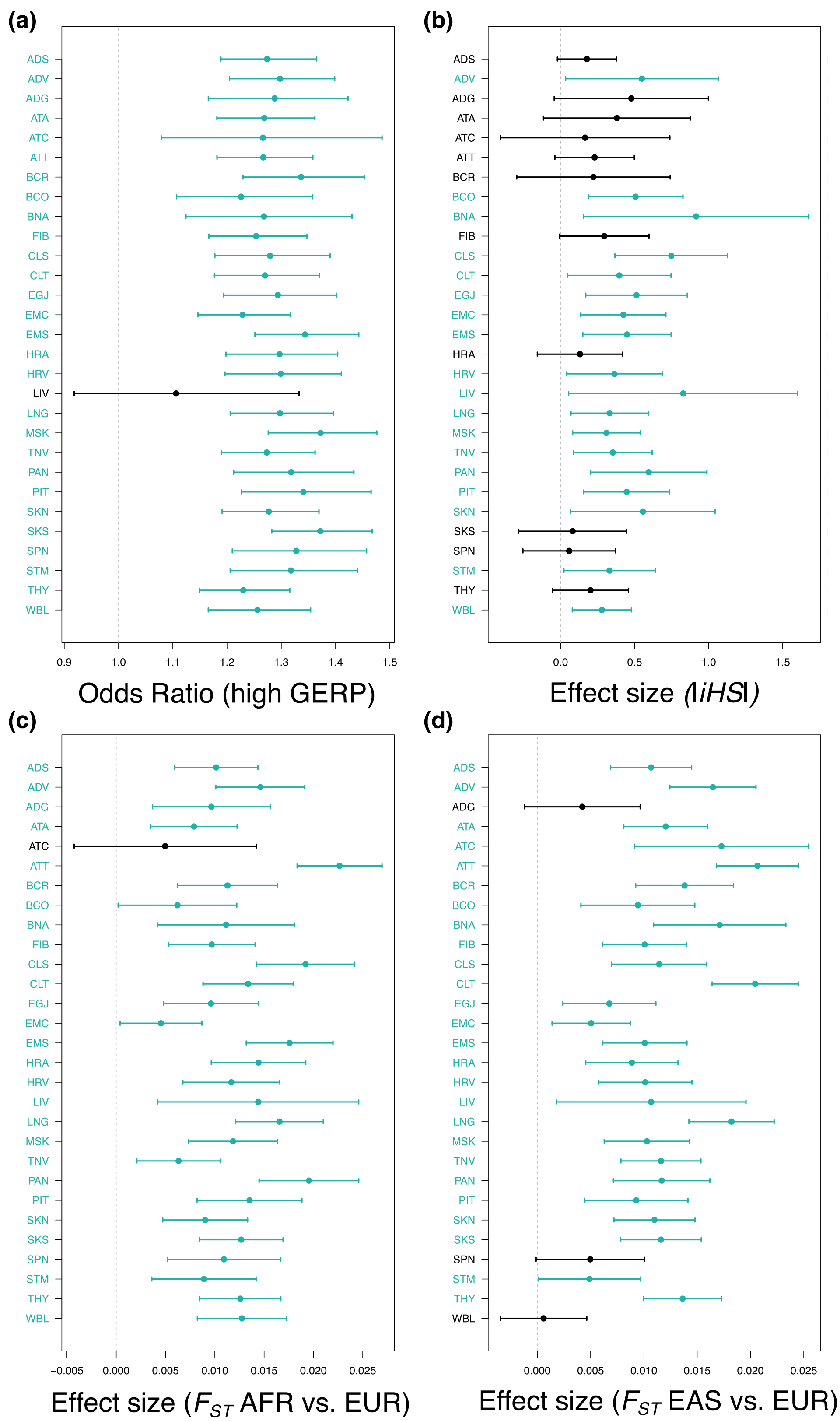

Fig. 4. While overall under negative selection, local hubs are enriched for positive selection signals in almost all tissue-specific eQTL networks. a) Odds ratio of being conserved when the SNP is a local hub vs. a leaf. Bars indicate confidence intervals. Confidence intervals were computed using a logistic regression. b) iHS that detect recent selective sweeps signals. c) F_{ST}(AFR, EUR) measuring population differentiation between African (AFR) and European (EUR) samples from 1000 genomes. d) F_{ST}(EAS, EUR) measuring population differentiation between East-Asian (EAS) and European (EUR) samples from 1000 genomes. b to d) Effect size of being a local hub vs a leaf on the value of several statistics sensitive to positive selection. Bars indicate 2 × SE. Standard error (SE) were computed using a linear regression. a to d) For each tissue-specific network, significant enrichment (an odds ratio significantly different from 1 or effect size significantly different from 0) are highlighted in blue. Significance was called when $P \leq 0.05$.

heritability across 29 tissue-specific eQTL networks. We found that although heritability is widely distributed across loci, as suggested by the omnigenic model, the distribution of heritability is far from homogeneous. Instead, there is an uneven distribution with the greatest heritability concentrated in network modules containing genes that represent trait-specific and biologically relevant functions. This makes sense as variations of a given polygenic trait arise through alterations of the expression of specific biological processes relevant to this trait. Further, we found that trait heritability was more likely to be explained by SNPs occupying key positions in the eQTL networks, especially among the "core SNPs" that are local hubs in their functional modules. The only exception are the artery coronary and liver networks, for which this enrichment is not observed. They also behave differently from the other tissue-specific networks in other aspects (tissue-specificity of modules, enrichment of local hubs in negative and polygenic selection signals). This is likely due the relatively small sample size for RNA-seq data in these tissues, leading to smaller eQTL networks, despite their high modularity. These same SNPs, which we had previously shown to be enriched in tissue-specific activated regulatory elements (Fagny et al. 2017), are thus likely to determine a significant proportion of the heritability of complex traits.

The clustering of heritability in modules is also not evenly distributed across traits and differs among tissues. We found that heritability in each phenotype tended to be clustered in a small number of biologically relevant modules within the highly modular eQTL networks that are relevant for understanding the phenotype in question. As previously noted, these modules tend to be trait-specific, even when traits are not genetically correlated. This module-dependent concentration of heritability, coupled with the enrichment of heritability in core SNPs of those same modules, could help explain why even highly connected regulatory networks (Boyle et al. 2017; Liu et al. 2019) are robust to the genetic perturbations of deleterious genetic variants. Indeed, the highly modular structure of eQTL networks provides a means by which the disruptive effect of regulatory mutations can be buffered in a tissue-specific manner against altering the broader functionality of the wider regulatory networks active in living cells.

We also found that global hubs are evolving under strong constraints, as observed in previous studies (Wollenberg Valero 2024). However, our results revealed that another category of nodes, the local hubs that regulate many genes involved in the same biological processes and articulate modules, have been preferential targets of past and ongoing polygenic selection. Because of their strategic position in the eQTL regulatory network, these loci greatly influence the regulation of biological processes in a tissue-specific manner, and, because their effect is limited beyond their "home" module, they are ideal candidates for adaptation. Finally, our results confirm the hypothesis, first presented in an opinion paper by Fagny and Austerlitz (Fagny and Austerlitz 2021), that local hubs are preferential targets for polygenic selection, and provide a means to understand how complex traits may evolve despite the strong pleiotropy that exists in biological networks.

All our results were obtained on bipartite eQTL networks inferred using statistical correlations between individual *cis* and *trans* SNPs to each gene independently. However, other relationships are hidden in our network, including spurious gene co-expression patterns, SNPs-SNPs interactions caused by linkage disequilibrium, and other SNP–SNP interactions such as epistasis. For the first part, we have demonstrated using comparison with co-expression networks that our approach manage to root out a lot of the spurious co-expression patterns observed in co-expression networks, focusing instead on the portion of gene expression correlated with genetic factors. We also controlled for linkage disequilibrium in our analyses, using two different strategies: (i) treating SNPs within a linkage block as one marker in our downstream analyses and (ii) using the LD-score software, a power tool to account for both SNP linkage and annotations overlaps. Finally, regularized regression models have been recently developed that take these interactions into account. However, the genome-wide application of such methods to a large genome like the human genome one is still limited by computation time, in particular when the eQTL detection is repeated across 29 tissue-specific datasets. Indeed, regularized regression models are generally applied on filtered datasets to limit the number of tests to perform and reduce the computational burde. They are either applied exclusively to *cis*-eQTLs (Marchetti-Bowick et al. 2019; Yan et al. 2020), or applied to heavily filtered and pruned data, such as functionally annotated regions with prior pruning of SNPs for linkage disequilibrium (Banerjee et al. 2021). This may lead to miss some important regulatory SNPs. We consequently arbitrated our pipeline choices in favor of increasing the number of candidates and to the detriment of the potential interactions, and choose a voluntarily loose *FDR* threshold to balance for the multiple testing issues of *trans*-eQTL detection.

Conclusions

Overall, our results demonstrate a synergy between the eQTL network and omnigenic models in explaining how genetic variants work together to influence traits while addressing their respective shortcomings. The value of such a synthesis can be seen in the results derived from the collection of complex traits we chose to analyze, including cancers, metabolic diseases, and auto-immune neurodegenerative disorders. In each of these traits we can see genetic risk factors identified through GWAS perturbing tissue-specific functional modules (while affecting others), while those modules are simultaneously affected by many other genetic variants of smaller overall effect size. These complex relationships define a conceptual framework for understanding disease risk, helps to define phenotypes for health and disease, and provide a means by which one could potentially prioritize therapeutic targets and design treatment protocols that account for the network architecture. Our findings also allow a better understanding of how complex traits can change as populations adapt to local changes through selection acting on traits mediated at the genetic and molecular level by key regulatory elements. This conceptual framework also provides an explanation, at the molecular level, of how polygenic selection of regulatory mutations beneficial in one environment could lead to the phenomenon of maladaptation that many suspect is the origin of many complex diseases.

Finally, the transferability of these results obtained on human data to other organisms is an open question. One main issue is the impact of the genome structure on the results. In plants genomes, there are many more structural variants than in the human genome (Marroni et al. 2014; Saxena et al. 2014). These variants are more likely to have a greater impact on the value of the phenotype and are also often associated to adaptation to the environment (Gui et al. 2022; Kang et al. 2023; Li et al. 2023). While they could be integrated in our eQTL network, it is not clear how their presence would affect the conclusions. Keeping in mind these caveats, studying the transferability of these results to other organisms could be of tremendous value to breeders by pointing out which loci are the more likely to improve the tolerance of crops or cattle to environmental stresses without jeopardizing their agricultural values.

Materials and Methods

GTEx Dataset

We used genotyping and gene expression level data from the NHGRI GTEx project version 8.0 (GTEx Consortium 2015). For appropriate statistical power in downstream analyses and network stability, we filtered out tissues for which the number of individuals with both genotyping and RNA-seq data available was less than 200; sex-specific tissues were not included. This left 29 tissues for analyses, as can be found in supplementary table S1, Supplementary Material online. Genotyping data were downloaded from the database of genotypes and phenotypes (dbGaP): phs000424.v8.p2. Genotyping data were preprocessed on the Bridges system at the Pittsburgh Supercomputing Center (PSC) and the Cannon cluster supported by the Faculty of Arts and Sciences Division of Science, Research Computing Group at Harvard University (see Gaynor et al. 2022).

The sequencing data were processed in plink 1.90 to retain only SNPs, and we removed variants with genotype missingness greater than 10% or minor allele frequency less than 0.1 (Purcell et al. 2007). SNP imputation was then performed using Eagle2 (Loh et al. 2016).

Fully processed, filtered, and normalized RNA-seq data were obtained from the GTEx Portal (www.gtexportal.org). The GENCODE 26 model was used to collapse transcripts and quantify expression using RNA-SeQC (https://www.gencodegenes.org/human/release_26.html#).

Bipartite eQTL Network Inference

eQTLs were obtained from Gaynor et al. (2022). Rapidly, with the R MatrixEQTL package (Shabalin 2012), the association between SNP genotypes and gene expression was modeled using linear regression (Eq. 1) that included potential confounding factors as covariates (Kendziorski et al. 2006): the two first principal components for population structure, sex, age and RIN that measures RNA quality. If **G** is an $r \times m$ matrix of gene expression and **S** is an $r \times n$ matrix of SNP genotypes, each with r rows representing observations and columns representing n SNPs and m genes, respectively, **X** is a covariate matrix, the eQTL of a particular SNP i on a locus's gene expression j is then :

$$\mathbf{G}_{\mathrm{j}} = \mathbf{X}^{\mathrm{T}}\boldsymbol{\alpha} + t_{ij}\mathbf{S}_{\mathrm{i}}. \tag{1}$$

Associations were evaluated for SNPs in both *cis*—SNPs within 1MB of a gene's transcription start site—and *trans*. The eQTL associations between all pairs of SNPs and genes were then represented as a sparse, weighted bipartite network (Fig. 5). Each SNP and gene was considered a node in the network. Using a fixed cutoff $q = 0.2$ on the FDR of the eQTL regression (Platig et al. 2016; Fagny et al. 2017), the edge weight $a_{i,j}$ between SNP i and gene j was defined by the function $I_{i,j}\{\mathrm{FDR} <= q\}|t_{ij}|$, where t_{ij} is the value of the t statistic computed by the MatrixEQTL package. Thus, when the estimated FDR of the eQTL regression was below the threshold of 0.2, then $a_{i,j} = |t_{ij}|$, indicating that there was an edge connecting the nodes, and $a_{i,j} = 0$ otherwise.

Overlap with Hi-C Data

3D chromatin loop data stemming from Hi-C experiments for 9 different tissues corresponding to nine of the 29 explored tissues were downloaded from the 3D genome website (see Data Availability). Loops had a resolution of 10 kb, with intrachromosomal loops length spanning between 70 and 2,880 kb. To match our *cis*-eQTL data, we retained only loops spanning a maximum of 1 Mb (on average 99.5% [97.1 to 99.9] of all loops, corresponding to 13,775 [7,812 to 18,749] loops), and for which one side of the loop's 10 kb block was containing a gene TSS. This filtered dataset corresponded to our "null" dataset.

For this analysis, we merged eQTLs by blocks of linkage disequilibrium (with an FDR cutoff of 0.2 for SNP–gene association significance, we retained an average of 749,672 [403,759 to 1,104,908] significant block–gene associations. We retrieved overlaps of Hi-C loops and eQTL data, keeping only cases where the 10-kb block containing the eQTL's LD block was on one side of one loop and the 10-kb block containing the TSS of the associated gene was on the other side of the same loop (on average 646 [304 to 1,068]). We then used a χ^2 test to compare proportions of Hi-C loops in LD blocks that carried significant eQTL and in LD blocks that did not.

Network Summary Statistics

The module structure of each tissue-specific network was determined using the bipartite modularity maximization approach (Platig et al. 2016) implemented in the netZooR Bioconductor package. This is a two-step approach : (i) project the eQTL network on the gene space and infer a first cluster structure using one of the three available algorithms : louvain clustering (LCS), leading eigenvector community detection (LEC), and the fastgreedy community detection (FG); and (ii) improve the clustering by adding SNPs back in the network and searching for improved modularity by reattributing SNPs to modules to obtain the best modularity possible and then genes. The second step is iterated as many times as necessary to reach convergence. We tested all three algorithms on step (i) to ensure that our results are robust to clustering method and implemented a new version of the second step of the algorithm (CONDOR:condorSplitMatrixModularity) that allows for a balance between computation time and memory usage (Fig. 5) and has been published in the R netzooR package fork of https://github.com/maudf/.

$$k_i = \sum_{j=1}^{g} I_{i,j}\{\mathrm{FDR} < q\}|t_{ij}| \tag{2}$$

The SNP core score was defined as the SNP's contribution to the modularity of its module, it measures the centrality of the SNP in the module (Platig et al. 2016). If $m = s \times g$ is the total number of possible edges in a network made of s SNPs and g genes, $\tilde{a}_{ij}$ is the observed edge value between SNP i and gene j. Here, d_j is the gene i indegree defined as $d_j = \sum_{i=1}^{s} I_{i,j}\{\mathrm{FDR} < q\}|t_{ij}|$, then, for SNP i in module h, its core score, Q_{ih}, is defined by Eq. 3:

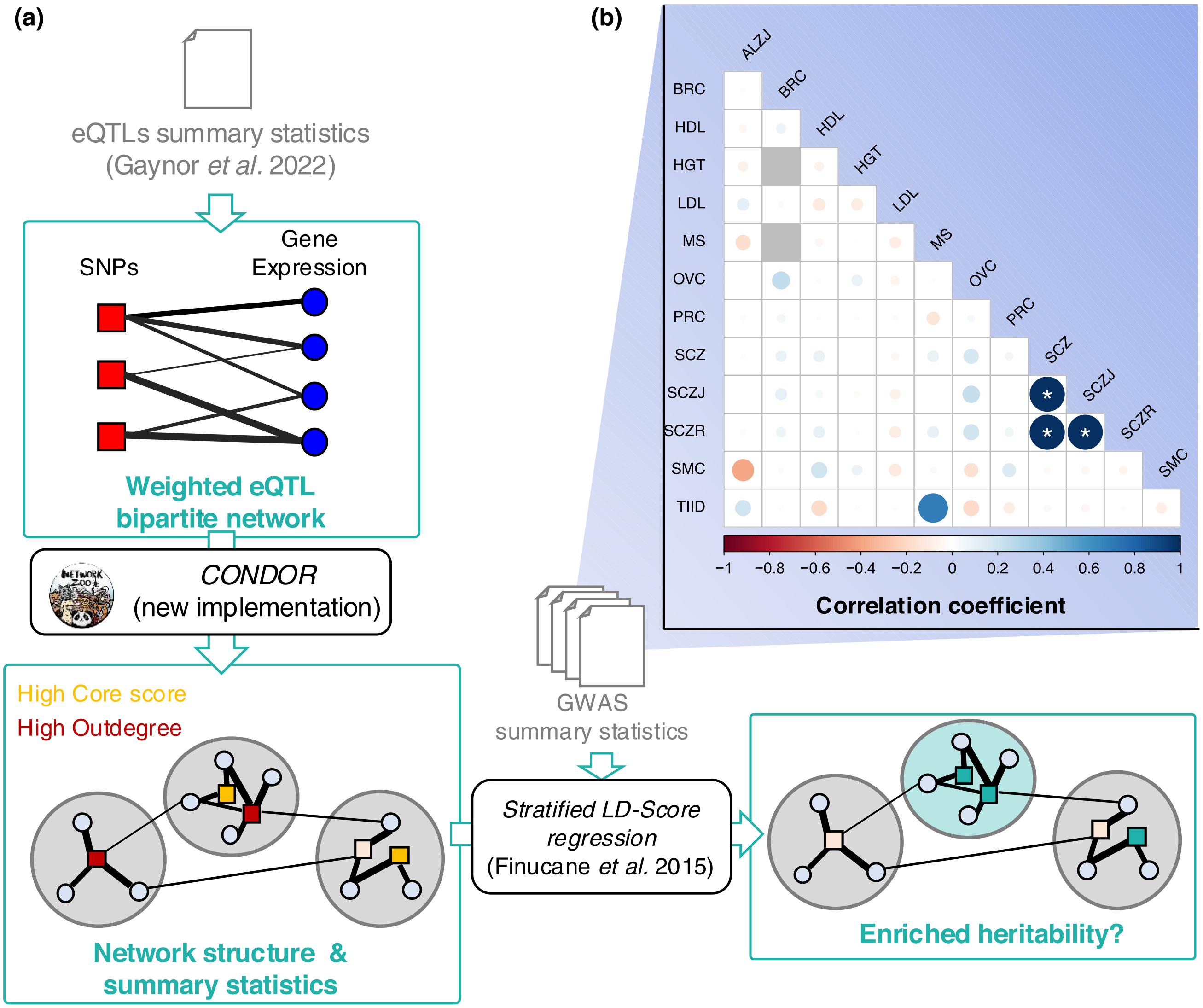

Fig. 5. Pipeline and traits correlation. a) Pipeline of data analyses. Input data are in grey. GTEx eQTL summary statistics were obtained from a previous study (Gaynor et al. 2022) and are described in Materials and Methods and supplementary table S1, Supplementary Material online. GWAS summary statistics were downloaded from https://console.cloud.google.com/storage/browser/broad-alkesgroup-public-requester-pays and are described in supplementary table S2, Supplementary Material online. eQTL network summary statistics are presented in supplementary fig. S1, Supplementary Material online. b) Pairwise genetic correlations between traits based on GWAS summary statistics.

$$Q_{ih} = \frac{1}{m}\sum_{j}\left(\tilde{a}_{ij} - \frac{k_i \times d_j}{m}\right)\delta(C_i, h)\delta(C_j, h) \qquad (3)$$

GWAS Data

GWAS data were obtained from the Alkes group (see supplementary table S2, Supplementary Material online). We considered a total of eleven traits and diseases presenting varying levels of estimated genetic heritability. For each trait or disease, we obtained summary statistics including SNP chromosome, position, alleles 1 and 2, χ^2, and Z-score. Here, the relationship between an individual's i genotype at SNP s and its phenotype Y, accounting for covariates M, is usually tested using the following regression: $Y_{i,s} = \beta_{1,s}X_{i,s} + \beta_2 M_i + \epsilon_{i,s}$, where $\epsilon_{i,s}$ is the error term. Then, $Z = \beta_{1,s}/\mathrm{se}(\beta_{1,s})$ and $\chi^2 = qnorm((P/2)^2)$ where $P = 2 \times pnorm(-|Z|)$.

We used Z^2 as a proxy for normalized heritability explained by each SNP. High heritability SNPs were defined as those with a Z^2 in the 95th percentile of the distribution.

GWAS Outdegree and Core Score Enrichment among High Heritability SNPs

We compared the distribution of SNP outdegrees and core scores between the high heritability SNPs and the rest of the SNPs in the network using a likelihood-ratio test (LRT), correcting for linkage disequilibrium. To control for LD between SNPs, we generated lists of SNPs falling into the same LD block, using the plink1.9—blocks option, a 5-Mb maximum block size, and an r^2 of 0.8. In each module, for each LD block, we extracted the median of either outdegrees (k_i) or core scores (Q_{ih}) for high and nonhigh heritability SNPs separately and used these values as input in the linear regressions.

The LRT we used assesses whether a linear model that includes GWAS status (Eq. (5)) fits the observed data better than a linear model that does not include this variable (Eq. (4)). As the distribution of SNP Q_{ih} is not uniform across modules, we added module identity as a covariate in the linear regression when computing LTR for core scores. In Eqs. (4) and (5), Score$_i$ is the score of SNP i

(either outdegree or core score), $I(GWAS = 1)$ is an indicator function equal to 1 if the SNP is a high heritability SNP and equal to 0 otherwise, and $I(C_k = 1)$ is an indicator function equal to 1 if the SNP belongs to module k and equal to 0 otherwise:

$$Score_i \sim \left[\sum_{k=1}^{n-1} I(C_k = 1)\right] + \epsilon \tag{4}$$

$$Score_i \sim I(GWAS = 1) + \left[\sum_{k=1}^{n-1} I(C_k = 1)\right] + \epsilon \tag{5}$$

Genetic Correlation Between Traits and Heritability Enrichment Analyses among Local and Global Hubs

We computed partitioned heritability with the 97 annotation baseline-LD model from Gazal et al. (2018) and Hujoel et al. (2019) for each of the eleven traits and diseases selected above (Fig. 5). This allowed us to estimate the enrichment and standardized effect size of the baseline annotations and three additional parameters on the heritability: belonging to the network (being an eQTL), having a high outdegree, and having a high core score (Gazal et al. 2018; Hujoel et al. 2019). We considered a binary annotation to ensure sufficiently stable estimates. For outdegrees and core scores, annotations were set equal to 1 if k_i and Q_{ih} were in the top quartile of the distribution and 0 otherwise.

Given that approximately 85% of the GTEx study population consists of individuals of European descent, we used LD scores computed from the 1000 Genomes Project data from individuals with European ancestry using the GRCh38 genome version and regression weights that exclude the HLA region; P-values were computed using a block-jacknife.

Meta-analyses were then performed across uncorrelated traits using the meta.summaries function from the R rmeta package v.3.0, with random-effect weights. To identify uncorrelated traits and diseases, pairwise genetic correlations between the eleven traits and diseases were first computed using stratified LD-score regression (S-LDSC). Traits and diseases with a pairwise genetic correlation of less than 0.3 were considered uncorrelated.

High Heritability SNP Enrichment Among Modules

High heritability SNP enrichment among modules was performed using a χ^2-test on data corrected for linkage disequilibrium. Using the same LD blocks as previously described, we considered that an LD block was a high heritability block if at least one SNP in this LD block had a Z^2 in the top 95th percentile. For each module, we counted the number of high and nonhigh heritability LD blocks in and outside of the modules and used these data to perform a χ^2-test.

Gene Ontology Enrichment Analyses and Identification of Tissue-Specific Modules

We performed Gene Ontology enrichment analyses using the Bioconductor R topGO package v.2.44 (Alexa and Rahnenfuhrer 2016), using the elim method; this method is more efficient than Fisher's exact test (Alexa et al. 2006). The GO categories are tested sequentially, following the GO tree structure from bottom to top: if one GO category is found to be significant, the genes involved are removed from the parent nodes before they are tested. The tests are thus not independent, and no multiple testing correction can be applied. Following the guidelines in the topGO users' manual, we filtered uncorrected P-values using a stringent threshold of 0.01. We also filtered out GO categories that did not include at least three genes in the gene set of interest. The gene ontology database used in this analysis was the one from the R bioconductor org.Hs.eg.db package v.3.13.0. For each test, the background gene set contains all the genes of the network and the gene set of interest contains all the genes from the module of interest.

We identified common and tissue-specific modules in the eQTL networks based on pairwise comparisons of GO term assignments. For each module of a first network, the GO ID enriched in the module was compared to the GO ID enriched in each of the modules in a second network using Jaccard Index. The best matching module was determined based on the highest Jaccard Index, and if the best Jaccard Index was ≥0.3, the modules were considered as similar and otherwise different. Then, for each module in each tissue-specific network, we counted the number of similar modules in the other 28 networks. If this number of similar functional modules was ≤3, we considered it a tissue-specific module. To evaluate the impact of the jaccard index threshold, we also investigated a threshold of 0.4.

Selection Scores Enrichment Analyses

Conserved SNPs were determined by intersecting the annotations of the SNPs present in our eQTL networks with conserved regions determined using GERP scores downloaded from Ensembl 111 release (https://ftp.ensembl.org/pub/release-111/bed/ensembl-compara/91_mammals.gerp_constrained_element/gerp_constrained_elements.homo_sapiens.bb Martin et al. 2023). Enrichment in conserved SNPs among local and/or global hubs were computed using the following logistic regression model, where $I(GERP = 1)$ is an indicator function equal to 1 if the SNP falls in a conserved region and equal to 0 otherwise, $I(Cat = 1)$ is an indicator function equal to 1 if the SNP belongs to a given category (local or global hubs) and equal to 0 otherwise, and $I(C_k = 1)$ is an indicator function equal to 1 if the SNP belongs to module k and equal to 0 otherwise:

$$\mathrm{Logit}(I(GERP_i = 1)) \sim I(Cat_i = 1) + \left[\sum_{k=1}^{n-1} I(C_{ik} = 1)\right] + \epsilon$$

iHS, which measures the differences in haplotype lengths between haplotypes carrying the ancestral and the derived SNPs, detect signatures of recent selective sweeps (Voight et al. 2005). We downloaded iHS standardized scores (Johnson and Voight 2018) computed for 26 populations for 1000 genomes project Phase 3 dataset from https://zenodo.org/records/7842512. We extracted scores computed on the CEU population (Utah residents (CEPH) with Northern and Western European ancestry) samples as being the closest population from the European descent samples that represent 85% of the GTEx dataset v8 release.

F_{ST} between African, Eurasian, and European samples from the 1000 genomes data were computed using plink2.0 (Purcell et al. 2007) on genotypic data for samples from the EUR, AFR, and EAS populations from the 1000 genomes Phase 3 dataset. Genotyping data were downloaded from ftp://ftp.1000genomes.ebi.ac.uk/vol1/ftp/data_collections/1000G_2504_high_coverage/working/20201028_3202_raw_GT_with_annot/.

Effect sizes of being a local or global hub on iHS or F_{ST} were computed using the following linear regression model, where $Score_i$ is the value of the statistics for SNP i, $I(\mathrm{Cat}=1)$ is an indicator function equal to 1 if the SNP belongs to a given category (local or global hubs) and equal to 0 otherwise, and $I(C_k = 1)$ is an indicator function equal to 1 if the SNP i belongs to module k and equal to 0 otherwise:

$$Score_i \sim I(Cat_i = 1) + \left[\sum_{k=1}^{n-1} I(C_{ik} = 1)\right] + \epsilon_i$$

The community ID terms were only added when local hubs were one of the two categories compared, as the q-score determining local hubs depends on community ID.

Supplementary Material

Supplementary material is available at *Molecular Biology and Evolution* online.

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Author Contributions

K.L.S., J.P., J.Q., and M.F. contributed to the conception of the work. K.L.S., J.P., J.Q., and M.F. contributed to the study design and method development. K.L.S., J.P., and M.F. contributed analysis, verified the data, and drafted the manuscript. All authors reviewed the manuscript and approved the submitted work.

Ethics Approval

This work was conducted under dbGaP-approved protocol # 9112.

Conflict of Interest

The authors declare no competing interests.

Data Availability

All the code used to analyze the data is available at https://github.com/maudf/heritability. The new CONDOR: condorSplitMatrixModularity is available at https://github.com/maudf/netZooR. The data used for the analyses described in this manuscript were obtained from the GTEx Portal on 17 December 2019 and dbGaP accession number phs000424.v8 on 12/17 19 for RNA-seq and Genotyping data, from https://console.cloud.google.com/storage/browser/broad-alkesgroup-public-requester-pays for GWAS data, and from https://3dgenome.fsm.northwestern.edu/downloads/loops-hg38.zip on 01/02/2025 for the Hi-C data.

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Functional Determinants and Evolutionary Consequences of Pleiotropy in Complex and Mendelian Traits

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Abstract

Pleiotropy, a phenomenon of multiple phenotypic effects of the same genetic alteration, is one of the most important features of genotype-to-phenotype networks. Over the last century, biologists have actively debated the prevalence, mechanisms, and consequences of pleiotropy. In this work, we employed data on genotype-to-phenotype associations from the Human Phenotype Ontology and Mouse Genome Database, as well as genome-wide associations from the UK Biobank and FinnGen cohorts to investigate the similarities and dissimilarities in the patterns of pleiotropy between species and different trait types (i.e. Mendelian traits and complex traits). We found that the pleiotropic effects of genes correlate well between species but have a much weaker correlation when comparing different types of traits for the same species. In all cases, however, highly pleiotropic genes possessed a common set of features, such as broad expression across tissues, involvement in many biological processes, or a high number of protein–protein interactions of the respective gene products. Furthermore, we observed a universal tendency of highly pleiotropic genes to be under greater negative selection pressure compared to other groups of genes, including genes essential for cell growth and development. Besides, highly pleiotropic genes also show a significant enrichment of recent positive selection signals. Taken together, our results pinpoint a common mechanism underlying pleiotropic effects in different trait domains and suggest that high degree of pleiotropy plays a role in adaptation, despite imposing additional constraint on genetic variation.

Keywords: genotype-to-phenotype map, pleiotropy, Human Phenotype Ontology, mammalian phenotype ontology

Introduction

Reconstruction of the complex network of genotype-to-phenotype relationships is a pivotal task of modern genetics. Over the years, the development and application of high-throughput methods have facilitated the construction of large-scale resources that aggregate data on genotype-to-phenotype associations, such as the Mouse Genome Database (MGD) (Blake et al. 2003) or the National Human Genome Research Institute (NHGRI) GWAS Catalog (Buniello et al. 2019). Construction of phenotype ontologies, such as the Human Phenotype Ontology (HPO) (Robinson et al. 2008) and the Mammalian Phenotype (MP) Ontology (Smith et al., 2005), has further enabled accurate integration and comparison of genotype-to-phenotype relationships, both within and between species.

Pleiotropy is a phenomenon in which a single mutation affects multiple independent traits in the phenotype (Wagner and Zhang 2011) and is thus one of the major forces that shape the genotype-to-phenotype networks. Since the advent of genomics, multiple studies have demonstrated the abundance of pleiotropic effects in different species, and different theoretical models of pleiotropy have been proposed. The modular pleiotropy model predicts that genes with similar functions should affect groups (or "modules") of related traits, while the universal pleiotropy model (also known as the omnigenic hypothesis) states that all genes can be expected to impact all traits (reviewed in Tyler et al. 2016). Recent studies favor the modular pleiotropy model, with two key pieces of supporting evidence: (i) the observed degree of pleiotropy for most mutations is modest, and (ii) the networks of gene–trait connections have a high degree of modularity. For example, both of these features of genotype-to-phenotype networks have been demonstrated in a comparative study of yeast, nematodes, and mice (Wang et al. 2010).

Genome-wide association studies (GWAS) provide a rich set of information for studying pleiotropy in the human genome, and many such studies have been performed (e.g. Pickrell et al. 2016; Chesmore et al. 2018). Emergence of large-scale deeply phenotyped cohorts, such as the UK Biobank (UKB) (Bycroft et al. 2018), has opened new prospects for pleiotropy research. Several efforts have been undertaken to study pleiotropy in UKB (Jordan et al. 2019; Watanabe et al. 2019), including a study performed by our group (Shikov et al. 2020). These studies have confirmed that pleiotropic effects are typical for a substantial proportion of genes, although the estimated proportion of pleiotropic loci varied widely between studies.

Estimation of the degree of pleiotropy from GWAS data is complicated by various factors, including (i) linkage disequilibrium (LD) between multiple genetic variants at a locus, leading to spurious pleiotropy (Chebib and Guillaume 2021); (ii) genetic and environmental correlations between traits that inflate the estimated level of pleiotropy; and (iii) the burden of multiple testing of association between millions of variants and thousands of traits (reviewed in Hackinger and Zeggini 2017). In line with these considerations, the estimated

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proportion of loci with pleiotropic effects in the UKB data was heavily dependent on the approach used to define independent phenotypes. For example, more than half of the human genome was reported to bear signals of pleiotropy by Watanabe et al. (2019) when using trait domain as a measure of trait independence; however, our analysis based on hierarchical clustering of traits by their phenotypic correlation yielded a much more conservative estimate of 5.4% of the genome (Shikov et al. 2020).

A particularly intriguing question that arises when analyzing pleiotropic effects of genetic variation is the evolutionary consequences of pleiotropy (reviewed in Zhang 2023). In 2000, Orr (2000) found that pleiotropy should naturally decrease the rate of adaptation within Fisher's geometric model framework. This model, however, assumes constant total effects of genetic variants and contradicts empirical observations (e.g. Wagner et al. 2008). Furthermore, Wang et al. (2010) have shown that pleiotropy could, to the contrary, increase the rate of adaptation if the effect size of a mutation is set to be dependent on the degree of pleiotropy. Further support for this model comes from studies that report an enrichment of pleiotropic loci in humans with signals of recent positive selection (e.g. Sakaue et al. 2021). The adaptive effect, however, is expected to be restricted to loci with a moderate degree of pleiotropy, as evidenced by studies in plants (Frachon et al. 2017).

In our earlier work, we have observed an unexpected enrichment of common variants at pleiotropic loci, an observation that challenges the concept of the evolutionary cost of pleiotropy and could possibly indicate an adaptive role of pleiotropic genetic variants. A later study by Novo et al. (2021) suggested that pleiotropic loci are subject to strong background selection; however, the significance of the trend varied for different source data used. Furthermore, the *B* statistic used by Novo et al. can deviate from expectation due to different selection types and may also be confounded by recombination rate. Taken together, results of previous studies do not allow a definitive conclusion regarding the evolutionary consequences of pleiotropy in the mammalian genome.

To carefully address these issues, as well as to provide a deeper insight into the mechanistic basis of pleiotropy in mammals, we have evaluated the pleiotropic effects of human and mouse genes using both Mendelian trait data (provided by the HPO and the MGD) and biobank-scale phenome-wide association studies for hundreds of complex traits (from the UKB and FinnGen projects; Kurki et al. 2023) to analyze functional properties of pleiotropic genes in various species and trait domains.

Results

Evaluating the Concordance of Pleiotropic Effects Between Species and Trait Domains

To compare the patterns of pleiotropy between species and trait domains, we first set off to construct a dataset based on publicly available information on genotype-to-phenotype relationships in humans and mice. To this end, we collected and merged data from four main sources: the HPO, the MGD, the pan-UKB, and FinnGen studies. The former two sources contain human (HPO) and murine (MGD) gene annotations with HPO or MP ontologies, respectively, while the latter two datasets represent a compendium of genome-wide association results for 480 quantitative traits and 2,493 complex diseases and disease-related endpoints, respectively (see Methods and Note S1 for details on data preprocessing). In total, the dataset contained 151,636 gene-to-trait relationships, with 52,849, 67,958, 12,065, and 18,764 coming from each of the aforementioned data sources. A total of 4,999, 12,834, 6,469, and 5,650 genes were associated with at least one trait in HPO, MGD, pan-UKB, and FinnGen data, respectively. Of note, only 592 genes had at least one gene–trait association across all four datasets, and trait associations from both trait domains (from either pan-UKB or FinnGen and either HPO or MGD) were available for 4,296 genes. Thus, more than half of the gene–trait connections were unique to each trait domain (55.4% for complex traits data and 68.3% for Mendelian trait data).

Before delving into the comparative analysis of the obtained data, we first examined the distribution of the degree of pleiotropy observed in each dataset. While all distributions had a characteristic left skew (Fig. 1a), the overall shape of the distribution varied. The distribution was much more asymmetric in the case of the genome-wide association data (Fig. 1a, two right subplots), with a median of 1 and 2 clusters of correlated complex traits (see Methods for details) associated with a gene. On the other hand, in both HPO- and MGD-based data, the median number of phenotype terms associated with each gene was much greater (11 and 4, respectively), corresponding to a lesser skew of the distribution (Fig. 1a, two left subplots). Only 287 out of 4,999 human genes (5.9%) were associated with exactly one upper-level HPO term, i.e. were nonpleiotropic (for comparison, 59.7% and 40.7% of genes were nonpleiotropic in the two complex trait datasets). In MGD, nonpleiotropic genes comprised a slightly greater fraction of genes (1,605 out of 12,834, 12.5%). In line with the differences in the distribution shape, the genotype-to-phenotype networks differed in the estimated degree of modularity (0.118 and 0.205 vs. 0.516 and 0.550 for HPO and MGD vs. pan-UKB and FinnGen data, respectively), with complex trait-based networks having by far the greatest level of modularity. In all cases, the observed modularity of the network was significantly higher compared to the expected value obtained using random permutation of edges (Extended Data Figure S1).

Having characterized the overall structure of the data, we next went on to compare the patterns of pleiotropy between the datasets. To this end, we used orthologous gene mapping between human and mouse to create a single merged dataset. Comparison of the degree of pleiotropy for single-copy orthologous gene pairs in mouse and human revealed a modest significant correlation (Spearman's $P = 0.26$, Fig. 1b). At the same time, the degree of pleiotropy estimated using complex trait data and Mendelian trait data showed much lower correlation. For instance, there was no significant correlation between the gene pleiotropy in HPO and pan-UKB or FinnGen (Fig. 1b). Remarkably, the degree of pleiotropy estimated for murine genes using MGD data significantly correlated with both GWAS-based estimates for human orthologs (Spearman's correlation $= 0.10$, $P < 0.001$ in both cases). The greater significance of the estimates for MGD data compared to HPO may be explained by the higher number of genes with nonzero degree of pleiotropy in mice.

Given the relatively low correlation between the degree of pleiotropy estimates, we next tested if a different result could be obtained when testing the overlap between sets of pleiotropic and nonpleiotropic genes without taking into account the numeric estimate of the degree of pleiotropy. To this end, we divided each dataset into four categories:

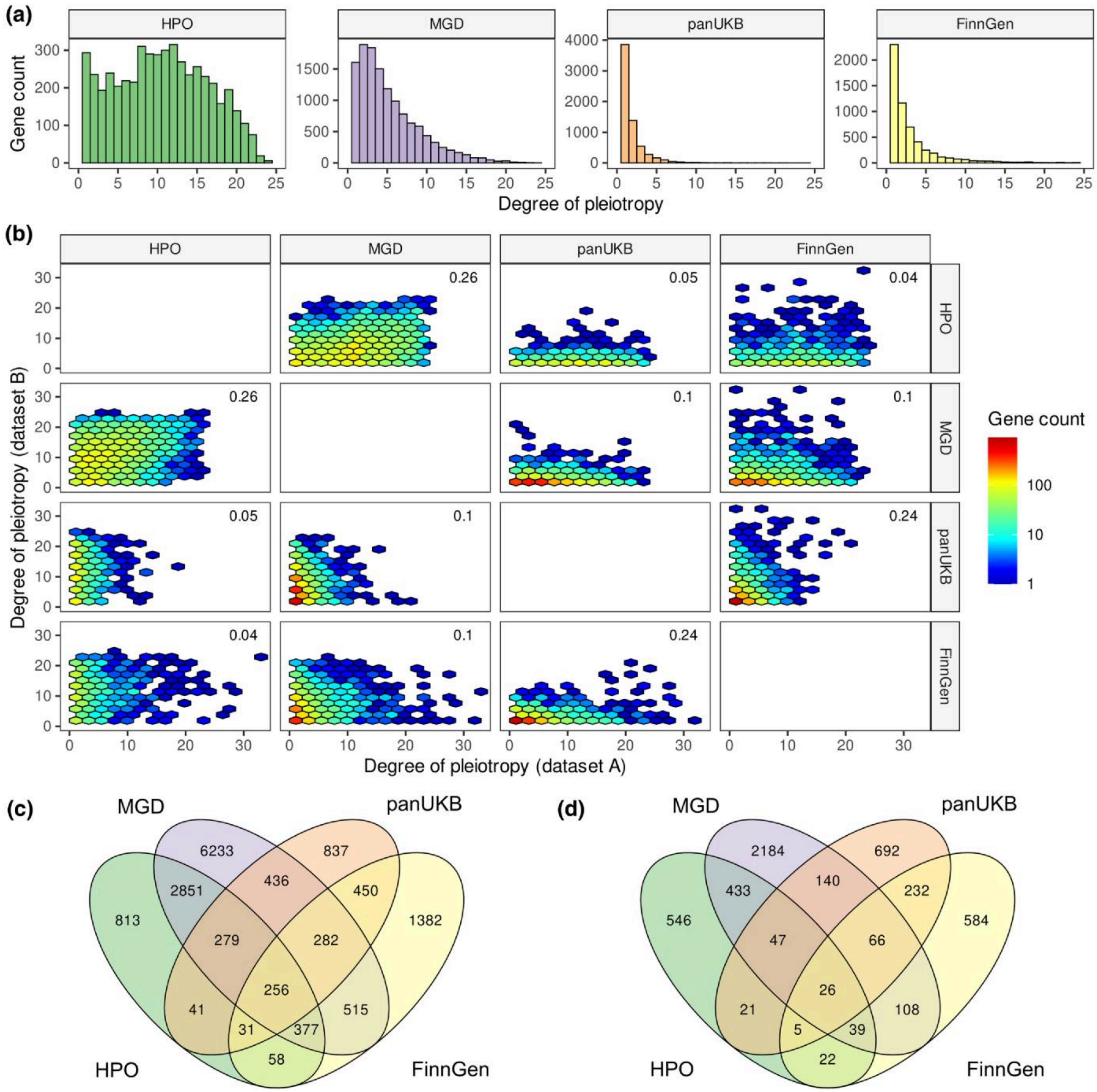

Fig. 1. Gene-level degrees of pleiotropy correlate between species and trait domains. a) Distribution of the degree of pleiotropy of human and mouse genes according to different sources of data (from left to right—Human Phenotype Ontology, Mouse Genome Database, pan-UK Biobank, and FinnGen GWAS data). b) Hexagonal scatter plots showing the correspondence between the degree of pleiotropy of genes between different data sources. c, d) Venn diagrams showing the overlaps between sets of all pleiotropic genes c) or highly pleiotropic genes d) between the three datasets (see text for details on cutoff selection).

(i) nonpleiotropic genes with no known phenotype (denoted as "NP"); (ii) nonpleiotropic genes associated with exactly one upper-level ontology term or one complex trait cluster (denoted "ST"); (iii) highly pleiotropic ("HP") genes (identified as top quartile of genes by degree of pleiotropy; the trait/term count cutoffs are 15 [HPO], 9 [MGD], 2 [pan-UKB], and 3 [FinnGen]); and (iv) modestly pleiotropic ("MP") genes (this group includes all of the remaining pleiotropic genes). The proportions of genes in different groups can be seen on Extended Data Figure S2 and are consistent with the shape of the respective distributions of the degree of pleiotropy (i.e. a greater fraction of genes in MP/HP groups, as well as a lower proportion of ST/NP genes, is observed for GPO and MGD).

We next compared the sets of pleiotropic genes between the dataset. When moderately pleiotropic and highly pleiotropic genes were combined, 2,275 genes were pleiotropic in both trait domains (hypergeometric test P-value << 0.011), and a total of 256 genes overlapped between all datasets (Fig. 1c). When only highly pleiotropic genes were considered, 26 genes emerged as overlapping between all data types (Fig. 1d). Importantly, as many as 2,851 genes were pleiotropic according to both HPO and MGD data but were not pleiotropic in GWAS, and a total of 9,897 genes were uniquely pleiotropic according to either HPO or MGD data. This amounts to 81.3% of all genes that were pleiotropic in either human or mouse based on Mendelian trait data. At the same time, only 54.0% (2,669) of genes pleiotropic in pan-UKB or FinnGen data were not pleiotropic in the Mendelian trait domain, with the vast majority of these genes (2,501) having no phenotype associations in either HPO or MGD. Of note, the percentages of shared highly pleiotropic genes was even lower than those observed for all pleiotropic genes (only 13.0% of genes highly pleiotropic in HPO or MGD were pleiotropic in GWAS).

One factor that may substantially affect the observed overlap between sets of pleiotropic genes is the ascertainment bias resulting from different methods of phenotyping used (such an effect is expected to be most pronounced when comparing HPO and MGD datasets, taking into account that phenotyping in controlled experimental conditions could be more thorough and unbiased compared to phenotyping based on self-reporting of patients with Mendelian disease). We hypothesized that, if the ascertainment bias is pronounced, a greater proportion of known essential genes should have a detailed phenotypic description in humans compared to mice (i.e. mild phenotypes should be underrepresented in HPO data). To test this hypothesis, we utilized genes from the full spectrum of intolerance to loss of function (FUSIL) (Cacheiro et al. 2020). In FUSIL, genes are divided into five categories—cellular lethal (CL), developmental lethal (DL), subviable (SV), viable with (VP) or without (VN) phenotype, according to the results of CRISPR screens in human cell lines and deep phenotyping of mice by the International Mouse Phenotype Consortium (IMPC). As expected, a much higher fraction of essential genes (from CL, DL, and SV categories) had associated Mendelian phenotypes; however, this enrichment was significant for both HPO and MGD. Thus, 51.4% of DL genes were associated with at least one term in HPO data (compared to only 17.7% among all human genes) (Extended Data Figure S2). For MGD, a staggering 94.8% of DL genes had phenotype annotations (compared to the baseline value of 44.3%). While these figures emphasize a much richer phenotypic description of murine genes, they do not indicate a substantial bias toward functionally important genes in HPO.

Taken together, these numbers suggest that, despite low correlation between the degrees of pleiotropy between complex and Mendelian traits, there is a significant overlap between the sets of pleiotropic genes between trait domains. It is also important to note that human orthologs of as many as 1,233 genes with pleiotropic effects in MGD, but not HPO data, were pleiotropic according to GWAS data. These findings may suggest that many of the genes that demonstrate detectable effects on the phenotype in mice have a subtle phenotypic manifestation in humans that is detectable only by genome-wide association analysis.

Pleiotropic Genes Are Characterized by Common Functional Properties

Having split the dataset into different groups of genes depending on their estimated degree of pleiotropy, we next sought to identify shared features of pleiotropic genes between species and trait domains. To this end, we have annotated our dataset with various gene-level properties, such as the summary statistics of gene expression profile (human gene data from GTEx were used), numbers of protein–protein interactions (according to the STRING database), and measures of haploinsufficiency and triplosensitivity (Collins et al. 2022).

Analysis of the expression pattern of genes in different pleiotropy groups showed that more pleiotropic genes tend to have broader expression profiles (as evidenced by the number of tissues with expression at the level of at least five transcripts per million [TPM]; Wilcoxon–Mann–Whitney test P-value < 0.001 when comparing nonpleiotropic and highly pleiotropic genes in all datasets; Fig. 2a) and greater mean/median expression across tissues (Wilcoxon–Mann–Whitney test P-value < 0.001, Extended Data Figure S3). Importantly, the number of tissues with detectable expression was much greater for highly pleiotropic genes in the Mendelian trait data (30 tissues for HPO and 27 for MGD), compared to those identified from GWAS data (9), and was closer to that of known essential genes (48 and 36 for CL and DL categories in FUSIL, respectively; Fig. 2a). It is also important to note that genes with no phenotypic associations showed the greatest tissue specificity across all data sources, except FinnGen (e.g. a gene with no phenotype was expressed in 1, 0, or 2 tissues for HPO, MGD, and pan-UKB, respectively). These numbers suggest that, despite differences in the depth of phenotyping between the datasets, few functionally important genes lack phenotype information across different types of data.

Besides the breadth of expression across tissues, we also observed a significant increase in the maximum expression level of genes with greater degrees of pleiotropy ($P < 0.001$ in all cases, Extended Data Figure S3). The difference between nonpleiotropic and highly pleiotropic genes was the most pronounced for genes at pleiotropic pan-UKB loci (median of highest expression values—22.2, 23.8, 30.0, and 35.6 TPM for NP, ST, MP, and HP groups, respectively). Similarly to the results of the breadth of expression profile, pleiotropic Mendelian trait genes had higher expression levels compared to genes at pleiotropic GWAS loci (55.3 TPM for HPO data and 54.7 for MGD vs. 36.3 for pan-UKB and 26.6 for FinnGen). Of note, murine genes with no phenotype had by far the lowest level of expression (median of the highest expression value = 3.9 TPM), emphasizing a much greater depth of phenotyping in mice.

In addition to broader expression, we also demonstrated that the genes with greater degree of pleiotropy encode proteins with a larger number of protein–protein interactions. This tendency, however, was more apparent in the Mendelian trait domain (median number of interactions for highly pleiotropic proteins = 127 and 101 compared to 42.5 and 45.0 for nonpleiotropic human and murine genes, respectively, $P < 0.001$) (Fig. 2b). For GWAS-based data, no statistically significant differences in the number of PPIs were detected (however, the difference could be seen with a different approach to GWAS data preprocessing, see Note S1, section 2). Again, the number of interactions for highly pleiotropic GWAS genes was much smaller compared to highly pleiotropic Mendelian trait genes, in line with our observations made during expression pattern analysis. For all datasets but HPO, however, highly pleiotropic genes had lower numbers of PPIs compared to essential genes, which was especially high for the CL category (Fig. 2b).

As a next step, we then tested whether the highly pleiotropic genes have broader sets of functions compared to nonpleiotropic ones. To this end, we have annotated each gene with

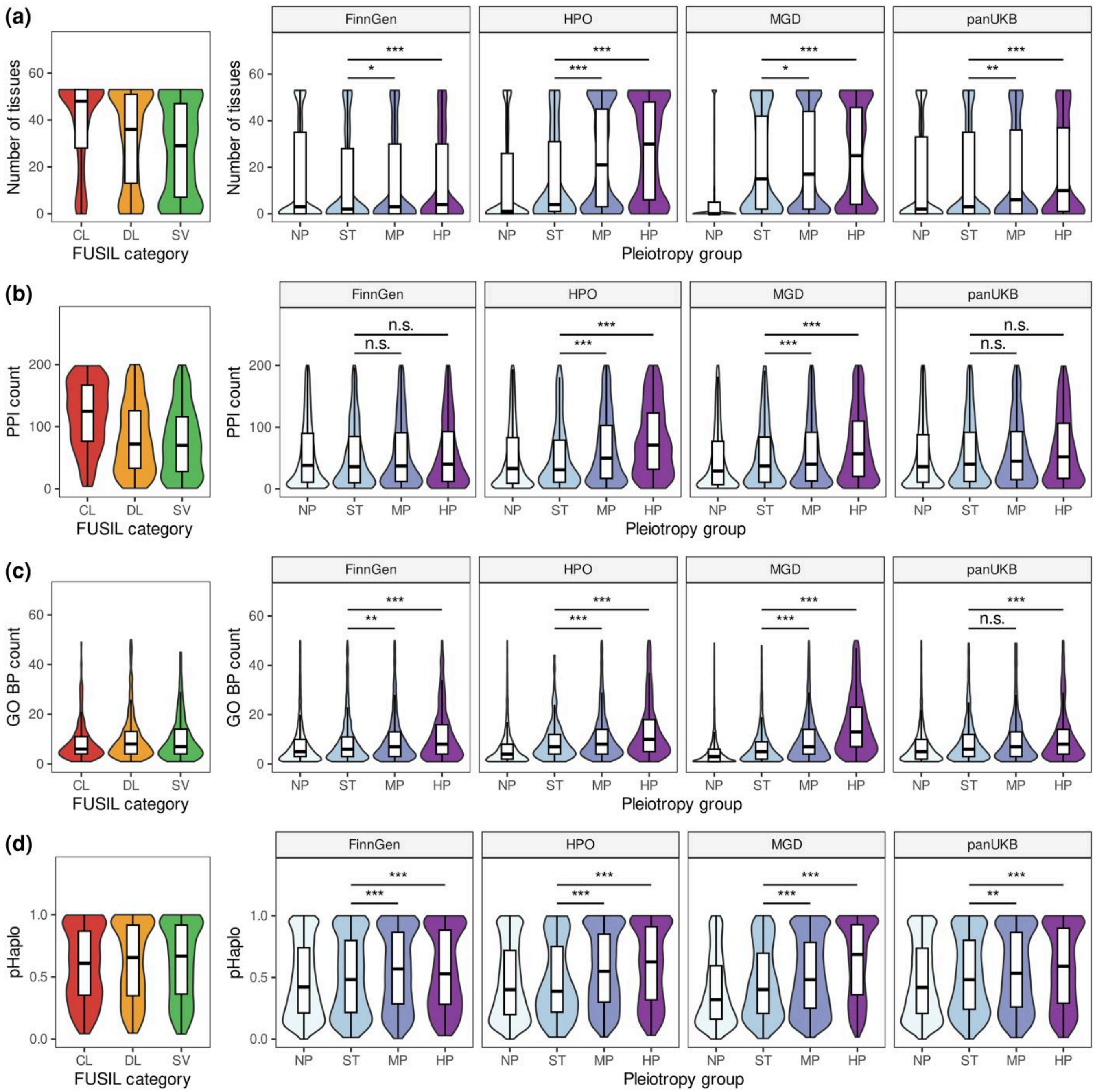

Fig. 2. Pleiotropic effects correlate with functional significance of genes in both trait domains. Shown are violin and box plots for a) number of tissues expressing a gene at the level of at least five TPM according to the Genotype Tissue Expression (GTEx) v8 data; b) number of protein–protein interactions according to the STRING database; c) number of biological process (BP) terms from gene ontology (GO); and d) pHaplo scores from Collins et al. 2022. The following abbreviations are used for pleiotropy groups: NP, genes with no associated phenotypes; ST, genes linked to a single trait or term; MP, moderately pleiotropic genes; HP, highly pleiotropic genes. Values for three categories of essential genes (CL, cellular lethal; DL, developmental lethal; SV, subviable) from the FUSIL study (Cacheiro et al. 2020) are shown on each panel. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, n.s., no significant differences in Wilcoxon–Mann–Whitney test with Benjamini–Hochberg FDR adjustment.

the number of corresponding gene ontology (GO) terms and compared these numbers between gene groups. Indeed, highly pleiotropic genes had higher numbers of associated terms in all three GO branches (biological process [BP], cellular component [CC], and molecular function [MF]) (Fig. 2c, Extended Data Figure S4). Thus, a median highly pleiotropic Mendelian trait gene was involved in 11 and 16 BPs (for human and mouse, respectively, according to HPO and MGD data), compared to only 7 and 5 for nonpleiotropic ones. In the complex trait domain, the difference was also significant (9 vs. 6 for FinnGen, 8 vs. 6 for pan-UKB; $P << 0.001$). In good concordance with the results obtained for other gene-level features, genes with no phenotype information universally had the lowest number of associated GO terms across all data sources (median number from 3 to 5 terms). More curiously, highly pleiotropic genes from all datasets were annotated with a larger number of BP and MF terms compared to essential genes (including CL, DL, and SV genes) (Fig. 2c).

Given the above observations, we also tested whether the highly pleiotropic genes have higher degree of dosage sensitivity. To perform such a test, we employed the pHaplo and pTriplo metrics calculated in a recent study of copy number variation in the human genome (Collins et al. 2022). Indeed, we observed that significantly more pleiotropic genes have higher pHaplo and pTriplo values, approaching and even surpassing those of known essential genes (Fig. 2d, Extended Data Figure S5). The difference in pHaplo and pTriplo was also weaker for complex trait data (HP vs. ST, pHaplo: 0.59 vs. 0.48 in pan-UKB data, 0.62 vs. 0.39 in HPO; pTriplo: 0.50 vs. 0.43 in pan-UKB; 0.54 vs. 0.40 in HPO).

Besides the analysis of various measurable properties of the genes, we also used gene set enrichment analysis to get insights into the function of the genes in different pleiotropy groups. Due to the inherent uncertainty of causal gene identification from GWAS data, we primarily focused this analysis on genes identified using Mendelian trait data (HPO and MGD). For highly pleiotropic genes in both humans and mice, analysis of canonical pathways (CPs) and hallmark gene sets demonstrated enrichment for pathways related to cell proliferation and DNA repair (e.g. KEGG's collection of pathways in cancer, UV response and TGF**β** signaling hallmarks) (Extended Data Figures S6 and S7). Similar results were obtained when analyzing GO term enrichment, with development, morphogenesis, and cell proliferation among the most enriched BP terms (Extended Data Figure S8). In line with these observations, MF term analysis identified terms related to chromatin and DNA binding, transcription factor binding, and protein complex formation to be the most overrepresented (Extended Data Figure S9). Again, these patterns were shared for highly pleiotropic human and murine genes, and both of these sets were highly enriched for CC terms associated with chromatin (Extended Data Figure S10).

Of note, we also found that highly pleiotropic GWAS genes tend to show enrichment for highly similar gene sets. Thus, 44 enriched MSigDB CPs were shared between highly pleiotropic genes from all four data sources, and only a minor fraction of pathways (66 out of 351, or 18.8%) was uniquely enriched among highly pleiotropic complex trait genes (Fig. 3a). The degree of overlap was even more significant for enriched GO terms, with 733 BP terms shared between all data sources, and only a handful of terms (50 out of 1,915, or 2.6%) were uniquely enriched among pleiotropic GWAS genes. In other GO branches, the proportion of terms specific to complex trait data was also small but slightly higher (17.5% for CC and 13.7% for MF, corresponding to 10 and 22 terms in each case, respectively) (Extended Data Figure S11). These results show that the overlap in the BPs and MFs of highly pleiotropic genes is greater compared to the overlap of the gene sets themselves (Fig. 1d).

We next questioned if the observed overlap in the sets of enriched gene sets is driven by genes that are pleiotropic in both trait domains. To answer this question, we performed gene set enrichment analysis on highly pleiotropic GWAS genes that had no phenotype associations in either HPO or MGD (and vice versa). Importantly, such an analysis identified a large number of terms and pathways enriched among highly pleiotropic genes specific to HPO and MGD data (Fig. 3b). For genes uniquely pleiotropic in GWAS, only a handful of hits were discovered (26 CPs, 5 GO BP terms, and 2 GO MF terms)—for GWAS data (Fig. 3b, Extended Data Figure S11). Of note, no processes or pathways shared between trait domains were identified when using the domain-specific genes for enrichment analysis. This result implies that, while there is a very substantial overlap in the set of BPs controlled by highly pleiotropic genes from Mendelian and complex trait domains, this overlap is likely driven by genes that demonstrate pleiotropic effects in both domains. Domain-specific pleiotropic genes, on the other hand, tend to be involved in a specific set of processes. For the complex trait domain, this set of processes predominantly included various pathways associated with xenobiotic and steroid metabolism.

Having demonstrated the similarity in the sets of processes controlled by pleiotropic genes, we next went on to identify the BPs, CCs, and MFs that are associated with the greatest proportion of highly pleiotropic genes across all datasets. To this end, we focused our analysis on a limited set of GO terms and pathways with large numbers of genes (see Methods for details) and then prioritized these gene sets by the minimum proportion of pleiotropic genes across datasets. Five out of 56 preselected CPs showed significant enrichment of highly pleiotropic genes in all three datasets. In good concordance with results described above, these pathways corresponded to cell proliferation-controlling signaling cascades (e.g. growth factor receptor signaling cascade), genes involved in hemostasis and immune system function (Fig. 3c). In line with these findings, the highest proportion of highly pleiotropic genes was found for GO BP terms related to development, immune cell differentiation and activity, hemostasis, and cell adhesion (Extended Data Figure S12a). Within the CC ontology, pleiotropic genes from both trait domains were involved in transcriptional regulation complexes and chromatin, as well as in cell membrane regions and vesicles (Extended Data Figure S12b). In good concordance with these observations, MF terms linked to transcription factor binding, protein kinase binding or activity, and DNA binding had the highest fraction of highly pleiotropic genes across all datasets (Extended Data Figure S12c). Of note, the proportions of genes with highly pleiotropic effects in mice were universally higher; thus, for the majority of the aforementioned terms, more than half of the corresponding genes had strong pleiotropic effects according to MGD data.

Shared Patterns of Selection at Pleiotropic Loci Across Datasets

Results of the functional property analysis suggested that genes at pleiotropic loci (at least, within the monogenic trait domain) correspond to broadly expressed genes with a large number of protein–protein interactions. This result predicts that these genes should experience greater pressure of negative natural selection constraining genetic variation within them. At the same time, it is not clear whether these patterns could be expected at pleiotropic GWAS loci. Hence, we next investigated the patterns of natural selection at pleiotropic genes and loci using several widely used measures of both negative (phastCons scores, loss-of-function [LoF] variation, and missense variation statistics from gnomAD) and positive (population-specific integrated haplotype scores [iHS] and the density of recent coalescence events [DRC150]) selection.

When phastCons scores were used as a measure of negative selection, we found no significant difference in the level of evolutionary conservation between pleiotropy groups (Extended Data Figures S13 and S14a,d). However, significant differences between gene groups were detected when using gnomAD-derived gene-level constraint measures (LoF

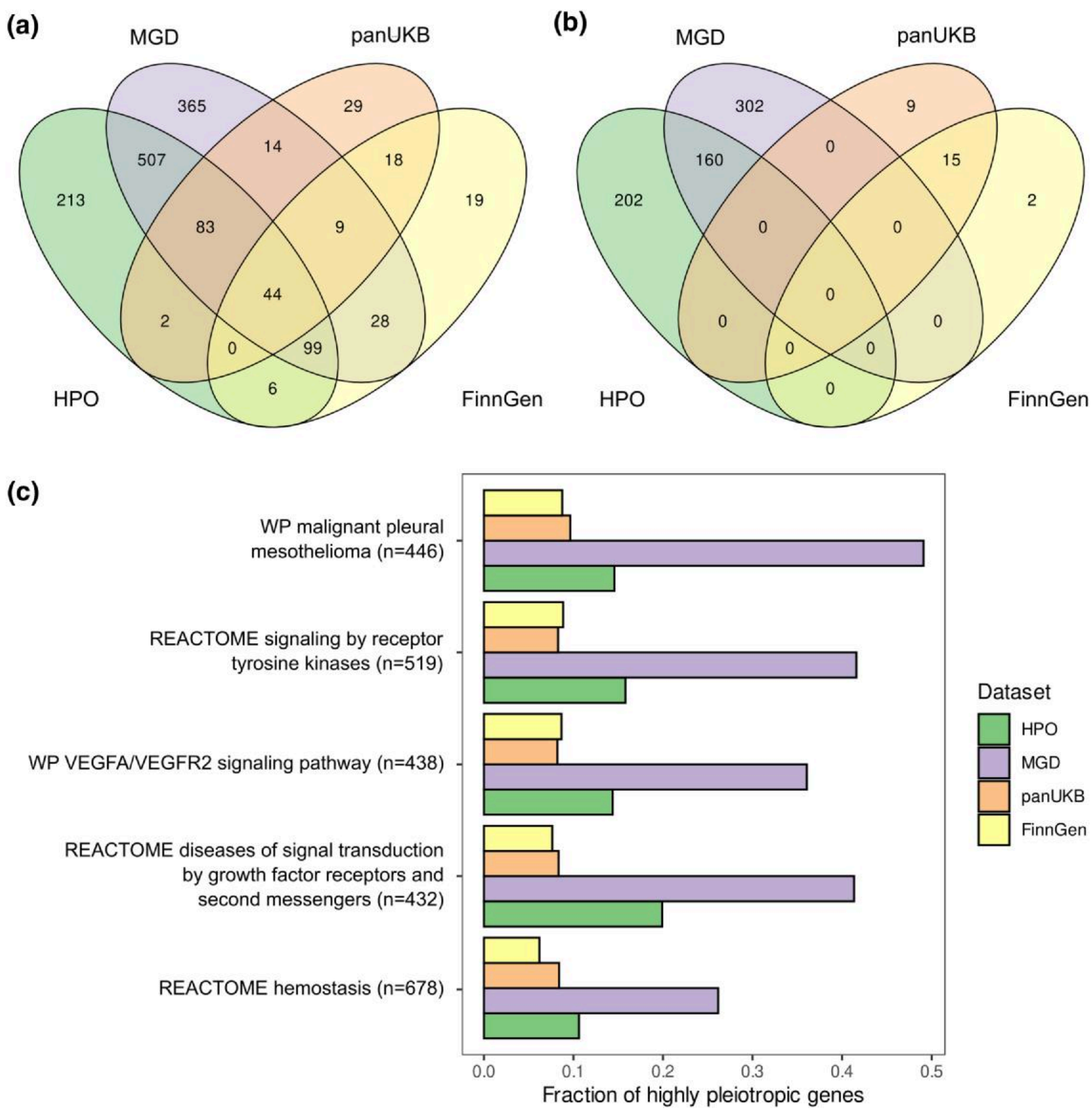

Fig. 3. Pleiotropic genes in complex and Mendelian traits are involved in many common processes. a, b) Venn diagrams showing the overlap between enriched GO terms identified using the highly pleiotropic gene sets from indicated data sources. All a) or dataset-specific b) highly pleiotropic genes were used for analysis. c) A bar plot showing the fraction of highly pleiotropic genes for the six large canonical pathways that are significantly enriched with highly pleiotropic genes from all three datasets.

observed-to-expected upper fraction [LOEUF] and probability of LoF intolerance [pLI]). Unlike phastCons scores, which are constructed using whole-genome alignments of multiple distantly related vertebrate species, LOEUF and pLI values reflect the LoF tolerance of human genes based on genetic variation within populations. With these measures being used, highly pleiotropic genes showed universally greater constraint (Fig. 4b, Extended Data Figure S15a). Thus, the median LOEUF value for ST genes was 0.91, 0.89, 0.77, and 0.77 for HPO, MGD, pan-UKB, and FinnGen data, respectively. For highly pleiotropic genes, the value dropped precipitously to 0.61, 0.56, 0.61, and 0.69, respectively. In concordance with some of the earlier observations, this level of constraint was comparable or even more substantial than that of the essential genes (0.61, 0.54, and 0.55 for CL, DL, and SV categories, respectively, from FUSIL).

A similarly pronounced difference was observed when using other population-based metrics of selective constraint, such as the selection coefficient against heterozygous protein-truncating variants such as s_{het} (proposed by Cassa et al. 2017, estimates taken from Zeng et al. 2024, Extended Data Figure S15a) or the Z-score of the observed-to-expected missense variant ratio (Extended Data Figure S15b). When genes were split into categories according to their pLI value, highly pleiotropic genes showed a universal and highly significant enrichment with LoF-intolerant ($pLI > 0.9$) genes across all data sources (30%, 31%, 27%, and 24% of highly pleiotropic genes were LoF intolerant compared to 14%, 13%, 20%, and 20% of nonpleiotropic ST genes for HPO, MGD, pan-UKB, and FinnGen data, respectively; $P << 0.001$ in chi-squared test in all cases). Of note, the proportions of LoF-intolerant genes were similarly high for all FUSIL categories (Extended Data Figure S15c). Taken together, these results indicate that pleiotropic genes are under greater negative selection pressure in both complex and Mendelian trait domains.

We next went on to test the enrichment of pleiotropic genes and loci with recent positive selection signals. As mentioned earlier, two different measures of recent positive selection, which capture different timescales of selection, were utilized. As recent positive selection is not expected to operate

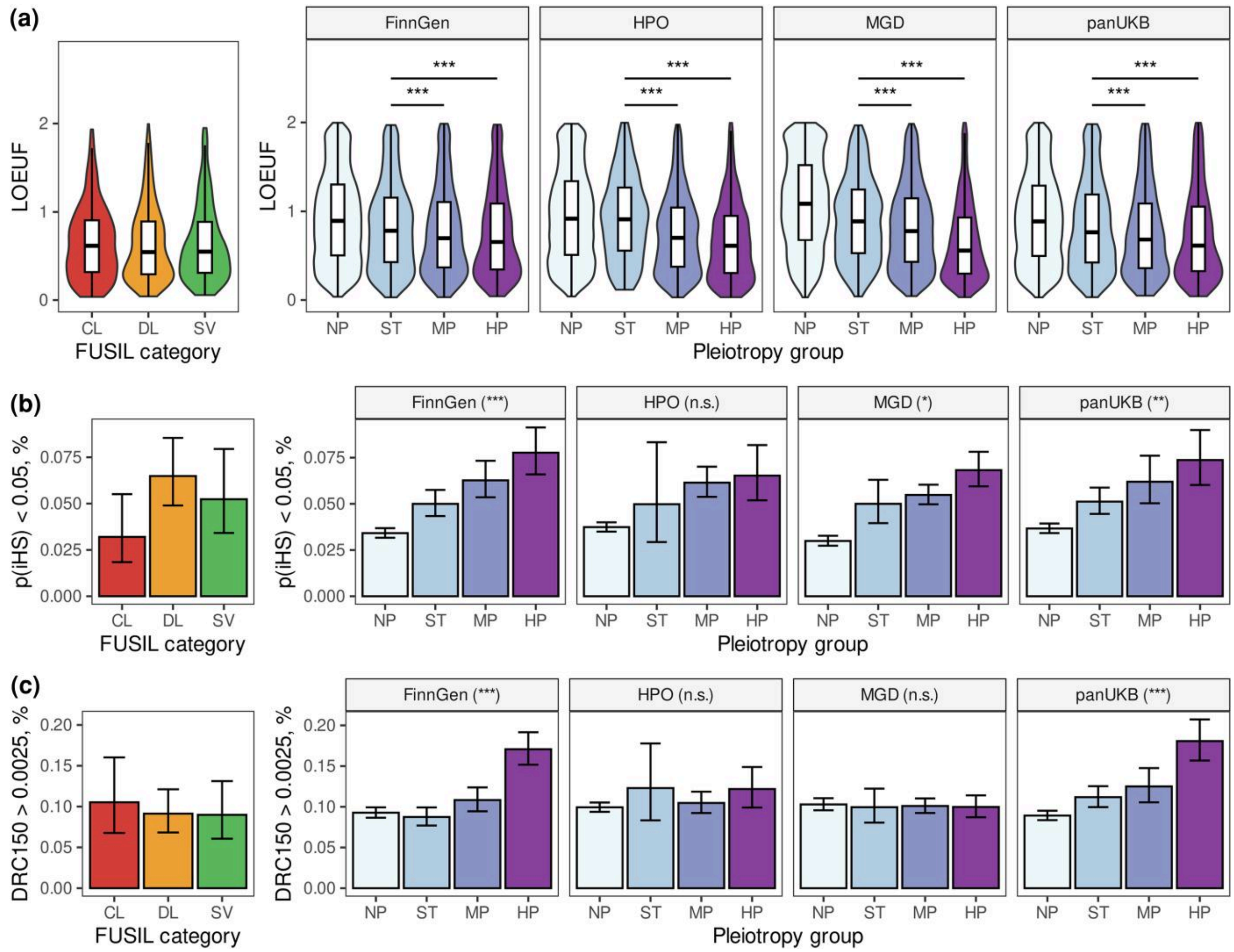

Fig. 4. Pleiotropic genes are under greater evolutionary constraint despite bearing signals of recent adaptation. a) Violin-and-box plots of the loss-of-function observed-to-expected upper fraction (LOEUF) values for genes in indicated pleiotropy groups. b, c) Bar plots showing the proportion of genes with statistically significant (DFR-adjusted $P < 0.01$, see Methods) gene-wise maximum iHS value b) or density of recent coalescent events (DRC150) score >0.0025 c) in each pleiotropy group. The following abbreviations are used for pleiotropy groups: NP, genes with no associated phenotypes; ST, genes linked to a single trait or term; MP, moderately pleiotropic genes; HP, highly pleiotropic genes. Values for three categories of essential genes (CL, cellular lethal; DL, developmental lethal; SV, subviable) from the FUSIL study (Cacheiro et al. 2020) are shown on each panel. $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, n.s., no significant differences in Wilcoxon–Mann–Whitney test (a) or chi-squared test (b, c) with Benjamini–Hochberg FDR adjustment. On (b, c) highly pleiotropic genes were compared to nonpleiotropic ones. Error bars on (b, c) correspond to the 95% confidence interval for the binomial proportion.

uniformly on all genes in a group or across the entire gene sequence, we focused on comparing the proportions of genes for which the maximum value of proximal single-nucleotide polymorphisms (SNPs) either exceeded a predefined threshold or significantly deviated from the gene length-adjusted expectation under the normal distribution (see Materials and Methods for details of the calculation procedure). As can be seen in Figs 4b, c, we observed a general tendency toward the enrichment of pleiotropic genes with signals of recent positive selection, though the strength of this trend varied between the datasets and metrics. For instance, 6.8% of highly pleiotropic murine gene orthologs had a significant maximum iHS value of neighboring SNPs (see Methods for details of the calculation), compared to 3.0% (for the NP group) or 5.0% (for the ST group) of nonpleiotropic ones (Fig. 4c). Similarly, as many as 7.3% and 7.9% of highly pleiotropic genes in the pan-UKB and FinnGen GWAS data bore a recent positive selection signal according to iHS values, compared to only 5.1% and 5.0% of genes with a single associated trait and 3.7% of genes with no associated phenotypes. A similar tendency was observed with using a hard cutoff of iHS > 4 (Extended Data Figure S16a) and was accompanied by a subtle shift in the median value (Extended Data Figure S16b). When the DRC150 value was used as a measure of positive selection, only highly pleiotropic complex trait genes showed a significant enrichment with large DRC150 values (18.0% and 19.4% compared to 11.1% and 9.3% for nonpleiotropic genes, according to pan-UKB and FinnGen data, Fig. 4c).

Given a general trend of highly pleiotropic genes to be longer, especially in GWAS data (Extended Data Figure S17a), we explicitly tested the impact of gene length on the enrichment of recent positive selection signals at pleiotropic GWAS loci. Two methods were employed for this purpose: (i) evaluating the proportion of loci (irrespective of their gene content) bearing a significant selection signal and (ii) comparing the observed proportion of genes with positive selection signals to random sets of genes with comparable length. Enrichment of recent positive selection at highly pleiotropic genes/loci was confirmed with both of these methods for the DRC150, but not iHS (Extended Data Figure S14, Extended Data Figure S17b–e). Of note, the effect of pleiotropy on DRC150 in both pan-UKB and FinnGen data was by far more pronounced when a maximum value of DRC150 for all variants in a 100,000 bp locus was evaluated (36.4% vs. 10.3% [pan-UKB], Extended Data Figure S14c,f).

Given the above observations, we next questioned if highly pleiotropic loci are enriched with genes that bear a combination of negative and positive selection signatures. To answer this question, we selected a subset of human genes with strong selective constraint against LoF variation (LOEUF < 0.6) and a nearby signal of recent positive selection (gene-level FDR-adjusted P-value[|iHS|] < 0.05, see Methods). A total of 406 (1.6%) genes met these conditions. The proportion of such genes among nonpleiotropic ones was similar (1.9% in HPO and MGD, 1.9 and 1.8% in GWAS data). Among the highly pleiotropic genes, however, this proportion was significantly higher for all datasets except HPO (4.4%, 3.4%, and 2.9% for MGD, pan-UKB, and FinnGen, respectively) (Extended Data Figure S18a). Similar results were also observed when using a combination of low LOEUF and high DRC150 values (Extended Data Figure S18b).

Finally, we asked if the observed overlap between signals of negative and positive selection can be explained by the general correlation between gene-level constraint (LOEUF) and iHS/DRC150 that may be driven by background selection. To answer this question, we compared the LOEUF values for genes with and without a neighboring positive selection signal (according to iHS or DRC150). This analysis showed that, while genes with a significant maximum iHS value indeed tend to be under greater selective constraint, an inverse tendency is observed for DRC150 (Extended Data Figure S19). Hence, we conclude that the observed enrichment of overlapping signals of negative and positive selection cannot be explained solely by the effects of background selection.

Taken together, these findings demonstrate a universal propensity of pleiotropic genes to be under increased selective constraint and validate earlier suggestions regarding a possible enrichment of pleiotropic loci with positive selection signals. Moreover, our results reveal a unique enrichment of pleiotropic genes with those bearing combined signals of positive and negative selection.

Discussion

The phenomenon of pleiotropy has attracted the attention of researchers since the early years of genetics. It is now well established that pleiotropy is abundant in natural systems, and the distribution of the degree of pleiotropy usually has a characteristic L shape (reviewed in Zhang 2023). All four data sources used in our analysis also showed a characteristic left skew of the distribution; however, we observed substantial differences in the median degree of pleiotropy as well as the estimated modularity of the underlying genotype-to-phenotype networks. Overall, gene-level pleiotropy estimated using HPO and MGD data was much higher compared to the genome-wide association data. Earlier studies based on mouse phenotype data also showed a higher median degree of pleiotropy (Wang et al. 2010; Muñoz-Fuentes et al. 2022). It is thus necessary to consider the possible sources of these differences.

The most natural explanation of the results is the different nature of traits and mutations used to construct the data. For HPO, the main source of gene annotation is the phenotype observed in patients with Mendelian disease caused by a pathogenic variant in a given gene, usually being a LoF allele (Köhler et al. 2021). Similarly, mouse phenotype data come from single gene or high-throughput knockout assays. Thus, genotype–phenotype relationships recorded in both HPO and MGD have two characteristic features. First, these relationships come from high-impact genetic variants which are commonly assumed to have higher degrees of pleiotropy compared to regulatory variation (Carroll 2005). Second, the phenotypic description in both HPO and MGD predominantly focuses on phenotypic abnormalities observed in affected individuals. In both of these regards, HPO and MGD substantially differ from complex trait data. For example, genetic variation used to conduct genome-wide association analysis is mostly regulatory or silent, though pleiotropic variants are known to be enriched for missense variants (Watanabe et al. 2019; Shikov et al. 2020). Another distinctive feature of the phenome-wide association data is that these datasets typically include a more diverse set of traits (including continuous ones). However, it is important to note that we did not find any marked differences in the patterns of pleiotropy observed in pan-UKB and FinnGen data, even though different types of traits were used for analysis from these two sources (continuous traits and complex diseases, respectively). Besides, murine quantitative trait loci (QTL) data from MGD also show a distribution of the degree of pleiotropy similar to those observed for different types of human complex traits (see Note S1, section 4).

While the trait and mutation type could explain the observed differences in the degree of pleiotropy, it is hard to rule out the possibility of a technical confounding. For example, upper-level MP ontology terms are not entirely independent, with possible implications for pleiotropy analysis (Muñoz-Fuentes et al. 2022). In our analysis, we chose upper-level MP terms over other ontology term-based metrics as they maximized the number of nonpleiotropic genes and minimized the median degree of pleiotropy (see Note S1, section 1), It is also noteworthy that the genotype-to-phenotype networks constructed using upper-level ontology terms show a significantly higher degree of modularity compared to random networks of similar size (Extended Data Figure S1), suggesting that much of the vertical pleiotropy (pleiotropic effects resulting from cause–effect relationships or shared mechanistic basis of the two traits) has been efficiently removed from the data. However, we believe that the phenotype definition still plays a role in inflating the degree of pleiotropy in certain cases.

For genome-wide association data, on the other hand, the estimate of the gene-level degree of pleiotropy could be impacted by the causal gene selection method based on summary statistics (see Note S1, section 2), as well as method for independent phenotype definition (Note S1, section 3). In our earlier work, we have used phenotypic correlation-based clustering to reduce the impact of vertical pleiotropy, as we have previously shown a positive impact of this procedure on the phenome-wide analysis results (Shikov et al. 2020). While this procedure efficiently reduces the observed degree of pleiotropy, it is still possible that many complex traits with relatively low observed phenotypic correlation could still fall under the same upper-level phenotype ontology term. Hence, we could expect that mapping complex traits to phenotype ontology terms could further reduce the observed statistical pleiotropy in GWAS data.

In our study, we showed that genes bearing highly pleiotropic variation are more broadly expressed (Fig. 2a, Extended Data Figure S3), have more protein–protein interactions (Fig. 2b) and a richer set of MFs (Extended Data Figure S4), participate in more BPs (Fig. 2c), are more dosage sensitive (Fig. 2d, Extended Data Figure S5), and, finally, are under greater selective constraint (Fig. 4a, Extended Data Figure S15). In many of these aspects, highly pleiotropic genes tend to be similar to the ones known to be essential for cellular viability and embryonic development. These results are in line with earlier observations made in model organisms (He and Zhang 2006) and in human complex traits (Shikov et al. 2020). A more intriguing finding, however, is the striking similarity in the properties of highly pleiotropic genes in complex traits and Mendelian traits, which is observed despite the aforementioned discrepancies in the degree of pleiotropy. Besides the consistent effect of pleiotropy on the various gene-level features across datasets, it is also important to emphasize that a highly pleiotropic gene according to GWAS data was more similar to a moderately pleiotropic or nonpleiotropic gene according to HPO or MGD data. These observations are in line with a lower median degree of pleiotropy observed in the genome-wide association data. Nevertheless, we can conclude that pleiotropic effects in both complex and Mendelian trait domains are associated with the same genic features and are hence likely to be guided by similar or overlapping mechanisms.

It is also worth noting that the differences in all gene-level metrics tested were usually much more pronounced in MGD than in HPO data, and certain trends (e.g. the enrichment of highly pleiotropic genes with recent positive selection signals) were significant in MGD, but not HPO. These observations suggest that the phenotype of high-impact genetic variation is much better described in mice than in humans. Indeed, large-scale phenotyping efforts such as the IMPC (Meehan et al. 2017; Cacheiro et al. 2019) have greatly enhanced the depth of mouse phenotype description. In humans, on the other hand, description of the phenotype is performed only by examining the patients with rare disease. Hence, many genes may lack phenotype annotation due to various factors, such as survival of the patient until the time of molecular diagnostics or the ability to clearly demonstrate the causal effect of a genetic variant observed in a patient. However, despite a potentially substantial role of the aforementioned phenotype ascertainment bias, two pieces of indirect evidence suggest that it does not interfere with the analysis of functional and evolutionary gene properties. First, genes with no associated Mendelian phenotypes are much more similar to nonpleiotropic than to pleiotropic genes and are typically showing lower levels of biological relevance and evolutionary constraint compared to single trait-associated genes (Figs 2 and 4). Second, while there is a notable overrepresentation of well-phenotyped genes among the ones known to be essential for development and cell viability (Extended Data Figure S2), this trend is observed for both human and murine genes.

Besides shared properties of pleiotropic genes, our analysis identified a set of processes that are enriched with pleiotropic genes across species and trait domains. This set of pathways includes ones involved in cell proliferation, signaling, immune system function, and extracellular matrix organization (Fig. 3, Extended Data Figure S11). These findings are in good concordance with our earlier analysis of pleiotropic gene functions in complex traits (Shikov et al. 2020). Besides, the present analysis indicates that the previously observed enrichment of highly pleiotropic loci with liver-specific genes is the unique feature of the complex trait domain. Thus, xenobiotic metabolism and related pathways were enriched among GWAS-specific highly pleiotropic genes, but not among highly pleiotropic Mendelian trait genes in either species.

Shared functions of highly pleiotropic genes across trait domains (Fig. 3, Extended Data Figure S10) suggest that a highly similar set of pathways is responsible for pleiotropic effects of genetic variation; however, the magnitude of these effects would be dependent on the type of genetic variant (e.g. null [LoF] mutation, change in the amino acid sequence of a protein, or a variant in the regulatory element) (Fig. 5). Thus, high-impact genetic variants in crucial pathway genes would have pronounced pleiotropic effects and manifest in Mendelian phenotypes (rare congenital disorders) and would be subject to enhanced purifying selection. Lower-impact variants in the same genes would also have pleiotropic effects, albeit with a lower magnitude, driving multiple complex traits rather than Mendelian phenotypes. Such variants will be enriched in adaptive alleles, in accordance with both our (Fig. 4) and earlier findings (e.g. Frachon et al. 2017; Sakaue et al 2021). In pathways with a restricted set of functions, on the other hand, all types of mutations would likely lack pleiotropic effects and would hence be both less adaptive and subject to lower negative selective pressure.

In our previous work, we observed an interesting tendency of pleiotropic genetic variants in GWAS to have higher allele frequency compared to their nonpleiotropic counterparts (Shikov et al. 2020). At the time, we have proposed three explanations for this observation: (i) removal of rare variants at pleiotropic loci due to strong purifying selection; (ii) greater power to detect genome-wide associations for common genetic variants; and (iii) spurious pleiotropy of common variants that tag different rare causal variants at the same locus. In addition to the three hypotheses mentioned above, one might also suggest that higher frequency of pleiotropic variants may be driven by an enrichment of loci with recent positive selection signals, an assumption that has been supported by evidence in some recent studies (Frachon et al. 2017; Sakaue et al. 2021). The results included in the present study indicate that pleiotropy correlates both with increased strength of purifying selection and higher frequency of adaptive alleles, favoring both the first and fourth hypotheses. Moreover, our results show that both negative selection strength and the enrichment with positive selection signals are highest for highly pleiotropic genes. This is especially true for genes with pleiotropic effects in mice, which showed both the strongest negative selection against LoF variants and the most pronounced enrichment with adaptive alleles (according to the iHS value). These observations contradict earlier findings and predictions that pleiotropy should increase the rate of adaptation only for moderate degrees of pleiotropy (Wang et al. 2010; Frachon et al. 2017). It is important to note, however, that the enrichment with more recent positive selection (detected using the DRC150 statistic) was observed only in GWAS data, where the cutoff for selection of highly pleiotropic genes was lower. Still, the proportion of genes with high DRC150 value increased when higher degree of pleiotropy was demanded (e.g. 32 out of 168 genes had a positive selection signal when requiring at least eight trait clusters). Hence, we believe that our results emphasize that a high degree of pleiotropy is, at the very least, compatible with active positive selection.

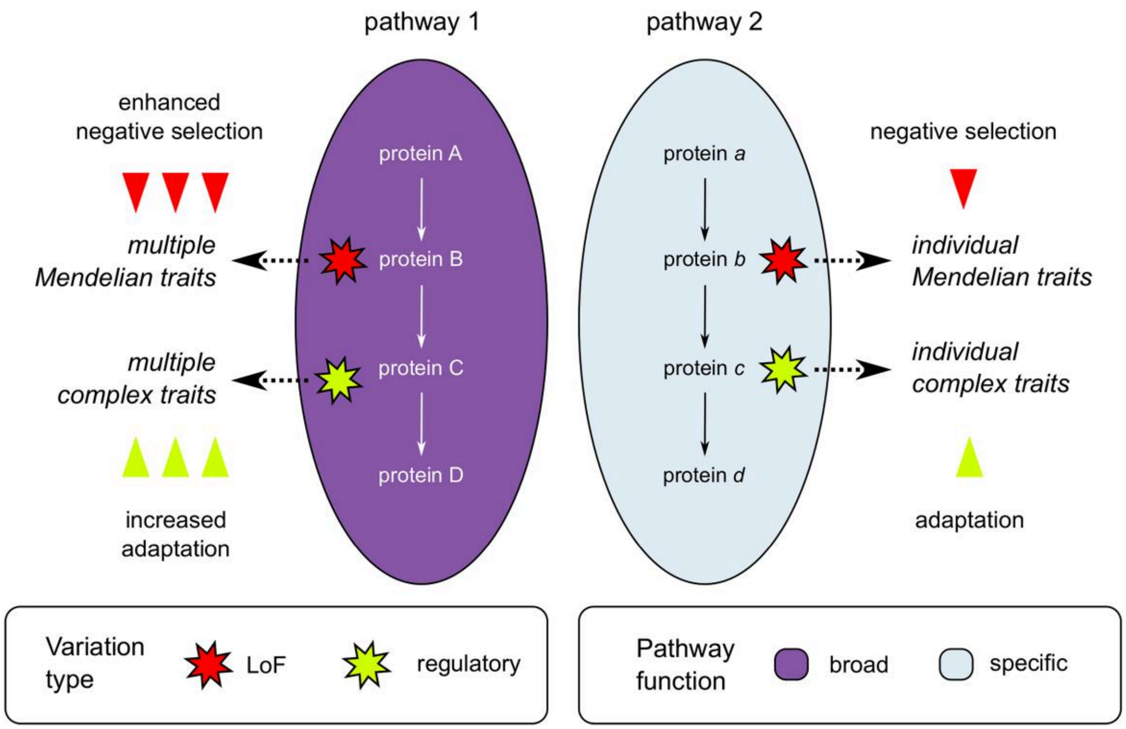

Fig. 5. A model for pleiotropic effects in different trait domains. Ellipses represent pathways colored by their functional relevance in an organism (i.e. broad or tissue/cell type specific); arrows connecting proteins within pathways represent the direction of signal transduction or position of genes in metabolic reaction chains. Polygonal stars represent genetic variants that are colored according to their type (LoF or regulatory). See text for further explanation of the model.

To sum up, pleiotropy is not only a basic feature of genotype-to-phenotype networks, but an inevitable consequence of the organismal complexity. Still, while pleiotropy imposes additional constraint on genetic variation, it appears to provide additional opportunities for adaptation of complex organisms to changing environments.

Materials and Methods

Genotype-to-Phenotype Data Collection

To compile a dataset consisting of genotype-to-phenotype relationships in the Mendelian trait domain, we obtained information about orthologous gene pairs between mice and humans from the MGD (the HMD_HumanPhenotype file was used; https://www.informatics.jax.org/downloads/reports/HMD_HumanPhenotype.rpt). For each ortholog, we included associations with phenotype terms from the MP ontology for mice and the HPO for humans. For human genes, gene annotations were downloaded directly from the HPO website (https://hpo.jax.org/data/annotations, accessed on 2024 July 01). For MP data, phenotypic information (excluding conditional mutations) was retrieved from the MGI_GenePheno file (https://www.informatics.jax.org/downloads/reports/MGI_GenePheno.rpt).

As a source of genotype-to-phenotype relationship information in complex trait domains, we utilized publicly available summary statistics of genome-wide association analysis by the pan-UKB study (Karczewski et al. 2025) and FinnGen project release 12 (Kurki et al. 2023). The pan-UKB manifest file with heritability estimates was preprocessed to select traits with $P(h^2 = 0) < 0.01$ according to the standard normal distribution (Z-score > 2.32). For traits with different encoding and preprocessing methods used in the original pan-UKB study, a single entry with the maximum heritability estimate was selected. Only continuous traits and biomarkers were included into the analysis. Summary statistics files were downloaded for the resulting set of 480 traits (Table S1). As the heritability estimates were not available for FinnGen, summary statistics for all traits were used, excluding quantitative endpoints (BMI, height, and weight).

Definition of Pleiotropy in Phenotype Ontology Data

To calculate the degree of pleiotropy of a gene, all phenotype ontology terms associated with a particular gene were converted to upper-level terms using the ontobio package (for HPO data, terms were converted to MP to simplify further cross-species comparison, official upper-level term mapping provided by MGI was used: https://github.com/mgijax/mammalian-phenotype-ontology/tree/main/mappings). Due to a strong bias in the number of genes annotated with the MP:0010768 ("mortality/aging") term, this term was removed from further analysis. Then, the number of remaining upper-level MP terms linked to each gene was used as a measure of its degree of pleiotropy. To identify highly pleiotropic genes, we split all genes into quartiles according to their degree of pleiotropy and then considered all genes in the fourth quartile as highly pleiotropic ones. Besides the aforementioned approach, clustering of all phenotype ontology terms was attempted as a measure of pleiotropy, with results of the analysis being similar to those obtained with upper-level terms (see Note S1, section 1 for details).

Definition of Pleiotropy in Genome-Wide Association Data

Summary statistics of pan-UKB GWAS were used to identify lead SNPs using the clumping procedure available in PLINK v. 1.9 (Purcell et al. 2007) and the 1000 Genomes phase 3 genotype data (Auton et al. 2015). Following clumping, the closest gene was retrieved for each of the identified lead SNPs using BEDTools (Quinlan and Hall 2010) and the GENCODE v19 human gene annotation file (Frankish et al. 2021). An alternative approach based on all genes within a 100,000 bp interval around each lead SNP was also attempted, with similar results (see Note S1, section 2).

A matrix of gene–trait associations was constructed using the inferred causal gene information. Results of hierarchical trait clustering using the phenotypic correlation estimates provided by the authors of the pan-UKB study were used to merge associations by cluster (for each cluster, a gene was considered to be associated with the cluster if it was identified as the closest gene for a lead SNP in GWAS results for at least one trait). The number of associated trait clusters was used as the resulting measure of the degree of pleiotropy, similar to our earlier work (Shikov et al. 2020). Similar to the analysis of phenotype ontology terms, the top 25% of all genes with the highest degree of pleiotropy were considered as highly pleiotropic, and all other genes with a degree of pleiotropy greater than 1 were considered as moderately pleiotropic.

For FinnGen data, the UCSC liftOver toolkit was used to convert genomic coordinates to the hg19 genome assembly. After that, genes closest to lead SNPs were identified using the same procedure as described above for the pan-UKB data, and the gene–trait association matrix was constructed. As an alternative way of causal gene identification, results of fine mapping by SuSIE (Zou et al. 2022) provided by the authors (https://docs.finngen.fi/finngen-data-specifics/green-library-data-aggregate-data/core-analysis-results-files/finemapping-results-format) were employed, and a gene corresponding to a variant with the most severe impact in each credible set was used for further processing. After construction of the gene–trait association matrix, traits were clustered using a distance metric based on reconstituted phenotypic correlation ($d = 1 - r_p$). The r_p values were obtained from variant effect sizes and standard errors using the PhenoSpD package for R (Zheng et al. 2018).

Analysis of Genotype-to-Phenotype Network Modularity

To perform an analysis of modularity of the bipartite gene–trait network, preprocessed gene-to-trait associations from each data source were converted into an incidence matrix. The matrix was then used to construct a bipartite graph using the igraph package (Csárdi and Nepusz 2006), followed by greedy community detection and calculation of network modularity score.

To obtain the distribution of expected modularity scores, each network was randomly shuffled, preserving the total number of edges for each node. This procedure was repeated multiple times, resulting in 100 reshuffled networks for each data source. Modularity was then computed for each such network, and the distribution of the resulting scores was compared with the true modularity of the real network.

Comparison of Functional Gene Properties

Summary statistics of gene expression profile (median expression across tissues, number of tissues with expression at five or more TPM, maximum expression across tissues) were calculated using the Genotype Tissue Expression (GTEx) v8 data (Lonsdale et al. 2013). The number of protein–protein interactions recorded for a gene was retrieved using the BioGRID database (Oughtred, et al. 2021). Additionally, gene-level probabilities of dosage sensitivity (pHaplo and pTriplo) were taken from a paper by Collins et al. (2022). Finally, the number of GO (Ashburner et al. 2000) terms associated with each gene was calculated using the org.Hs.eg.db and org.Mm.eg.db packages for R.

Besides the analysis of numeric features, a gene set enrichment analysis using the GO term enrichment was performed for highly pleiotropic genes using the clusterProfiler package for R (Yu et al. 2012). In addition to GO terms, hallmark gene sets and CPs from the Molecular Signatures Database (MSigDB) (Liberzon et al. 2011, 2015) were used in gene set enrichment analysis.

For the analysis of gene sets with the most consistent enrichment across datasets, we merged the results of enrichment analysis for the three data sources into a single dataset (per each gene set group: hallmarks, CPs, BP, CC, and MF branches of GO) and then ordered the gene sets by the minimum percentage of highly pleiotropic genes across the data sources. For GO terms, only large (>500 genes) terms were considered. For a largely redundant CP collection, a set of mostly nonoverlapping pathways was constructed as follows: (1) all pathways were ordered by the number of genes; (2) the largest pathway was selected, and all genes corresponding to this pathway were removed; and(3) the procedure was repeated with the remaining genes until at least one pathway with >25 genes remained. This procedure resulted in a set of 56 (out of initial 2,982) mostly nonredundant pathways covering more than 87% (11,681/13,343) of all human genes with known function (Table S2).

To provide a broader context in functional gene property analysis, results of the FUSIL study were employed (Cacheiro et al. 2020). Distributions of the functional metrics were evaluated for three groups of essential genes—CL, DL, and SV. Proportions of pleiotropic genes in different FUSIL categories were also evaluated to assess the strength of phenotype ascertainment bias.

Signals of Positive and Negative Selection

For the analysis of purifying selection for pleiotropic and nonpleiotropic loci, gene-level constraint measures for human genes were retrieved from the Genome Aggregation Database (gnomAD) v.2.1. (Karczewski et al. 2020). LOEUF, Z-score for observed-to-expected missense variant ratio, and the pLI (Lek et al. 2016) were used for gene annotation and comparison. Additionally, we used phastCons (Siepel et al. 2005; 10.1101/gr.3715005) conservation scores computed using 100 vertebrate species obtained from the UCSC Table Browser (https://genome.ucsc.edu/cgi-bin/hgTables, accessed on 2024 July 10). The phastCons scores in bigWig format were converted to bedGraph format using the bigWigToBedGraph tool (https://github.com/ENCODE-DCC/kentUtils, accessed on 2024 July 20) and then averaged over the gene or locus interval.

To assess the overlap with recent positive selection signals, two main metrics were employed that capture different time periods: (i) the integrated haplotype score (iHS) and (ii) density of recent (over the last 150 generations) coalescence events (DRC150). The iHS values for the British (GBR) subpopulation from 1000 Genomes project (Auton et al. 2015) were taken from a recent analysis by Johnson and Voight (2018). DRC150 were taken from the original study by Palamara et al. (2018). For both metrics, gene-wise iHS and DRC150 value were calculated by intersecting the gene interval (for HPO/MP data) or the 100,000 bp interval around the lead SNP with SNP coordinates (annotated with respective iHS and DRC150 values), and the maximum absolute value of each metric was retrieved for a gene or locus in question.

To determine the statistical significance of the gene-wise maximum iHS value, a *P*-value was computed assuming that the iHS value for each of the *n* SNPs in a gene is normally distributed, as follows:

$$p(x|n) = 1 - \prod_{i=1}^{n} N(x),$$

where x is the observed maximum iHS value and $N(x)$ is the normal cumulative distribution function.

Supplementary material

Supplementary material is available at *Molecular Biology and Evolution* online.

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Data Availability

All data and code pertinent to the analysis presented in this work are available through GitHub: https://github.com/ibre-research/pleiotropy_analysis/.

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